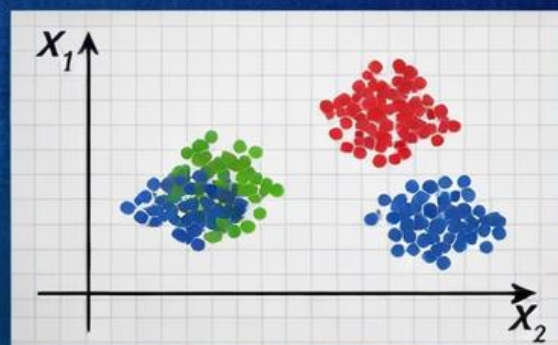
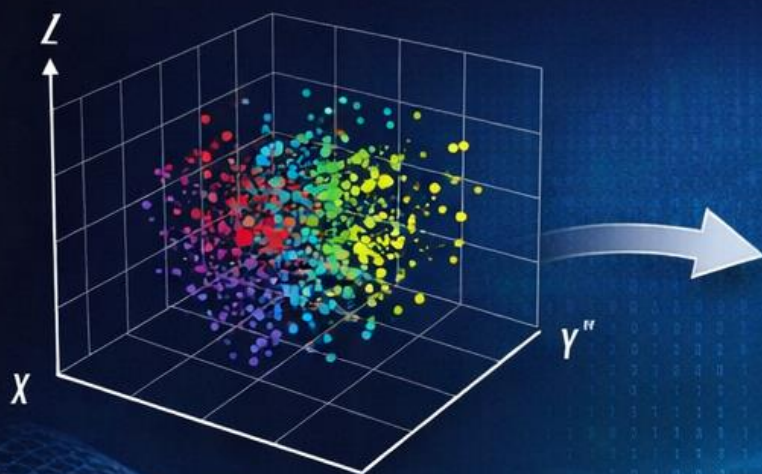




Dimensionality Reduction Techniques

Design, Theory, and Applications

A Unified and Scenario-Driven Framework

Aderval S. Luna & Nathalia Alvim de Souza

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First edition

Published by the authors (self-published)

Place of publication: Rio de Janeiro, Brazil

Year of publication: 2026

ISBN: not assigned — this work is distributed free of charge in digital form

DOI: <https://doi.org/10.5281/zenodo.22221611>

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The datasets analysed in this book are distributed by the UCI Machine Learning Repository and are cited individually in the reference list. The R code and the data required to reproduce every result are available at <https://github.com/Laboratorio-de-Tecnologia-Computacional/Dimensionality-Reduction-Techniques-An-Innovative-Approach>

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Abstract

High-dimensional data are now routine across the physical, biological and engineering sciences, and dimensionality reduction has become a standard step in extracting interpretable structure from them. The difficulty is that no single technique is universally preferable: each method optimises a different objective, and a projection that is excellent for visualising clusters may be poor at preserving global geometry or at supporting reconstruction.

This book addresses that problem with a unified, scenario-driven framework for comparing dimensionality reduction techniques. Fourteen convex and nonconvex methods — among them Principal Component Analysis, kernel and L1-norm PCA, classical and nonmetric multidimensional scaling, Isomap, diffusion maps, independent component analysis, Hessian locally linear embedding, three graph-layout algorithms and t-SNE — are presented with their mathematical formulation, assumptions, computational cost and known limitations.

Their behaviour is then assessed with thirteen normalised quality measures spanning local neighbourhood fidelity, global geometry, reconstruction error and class separability. These measures are combined into four composite indices and weighted according to four analytical objectives — clustering, visualisation, reconstruction and machine-learning pipeline integration — so that the choice of method follows from the purpose of the analysis rather than from convention. A robustness analysis based on $\pm 10\%$ perturbation of the weights quantifies how stable each ranking is.

The framework is demonstrated on seven benchmark datasets from the UCI Machine Learning Repository, covering sample sizes from 88 to 13,611 observations. All results are reproducible: the R code, the datasets and the random seeds are available in the repository cited in the Data Availability section.

Keywords

Dimensionality reduction; manifold learning; quality assessment; multi-criteria benchmarking; principal component analysis; t-SNE; multidimensional scaling; chemometrics; reproducible research; R

Authors

Aderval S. Luna is a Full Professor at the Rio de Janeiro State University (UERJ) and a permanent member of its Graduate Program in Chemical Engineering. He holds a B.Sc. in Industrial Chemistry from UERJ, an MBA in Data Science from the University of São Paulo (São Carlos-SP), and both an M.Sc. and a D.Sc. in Chemistry from the Pontifical Catholic University of Rio de Janeiro. His doctoral research included a collaborative period at the National Research Council of Canada, Ottawa, Canada. He also completed postdoctoral training in chemometrics with the Research Group of Qualimetry and Chemometrics at the Universitat Rovira i Virgili, Tarragona, Catalunya, Spain.

Nathalia Alvim de Souza is a D.Sc. student in the Graduate Program in Chemical Engineering at Rio de Janeiro State University. She holds a degree in Chemical Engineering from the Faculty of Technology (UERJ–Resende) and an M.Sc. in Chemical Engineering from UERJ.

Declaration of generative AI and AI-assisted technologies in the book preparation process

This book was developed with the assistance of generative IA as a supporting tool for text generation, idea organization, and coding tasks. The use of this technology aimed to enhance productivity and facilitate the structuring of complex technical content throughout the manuscript. All content generated with the aid of IA was carefully reviewed, edited, and validated by the authors to ensure scientific accuracy, coherence, and alignment with the work's objectives. The authors take full responsibility for the final content presented in this book, including any remaining errors, omissions, or inaccuracies.

In addition, all R code included in this book was generated with the assistance of generative IA and subsequently verified, tested, and, when necessary, adapted by the authors to ensure correctness, reproducibility, and applicability within the proposed methodological framework.

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List of Abbreviations

AUC	Area Under the Curve
AUC R(NX)	Area under the rank-based neighbourhood-preservation curve
cMDS	Classical Multidimensional Scaling
CRAN	Comprehensive R Archive Network
DR	Dimensionality Reduction
DrL	Distributed Recursive (Graph) Layout
DRR	Dimensionality Reduction via Regression
EDXRF	Energy-Dispersive X-Ray Fluorescence
FastICA	Fast Independent Component Analysis
FNA	Fine Needle Aspirate
GCE	Gabriel Classification Error
HLLE	Hessian Locally Linear Embedding
ICA	Independent Component Analysis
IR	Infrared
JADE	Joint Approximate Diagonalization of Eigenmatrices
kNN, k-NN	k-Nearest Neighbours
kPCA	Kernel Principal Component Analysis
KK	Kamada–Kawai
KL	Kullback–Leibler
L1-PCA	L1-Norm Principal Component Analysis
LLE	Locally Linear Embedding
MDS	Multidimensional Scaling
ML	Machine Learning
NIR	Near-Infrared
nMDS	Nonmetric Multidimensional Scaling
PCA	Principal Component Analysis
RBF	Radial Basis Function
RMSE	Root-Mean-Square Error
SVD	Singular Value Decomposition
t-SNE	t-Distributed Stochastic Neighbour Embedding
UCI	University of California, Irvine (Machine Learning Repository)
UMAP	Uniform Manifold Approximation and Projection

List of Symbols

n	number of samples (observations) in the dataset
p	number of original variables (features)
k	neighbourhood size used by the rank-based quality measures
X	data matrix in the original high-dimensional space
Y	embedding produced by the dimensionality reduction method
d	dimensionality of the embedding ($d = 2$ throughout this book)
Q_{local}	local quality: fraction of nearest neighbours preserved
Q_{global}	global quality: preservation of large-scale arrangement
R(NX)	neighbourhood preservation corrected for random agreement
τ	Kendall's rank correlation coefficient
ρ	Spearman rank correlation coefficient
CCC	cophenetic correlation coefficient
Index_{Local}	composite index of local structure preservation
Index_{Global}	composite index of global structure preservation
Index_{Geom}	composite index of geometric fidelity
Index_{Class}	composite index of class separability
Final_Score	objective-weighted combination of the four composite indices

Preface

The rapid expansion of high-dimensional data across scientific and engineering disciplines has intensified the need for robust methods to extract meaningful structure from complex datasets. Dimensionality reduction (DR) has become a central component of modern data analysis, providing tools to simplify data representation while preserving relevant information for interpretation, visualization, and modeling.

This book presents a systematic and unified treatment of dimensionality reduction techniques, combining theoretical foundations, algorithmic perspectives, and practical evaluation strategies. In contrast to approaches that treat methods in isolation, the present work adopts a comparative, application-oriented framework, enabling consistent analysis across a broad range of techniques.

A key premise underlying this work is that the performance of dimensionality reduction methods is inherently context-dependent. No single technique is universally optimal, and method selection should be guided by the specific objectives of the analysis. To address this, the book introduces a scenario-driven evaluation framework supported by a multi-criteria benchmarking strategy. This framework integrates complementary performance measures that capture preservation of local structure, global geometry, reconstruction fidelity, and class separability.

The text covers a wide spectrum of methods, including classical linear techniques such as Principal Component Analysis and Independent Component Analysis, as well as nonlinear approaches including Kernel PCA, multidimensional scaling, manifold learning methods, and probabilistic and graph-based techniques. Each method is presented with its mathematical formulation, underlying assumptions, computational aspects, and known limitations.

Particular emphasis is placed on evaluating the results of dimensionality reduction. The proposed benchmarking framework incorporates multiple quantitative metrics, enabling a more comprehensive and reproducible assessment than single-measure approaches. In addition, robustness analysis based on controlled perturbations is introduced to evaluate the stability of method rankings under varying conditions.

The methodologies discussed in this book are applicable to a wide range of domains, including chemometrics, spectroscopy, machine learning, and process analytical technology. Practical considerations, including implementation aspects and integration into analytical workflows, are addressed to support real-world applications.

This book is intended for researchers, graduate students, and practitioners seeking a rigorous and structured understanding of dimensionality reduction methods. It may also serve as a reference for advanced courses in multivariate analysis, machine learning, and data-driven modeling.

The authors hope that this work contributes to a more systematic and transparent use of dimensionality reduction techniques and supports informed methodological choices in high-dimensional data analysis.

Acknowledgements

A. S. Luna thanks CNPq (304581/2022-4) and FAPERJ (E-26/204.174/2024) for the scholarship, and UERJ for the PAPD Senior Program.

This study has been financially supported by the Human Resources Training Program of the Brazilian National Agency for Petroleum, Natural Gas and Biofuels – PRH-ANP, under the management of the São Paulo Research Foundation (FAPESP), Brazil. Process N. 2025/03791-6.

1. INTRODUCTION

Real-world data are often high-dimensional. Dimensionality reduction (DR) is the process of transforming data into a lower-dimensional, more meaningful representation. Ideally, this reduced representation should match the intrinsic dimensionality of the original data, i.e., the smallest number of parameters required to explain the observed data properties [1–2]. DR is critical in many areas because it mitigates the curse of dimensionality and other undesirable properties of high-dimensional spaces [3]. As a result, DR facilitates, among other things, the classification, visualization, and compression of high-dimensional data. The primary difficulty encountered when applying DR techniques is that each approach is designed to preserve certain aspects of the original method and, therefore, may be appropriate for one task but inappropriate for another. Most methods have tunable parameters and follow different assumptions. The quality of the obtained outcome may strongly depend on their tuning, which adds additional complexity [4].

Addressing this gap, the novelty of this work lies in establishing the first comprehensive, scenario-driven, and quantitatively unified framework for rigorously comparing both convex and nonconvex dimensionality reduction techniques. While previous studies typically evaluated DR algorithms using isolated metrics or limited experimental conditions, this research integrates thirteen normalized quality measures—spanning local, global, geometric, and class-preservation perspectives—into a systematic multicriteria decision model that reflects fundamental analytical objectives such as clustering, visualization, reconstruction, and ML pipeline performance. Additionally, the introduction of a robustness analysis via controlled weight perturbations addresses a long-standing limitation in DR benchmarking by quantifying the stability of algorithm rankings and revealing their sensitivity to metric variability, an area that has rarely been

explored in the literature. This unified approach transforms the DR assessment process from a collection of ad hoc practices into a transparent, reproducible, and generalizable evaluation paradigm, paving the way for more consistent methodological choices across disciplines that rely on high-dimensional data. By closing the gap between the theoretical properties of DR methods and their practical suitability, the framework establishes a new benchmark for manifold learning algorithms. It serves as a valuable reference model for future developments in the field. Therefore, this study proposes an innovative and comprehensive framework for systematically selecting the most suitable dimensionality reduction technique by rigorously evaluating a broad set of performance metrics reported in the literature.

2. THEORY

2.1 Criteria for Selecting Dimensionality Reduction Algorithms

The selection of an appropriate dimensionality reduction (DR) technique should be guided by four fundamental considerations:

- (i) the primary objective of the analysis,
- (ii) the evaluation metrics used to judge performance,
- (iii) the intrinsic nature of the given dataset, and
- (iv) the expected risk of geometric distortions or artifacts.

As an initial step, clearly defining the type of structure to be preserved—a local neighborhood, a global manifold geometry, smoothness and continuity, or a primarily aesthetic two-dimensional separation—is essential for visualization. Table 1 synthesizes these objectives, establishes the associated priorities, and identifies the most informative quantitative metrics for each class of analytical goals.

Table 1 - Objectives and Recommended Evaluation Metrics for DR Techniques

Objective	Priority	Key metrics
Preserve local neighborhood (for clustering, maps, fine topology)	Locality	Q_{local} , Trustworthiness, Precision/Recall, k-NN Accuracy
Preserve global manifold structure	Globality	Q_{global} , AUC_R(NX), Cophenetic, Kendall Tau
Stable and continuous geometric representation	Smoothness and low distortion	RMSE, GCE, C measure, Discontinuity
Beautiful 2D visualization with group separation	Layout	t-SNE, Kamada–Kawai, Fruchterman–Reingold,

2.2 Classification of Dimensionality Reduction Techniques

DR techniques can be split into convex and nonconvex techniques. Convex techniques optimize an objective function that does not contain any local optima, whereas nonconvex techniques optimize an objective function that does contain local optima. Convex DR techniques are split into techniques that perform eigendecomposition on a full matrix and those that perform eigendecomposition on a sparse matrix. Table 2 shows the taxonomy of DR techniques. Detailed descriptions of all the techniques mentioned below can be found in the literature, and a short summary will also be presented [5].

Table 2 – Taxonomy of dimensionality reduction techniques.

Dimensionality reduction	Convex	Full spectra	Euclidean distance	PCA, Classical Scaling
			Geodesic distance	Isomap
			Kernel-based	Kernel PCA, MVU
			Diffusion distance	Diffusion maps
		Sparse spectral	Reconstruction weights	LLE
			Neighborhood graph Laplacian	Laplacian Eigenmaps
			Local tangent space	Hessian LLE, LTSA
	Nonconvex		Weighted Euclidean distances	Sammon mapping
			Alignment of local linear models	LLC, Manifold charting
			Neural network	Autoencoder

*Adapted from van der Maaten *et al.* [5].

2.2.1 Convex DR Techniques

2.2.1.1 DR Techniques Based on a Full Matrix

2.2.1.1.1 Principal Component Analysis

1. Introduction

Principal Component Analysis (PCA) is a linear dimensionality reduction technique that transforms a dataset into a new coordinate system where the directions (principal components) maximize the variance of the data.

Figure 1 provides an intuitive representation of PCA as a projection-based dimensionality reduction technique. The original dataset, depicted as a cloud of points in a high-dimensional space, is transformed into a lower-dimensional representation by projecting it onto a new coordinate system defined by the principal components. These components are oriented along directions of maximum variance, ensuring that the most informative structure of the data is retained. As illustrated, the reduced representation preserves the relative arrangement of the samples while significantly simplifying the data structure. This visual abstraction highlights PCA's dual role in both data compression and structure preservation, making it particularly valuable for exploratory analysis and visualization of complex datasets.

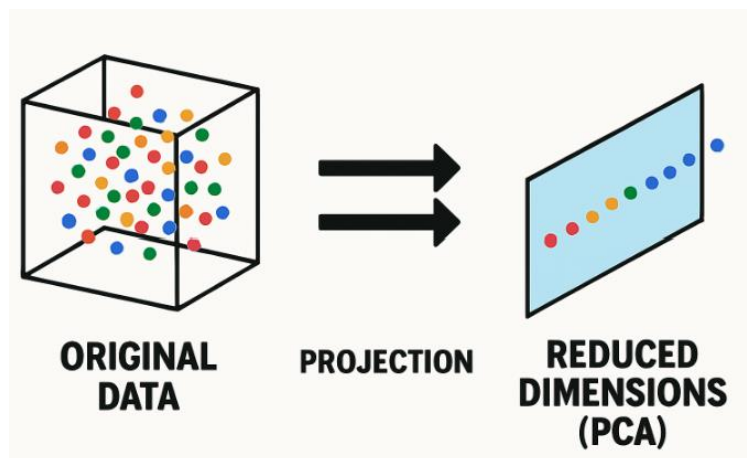


Figure 1 – Schematic illustration of Principal Component Analysis (PCA).

High-dimensional data (left) are projected onto a lower-dimensional subspace (right) defined by principal components that maximize the variance of the data. The projection transforms the original variables into a new set of orthogonal axes, enabling dimensionality reduction while preserving the most relevant information.

Table 3 summarizes the fundamental properties that define Principal Component Analysis (PCA) as a dimensionality reduction technique. PCA is inherently linear, projecting the data onto linear combinations of the original variables, while ensuring that the resulting components are orthogonal and therefore uncorrelated. The method is variance-based, seeking directions that maximize explained variance in the dataset, making it highly effective at capturing dominant data structures. Additionally, PCA is approximately invertible, allowing reconstruction of the original data from the reduced representation with controlled information loss. Finally, PCA operates as a global method, meaning that the transformation is derived from the entire dataset rather than local neighborhoods. Together, these properties explain its robustness, interpretability, and widespread use across scientific and engineering applications.

Table 3 – Principal Component Analysis Key Properties

Property	Description
Linear	Projects data onto linear combinations of features
Orthogonal	Principal components are uncorrelated
Variance-based	Maximizes variance
Invertible	Allows reconstruction (approximate)
Global	Considers the entire dataset

In machine learning, PCA is used for:

- Feature extraction
- Noise reduction
- Data compression
- Visualization of high-dimensional data

2. Mathematical Formulation

1. Data Centering

Given a dataset:

$$\mathbf{X} \in \mathbb{R}^{n \times p} \quad (1)$$

where:

n = number of samples

p = number of features

The centered data matrix is:

$$\mathbf{X}_c = \mathbf{X} - \mathbf{1}\boldsymbol{\mu}^T \quad (2)$$

where $\boldsymbol{\mu}$ is the mean vector.

3. Covariance Matrix

$$\boldsymbol{\Sigma} = \frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c \quad (3)$$

4. Eigenvalue Decomposition

$$\boldsymbol{\Sigma} = \lambda_i \mathbf{v}_i \quad (4)$$

where:

λ_i : eigenvalues (variance explained)

$\mathbf{v}_i$: eigenvectors (principal directions)

5. Projection (Dimensionality Reduction)

Select the top k eigenvectors:

$$\mathbf{W}_k = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k] \quad (5)$$

Project data:

$$\mathbf{Z} = \mathbf{X}_c \mathbf{W}_k \quad (6)$$

6. Reconstruction (Invertibility Property)

PCA is invertible (approximate):

$$\hat{\mathbf{X}} = \mathbf{Z} \mathbf{W}_k^T + \boldsymbol{\mu} \quad (7)$$

7. Variance Maximization Interpretation

PCA solves:

$$\max_{\mathbf{w}} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to } \|\mathbf{w}\| = 1 \quad (8)$$

This ensures each component captures the maximum possible variance.

8. Machine Learning Interpretation

PCA finds a low-dimensional subspace that preserves maximum information (variance)

It is:

- Unsupervised
- Orthogonal transformation
- Global linear method

9. Geometric Interpretation

- Data is projected onto the directions of maximum spread
- Principal components are orthogonal axes
- The first component captures the largest variance, the second the next, etc.

Therefore, PCA provides an optimal linear embedding with respect to variance preservation, offering a mathematically elegant and computationally efficient foundation for dimensionality reduction in machine learning [6].

10. Limitations

- Captures only linear relationships

- Sensitive to scaling
- Does not consider class labels
- May fail in nonlinear manifolds

11. Applications

Principal Component Analysis (PCA) has been widely applied across scientific and engineering domains, particularly for high-dimensional data that require simplification, interpretation, and noise reduction. Its linear nature, computational efficiency, and interpretability make it one of the most widely used dimensionality reduction techniques.

11.1 Chemometrics and Spectroscopy

- Extraction of dominant variance patterns in Raman, NIR, IR, and UV–Vis spectra
- Detection of chemical variability, including batch effects and compositional differences
- Spectral pre-processing and noise reduction (e.g., baseline correction support)
- Data compression for high-dimensional spectral datasets while preserving relevant chemical information
- Exploratory analysis for outlier detection and sample classification

11.2 Image Processing

- Face recognition using Eigenfaces (classical PCA-based approach)
- Image compression by reducing redundancy in pixel space
- Noise filtering through reconstruction using principal components
- Feature extraction for downstream tasks such as classification and clustering

11.3 Bioinformatics

- Analysis of gene expression data (e.g., microarray and RNA-seq)
- Identification of dominant biological patterns and sample grouping

- Reduction of dimensionality in high-throughput omics datasets
- Visualization of population structure and genetic variability

11.4 Fault Detection and Process Monitoring

- Multivariate statistical process control (MSPC) in chemical engineering
- Detection of abnormal operating conditions via score and residual analysis (Hotelling's (T^2) and SPE/Q statistics)
- Monitoring of industrial processes with correlated variables
- Early fault detection in continuous and batch processes

11.5 Machine Learning Pipelines

- Preprocessing step to reduce dimensionality before classification or regression (e.g., SVM, k-NN, PLS-DA)
- Mitigation of multicollinearity in high-dimensional datasets
- Acceleration of training by reducing feature space
- Visualization of complex datasets in 2D or 3D latent spaces

12. Final Remarks

Principal Component Analysis (PCA) remains one of the most fundamental and enduring techniques in data analysis, serving as a cornerstone for dimensionality reduction, exploratory data analysis, and preprocessing in machine learning pipelines. Its strength lies in its solid mathematical foundation, computational efficiency, and interpretability, which together enable the transformation of complex, high-dimensional datasets into a reduced set of orthogonal latent variables that capture the dominant sources of variance.

From a practical perspective, PCA provides a powerful framework for uncovering hidden structures in data, identifying patterns, detecting outliers, and compressing information with minimal loss. In fields such as chemometrics, spectroscopy,

bioinformatics, and process monitoring, PCA has become indispensable for interpreting multivariate data and supporting decision-making processes. Its integration into multivariate statistical process control (MSPC) and its role in feature extraction for machine learning models further highlight its versatility.

However, despite its widespread applicability, PCA is not without limitations. As a linear method, it may fail to capture complex nonlinear relationships inherent in many real-world datasets. Additionally, because PCA maximizes variance rather than class separability, the resulting components are not necessarily optimal for classification tasks. The method is also sensitive to data scaling and outliers, which can distort the principal components if not properly addressed through preprocessing.

In modern data science and machine learning contexts, PCA should therefore be viewed not as a universal solution, but as a robust baseline and an essential component within a broader methodological toolkit. It often serves as a first step in data analysis, providing insights that guide the selection of more advanced techniques such as Kernel PCA, manifold learning methods, or deep learning-based representations when nonlinear structures are present.

Importantly, the interpretability of PCA—through loadings, scores, and explained variance—remains a key advantage, particularly in scientific domains where understanding the relationship between variables is as important as predictive performance. In this sense, PCA bridges the gap between statistical rigor and practical applicability, making it especially valuable in disciplines that demand both transparency and efficiency.

In conclusion, PCA remains a highly relevant and reliable method in the era of big data and artificial intelligence. Its role as a foundational technique ensures that it will

remain a critical tool for both researchers and practitioners, whether used independently or as part of more sophisticated analytical frameworks.

2.2.1.1.2 Kernel Principal Component Analysis

Kernel Principal Component Analysis (Kernel PCA) is a nonlinear extension of classical Principal Component Analysis (PCA), designed to capture complex structures in data that cannot be adequately represented in a linear subspace. By implicitly mapping the original data into a high-dimensional feature space through a kernel function, Kernel PCA enables linear operations in that space while preserving nonlinear relationships in the input domain.

Figure 2 illustrates the fundamental principle underlying Kernel PCA: transforming nonlinear relationships into linear ones via an implicit mapping to a high-dimensional feature space. In the original space, the data exhibit a structure that cannot be adequately captured by linear projections, as evidenced by the circular arrangement of the samples. By applying the kernel trick, the data are lifted into a higher-dimensional space where their structure becomes linearly separable. PCA is then applied in this transformed space, enabling the extraction of principal components that reflect the intrinsic geometry of the data. The final projection demonstrates how Kernel PCA preserves nonlinear patterns in a compact, low-dimensional representation. This capability makes Kernel PCA particularly suitable for complex datasets where linear methods such as classical PCA fail to capture essential structures.

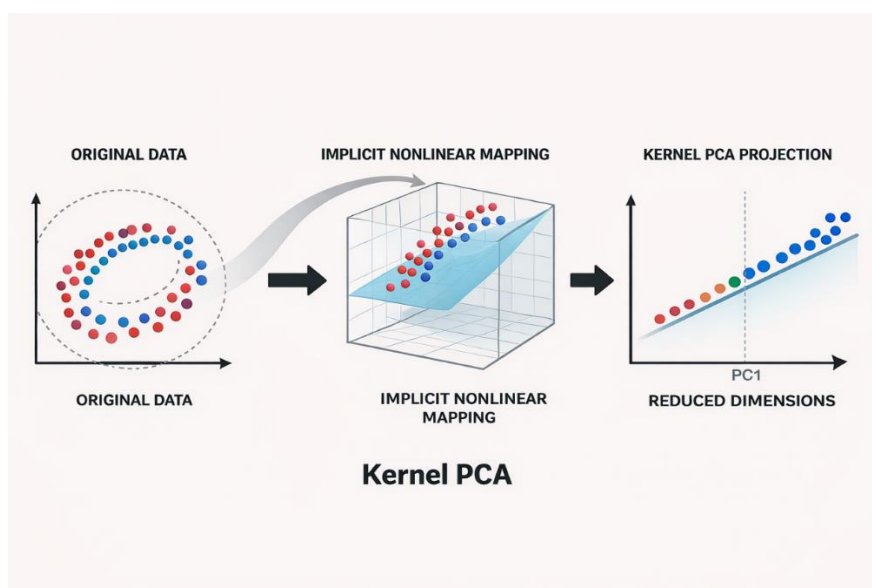


Figure 2 - Conceptual illustration of Kernel Principal Component Analysis (Kernel PCA).

The original dataset (left), characterized by a nonlinear structure in the input space, is implicitly mapped into a higher-dimensional feature space through a kernel function (center), where the data become approximately linearly separable. PCA is then performed in this transformed space, and the data are projected onto the leading principal component(s), resulting in a low-dimensional representation (right) that captures the underlying nonlinear structure.

Table 4 highlights the fundamental trade-offs between Principal Component Analysis (PCA) and Kernel PCA, emphasizing the balance between interpretability, computational efficiency, and modeling flexibility. PCA, being a linear method, offers high interpretability through explicit loadings and straightforward reconstruction, making it particularly attractive in scientific domains such as chemometrics, where understanding variable contributions is essential. Its good scalability further supports its application to large datasets.

Table 4 – Comparison of the Properties of PCA with Kernel PCA.

Feature	PCA	Kernel PCA
Linearity	Linear	Nonlinear
Interpretability	High	Low
Scalability	Good	Limited
Feature space	Explicit	Implicit
Reconstruction	Straightforward	Difficult

In contrast, Kernel PCA extends PCA to nonlinear feature spaces, enabling the capture of complex structures that cannot be represented by linear projections. However, this increased flexibility comes at the cost of reduced interpretability, since the transformation takes place in an implicit, high-dimensional space defined by a kernel function. Additionally, Kernel PCA is computationally more demanding and presents challenges in data reconstruction, which can limit its practical use in large-scale or real-time applications.

Overall, the table underscores that PCA and Kernel PCA should not be viewed as competing methods, but rather as complementary tools. The choice between them depends on the nature of the data and the analytical objective: PCA is preferred for transparency and efficiency, while Kernel PCA is more suitable when nonlinear patterns are dominant, and interpretability is less critical.

1. Theoretical Foundations

Let $\{x_i\}_{i=1}^n$, with $x_i \in \mathbb{R}^d$, be a dataset. In classical PCA, we seek directions that maximize the variance of the projected data. Kernel PCA extends this idea by first

mapping the data into a high-dimensional (possibly infinite-dimensional) feature space $\mathcal{F}$, using a nonlinear mapping:

$$\phi : \mathbb{R}^d \rightarrow \mathcal{F}, \quad \mathbf{x} \mapsto \phi(\mathbf{x})$$

Instead of explicitly computing $\phi(x)$, Kernel PCA relies on the kernel trick, where the inner product in $\mathcal{F}$ is computed using a kernel function:

$$k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle \quad (9)$$

Common kernel functions include:

- Linear kernel: $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y}$
- Polynomial kernel: $k(\mathbf{x}, \mathbf{y}) = (\mathbf{x}^T \mathbf{y} + \mathbf{c})^p$
- Gaussian (RBF) kernel: $k(\mathbf{x}, \mathbf{y}) = \exp\left(-\frac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right)$

2. Kernel Matrix and Centering

Define the kernel (Gram) matrix:

$$\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) \quad (10)$$

To perform PCA in feature space, the kernel matrix must be centered:

$$\mathbf{K}_c = \mathbf{K} - \mathbf{1}_n \mathbf{K} - \mathbf{K} \mathbf{1}_n + \mathbf{1}_n \mathbf{K} \mathbf{1}_n \quad (11)$$

where $\mathbf{1}_n$ is an $n \times n$ matrix with all entries equal to $(1/n)$.

3. Eigenvalue Problem

Kernel PCA solves the eigenvalue problem:

$$\mathbf{K}_c \alpha = \lambda \alpha \quad (12)$$

The eigenvectors α correspond to principal components in feature space. The projection of a new data point $\mathbf{x}$, onto the k -th principal component is given by:

$$\mathbf{y}_k(\mathbf{x}) = \sum_{i=1}^n \alpha_i^{(k)} k(\mathbf{x}_i, \mathbf{x}) \quad (13)$$

4. Interpretation

Unlike linear PCA, Kernel PCA does not yield explicit directions in the original input space. Instead, it provides projections in an implicit feature space where nonlinear patterns—such as clusters lying on curved manifolds—become linearly separable.

This makes Kernel PCA particularly powerful for:

- Unfolding nonlinear manifolds (e.g., Swiss roll)
- Capturing nonlinear correlations
- Enhancing separability in classification tasks

5. Pre-image Problem

A key challenge in Kernel PCA is the pre-image problem, i.e., reconstructing the original input $\mathbf{x}$ from its feature space representation. Since $\phi(\cdot)$ is typically non-invertible or unknown, approximate methods are required:

$$\mathbf{x}^* \approx \arg \max_{\mathbf{x}} \|\phi(\mathbf{x}) - \sum_{k=1}^m y_k \mathbf{v}_k\|^2 \quad (14)$$

This limitation complicates interpretability and reconstruction tasks.

6. Computational Complexity

Kernel PCA has a higher computational cost than linear PCA:

- Kernel matrix computation: On^2
- Eigen-decomposition: On^3

This makes it less suitable for very large datasets, although approximations such as:

- Nyström method
- Random Fourier Features

can mitigate this issue.

7. Limitations

Despite its flexibility, Kernel PCA presents several challenges:

1. Scalability: Poor performance for large datasets due to the kernel matrix.
2. Kernel selection: Performance is highly dependent on the choice of kernel and hyperparameters (e.g., σ in RBF).
3. Lack of interpretability: Results are expressed in an implicit feature space.
4. Pre-image difficulty: Reconstruction is nontrivial and often approximate.
5. Sensitivity to noise: Kernel methods may amplify noise depending on the kernel.

8. Applications

Kernel PCA has been successfully applied across various domains:

8.1 Chemometrics and Spectroscopy

- Nonlinear feature extraction from Raman, NIR, and IR spectra
- Detection of subtle chemical variations
- Data compression for high-dimensional spectral datasets

8.2 Image Processing

- Face recognition (Eigenfaces vs. Kernel Eigenfaces)
- Image denoising and compression
- Nonlinear feature extraction

8.3 Bioinformatics

- Gene expression analysis
- Protein structure classification

8.4 Fault Detection and Process Monitoring

- Nonlinear process control
- Detection of anomalies in chemical engineering systems

8.5 Machine Learning Pipelines

- Preprocessing step for classification (e.g., SVM)
- Visualization of complex datasets

9. Final Remarks

Kernel PCA represents a fundamental advancement in dimensionality reduction by enabling the analysis of nonlinear structures through implicit feature mappings. Its strength lies in flexibility and expressive power, while its limitations motivate ongoing research in scalable kernel methods, interpretable embeddings, and hybrid approaches combining deep learning and kernel techniques.

For modern applications—particularly in chemometrics, materials science, and spectroscopy—Kernel PCA remains a valuable tool when nonlinear relationships dominate the data structure [7].

2.2.1.1.3 L1-Norm Principal Component Analysis

L1-norm Principal Component Analysis (L1-PCA), also known as robust PCA based on the L1 norm, is an alternative formulation of classical PCA that replaces the conventional L2-norm (least-squares) criterion with an L1-norm objective. This modification significantly enhances robustness against outliers and heavy-tailed noise, making L1-PCA particularly suitable for real-world datasets where deviations from Gaussian assumptions are common.

Figure 3 highlights the key advantage of L1-norm PCA over classical PCA: robustness to outliers. While traditional PCA is based on the L2 norm and therefore disproportionately influenced by extreme observations, L1-PCA relies on the L1 norm, which reduces the impact of large deviations. As a result, the principal direction obtained by L1-PCA better reflects the intrinsic structure of the inlier data. This property makes L1-PCA particularly suitable for real-world datasets in chemometrics, spectroscopy, and other applications where noise and outliers are common, providing more stable and reliable dimensionality reduction.

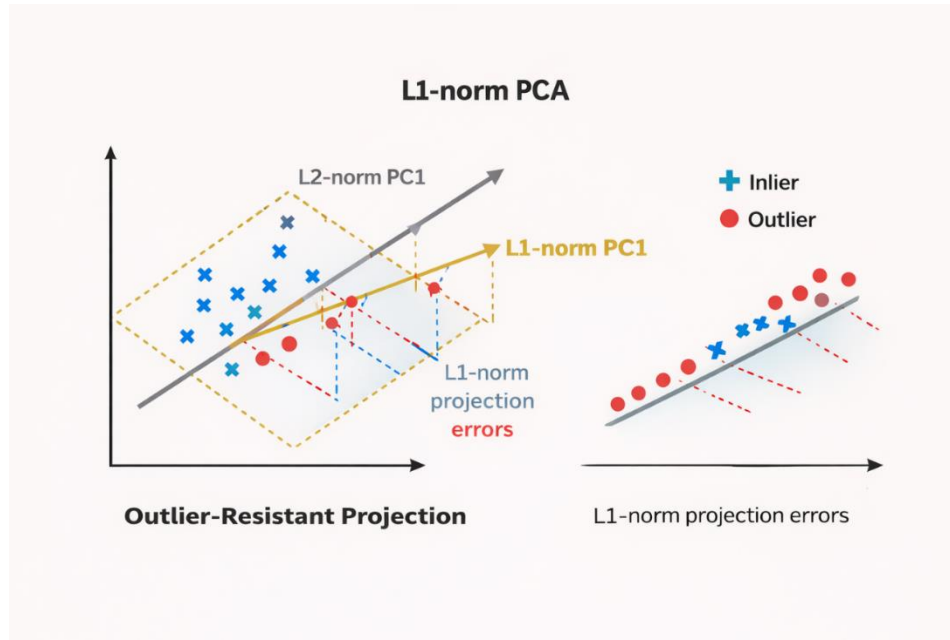


Figure 3 - Illustration of *L1-norm Principal Component Analysis (L1-PCA)* compared to *classical L2-norm PCA*.

The dataset contains inliers (blue crosses) and outliers (red circles). The classical PCA direction (L2-norm PC1, gray arrow) is strongly influenced by outliers due to the squared-error minimization. In contrast, L1-PCA (yellow arrow) minimizes the sum of absolute projection errors, resulting in a principal direction that is more robust to outliers and better aligned with the majority of the data. Dashed lines represent projection errors under the L1 norm.

Table 5 summarizes the key distinctions between classical PCA and its L1-norm variant, underscoring the impact of the underlying optimization criterion. While L2-norm PCA maximizes variance and is computationally efficient, it is highly sensitive to outliers because of its squared-error formulation. In contrast, L1-norm PCA replaces the quadratic objective with an absolute deviation criterion, significantly improving robustness to anomalous observations. However, this robustness comes at the cost of increased computational complexity and reduced analytical tractability. Overall, the table reinforces that L1-PCA is particularly advantageous in noisy or contaminated datasets, whereas L2-

PCA remains preferable when computational efficiency and interpretability are primary concerns.

Table 5 – Comparison of the fundamental properties of L2-norm PCA (classical PCA) and L1-norm PCA.

Feature	L2-PCA	L1-PCA
Objective	Variance maximization	Absolute deviation maximization
Sensitivity to outliers	High	Low
Optimization	Closed-form (SVD)	Iterative/combinatorial
Robustness	Low	High
Computational cost	Low	Higher

1. Motivation and Background

Classical PCA maximizes the variance of projected data or, equivalently, minimizes the squared reconstruction error:

$$\min_{\mathbf{W}} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{W}\mathbf{W}^T \mathbf{x}_i\|_2^2 \quad (15)$$

This formulation is optimal under Gaussian noise assumptions but is highly sensitive to outliers, since squared errors amplify large deviations.

In contrast, L1-PCA replaces the L2 norm with the L1 norm, reducing the influence of extreme observations:

$$\min_{\mathbf{W}} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{W}\mathbf{W}^T \mathbf{x}_i\|_1 \quad (16)$$

or, in projection form, maximizing the L1 dispersion:

$$\max_{\mathbf{x}} \sum_{i=1}^n |\mathbf{w}^T \mathbf{x}_i| \quad \text{subject to} \quad \|\mathbf{w}\|_2 = 1 \quad (17)$$

This formulation seeks directions that maximize the sum of absolute deviations rather than the sum of squared deviations.

2. Mathematical Formulation

Given a centered dataset $\mathbf{X} \in \mathbb{R}^{d \times n}$, the first L1 principal component can be defined as:

$$\mathbf{w}^* = \arg \max_{\|\mathbf{w}\|_2=1} \sum_{i=1}^n |\mathbf{w}^T \mathbf{x}_i| \quad (18)$$

For multiple components, the problem becomes:

$$\max_{\mathbf{W}^T \mathbf{W} = \mathbf{I}} \sum_{i=1}^n \|\mathbf{W}^T \mathbf{x}_i\|_1 \quad (19)$$

Unlike L2-PCA, this optimization problem is non-convex and non-smooth, making it computationally more challenging.

3. Geometric Interpretation

L1-PCA seeks projection directions that maximize the sum of absolute projections, which corresponds to aligning with the data's median structure rather than its mean.

Key differences:

- L2-PCA: Sensitive to variance $\rightarrow$ influenced by outliers
- L1-PCA: Sensitive to median structure $\rightarrow$ robust to outliers

Thus, L1-PCA is better suited for datasets with:

- Sparse noise
- Impulsive disturbances
- Heavy-tailed distributions

4. Algorithms for L1-PCA

Due to the lack of a closed-form solution, several algorithms have been proposed:

4.1 Iterative Reweighting Methods

Approximate the L1 norm via weighted L2 norms:

$$\sum |x| \approx \sum \mathbf{w}_i \mathbf{x}_i^2$$

4.2 Bit-Flipping Algorithm

Efficient method for computing exact L1 principal components by optimizing sign vectors.

4.3 Convex Relaxation Approaches

Transform the problem into a tractable convex optimization problem.

4.4 Alternating Optimization

Iteratively updates projection vectors and weights.

5. Robustness Properties

The key advantage of L1-PCA lies in its robustness:

- Breakdown point: Higher than L2-PCA
- Influence function: Bounded
- Outlier resistance: Strong

Because the L1 norm grows linearly rather than quadratically, extreme values have limited impact on the solution.

6. Limitations

Despite its robustness, L1-PCA has several drawbacks:

1. Computational complexity
 - Non-convex optimization
 - Often requires iterative or combinatorial methods
2. Non-uniqueness of solutions
 - Multiple optimal directions may exist
3. Lack of orthogonality guarantees
 - Components may not be strictly orthogonal
4. Scalability issues
 - Less efficient than standard PCA for large datasets
5. Interpretability challenges

- Results may be less intuitive compared to variance-based PCA

7. Applications

L1-PCA has been successfully applied in various domains:

7.1 Chemometrics and Spectroscopy

- Robust feature extraction from Raman, NIR, and IR data
- Handling spectral outliers due to instrumental noise

7.2 Image Processing

- Face recognition under occlusion
- Background subtraction
- Robust image compression

7.3 Bioinformatics

- Gene expression analysis with noisy measurements

7.4 Signal Processing

- Impulsive noise filtering
- Fault detection in industrial systems

7.5 Financial Data Analysis

- Handling extreme events (e.g., market shocks)

8. Extensions and Variants

Several extensions of L1-PCA have been proposed:

- Sparse L1-PCA: Adds sparsity constraints
- Kernel L1-PCA: Combines robustness with nonlinear mapping
- Robust PCA (RPCA): Decomposes data into low-rank + sparse components
- Online L1-PCA: Handles streaming data

9. Final Remarks

L1-norm PCA represents a powerful alternative to classical PCA in scenarios where robustness is critical. By focusing on absolute deviations rather than squared errors, it is more resilient to outliers and non-Gaussian noise. However, this robustness comes at the cost of increased computational complexity and more challenging optimization.

In modern data science and chemometrics, L1-PCA is particularly relevant for high-dimensional, noisy datasets—such as spectroscopic measurements—where traditional PCA may fail to capture the true underlying structure [8–10].

2.2.1.1.4 Classical Multidimensional Scaling

1. Introduction

Classical Multidimensional Scaling (cMDS) is a geometric method that aims to represent objects as points in a low-dimensional Euclidean space such that the pairwise distances between points approximate the original dissimilarities. It is particularly useful when the original data are not explicitly available, and only a distance or dissimilarity matrix is known [11–12].

cMDS has strong theoretical connections to linear algebra and spectral decomposition, and under Euclidean conditions, it is mathematically equivalent to Principal Component Analysis (PCA).

Figure 4 illustrates the fundamental principle of Classical MDS: the transformation of pairwise distances into a geometric representation in a reduced-dimensional space. By preserving inter-point distances, MDS provides a faithful visualization of the data's global structure, making it particularly suitable for exploratory analysis and similarity-based modeling.

Classical Multidimensional Scaling (MDS)

Dimensionality Reduction

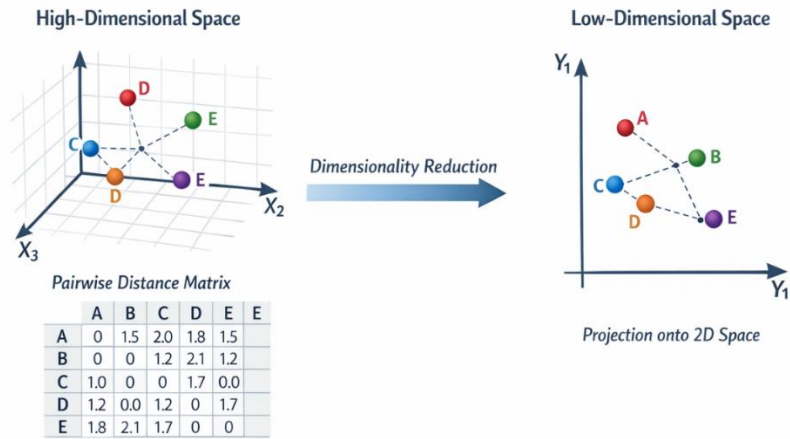


Figure 4 - Schematic representation of Classical Multidimensional Scaling (MDS) applied to dimensionality reduction.

On the left, a set of points is represented in a high-dimensional space, with pairwise relationships encoded in a distance matrix. These distances summarize the dissimilarities among observations. Classical MDS transforms this distance matrix into an inner-product (Gram) matrix via double centering, followed by eigenvalue decomposition. On the right, the data are projected into a low-dimensional (typically 2D) space such that the pairwise Euclidean distances between points are preserved as closely as possible. The resulting configuration maintains the global geometric structure of the original dataset.

Table 6 summarizes the main theoretical and practical characteristics of Classical Multidimensional Scaling (MDS). The method is fundamentally distance-based, operating on a pairwise dissimilarity matrix and aiming to preserve Euclidean distances in a reduced-dimensional representation. Its mathematical formulation relies on double-centering of the squared distance matrix followed by eigenvalue decomposition, making

it closely related to Principal Component Analysis (PCA) when Euclidean distances are used. Classical MDS is effective in preserving global data structure but is less capable of capturing local or nonlinear relationships. While it provides a meaningful geometric embedding, interpretability and invertibility are limited, as the resulting coordinates are not directly associated with the original variables. Additionally, the method can be sensitive to noise and computationally demanding for large datasets due to its cubic complexity. Overall, Classical MDS is a powerful tool for exploratory analysis and visualization when global structure preservation is the primary objective.

Table 6 – Key Properties of Classical Multidimensional Scaling (MDS)

Properties	Description
Type of method	Distance-based dimensionality reduction technique
Input data	Pairwise distance (or dissimilarity) matrix
Objective	Preserve pairwise Euclidean distances in a lower-dimensional space
Mathematical foundation	Double-centering of the squared distance matrix followed by eigenvalue decomposition
Linearity	Linear (equivalent to PCA when distances are Euclidean)
Output	Low-dimensional coordinates (embedding of the data)
Metric assumption	Assumes Euclidean distances (metric MDS)
Global structure preservation	Strong (focuses on preserving overall geometry)
Local structure preservation	Limited compared to nonlinear methods
Interpretability	Moderate (coordinates are not directly linked to original variables)
Invertibility	Limited (original variables cannot be fully recovered)
Sensitivity to noise	Moderate to high (depends on the quality of the distance matrix)

Properties	Description
Computational complexity	$O(n^3)$ due to eigen decomposition
Scalability	Limited for large datasets
Relation to PCA	Equivalent to PCA when applied to Euclidean distance matrix
Handling nonlinearity	Limited (does not capture nonlinear manifolds)
Eigenvalues interpretation	Positive eigenvalues correspond to meaningful dimensions; negative values indicate non-Euclidean structure

2. Mathematical Formulation

Let $\mathbf{D} = [d_{ij}]$ be a matrix of pairwise distances between n objects. The goal of cMDS is to find a configuration of points $\mathbf{X} \in \mathbb{R}^{n \times p}$ such that:

$$d_{ij}^2 \approx \|\mathbf{x}_i - \mathbf{x}_j\|^2 \quad (20)$$

2.1 Double Centering

The method begins by transforming the distance matrix into a Gram (inner product) matrix. Define the matrix:

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J} \quad (21)$$

where:

- $\mathbf{D}^{(2)}$ is the matrix of squared distances,
- $\mathbf{J} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the centering matrix.

This operation is known as double centering, and it converts distances into scalar products.

2.2 Eigenvalue Decomposition

The matrix $\mathbf{B}$ is symmetric and can be decomposed as:

$$\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T \quad (22)$$

where:

- $\mathbf{V}$ contains eigenvectors,
- $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues.

The low-dimensional embedding is obtained as:

$$\mathbf{X}_p = \mathbf{V}_p \mathbf{\Lambda}_p^{1/2} \quad (23)$$

where only the top p positive eigenvalues are retained.

3. Geometric Interpretation

cMDS reconstructs a configuration of points whose inner products match those derived from the original distance matrix. Thus, it finds a Euclidean embedding that best preserves pairwise distances in a least-squares sense.

Importantly:

- If distances are Euclidean $\rightarrow$ exact reconstruction is possible.
- If not $\rightarrow$ negative eigenvalues appear, indicating a non-Euclidean structure.

4. Relationship with PCA

When the distance matrix is computed from centered data:

$$d_{ij}^2 \approx \|\mathbf{x}_i - \mathbf{x}_j\|^2 \quad (24)$$

then cMDS becomes equivalent to PCA:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^T \quad (25)$$

Thus:

- PCA operates on the covariance structure,
- cMDS operates on distance structure.

5. Advantages

5.1 Works with distance matrices

- Does not require raw data
- Applicable to arbitrary dissimilarities

5.2 Strong mathematical foundation

- Based on spectral decomposition
- Deterministic and non-iterative

5.3 Global structure preservation

- Preserves large-scale geometry
- Effective for visualization of overall structure

5.4 Interpretability

- Eigenvalues indicate variance explained (analogous to PCA)

6. Stress and Reconstruction Error

Although Classical MDS does not explicitly minimize a stress function, the quality of the embedding can be assessed by:

$$Stress = \sqrt{\frac{\sum_{i < j} (d_{ij} - \|y_i - y_j\|)^2}{\sum_{i < j} d_{ij}^2}} \quad (26)$$

- Low stress indicates good preservation of distances.

7. Weaknesses and Limitations

7.1 Sensitivity to non-Euclidean distances

- Negative eigenvalues may arise
- Embedding may not be physically interpretable

7.2 Poor local structure preservation

- Does not explicitly preserve neighborhoods
- Inferior to methods like t-SNE or UMAP for clustering

7.3 Computational complexity

- Requires eigendecomposition of an $n \times n$ matrix

- Complexity: $O(n^3)$

7.4 Sensitivity to noise

- Noise in distances propagates globally

7.5 Not inherently nonlinear

- Cannot capture complex manifolds

8. Applications

8.1 Chemometrics

- Spectral similarity analysis (Raman, NIR, IR)
- Sample clustering and visualization

8.2 Bioinformatics

- Genetic distance analysis
- Phylogenetic visualization

8.3 Ecology

- Species dissimilarity mapping
- Community structure analysis

8.4 Social sciences

- Perceptual mapping
- Psychometrics

8.5 Materials science

- Similarity of materials based on descriptors

9. Practical Considerations

- Always inspect eigenvalues:
 - Negative values $\rightarrow$ non-Euclidean distances
- Consider dimensionality using cumulative variance
- Preprocess distances (e.g., normalization)

- Use stress or reconstruction error for evaluation

10. Final Remarks

Classical Multidimensional Scaling (cMDS) remains one of the most elegant and theoretically grounded methods in dimensionality reduction. Its formulation, based on reconstructing inner products from pairwise distances and then performing a spectral decomposition, provides a clear geometric interpretation of the embedding process. This strong mathematical foundation ensures determinism, reproducibility, and interpretability at the level of global structure, making cMDS an essential reference method in both theoretical and applied contexts.

A key strength of cMDS lies in its ability to operate directly on dissimilarity matrices, which is particularly advantageous in domains such as chemometrics, spectroscopy, and bioinformatics, where distance measures often carry more meaningful information than raw variables. Furthermore, its equivalence to PCA under Euclidean conditions highlights its central role within the broader family of spectral methods, reinforcing its conceptual importance as a bridge between variance-based and distance-based approaches.

However, cMDS is not without limitations. Its reliance on Euclidean assumptions limits its applicability to non-Euclidean or highly nonlinear data structures, often manifesting as negative eigenvalues and distorted embeddings. Additionally, its emphasis on preserving global distance comes at the expense of local neighborhood fidelity, making it less suitable for tasks such as clustering or manifold learning on complex datasets. Computational scalability is another constraint, particularly for large datasets requiring full eigendecomposition.

In modern data analysis pipelines, cMDS should therefore be viewed not as a universal solution, but as a baseline and interpretative tool. It provides valuable insights

into the intrinsic geometry of data and serves as a benchmark against which more advanced nonlinear techniques—such as Isomap, t-SNE, and UMAP—can be evaluated. When used appropriately, especially in conjunction with complementary methods, cMDS continues to offer significant analytical value, combining simplicity, rigor, and interpretability in a single coherent framework [11–12].

2.2.1.1.5 Isomap

Isomap (Isometric Mapping) is a nonlinear dimensionality reduction technique that extends Classical Multidimensional Scaling (MDS) to preserve geodesic distances rather than Euclidean distances. Introduced in 2000, Isomap is one of the earliest and most influential manifold learning methods, designed to recover the intrinsic geometry of data lying on a nonlinear low-dimensional manifold embedded in a high-dimensional space [13–14].

Figure 5 illustrates how Isomap extends Classical MDS to nonlinear settings by replacing Euclidean distances with geodesic distances computed on a neighborhood graph. By approximating the true manifold structure, Isomap effectively unfolds nonlinear data into a lower-dimensional space while preserving global relationships. This makes it particularly suitable for datasets that lie on curved manifolds, where linear techniques such as PCA or Classical MDS may fail to capture the underlying geometry.

Isomap (Isometric Mapping)

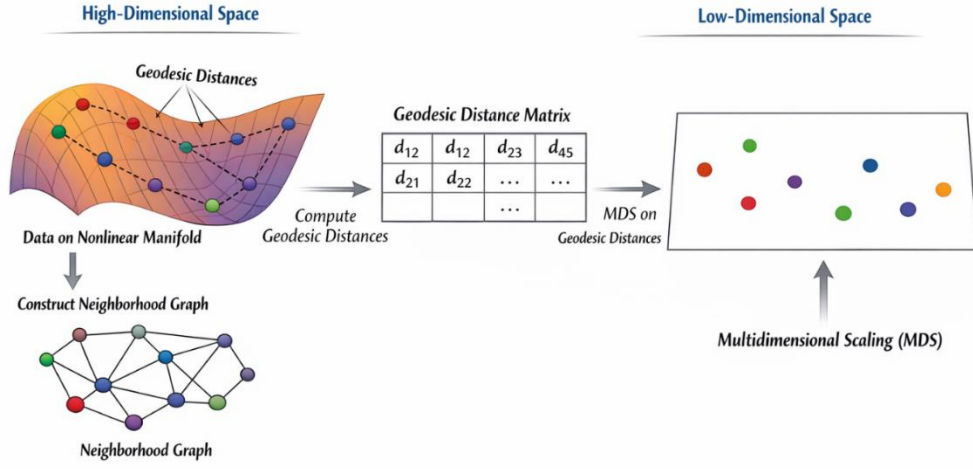


Figure 5 - Schematic illustration of the Isomap algorithm for nonlinear dimensionality reduction.

On the left, data points lie on a curved high-dimensional manifold, where Euclidean distances do not capture true relationships. A neighborhood graph is first constructed by connecting nearby points (e.g., k -nearest neighbors) to capture the local structure of the manifold. Geodesic distances between all pairs of points are then approximated by computing shortest paths along this graph. These distances are assembled into a geodesic distance matrix, which is subsequently used as input to Classical Multidimensional Scaling (MDS). On the right, the resulting low-dimensional embedding preserves the intrinsic geometry of the manifold by maintaining geodesic distances rather than straight-line Euclidean distances.

Table 7 summarizes the main characteristics of Isomap as a nonlinear dimensionality reduction technique grounded in manifold learning. Unlike linear methods, Isomap preserves geodesic distances computed over a neighborhood graph, allowing it to capture the intrinsic geometry of curved data structures. By combining

graph-based distance estimation with Classical MDS, the method achieves strong preservation of global structure while maintaining local relationships. However, its performance depends critically on the choice of neighborhood parameters and the quality of the graph, making it sensitive to noise, outliers, and disconnected components. Additionally, the lack of an explicit inverse mapping and its high computational cost limit scalability. Overall, Isomap is a powerful approach for unfolding nonlinear manifolds when preserving global geometric relationships is essential.

Table 7 – Key properties of the Isomap

Property	Description
Type of method	Nonlinear, manifold learning dimensionality reduction technique
Input data	High-dimensional data or pairwise distances
Objective	Preserve geodesic (manifold) distances in a low-dimensional embedding
Mathematical foundation	Construction of neighborhood graph + shortest path (geodesic distances) + Classical MDS
Linearity	Nonlinear
Output	Low-dimensional coordinates preserving intrinsic geometry
Metric assumption	Assumes data lie on a smooth manifold; uses geodesic distances
Global structure preservation	Strong (better than many nonlinear methods)
Local structure preservation	Good (via neighborhood graph construction)
Interpretability	Moderate (embedding not directly linked to original variables)
Invertibility	Limited (no explicit inverse mapping)
Sensitivity to noise	High (sensitive to noise and outliers affecting graph construction)

Property	Description
Computational complexity	High (graph construction + shortest path + eigen decomposition; typically $O(n^3)$)
Scalability	Limited for large datasets
Relation to MDS	Extension of Classical MDS using geodesic distances instead of Euclidean distances
Handling nonlinearity	Strong (captures nonlinear manifold structure)
Key parameter	Number of neighbors (k) or neighborhood radius (ϵ)
Failure modes	Short-circuit errors in the graph, disconnected components, and poor neighborhood selection
Eigenvalues interpretation	Leading eigenvalues define intrinsic dimensionality; negative values may indicate distortions

1. Conceptual Motivation

Many real-world datasets—such as spectroscopic signals, images, and molecular descriptors—lie on curved manifolds rather than linear subspaces. Linear techniques like PCA fail to capture these nonlinear structures.

Isomap addresses this limitation by:

1. Approximating the geodesic (manifold) distance between data points
2. Embedding the data in a low-dimensional space that preserves these distances

Thus, Isomap can be seen as a nonlinear generalization of Classical MDS.

2. Algorithmic Steps

Given a dataset $\{x_i\}_{i=1}^n$, the Isomap algorithm consists of three main steps:

2.1 Neighborhood Graph Construction

Construct a graph $G = (V, E)$, where each node represents a data point. Edges are defined using:

- k-nearest neighbors (k-NN), or
- ε -neighborhoods

Edges are weighted by Euclidean distances:

$$\mathbf{w}_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\| \quad (27)$$

2.2 Geodesic Distance Estimation

Compute shortest path distances between all pairs of nodes in the graph:

$$d_{ij}^{(G)} = \text{shortest path distance between } i \text{ and } j \quad (28)$$

This is typically done using:

- Dijkstra's algorithm
- Floyd–Warshall algorithm

These distances approximate the geodesic distances along the manifold.

2.3 Low-Dimensional Embedding (MDS Step)

Apply Classical MDS to the geodesic distance matrix $\mathbf{D}^{(G)}$:

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}(\mathbf{D}^{(G)})^2\mathbf{J} \quad (29)$$

Then compute the embedding via eigen-decomposition:

$$\mathbf{Y}_p = \mathbf{V}_p\mathbf{\Lambda}_p^{1/2} \quad (30)$$

3. Theoretical Interpretation

Isomap can be interpreted as:

- A global manifold learning method
- A spectral technique combining graph theory and linear algebra
- A nonlinear extension of MDS that preserves intrinsic geometry

If the manifold is sampled densely and the neighborhood graph is correctly constructed,

Isomap can recover the true low-dimensional structure.

4. Advantages

- Captures nonlinear global structure
- Preserves geodesic distances rather than Euclidean distances
- Provides interpretable embeddings
- Works well for smooth manifolds (e.g., Swiss roll)

5. Limitations and Weaknesses

Despite its strengths, Isomap presents several important limitations:

5.1 Sensitivity to Neighborhood Size

- Too small k : graph becomes disconnected
- Too large k : geodesic distances approximate Euclidean distances
- Requires careful tuning

5.2 Sensitivity to Noise and Outliers

- Noise can distort the neighborhood graph
- Outliers may create shortcuts, corrupting geodesic distances

5.3 Computational Complexity

- Graph construction: $O(n^2)$
- Shortest paths: $O(n^2 \log n)$ or $O(n^3)$
- MDS step: $O(n^3)$

Not suitable for very large datasets without approximations.

5.4 Assumption of a Single Connected Manifold

- Fails if data lies on multiple disconnected manifolds
- Cannot handle complex topologies well

5.5 Lack of Explicit Mapping

- No direct function $f: \mathbb{R}^d \rightarrow \mathbb{R}^p$

- Difficult to project new data points (out-of-sample problem)

5.6 Short-Circuit Errors

- Graph shortcuts can occur when distant points are incorrectly connected
- Leads to severe distortion of the embedding

6. Extensions and Improvements

Several variants have been proposed to address Isomap limitations:

- Landmark Isomap: reduces computational cost
- C-Isomap: improves robustness to noise
- Out-of-sample extensions: Nyström-based approximations
- Hessian LLE and diffusion maps: alternative manifold learning approaches

7. Applications

7.1 Chemometrics and Spectroscopy

- Nonlinear analysis of Raman, NIR, and IR datasets
- Discovery of hidden chemical structures

7.2 Image Processing

- Face recognition and pose estimation
- Nonlinear feature extraction

7.3 Bioinformatics

- Gene expression analysis
- Protein folding pathways

7.4 Materials Science

- Exploration of structure–property relationships
- Visualization of high-dimensional descriptors

7.5 Machine Learning Pipelines

- Preprocessing for classification/regression

- Visualization of complex datasets

8. Comparison with Other Methods

Table 8 provides a comparative overview of four widely used dimensionality reduction techniques, highlighting their fundamental differences in terms of method type, structural preservation, and mapping characteristics. Linear methods such as Principal Component Analysis (PCA) focus on preserving global variance with an explicit mapping, ensuring interpretability and reconstruction capabilities. In contrast, nonlinear approaches like Kernel PCA and Isomap operate in implicit feature spaces, capturing more complex data structures—Kernel PCA through kernel-induced transformations and Isomap through geodesic distances on a manifold. Classical MDS occupies an intermediate position among distance-based methods that preserve Euclidean distances but rely on an implicit mapping. Overall, the table emphasizes the trade-off between interpretability and the ability to model nonlinear structures, guiding the selection of methods based on the nature of the data and the analysis objective.

Table 8 – Comparison of the characteristics of DRs.

Method	Type	Structure Preserved	Mapping
PCA	Linear	Global variance	Explicit
Kernal PCA	Nonlinear	Implicit feature space	Implicit
MDS	Distance based	Euclidean distance	Implicit
Isomap	Nonlinear	Geodesic distance	Implicit

9. Final Remarks

Isomap is a milestone in nonlinear dimensionality reduction, introducing the concept of preserving geodesic distances. It provides a powerful framework for uncovering intrinsic data structures beyond linear assumptions. However, its reliance on graph construction and sensitivity to noise and parameter choices limit its applicability in complex or large-scale scenarios. In modern machine learning, Isomap remains a foundational method, often complemented or replaced by more scalable and robust techniques such as UMAP and diffusion maps.

2.2.1.1.6 Diffusion Map

Diffusion Maps is a nonlinear dimensionality reduction technique grounded in spectral graph theory and stochastic processes. It provides a powerful framework for uncovering the intrinsic geometry of data by modeling the connectivity between data points as a diffusion (random walk) process. Unlike methods such as PCA or Classical MDS, which rely on global Euclidean structure, Diffusion Maps capture the connectivity and multi-scale structure of the data manifold, making them particularly effective for complex, noisy, and nonlinear datasets [15].

Figure 6 illustrates how Diffusion Maps leverage a stochastic diffusion process to uncover the intrinsic geometry of high-dimensional data. By constructing a similarity graph and analyzing random walks on it, the method defines diffusion distances that are robust to noise and sensitive to the data's global connectivity. The resulting embedding, obtained from the leading eigenvectors of the diffusion operator, provides a meaningful low-dimensional representation of nonlinear structures, making Diffusion Maps particularly effective for complex datasets with manifold organization.

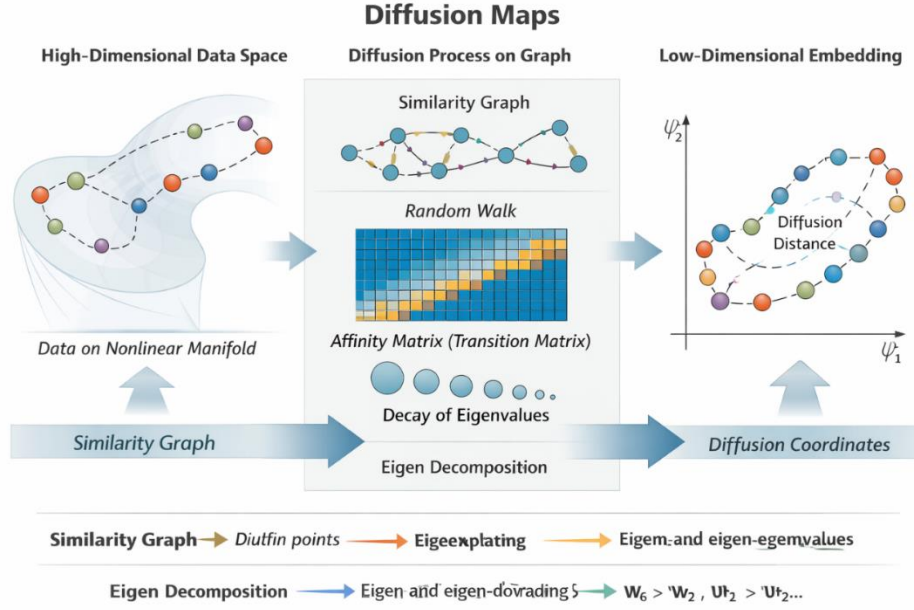


Figure 6 - Schematic representation of the Diffusion Maps algorithm for nonlinear dimensionality reduction.

On the left, data points are distributed on a nonlinear manifold in a high-dimensional space. A similarity graph is constructed using a kernel function (typically a Gaussian), which encodes local affinities between neighboring points. In the central panel, the similarity matrix is normalized to form a Markov transition (diffusion) matrix that models a random walk over the data. The diffusion process captures connectivity patterns over multiple steps, defining diffusion distances that reflect both local and global structure. Eigenvalue decomposition of the transition matrix yields eigenvalues and eigenvectors, whose leading components define the diffusion coordinates. On the right, the data are embedded into a low-dimensional space where distances correspond to diffusion distances, preserving the intrinsic geometry of the manifold.

Table 9 summarizes the main characteristics of Diffusion Maps as a nonlinear dimensionality reduction method grounded in manifold learning and spectral analysis. By constructing a kernel-based similarity matrix and modeling a Markov diffusion process, the method defines diffusion distances that effectively capture both local and global

structures of the data. This dual preservation makes Diffusion Maps particularly powerful for representing complex manifolds while maintaining robustness to noise. The embedding is obtained from the leading eigenvectors of the diffusion operator, which encode dominant modes of data connectivity. However, the method has limited interpretability, lacks an explicit inverse mapping, and can be computationally expensive for large datasets. Overall, Diffusion Maps provide a flexible and theoretically grounded approach for uncovering intrinsic geometric structures in high-dimensional data.

Table 9 – Key Properties of Diffusion Maps

Property	Description
Type of method	Nonlinear, manifold learning dimensionality reduction technique
Input data	High-dimensional data or pairwise similarities
Objective	Preserve diffusion distances that capture intrinsic data geometry
Mathematical foundation	Kernel-based similarity matrix + Markov transition matrix + eigenvalue decomposition
Linearity	Nonlinear
Output	Low-dimensional diffusion coordinates (embedding)
Metric assumption	Based on diffusion (connectivity) distances rather than Euclidean distances
Global structure preservation	Strong (captures long-range connectivity through diffusion process)
Local structure preservation	Strong (via local kernel-based similarities)
Interpretability	Moderate (eigenvectors correspond to diffusion modes)
Invertibility	Limited (no explicit inverse mapping)
Sensitivity to noise	Low to moderate (robust due to diffusion process)

Property	Description
Computational complexity	High (matrix construction + eigen decomposition; typically $O(n^3)$)
Scalability	Limited for large datasets (can be improved with approximations)
Relation to spectral methods	Closely related to spectral clustering and Laplacian eigenmaps
Handling nonlinearity	Strong (captures complex manifold structures)
Key parameter	Kernel bandwidth (ε) and number of diffusion steps (t)
Eigenvalues interpretation	Decay of eigenvalues reflects intrinsic dimensionality and data connectivity

1. Conceptual Motivation

In many real-world problems—such as spectroscopy, molecular dynamics, and biological systems—data points lie on a nonlinear manifold embedded in high-dimensional space. Instead of focusing solely on pairwise distances, Diffusion Maps analyze how information propagates through the dataset via local similarities.

The central idea is:

Points are considered close if there are many short paths connecting them. This notion leads to the concept of diffusion distance, which more robustly reflects the intrinsic geometry of the data than Euclidean or geodesic distances.

2. Construction of the Diffusion Operator

Given a dataset Let $\{x_i\}_{i=1}^n$, the method begins by constructing a similarity (kernel) matrix:

$$k(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{\varepsilon}\right) \quad (31)$$

where ε is a scale parameter controlling neighborhood size.

2.1 Normalization

Define the degree function:

$$\mathbf{d}(\mathbf{x}_i) = \sum_{j=1}^n \mathbf{k}(\mathbf{x}_i, \mathbf{x}_j) \quad (32)$$

The kernel is normalized to construct a Markov transition matrix:

$$\mathbf{p}_{ij} = \frac{\mathbf{k}(\mathbf{x}_i, \mathbf{x}_j)}{\mathbf{d}(\mathbf{x}_i)} \quad (33)$$

This defines a random walk on the data graph.

3. Diffusion Process

The transition matrix $\mathbf{P}$ describes a Markov process. After t steps, the diffusion probabilities are given by:

$$\mathbf{P}^t \quad (34)$$

The diffusion distance between points x_i and x_j is defined as:

$$D_i^2(i, j) = \sum_{k=1}^n \frac{(p_{ik}^{(t)} - p_{jk}^{(t)})^2}{\phi_0(k)} \quad (35)$$

where ϕ_0 is the stationary distribution.

4. Spectral Decomposition and Embedding

The key step is the eigen-decomposition of $\mathbf{P}$:

$$\mathbf{P}\psi_1 = \lambda_1\psi_1 \quad (36)$$

The embedding is defined as:

$$\psi_t(x_i) = (\lambda_1^t \psi_1(i), \lambda_2^t \psi_2(i), \dots, \lambda_m^t \psi_m(i)) \quad (37)$$

This mapping preserves diffusion distances:

$$D_t(i, j)^2 \approx \sum_{l=1}^m \lambda_l^{2t} (\psi_l(i) - \psi_l(j))^2 \quad (38)$$

5. Interpretation

Diffusion Maps can be interpreted as:

- A spectral embedding method
- A multi-scale analysis tool (controlled by t)

- A method that captures connectivity rather than direct distances

Key properties:

- Robust to noise
- Captures both local and global structure
- Reveals intrinsic manifold coordinates

6. Advantages

- Noise robustness: averaging over multiple paths reduces noise effects
- Multi-scale representation: parameter t controls resolution
- Nonlinear capability: captures complex manifolds
- Theoretical foundation: linked to Laplace–Beltrami operators

7. Limitations and Weaknesses

Despite its strengths, Diffusion Maps has several limitations:

7.1 Parameter Sensitivity

- Choice of kernel bandwidth ε is critical
- Poor choice leads to disconnected or overly dense graphs

7.2 Computational Cost

- Kernel matrix construction: $O(n^2)$
- Eigen-decomposition: $O(n^3)$

7.3 Scalability

- Not suitable for very large datasets without approximations
- Requires techniques like Nyström or sparse kernels

7.4 Lack of Explicit Mapping

- No direct mapping function for new data points
- Requires out-of-sample extensions

7.5 Interpretation Complexity

- Eigenvectors may not be easily interpretable
- Embedding is less intuitive than PCA

8. Comparison with Other Methods

Table 10 highlights the distinctive characteristics of Diffusion Maps compared to other dimensionality reduction techniques. While classical methods such as PCA and Kernel PCA focus on variance (linear and nonlinear, respectively), and MDS and Isomap emphasize distance preservation (Euclidean and geodesic), Diffusion Maps uniquely capture the intrinsic connectivity of the data through diffusion distances. By modeling the dataset as a Markov random walk, this approach effectively reveals the underlying manifold structure, making it particularly robust to noise and well-suited for complex, nonlinear datasets where connectivity is more informative than direct distances.

Table 10 – Comparison of the Diffusion Maps with related methods

Method	Structure Preserved	Key Idea
PCA	Global variance	Linear projection
Kernel PCA	Nonlinear variance	Kernel trick
MDS	Pairwise distances	Distance preservation
Isomap	Geodesic distances	Manifold unfolding
Diffusion Maps	Connectivity (diffusion distance)	Random walks

Diffusion Maps differ from Isomap in that they avoid shortest-path instability and instead rely on probabilistic connectivity.

9. Applications

9.1 Chemometrics and Spectroscopy

- Analysis of Raman, NIR, and IR data
- Extraction of nonlinear chemical features
- Noise-resistant dimensionality reduction

9.2 Molecular Dynamics

- Identification of slow collective variables
- Analysis of conformational landscapes

9.3 Bioinformatics

- Gene expression analysis
- Cell differentiation trajectories

9.4 Image Processing

- Shape analysis
- Nonlinear feature extraction

9.5 Materials Science

- Exploration of structure–property relationships
- Dimensionality reduction of descriptors

10. Final Remarks

Diffusion Maps represent a major advancement in nonlinear dimensionality reduction by introducing a probabilistic perspective on data geometry. By leveraging diffusion processes, the method provides a robust and flexible framework for uncovering intrinsic structures in complex datasets. In modern machine learning and chemometrics, Diffusion Maps are particularly valuable for noisy, nonlinear, and high-dimensional data, complementing methods such as Isomap and Kernel PCA. Their multi-scale nature and

strong theoretical foundation make them a cornerstone technique in manifold learning [15].

2.2.1.1.7 Dimensionality Reduction via Regression

Dimensionality Reduction via Regression (DRR) is a nonlinear dimensionality reduction framework that extends classical Principal Component Analysis (PCA) by modeling and removing nonlinear dependencies between principal components through regression. Unlike manifold learning methods such as Isomap or Diffusion Maps, which rely on graph-based representations, DRR remains grounded in a coordinate-based transformation, preserving interpretability while capturing nonlinear structure.

DRR is particularly relevant in chemometrics, spectroscopy, and high-dimensional data analysis, where variables often exhibit structured nonlinear dependencies that linear methods may not fully capture [16–17].

Figure 7 illustrates the core idea of Dimensionality Reduction via Regression, in which high-dimensional data are progressively modeled by a sequence of regression functions to extract latent variables. Unlike purely geometric methods, DRR leverages predictive relationships among variables to construct a low-dimensional embedding. By combining regression modeling with interpolation over the data space, the method can capture nonlinear dependencies while preserving relevant structural information. This makes DRR a flexible and powerful approach for dimensionality reduction, particularly in scenarios where functional relationships between variables play a central role.

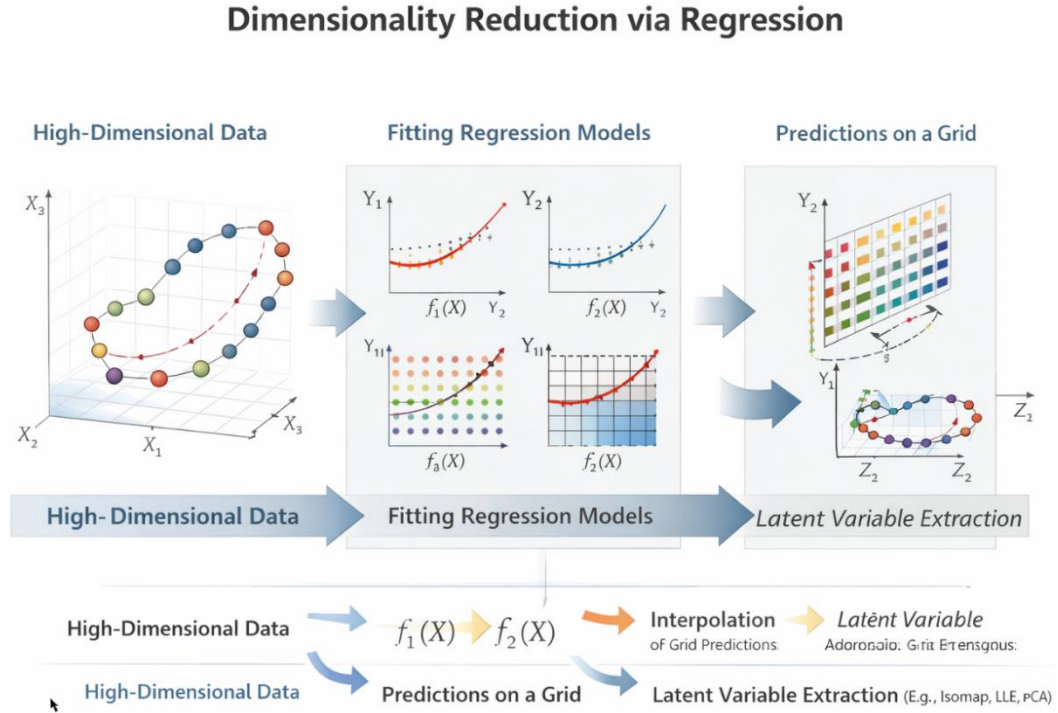


Figure 7 - Schematic illustration of Dimensionality Reduction via Regression (DRR).

On the left, high-dimensional data are represented in the original feature space, where nonlinear relationships among variables may exist. In the central panel, a sequence of regression models $f_1(X), f_2(X), \dots, Z_1, Z_2$ is fitted to capture dependencies between variables or components, typically in a sequential or hierarchical manner. These models approximate the structure of the data by learning functional relationships that describe the variance not explained by previous components. Predictions are then evaluated over a grid in the feature space, enabling interpolation of the learned structure. On the right, latent variables are extracted from the regression outputs, forming a low-dimensional representation that preserves the intrinsic structure of the data. This approach combines regression modeling with projection, enabling flexible handling of nonlinear relationships.

Table 11 summarizes the main characteristics of Dimensionality Reduction via Regression (DRR) as a flexible, model-driven approach to dimensionality reduction.

Unlike traditional distance- or variance-based methods, DRR captures the intrinsic structure of data by modeling functional dependencies between variables through sequential regression. This allows the method to effectively handle nonlinear relationships while providing a potentially interpretable framework, depending on the chosen regression model. The approach generalizes PCA by replacing linear projections with regression functions and iteratively extracting components from the residual information. However, its performance is strongly dependent on model selection, parameter tuning, and data quality, which can influence robustness and computational cost. Overall, DRR offers a powerful alternative for scenarios where capturing complex variable relationships is more important than preserving geometric distances.

Table 11 – Key Properties of Dimensionality Reduction via Regression (DRR)

Properties	Description
Type of method	Nonlinear, regression-based dimensionality reduction technique
Input data	High-dimensional data matrix
Objective	Capture intrinsic structure by modeling dependencies between variables via regression
Mathematical foundation	Sequential regression modeling (e.g., $(f_1(X), f_2(X), \dots)$) combined with latent variable extraction
Linearity	Nonlinear (depending on the regression model used)
Output	Low-dimensional latent variables (embedding)
Metric assumption	No explicit distance preservation; based on functional relationships
Global structure preservation	Moderate (depends on regression model quality)
Local structure preservation	Moderate to strong (captures local dependencies through regression)

Properties	Description
Interpretability	Moderate to high (regression functions can be interpreted)
Invertibility	Possible (approximate inverse via regression models)
Sensitivity to noise	Moderate (depends on robustness of regression models)
Computational complexity	Moderate to high (depends on regression method and number of components)
Scalability	Moderate (can scale with efficient regression algorithms)
Relation to PCA	Generalizes PCA by replacing linear projections with regression functions
Handling nonlinearity	Strong (flexible through nonlinear regression models)
Key parameter	Choice of regression model (linear, kernel, neural networks, etc.)
Model dependency	Strong (performance depends on regression model selection and tuning)
Sequential nature	Components extracted iteratively, modeling residual variance

1. Conceptual Motivation

Classical PCA transforms the data into orthogonal components:

$$\mathbf{z} = \mathbf{W}^T \mathbf{x} \quad (39)$$

where $\mathbf{z} = (z_1, z_2, \dots, z_d)$ are uncorrelated principal components. However, uncorrelated does not imply independent. In many real-world datasets, higher-order components exhibit nonlinear dependencies on lower-order components.

DRR addresses this limitation by:

- Sequentially modeling each principal component as a nonlinear function of the preceding components and removing this dependency.

This results in a representation where both linear and nonlinear redundancies are reduced.

2. Mathematical Formulation

Let $\mathbf{z} = (z_1, z_2, \dots, z_d)$ be the PCA-transformed data. DRR models each component z_k as:

$$z_k = f_k(z_1, z_2, \dots, z_{k-1}) + \varepsilon \quad (40)$$

where:

- $f_k(\cdot)$ is a nonlinear regression function
- ε is the residual (innovation term)

The transformed DRR representation is:

$$\tilde{z}_k = z_k - \hat{f}_k(z_1, \dots, z_{k-1}) \quad (41)$$

for $k = 2, \dots, d$, with $\tilde{z}_1 = z_1$.

Thus, DRR produces a new representation:

$$\tilde{\mathbf{z}} = (\tilde{z}_1, \tilde{z}_2, \dots, \tilde{z}_d) \quad (42)$$

where components are not only decorrelated but also approximately independent.

3. Regression Models

The functions f_k can be modeled using various regression techniques:

- Kernel regression
- Gaussian Process regression
- Neural networks
- Polynomial regression
- Support Vector Regression (SVR)

The choice of regression model determines the flexibility and complexity of the DRR transformation.

4. Invertibility and Reconstruction

One of the key advantages of DRR is that it provides an invertible transformation (under mild conditions), unlike many manifold learning methods.

Reconstruction proceeds sequentially:

$$z_k = \tilde{z}_k + \hat{f}_k(z_1, \dots, z_{k-1}) \quad (43)$$

Then the original data can be reconstructed using the inverse PCA transformation:

$$\mathbf{x} \approx \mathbf{W}\mathbf{z} \quad (44)$$

This property makes DRR particularly useful in applications requiring:

- Data compression
- Signal reconstruction
- Generative modeling

5. Interpretation

DRR can be interpreted as:

- A nonlinear extension of PCA
- A hierarchical regression model in latent space
- A method for removing nonlinear redundancy

Unlike Kernel PCA, which maps data to an implicit space, DRR operates in an explicit coordinate system, improving interpretability.

6. Advantages

- Captures nonlinear dependencies between components
- Invertible transformation, enabling reconstruction
- Improved compression efficiency compared to PCA
- Maintains explicit representation (interpretable)
- Flexible via choice of regression model

7. Limitations and Weaknesses

Despite its advantages, DRR has several limitations:

7.1 Model Dependency

- Performance depends heavily on the regression model used

- Risk of underfitting or overfitting

7.2 Computational Cost

- Requires fitting multiple regression models
- Complexity increases with dimensionality

7.3 Error Propagation

- Errors in early components affect subsequent regressions
- Sequential structure may amplify inaccuracies

7.4 Lack of Global Manifold Guarantees

- Does not explicitly preserve distances or topology
- Unlike Isomap or Diffusion Maps, it does not model geometry directly

7.5 Hyperparameter Sensitivity

- Requires tuning of regression parameters
- Model selection can be challenging

8. Comparison with Other Methods

Table 12 presents a comparative overview of major dimensionality reduction (DR) techniques, highlighting key differences in terms of method type, ability to model nonlinearity, mapping characteristics, and reconstruction capability. PCA stands out as a linear method with explicit mapping and straightforward reconstruction, making it highly interpretable and computationally efficient. In contrast, nonlinear methods such as Kernel PCA, Isomap, and Diffusion Maps provide enhanced capability to capture complex data structures but rely on implicit mappings and generally lack direct reconstruction. Dimensionality Reduction via Regression (DRR) combines the strengths of nonlinear modeling with an explicit mapping, allowing both flexibility in capturing nonlinear relationships and the possibility of reconstruction. Overall, the table emphasizes the trade-

offs between interpretability, flexibility, and reconstructive capability, guiding the selection of appropriate techniques based on the analytical objective.

Table 12 – Comparison of the DR Techniques

Method	Type	Nonlinearity	Mapping	Reconstruction
PCA	Linear	No	Explicit	Yes
Kernel PCA	Nonlinear	Yes	Implicit	Difficult
Isomap	Nonlinear	Yes	Implicit	No
Diffusion Maps	Nonlinear	Yes	Implicit	No
DRR	Nonlinear	Yes	Explicit	Yes

Therefore, DRR uniquely combines:

- Nonlinear modeling
- Explicit mapping
- Reconstruction capability

9. Applications

9.1 Chemometrics and Spectroscopy

- Compression of spectral data (Raman, NIR, IR)
- Removal of nonlinear redundancies
- Preprocessing for classification models

9.2 Image Processing

- Nonlinear feature extraction
- Image compression and reconstruction

9.3 Signal Processing

- Noise reduction
- Efficient representation of signals

9.4 Machine Learning Pipelines

- Feature engineering
- Preprocessing for regression and classification

9.5 Materials Science

- Dimensionality reduction of descriptor spaces
- Structure–property modeling

10. Final Remarks

Dimensionality Reduction via Regression represents a powerful and conceptually elegant extension of PCA, bridging the gap between linear methods and fully nonlinear manifold learning techniques. By explicitly modeling nonlinear dependencies among principal components, DRR improves data compression and representation while preserving interpretability and invertibility.

In modern applications—particularly in chemometrics and spectroscopy—DRR provides a valuable alternative to kernel and graph-based methods, especially when reconstruction and model transparency are essential.

2.2.1.1.8 Independent Component Analysis

Independent Component Analysis (ICA) is a statistical and computational technique for dimensionality reduction and blind source separation that aims to represent multivariate data as a linear combination of statistically independent components. Unlike PCA, which seeks uncorrelated directions maximizing variance, ICA goes further by identifying components that are maximally independent, typically exploiting higher-order statistical structure.

ICA has become a fundamental tool in machine learning, signal processing, and chemometrics, particularly in applications where observed data arise from mixtures of latent sources [18–20].

Figure 8 illustrates the fundamental principle of Independent Component Analysis, in which observed data are modeled as linear mixtures of statistically independent sources. By estimating an unmixing transformation, ICA separates mixed signals into independent components, revealing latent structures that are not accessible to variance-based methods such as PCA. This makes ICA particularly effective for applications involving signal separation, feature extraction, and analysis of complex datasets where independence, rather than variance, is the primary criterion.

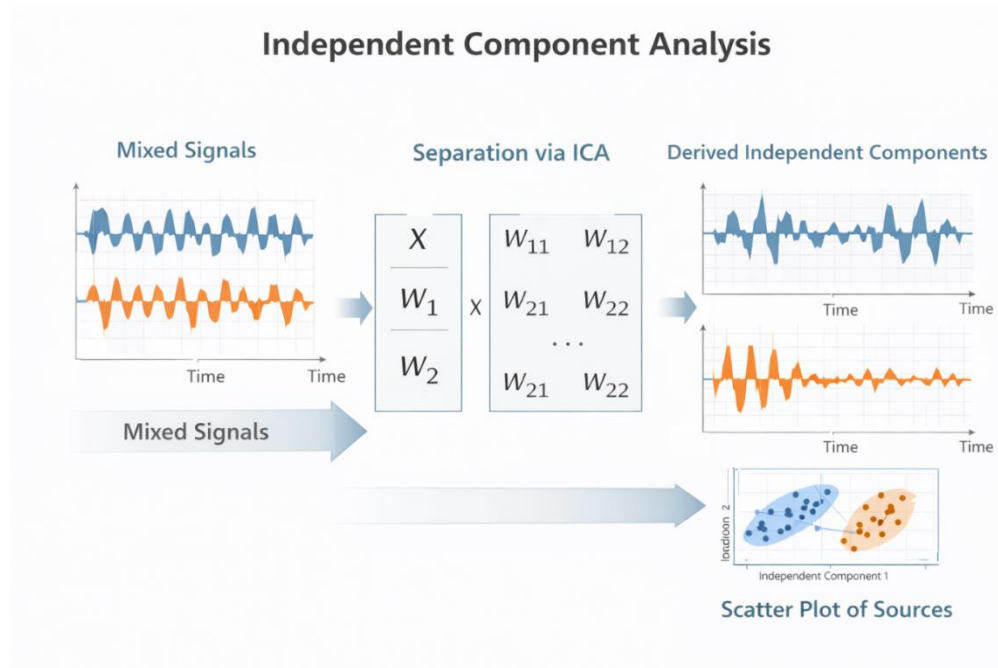


Figure 8 - Schematic representation of Independent Component Analysis (ICA) for blind source separation and dimensionality reduction.

On the left, observed signals (mixed signals) are shown as combinations of underlying independent sources, typically represented as time series. In the central panel, ICA estimates an unmixing matrix W that transforms the observed data matrix X into

statistically independent components, such that $\mathbf{S} = \mathbf{W}\mathbf{X}$. This transformation is achieved by maximizing statistical independence, often through measures of non-Gaussianity. On the right, the recovered independent components are displayed as separated signals, along with a scatter plot illustrating their statistical independence. The resulting components represent latent sources that are assumed to be non-Gaussian and mutually independent.

1. Conceptual Motivation

In many real-world scenarios, observed signals are mixtures of underlying independent sources. A classic example is the cocktail party problem, in which multiple audio signals are recorded simultaneously and must be separated.

Let $\mathbf{x} \in \mathbb{R}^d$ denote the observed data vector. ICA assumes:

$$\mathbf{x} = \mathbf{A}\mathbf{s} \tag{45}$$

where:

- $\mathbf{s} = (s_1, s_2, \dots, s_d)$ are latent independent components
- $\mathbf{A}$ is an unknown mixing matrix

The goal of ICA is to estimate both $\mathbf{A}$ and $\mathbf{s}$, given only observations $\mathbf{x}$.

2. Mathematical Formulation

ICA seeks an unmixing matrix $\mathbf{W}$ such that:

$$\mathbf{s} = \mathbf{W}\mathbf{x} \tag{46}$$

where the components s_i are as statistically independent as possible.

3. Independence and Non-Gaussianity

A key principle in ICA is that independence can be achieved by maximizing non-Gaussianity, based on the Central Limit Theorem: A sum of independent variables tends to be more Gaussian than the original variables. Thus, ICA identifies directions in which the projected data are maximally non-Gaussian.

4. Measures of Non-Gaussianity

Several criteria are used to quantify non-Gaussianity:

4.1 Kurtosis

$$\text{kurt}(y) = \mathbb{E}[y^4] - 3(\mathbb{E}[y^2])^2 \quad (47)$$

- Zero for Gaussian distributions
- Positive or negative for non-Gaussian signals

4.2 Negentropy

$$J(y) = H(y_{\text{gauss}}) - H(y) \quad (48)$$

where $H(\cdot)$ is entropy. Negentropy is always non-negative and zero only for Gaussian variables.

4.3 Contrast Functions

Approximations of negentropy are often used:

$$J(y) \approx (\mathbb{E}[G(y)] - \mathbb{E}[G(v)])^2 \quad (49)$$

where (v) is a Gaussian variable and $G(\cdot)$ is a nonlinear function.

5. Preprocessing Steps

Before applying ICA, data are typically preprocessed:

5.1 Centering

$$\mathbf{x} \leftarrow \mathbf{x} - \mathbb{E}[\mathbf{x}] \quad (50)$$

5.2 Whitening

$$\mathbf{x}_{\text{white}} = \mathbf{D}^{-1/2} \mathbf{E}^T \mathbf{x} \quad (51)$$

where $\mathbf{E}$ and $\mathbf{D}$ come from the eigen-decomposition of the covariance matrix.

Whitening ensures:

$$\mathbb{E}[\mathbf{x}_{\text{white}} \mathbf{x}_{\text{white}}^T] = \mathbf{I}$$

6. Algorithms for ICA

Several algorithms exist for solving the ICA problem:

6.1 FastICA

- Fixed-point iteration
- Maximizes non-Gaussianity efficiently

6.2 Infomax ICA

- Maximizes mutual information
- Based on neural network learning

6.3 JADE (Joint Approximate Diagonalization)

- Uses higher-order cumulants

7. Interpretation

ICA provides:

- Components that are statistically independent, not just uncorrelated
- A representation aligned with latent sources
- A decomposition driven by higher-order statistics

Compared to PCA:

- PCA captures variance
- ICA captures independence

8. Advantages

- Effective for blind source separation
- Captures non-Gaussian structure
- Provides interpretable latent variables
- Useful for signal decomposition

9. Limitations and Weaknesses

9.1 Independence Assumption

- Assumes true sources are independent
- Violations degrade performance

9.2 Non-Gaussianity Requirement

- At most one component can be Gaussian
- Otherwise, separation is not identifiable

9.3 Sensitivity to Noise

- Performance decreases with strong noise
- Noise may violate independence assumptions

9.4 Scaling and Permutation Ambiguity

- Order and scale of components are not identifiable

9.5 Computational Complexity

- Iterative algorithms may be expensive

9.6 Linear Mixing Assumption

- Assumes linear mixtures
- Cannot directly handle nonlinear mixtures

10. Comparison with Other Methods

Table 13 compares key dimensionality reduction methods based on their objectives, defining properties, and mapping characteristics. PCA and L1-PCA focus on variance-based representations, with L1-PCA offering improved robustness to outliers. Kernel PCA extends this concept to nonlinear structures through implicit mappings in high-dimensional feature spaces. In contrast, ICA shifts the objective from variance to statistical independence, leveraging higher-order statistics to extract meaningful latent sources. Dimensionality Reduction via Regression (DRR) further generalizes these approaches by modeling nonlinear dependencies with regression, enabling flexible, data-driven representations. Overall, the table highlights how different methods prioritize

distinct statistical principles—variance, robustness, independence, and functional relationships—guiding their applicability across diverse data scenarios.

Table 13 – Comparison of DR Methods

Method	Objective	Key Property	Mapping
PCA	Variance maximization	Uncorrelated components	Linear
L1-PCA	Robust variance	Outlier resistance	Linear
Kernel PCA	Nonlinear variance	Implicit mapping	Nonlinear
ICA	Independence	Higher-order statistics	Linear
DRR	Nonlinear decorrelation	Regression-based	Nonlinear

Therefore, ICA is unique in that it focuses on statistical independence rather than geometry or variance.

11. Applications

11.1 Signal Processing

- Audio source separation
- EEG/MEG signal analysis
- Noise removal

11.2 Chemometrics

- Decomposition of spectral mixtures
- Resolution of overlapping signals
- Feature extraction in spectroscopy

11.3 Image Processing

- Feature extraction
- Blind image separation
-

11.4 Finance

- Identification of latent factors in time series

11.5 Machine Learning

- Preprocessing step for classification
- Feature engineering

12. Final Remarks

Independent Component Analysis is a fundamental paradigm for dimensionality reduction that focuses on statistical independence rather than on preserving variance or distance. Its ability to separate latent sources makes it particularly valuable in signal processing and chemometrics. However, its reliance on strong assumptions—such as independence and non-Gaussianity—limits its applicability in more complex or nonlinear scenarios. In modern machine learning, ICA is often used alongside methods such as PCA, Kernel PCA, and Diffusion Maps to provide complementary insights into data structure [18–20].

2.2.1.2 DR Techniques Based on a Sparse Matrix

2.2.1.2.1 Hessian Locally Linear Embedding

Hessian Locally Linear Embedding (Hessian LLE or HLLE) is a nonlinear dimensionality reduction (DR) technique proposed by Donoho, David L., and Grimes, Carrie (2003) as an extension of the classical Locally Linear Embedding (LLE). The method is designed to recover low-dimensional manifold structure by explicitly incorporating second-order local geometry through estimation of the Hessian operator. Unlike standard LLE, which preserves first-order neighborhood relationships, HLLE aims to better capture the intrinsic curvature of the manifold [21–23].

Figure 9 illustrates the HLLE workflow, emphasizing its key distinction from first-order methods: the incorporation of curvature via local Hessian estimation. By modeling each neighborhood with a quadratic approximation, HLLE captures second-order geometric structure, enabling a more faithful embedding of smoothly curved manifolds compared to linear or first-order nonlinear techniques.

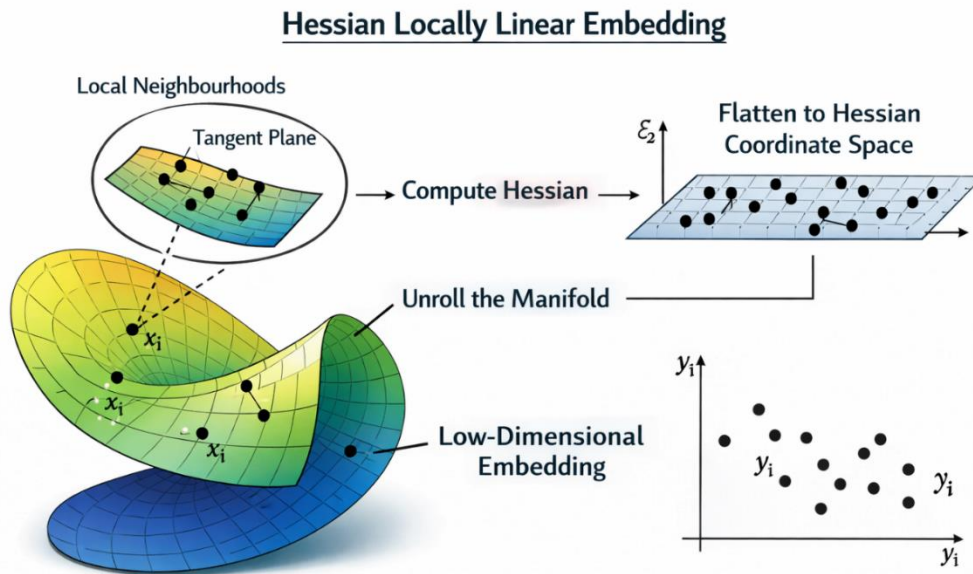


Figure 9 - Schematic representation of the Hessian Locally Linear Embedding (HLLE) algorithm.

Starting from a high-dimensional nonlinear manifold (bottom left), local neighborhoods around each data point x_i are identified. Within each neighborhood, a local tangent space is estimated to approximate the manifold geometry. The Hessian operator is then computed to capture second-order (curvature) information of the manifold. These local Hessian estimates are assembled into a global optimization problem, whose solution produces a low-dimensional embedding y_i that preserves the intrinsic curvature of the data. The final embedding (bottom right) represents an “unrolled” version of the original manifold, where nonlinear structures are preserved in a lower-dimensional space.

Table 14 summarizes the main characteristics of Hessian Locally Linear Embedding (HLLE) as a nonlinear manifold learning method that extends local linear approaches by incorporating second-order geometric information. The method relies on local neighborhood analysis to estimate tangent spaces and uses the Hessian operator to capture curvature, enabling a more faithful representation of the intrinsic manifold structure. Although HLLE provides improved unfolding of nonlinear data compared to first-order methods such as LLE, it involves solving an eigenvalue problem and requires careful selection of neighborhood size. The method is computationally demanding, sensitive to noise, and not inherently reconstructive, which may limit its scalability for large datasets.

Table 14 – Key Properties of Hessian Locally Linear Embedding (HLLE)

Property	Description
Method Type	Nonlinear dimensionality reduction
Theoretical Basis	Manifold learning using second-order (Hessian) geometry
Local Approximation	Local tangent space estimation via neighborhood analysis
Order of Approximation	Second-order (captures curvature)
Geometry Preserved	Local curvature and intrinsic manifold structure
Neighborhood Requirement	k-nearest neighbors (k-NN)
Mapping Type	Implicit (no explicit transformation function)
Reconstruction Ability	Not directly reconstructive
Optimization Problem	Eigenvalue problem based on Hessian functional minimization

Property	Description
Sensitivity to Noise	High (due to second-order derivative estimation)
Computational Cost	Moderate to high (depends on neighborhood size and data dimension)
Scalability	Limited for very large datasets
Dimensionality Constraint	Requires sufficient neighbors ($k >$ intrinsic dimension)
Advantages	Captures nonlinear curvature; improved manifold unfolding compared to LLE

1. Theoretical Foundations

Consider a dataset $\{x_i \in \mathbb{R}^D\}_{i=1}^n$ assumed to lie on or near a smooth d -dimensional manifold M , where $d \ll D$. The goal is to find embeddings $\{y_i \in \mathbb{R}^d\}$ that preserve local geometry.

HLLE is based on the assumption that, locally, the manifold can be approximated by a quadratic function. Thus, instead of preserving linear reconstruction weights (as in LLE), HLLE estimates the Hessian functional of the embedding coordinates.

2. Local Tangent Space Estimation

For each point x_i , define a neighborhood $N(i)$ of size k . A local tangent space is estimated using PCA:

$$\mathbf{X}_i = [\mathbf{x}_{i_1} - \mathbf{x}_i, \dots, \mathbf{x}_{i_k} - \mathbf{x}_i] \quad (52)$$

Perform singular value decomposition (SVD):

$$\mathbf{X}_i = \mathbf{U}_i \mathbf{\Sigma}_i \mathbf{V}_i^T \quad (53)$$

The first d columns of $\mathbf{U}_i$ define the local tangent coordinates:

$$\mathbf{Z}_i = \mathbf{U}_i^{(d)} \mathbf{\Sigma}_i^{(d)} \quad (54)$$

3. Hessian Estimation

The key innovation in HLLE is estimating the Hessian operator $\mathcal{H}$ of the manifold. Locally, the embedding coordinates $\mathbf{y}$ are assumed to satisfy:

$$\mathcal{H}(f) = \sum_{p,q} \frac{\delta^2 f}{\delta z_p \delta z_q} \quad (55)$$

HLLE constructs a set of basis functions over the local coordinates $\mathbf{Z}_i$, typically including quadratic terms:

$$\phi(z) = [z_1, \dots, z_d, z_1^2, z_1 z_2, \dots, z_d^2] \quad (56)$$

The Hessian estimator is derived by fitting these local quadratic functions and extracting the second-order coefficients. This leads to a local Hessian matrix $\mathbf{H}_i$, whose null space encodes directions of minimal curvature.

4. Global Embedding

The global embedding is obtained by assembling a sparse matrix $\mathbf{H}$ from all local Hessian estimators and solving:

$$\min_Y \sum_i \|\mathbf{H}_i \mathbf{Y}\|^2 \quad (57)$$

This reduces to an eigenvalue problem:

$$\mathbf{H}^T \mathbf{H} \mathbf{Y} = \lambda \mathbf{Y} \quad (58)$$

The embedding is given by the eigenvectors corresponding to the smallest non-zero eigenvalues (excluding trivial solutions).

5. Algorithm Summary

1. Identify k -nearest neighbors for each point
2. Estimate local tangent coordinates via PCA
3. Construct local Hessian estimators
4. Assemble a global sparse matrix
5. Solve the eigenvalue problem for embedding

6. Key Properties

- Nonlinear method: captures curved manifolds
- Second-order accuracy: incorporates curvature via Hessian
- Implicit mapping: no explicit transformation function
- Global consistency: derived from local Hessian constraints

7. Advantages

- Improved representation of curved manifolds compared to LLE
- Higher fidelity in geometric structure preservation, especially for smoothly varying manifolds
- Theoretically grounded in differential geometry
- Better handling of nonlinearity than first-order methods

8. Weaknesses and Limitations

Despite its theoretical elegance, HLLE presents several practical challenges:

8.1 Sensitivity to Noise

The estimation of second-order derivatives makes HLLE highly sensitive to noise.

Small perturbations in data can significantly distort the Hessian estimation.

8.2 Neighborhood Size Dependency

The choice of k is critical:

- Small k : unstable Hessian estimation
- Large k : violation of the local linearity assumption

8.3 Computational Complexity

HLLE requires:

- PCA for each neighborhood
- Construction of large sparse matrices
- Eigen decomposition of $n \times n$ matrices

This leads to scalability issues for large datasets.

8.4 Poor Performance on Non-Smooth Manifolds

HLLE assumes smooth manifolds. It performs poorly when:

- The manifold has discontinuities
- Data lies on intersecting manifolds
- There are sharp edges or corners

8.5 No Explicit Mapping (Out-of-Sample Problem)

Like LLE, HLLE lacks a parametric mapping:

- Cannot easily project new data points
- Requires extensions such as Nyström approximation

9. Applications

HLLE has been applied in several domains where manifold structure is smooth and nonlinear:

- Image processing: face recognition, pose estimation
- Computer vision: object tracking, motion analysis
- Bioinformatics: gene expression data visualization
- Spectroscopy and chemometrics: nonlinear structure in spectral datasets
- Signal processing: feature extraction from nonlinear signals

In chemometric contexts—aligned with your work in vibrational spectroscopy—HLLE can be particularly useful for capturing nonlinear relationships in Raman or NIR datasets, especially when curvature effects are relevant.

10. Comparison with Related Methods

Table 15 summarizes the fundamental differences between HLLE and related dimensionality reduction methods with respect to geometric representation and mapping properties. While LLE captures only first-order local linear relationships, HLLE extends

this framework by incorporating second-order information, thereby preserving local curvature. In contrast, Isomap focuses on global geometry via geodesic distances, whereas PCA represents global variance via a linear, explicit mapping with reconstruction capability. Notably, both LLE and HLLE rely on implicit mappings and do not support direct reconstruction, highlighting a trade-off between geometric fidelity and interpretability across methods.

Table 15 – Comparison of the HLLE with Related Methods

Method	Order	Geometry Captured	Mapping	Reconstruction
LLE	First-order	Local linear	Implicit	No
HLLE	Second-order	Local curvature	Implicit	No
Isomap	Global	Geodesic distances	Implicit	No
PCA	Linear	Global variance	Explicit	Yes

HLLE can be interpreted as a second-order extension of LLE, bridging local linear methods and curvature-aware manifold learning.

11. Final Remarks

Hessian LLE represents an important theoretical advancement in manifold learning by incorporating second-order differential structure. However, its practical use is often limited by noise sensitivity and computational cost. In modern applications, it is frequently outperformed by more robust methods such as t-SNE and UMAP, although HLLE remains valuable for understanding curvature-preserving embeddings in controlled settings [21–23].

2.2.2 Nonconvex DR Techniques

2.2.2.1 Dimensionality Reduction Approach for Landscapes

1. Introduction

Dimensionality Reduction by Learning a Landscape (DrL) is a nonlinear dimensionality reduction technique that preserves both local neighborhoods and global structure by modeling the data as a physical system evolving on an energy landscape. Originally inspired by force-directed graph drawing algorithms and statistical physics, DrL treats data points as interacting particles whose positions are iteratively optimized to minimize an energy function [24–25].

Figure 10 illustrates how DrL transforms high-dimensional data into a low-dimensional representation by modeling the system as an energy landscape. Through iterative optimization driven by attractive and repulsive forces, the method produces an embedding that preserves both local neighborhoods and global structure.

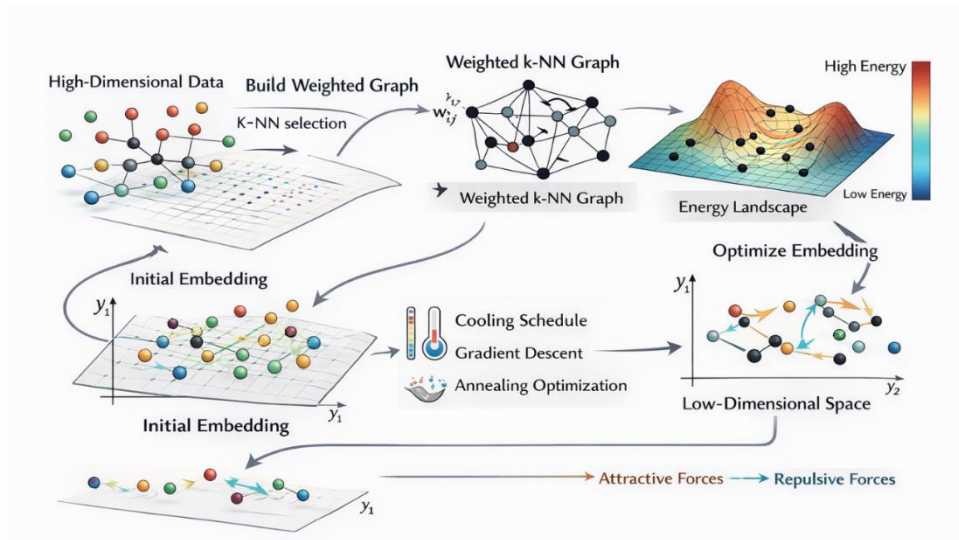


Figure 10 - Schematic representation of the Dimensionality Reduction by Learning a Landscape (DrL) method.

The process begins with high-dimensional data, from which a weighted k-nearest neighbors (k-NN) graph is constructed to encode pairwise similarities. This graph defines an energy landscape where attractive forces preserve local relationships and repulsive forces prevent point collapse. An initial low-dimensional embedding is iteratively optimized using gradient descent combined with a cooling (annealing) schedule, allowing the system to converge toward a stable configuration. The final low-dimensional embedding reflects a balance between local cohesion and global separation, preserving the intrinsic structure of the original data.

Unlike classical methods such as PCA or manifold-based approaches like Isomap and LLE, DrL provides a flexible framework that balances attraction and repulsion forces, enabling robust visualization of complex, high-dimensional datasets.

2. Conceptual Framework

The central idea of DrL is to construct a graph representation of the dataset and then embed this graph into a low-dimensional space by minimizing an energy function.

Given a dataset:

$$\mathbf{X} = \{x_1, x_2, \dots, x_n\}, \quad x_i \in \mathbb{R}^D \quad (59)$$

DrL first defines a weighted graph:

- Nodes represent data points
- Edges represent similarity (e.g., k-nearest neighbors)

The goal is to find low-dimensional embeddings:

$$\mathbf{Y} = \{y_1, y_2, \dots, y_n\}, \quad y_i \in \mathbb{R}^d, \quad d \ll D \quad (60)$$

such that the energy of the system is minimized.

3. Mathematical Formulation

3.1 Energy Function

DrL is based on an energy function combining attractive and repulsive forces:

$$E(Y) = \sum_{(i,j) \in \epsilon} w_{ij} \|y_i - y_j\|^2 + \lambda \sum_{i \neq j} \frac{1}{\|y_i - y_j\|^\alpha} \quad (61)$$

where:

- w_{ij} = similarity weight between points (i) and (j)
- ϵ = set of edges (typically k-NN graph)
- First term = attractive force (preserves local structure)
- Second term = repulsive force (prevents collapse)
- λ = regularization parameter
- α = controls repulsion decay

3.2 Optimization

The embedding is obtained by minimizing:

$$Y^* = \arg \min_Y E(Y) \quad (62)$$

This is typically solved using iterative optimization, such as:

- Gradient descent
- Force-directed simulation
- Multi-scale annealing strategies

The update rule can be expressed as:

$$y_i^{(t+1)} = y_i^t - \eta \frac{\delta E}{\delta y_i} \quad (63)$$

where η is the learning rate.

3.3 Force Interpretation

The gradient of the energy defines forces:

- Attractive force:

$$F_{ij}^{(att)} = -w_{ij}(y_i - y_j) \quad (64)$$

- Repulsive force:

$$F_{ij}^{(rep)} = \lambda \frac{y_i - y_j}{\|y_i - y_j\|^{\alpha+2}} \quad (65)$$

Thus, DrL can be interpreted as a dynamic system of interacting particles, similar to:

- Force-directed graph drawing
- Energy minimization

4. Algorithmic Steps

1. Construct k -NN graph from data
2. Assign edge weights w_{ij}
3. Initialize low-dimensional coordinates (random or PCA)
4. Iteratively update positions using force balance
5. Apply cooling/annealing to stabilize the solution
6. Output final embedding

5. Advantages

- Captures both local and global structure
- Flexible energy formulation
- Suitable for complex nonlinear manifolds
- Intuitive physical interpretation
- Produces visually meaningful embeddings

6. Weaknesses and Limitations

Despite its flexibility, DrL presents several challenges:

6.1 Computational Complexity

- Repulsive term scales as: $O(n^2)$
- Requires approximations (e.g., Barnes–Hut methods)

6.2 Parameter Sensitivity

- Sensitive to:
 - k - neighbors
 - λ

(66)

$$\circ \alpha \tag{67}$$

- Poor tuning may distort the structure

6.3 Lack of Explicit Mapping

- No function:

$$f: \mathbb{R}^D \rightarrow \mathbb{R}^d \tag{68}$$

- Cannot directly embed new points (out-of-sample problem)

6.4 Convergence Issues

- Non-convex optimization
- Risk of local minima

6.5 Noise Sensitivity

- Graph construction may be unstable under noise
- Affects neighborhood reliability

7. Comparison with Related Methods

Table 16 compares Dimensionality Reduction by Learning a Landscape (DrL) with classical dimensionality reduction methods, highlighting their underlying principles and structural preservation capabilities. While PCA focuses on global variance and LLE emphasizes local reconstruction, Isomap attempts to balance both through geodesic distances. In contrast, DrL stands out by simultaneously preserving local and global structures through an energy-landscape optimization framework, offering a more flexible and comprehensive representation of complex nonlinear data, albeit with an implicit mapping.

Table 16 – Comparison of DrL with related methods.

Method	Key Idea	Global Structure	Local Structure	Mapping
PCA	Variance maximization	Yes	No	Explicit
LLE	Linear reconstruction	No	Yes	Implicit
Isomap	Geodesic distances	Yes	Moderate	Implicit
DrL	Energy landscape optimization	Yes	Yes	Implicit

8. Applications

DrL has been widely used in:

8.1 Data Visualization

- High-dimensional datasets (omics, spectroscopy)
- Cluster discovery

8.2 Network Analysis

- Graph embedding
- Social networks

8.3 Bioinformatics

- Gene expression analysis
- Protein interaction networks

8.4 Chemometrics

Given your research context, DrL is particularly useful for:

- Spectroscopic data (Raman, NIR, IR)
- Exploratory analysis of complex chemical systems
- Nonlinear pattern discovery

8.5 Machine Learning Pipelines

- Preprocessing for classification/regression
- Feature extraction

9. Final Remarks

Dimensionality Reduction by Learning a Landscape (DrL) represents a powerful and flexible framework for nonlinear embedding, bridging ideas from graph theory, physics, and optimization. By explicitly modeling attraction–repulsion dynamics, DrL achieves a balanced preservation of local and global structures. However, its computational cost, sensitivity to parameters, and lack of explicit mapping limit its scalability and practical deployment in large-scale or real-time applications [24–25].

2.2.2.2 Kamada–Kawai Algorithm for Dimensionality Reduction and Graph Layout

1. Introduction

The Kamada–Kawai (KK) algorithm is a classical force-directed graph layout method that can also be interpreted as a nonlinear dimensionality reduction technique. Originally proposed for graph visualization, the method seeks to position nodes in a low-dimensional space such that the geometric distances between nodes reflect their graph-theoretic distances as faithfully as possible.

Unlike purely statistical approaches such as PCA or manifold learning techniques such as Isomap, the KK algorithm is grounded in a spring-embedding model, where nodes are connected by virtual springs whose natural lengths correspond to shortest-path distances in the graph. This formulation allows the method to preserve global structure while maintaining a coherent spatial representation [26].

Figure 11 illustrates how the Kamada–Kawai algorithm transforms a graph into a low-dimensional representation by modeling it as a system of springs. By minimizing an

energy function derived from graph-theoretic distances, the method produces an embedding that preserves global relationships while maintaining a coherent and interpretable spatial layout.

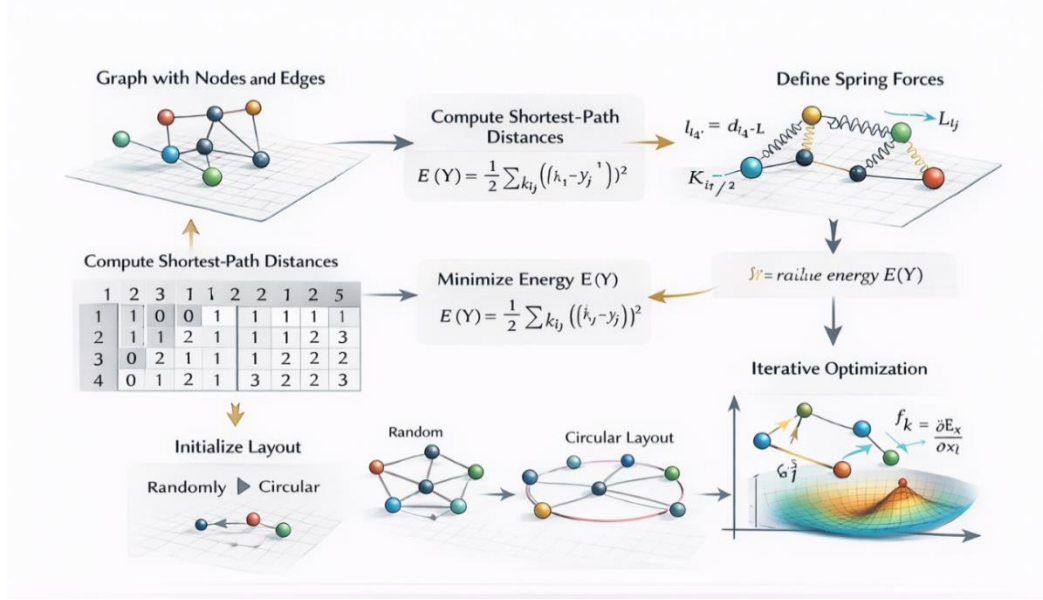


Figure 11 - Schematic illustration of the Kamada–Kawai (KK) algorithm for graph layout and dimensionality reduction.

The process begins with a graph composed of nodes and edges, from which shortest-path distances between all node pairs are computed. These distances define the ideal spring lengths l_{ij} , while spring constants k_{ij} are assigned to control the strength of interactions. An initial layout (random or circular) is then generated, and the system iteratively minimizes an energy function that measures the discrepancy between embedded distances and ideal graph distances. Through successive updates based on gradient or Newton–Raphson optimization, the nodes converge to a configuration where the global structure of the graph is preserved in a low-dimensional space.

Table 17 summarizes the key properties of the Kamada–Kawai algorithm, highlighting its foundation as a force-directed, graph-based dimensionality reduction method. By modeling nodes as a system of springs and minimizing an energy function

based on shortest-path distances, the algorithm effectively preserves the global structure of the data. The table outlines its theoretical basis, computational characteristics, and practical considerations, emphasizing both its strengths in producing interpretable layouts and its limitations related to computational cost, scalability, and sensitivity to initialization.

Table 17 – Key Properties of the Kamada–Kawai Algorithm

Property	Description
Method Type	Graph-based nonlinear dimensionality reduction / force-directed layout
Theoretical Basis	Spring-embedding model with energy minimization
Core Principle	Preserve graph-theoretic (shortest-path) distances
Geometry Preserved	Global structure of the graph
Distance Metric	Shortest-path distances (geodesic on graph)
Energy Function	Sum of squared differences between embedded and ideal distances
Optimization Method	Newton–Raphson or gradient-based iterative minimization
Mapping Type	Implicit (no explicit transformation function)
Reconstruction Ability	Not reconstructive
Computational Complexity	High $O(n^3)$ for shortest paths + $O(n^2)$ optimization)
Scalability	Limited to small and medium-sized datasets
Sensitivity to Initialization	Moderate to high (may converge to local minima)
Sensitivity to Noise	Moderate (depends on graph construction)
Dimensionality	Typically 2D or 3D embedding
Advantages	Good global structure preservation; intuitive physical interpretation

Property	Description
Limitations	Computationally expensive; poor scalability; weak local structure preservation

2. Conceptual Framework

Let a graph be defined as:

$$G = (V, E) \quad (69)$$

where:

- $V = \{1, 2, \dots, n\}$ is the set of nodes
- E is the set of edges

The goal is to find an embedding:

$$\mathbf{Y} = \{y_1, y_2, \dots, y_n\}, \quad y_i \in \mathbb{R}^d \quad (70)$$

such that distances in the embedding space approximate graph-theoretic distances.

3. Mathematical Formulation

3.1 Graph-Theoretic Distances

First, compute the shortest-path distances:

$$d_{ij} = \text{shortest path between nodes } i \text{ and } j \quad (71)$$

using algorithms such as Dijkstra or Floyd–Warshall.

3.2 Spring Model

Each pair of nodes i, j is connected by a virtual spring with:

- Ideal length:

$$l_{ij} = L \cdot d_{ij} \quad (72)$$

- Spring constant:

$$k_{ij} = \frac{K}{d_{ij}^2} \quad (73)$$

where:

- L is a scaling factor
- K is a global stiffness parameter

3.3 Energy Function

The KK algorithm minimizes the total system energy:

$$E(Y) = \frac{1}{2} \sum_{i \neq j} k_{ij} (\|y_i - y_j\| - l_{ij})^2 \quad (74)$$

This energy function penalizes discrepancies between embedded distances and ideal graph distances.

3.4 Optimization

The optimal embedding is obtained by:

$$Y^* = \arg \min_Y E(Y) \quad (75)$$

The optimization is typically performed using:

- Newton–Raphson method
- Gradient-based iterative updates

For each node (i), the partial derivatives are:

$$\frac{\delta E}{\delta x_i}, \frac{\delta E}{\delta y_i} \quad (76)$$

and the node with the largest gradient is iteratively adjusted.

4. Interpretation

The KK algorithm can be interpreted as:

- A multidimensional scaling (MDS)-like method, since it preserves pairwise distances
- A force-directed system, where:
 - Springs enforce global structure
 - Energy minimization yields equilibrium

It is closely related to stress minimization:

$$Stress = \sum_{i < j} w_{ij} (\|y_i - y_j\| - d_{ij})^2 \quad (77)$$

5. Algorithmic Steps

1. Compute shortest-path distances d_{ij}
2. Define l_{ij} and k_{ij}
3. Initialize node positions (random or circular layout)
4. Compute energy gradients
5. Select node with maximum energy contribution
6. Update its position using Newton–Raphson
7. Repeat until convergence

6. Advantages

- Preserves global graph structure
- Produces visually interpretable layouts
- Theoretically grounded in energy minimization
- Suitable for small to medium-sized graphs

7. Weaknesses and Limitations

7.1 Computational Complexity

- Requires all-pairs shortest paths: $\mathcal{O}(n^3)$
- Energy minimization: $\mathcal{O}(n^2)$

7.2 Scalability Issues

- Not suitable for large graphs
- Becomes slow for $(n > 10^3)$

7.3 Sensitivity to Initialization

- Different initial layouts may lead to different local minima

7.4 Local Minima

- Non-convex optimization problem
- No guarantee of global optimum

7.5 Overemphasis on Global Structure

- May distort local neighborhoods
- Less effective than LLE or t-SNE for local structure preservation

8. Comparison with Related Methods

Table 18 compares the Kamada–Kawai algorithm with classical dimensionality reduction methods in terms of their underlying principles and structural preservation. While PCA and MDS emphasize global structure by preserving variance and distances, and LLE focuses on local relationships, Isomap provides a balance by preserving geodesic distances. In contrast, Kamada–Kawai employs a spring energy minimization framework that effectively preserves global structure but offers limited local fidelity, positioning it as a visually intuitive yet computationally demanding approach with an implicit mapping.

Table 18 – Comparison of the Kamada-Kawai with related methods.

Method	Principle	Global	Local	Mapping
PCA	Variance maximization	Yes	No	Explicit
MDS	Distance preservation	Yes	Moderate	Explicit
Isomap	Geodesic distances	Yes	Moderate	Implicit
LLE	Local reconstruction	No	Yes	Implicit
Kamada–Kawai	Spring energy minimization	Yes	Limited	Implicit

9. Applications

9.1 Graph Visualization

- Social networks
- Biological networks
- Communication systems

9.2 Chemometrics and Spectroscopy

- Visualization of similarity graphs
- Cluster exploration in spectral datasets

9.3 Bioinformatics

- Protein interaction networks
- Gene regulatory networks

9.4 Machine Learning

- Visualization of high-dimensional similarity graphs
- Preprocessing for clustering

9.5 Network Science

- Structural analysis of complex systems

10. Final Remarks

The Kamada–Kawai algorithm represents a foundational approach in graph-based dimensionality reduction, offering a physically interpretable framework for embedding data. While it excels at preserving global relationships, its computational cost and sensitivity to local minima limit its application to smaller datasets. Nevertheless, it remains an important reference method and a conceptual bridge between graph theory, optimization, and modern embedding techniques [26].

2.2.2.3 Fruchterman–Reingold

1. Introduction and Conceptual Overview

The Fruchterman–Reingold algorithm is a force-directed method for graph layout and nonlinear dimensionality reduction, originally proposed by Thomas M. J. Fruchterman and Edward M. Reingold in 1991. It belongs to the class of energy-based

embedding techniques, in which the nodes of a graph are positioned in a low-dimensional space (typically 2D or 3D) to reflect the graph's underlying structure.

The method models the graph as a physical system in which nodes repel each other like charged particles, while edges act as springs that attract connected nodes. The equilibrium configuration of these competing forces yields an embedding that reveals clusters, connectivity patterns, and global organization [27–29].

Figure 12 demonstrates how the Fruchterman–Reingold algorithm transforms an initial random graph layout into a structured representation by iteratively balancing attractive and repulsive forces. The resulting embedding highlights relational patterns and network topology, making the method particularly effective for visualizing complex graph structures.

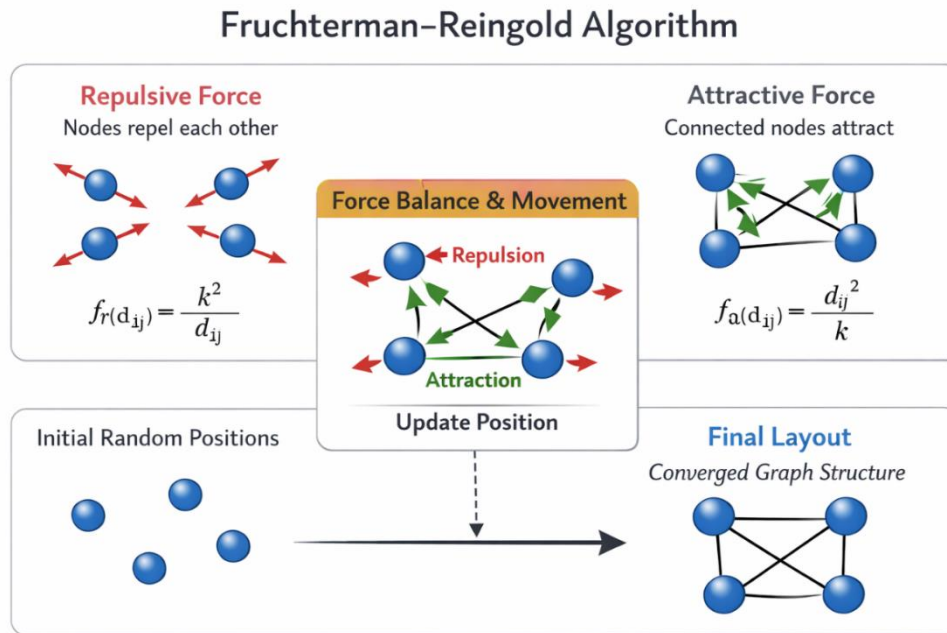


Figure 12 – Schematic illustration of the Fruchterman–Reingold force-directed layout algorithm.

Starting from an initial random placement of the nodes (left), attractive forces along the edges and repulsive forces between every pair of nodes are iteratively balanced until the system reaches a minimum-energy configuration (right), in which clusters and connectivity patterns become visually apparent.

Initially, nodes are randomly distributed in the embedding space (Initial Random Positions). Two types of forces are then iteratively applied:

- (i) repulsive forces between all node pairs, proportional to $\frac{k^2}{d_{ij}}$, which push nodes apart to avoid overlap, and
- (ii) attractive forces along graph edges, proportional to $\frac{d_{ij}^2}{k}$, which pull connected nodes together.

The central panel (*Force Balance & Movement*) shows how these competing forces are combined to update node positions at each iteration. Through a cooling process that limits movement over time, the system converges to a stable configuration. The final result (*Final Layout*) reveals the intrinsic structure of the graph, where clusters and connectivity patterns are visually emphasized.

Table 19 summarizes the main characteristics of the Fruchterman–Reingold algorithm, highlighting its physical force-based formulation, implicit mapping, and suitability for visualizing graph structures rather than preserving precise metric relationships.

Table 19 — Key Properties of the Fruchterman–Reingold Algorithm

Property	Description
Method Type	Graph-based nonlinear dimensionality reduction / force-directed layout
Theoretical Basis	Physical analogy using attractive (spring) and repulsive (electrostatic) forces
Core Principle	Balance between node repulsion and edge-based attraction
Geometry Preserved	Global graph structure and connectivity patterns
Distance Concept	Implicit; based on graph topology rather than explicit metric distances
Forces Modeled	Attractive force $f_a(d_{ij}) = \frac{d_{ij}^2}{k}$; Repulsive force $f_r(d_{ij}) = \frac{k^2}{d_{ij}}$
Energy Interpretation	Minimization of a global energy function combining attraction and repulsion
Optimization Method	Iterative force updates with cooling schedule (simulated annealing-like)
Mapping Type	Implicit (no explicit transformation function)
Reconstruction Ability	Not reconstructive (non-invertible)
Scalability	Limited; $O(n^2)$ per iteration (can be improved with approximations)
Sensitivity to Initialization	High; different initial layouts may lead to different solutions
Parameter Dependence	Requires tuning of optimal distance k , temperature, and iteration count
Robustness to Noise	Moderate; preserves connectivity but may distort metric relationships
Typical Output Dimension	2D or 3D visualization space

Property	Description
Main Applications	Network visualization, cluster detection, social and biological graph analysis
Interpretability	High (visual interpretation of clusters, hubs, and graph structure)

2. Mathematical Formulation

Let $G = (V, E)$ be a graph with $n = |V|$ vertices. Each node $i \in V$ is assigned a position vector $x_i \in \mathbb{R}^d$, typically with $d = 2$.

A key parameter of the model is the optimal pairwise distance:

$$k = C \sqrt{\frac{A}{n}}$$

where:

- A is the area of the layout space,
- C is a constant $C \approx 1$.

2.1 Attractive Force (Edges)

For two nodes i and j connected by an edge:

$$f_a(d_{ij}) = \frac{d_{ij}^2}{k}$$

2.2 Repulsive Force (All Node Pairs)

For all pairs of nodes:

$$f_r(d_{ij}) = \frac{k^2}{d_{ij}}$$

where:

$$d_{ij} = \|x_i - x_j\|$$

2.3 Force Update

The displacement of node i is computed as:

$$\Delta x_i = \sum_{j \neq i} \frac{x_i - x_j}{d_{ij}} f_r(d_{ij}) + \sum_{j \in \mathcal{N}(i)} \frac{x_j - x_i}{d_{ij}} f_a(d_{ij})$$

A temperature parameter T is used to limit movement and ensure convergence:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \min(\|\Delta \mathbf{x}_i\|, T) \cdot \frac{\Delta \mathbf{x}_i}{\|\Delta \mathbf{x}_i\|}$$

The temperature is gradually reduced (cooling schedule), analogous to simulated annealing.

3. Energy Interpretation

The algorithm can be interpreted as minimizing an energy function of the form:

$$E = \sum_{(i,j) \in E} \frac{d_{ij}^2}{k} - \sum_{i \neq j} k^2 \ln(d_{ij})$$

This formulation reflects a balance between:

- Attractive energy: pulls connected nodes together,
- Repulsive energy: spreads nodes apart globally.

4. Algorithmic Procedure

1. Initialize node positions randomly
2. Compute pairwise repulsive forces
3. Compute attractive forces along edges
4. Update node positions with temperature constraint
5. Reduce temperature (cooling)
6. Repeat until convergence or maximum iterations

5. Properties

- Method Type: Graph-based nonlinear embedding
- Mapping: Implicit (no explicit transformation function)

- Geometry Preserved: Global graph structure and connectivity
- Distance Concept: Graph topology rather than Euclidean or geodesic distances
- Reconstruction: Not invertible
- Interpretability: Visual and structural (cluster separation, hubs, communities)

6. Weaknesses and Limitations

Despite its intuitive formulation, the Fruchterman–Reingold algorithm has several limitations:

6.1 Computational Complexity

- The naive implementation requires $O(n^2)$ computations per iteration due to pairwise repulsion.
- This makes it unsuitable for large-scale datasets without approximations (e.g., Barnes–Hut).

6.2 Local Minima

- The energy landscape is highly non-convex.
- The final layout depends on initialization and may converge to suboptimal configurations.

6.3 Lack of Metric Preservation

- Unlike MDS or Isomap, it does not explicitly preserve distances (Euclidean or geodesic).
- The embedding is qualitative rather than quantitatively faithful.

6.4 Parameter Sensitivity

- The choice of k , temperature schedule, and number of iterations affects the final result.
- No universal parameter setting exists.

6.5 Scalability Issues

- Performance degrades with dense graphs or very large node sets.
- Requires optimization strategies (multilevel layouts, GPU acceleration).

7. Applications

The Fruchterman–Reingold algorithm is widely used in scenarios where relational structure is more important than metric accuracy:

7.1 Network Visualization

- Social networks
- Biological networks (protein–protein interaction, gene networks)
- Communication and transportation networks

7.2 Data Exploration

- Visualization of similarity graphs (e.g., k-NN graphs)
- Cluster detection and community structure analysis

7.3 Chemometrics and Scientific Data

- Visualization of similarity relationships between spectra
- Graph-based representation of high-dimensional chemical datasets

7.4 Machine Learning Pipelines

- Preprocessing for graph-based learning
- Visualization of embeddings derived from manifold learning methods

8. Extensions and Variants

Several improvements have been proposed:

- Barnes–Hut approximation → reduces repulsion computation to ($O(n \log n)$)
- Multilevel methods → hierarchical graph coarsening for scalability
- ForceAtlas (Gephi) → continuous and adaptive force-directed layout
- Kamada–Kawai algorithm → energy based on shortest-path distances

9. Final Remarks

The Fruchterman–Reingold algorithm remains one of the most influential force-directed methods due to its simplicity and intuitive physical analogy. While it does not provide a rigorous geometric embedding in the classical sense, it excels in revealing structural patterns, making it a powerful tool for exploratory data analysis and network visualization [27–29].

2.2.2.4 Nonmetric Multidimensional Scaling

1. Introduction and Conceptual Framework

Nonmetric Multidimensional Scaling (nMDS) is a nonlinear dimensionality reduction technique that represents high-dimensional data in a low-dimensional space while preserving the rank order of dissimilarities rather than their absolute magnitudes. Unlike classical or metric MDS, which attempts to preserve actual distances, nMDS focuses on maintaining the **monotonic relationship** between dissimilarities and distances in the embedding space.

This characteristic makes nMDS particularly suitable for datasets in which only ordinal information is reliable, or in which the scale of dissimilarities is unknown or non-Euclidean [30–31].

Figure 13 illustrates the fundamental principle of Nonmetric Multidimensional Scaling (nMDS): transforming high-dimensional dissimilarity data into a low-dimensional representation while preserving the relative ordering of pairwise relationships. Unlike metric approaches, nMDS does not attempt to reproduce exact distances; instead, it ensures that objects with smaller dissimilarities remain closer in the embedded space than those with larger dissimilarities. This rank-based preservation

enables meaningful visualization of complex, nonlinear structures, even when the original distance scale is unknown or non-Euclidean.

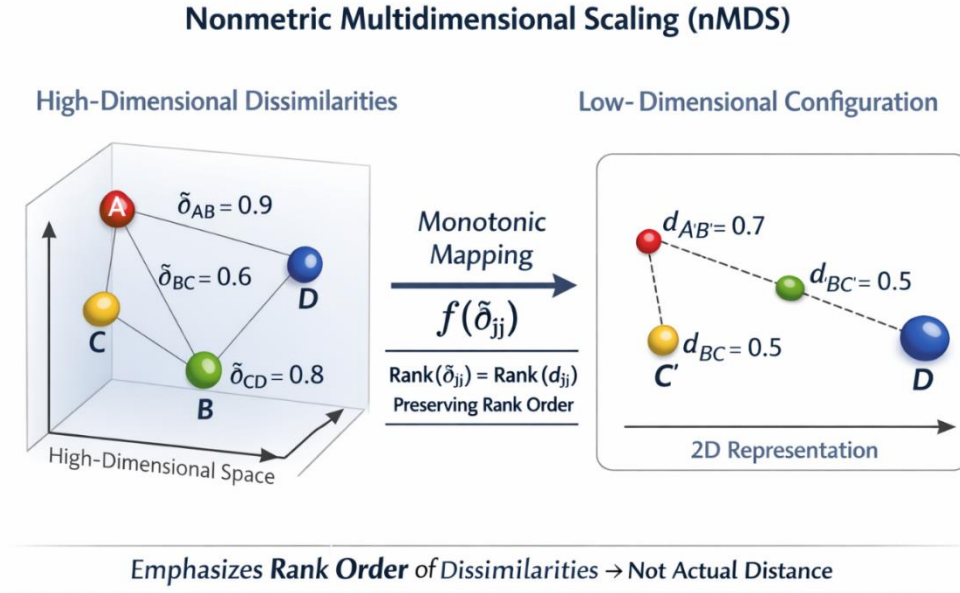


Figure 13 – Schematic illustration of the Nonmetric Multidimensional Scaling (nMDS)

The left panel represents objects (A–D) in a high-dimensional space, where pairwise dissimilarities (δ_{ij}) are defined and quantified. These dissimilarities do not necessarily correspond to Euclidean distances but encode relative relationships among the objects. The central step illustrates the application of a monotonic transformation $f(\delta_{ij})$, which preserves the rank order of dissimilarities rather than their absolute values. The right panel shows the resulting configuration in a low-dimensional (2D) space, where the distances (d_{ij}) approximate the transformed dissimilarities such that:

$$\text{Rank}(\delta_{ij}) = \text{Rank}(d_{ij})$$

Dashed lines indicate distances in the embedded space, emphasizing that nMDS prioritizes ordinal consistency over metric fidelity.

The key properties of Nonmetric Multidimensional Scaling (nMDS) are summarized in Table 20, highlighting its conceptual foundation, computational characteristics, and practical implications. The method is fundamentally based on the preservation of rank-order relationships among dissimilarities through a monotonic transformation, distinguishing it from metric approaches that aim to preserve absolute distances. The table outlines essential aspects such as the stress-based optimization framework, implicit mapping nature, and its robustness to non-metric data, while also emphasizing limitations related to scalability, interpretability, and lack of an explicit projection function. Together, these properties provide a comprehensive overview of nMDS as a flexible yet computationally demanding technique primarily suited for exploratory data analysis and visualization.

Table 20 – Key Properties of Nonmetric Multidimensional Scaling (nMDS)

Property	Description
Method Type	Nonlinear dimensionality reduction/ordination technique
Theoretical Basis	Rank-order preservation via monotonic transformation of dissimilarities
Core Principle	Preserve the ordering of pairwise dissimilarities rather than their exact values
Input Data	Dissimilarity (or distance) matrix (metric or non-metric)
Distance Representation	Euclidean distances in the low-dimensional space
Mapping Type	Implicit (no explicit transformation function)
Transformation Function	Monotonic function $f(\delta_{ij})$ estimated via isotonic regression
Objective Function	Minimization of stress (e.g., Kruskal's Stress-1)
Optimization Method	Iterative (e.g., SMACOF, gradient-based methods)
Geometry Preserved	Ordinal (rank-based) structure of dissimilarities

Property	Description
Local Structure Preservation	Moderate to high (depends on data and configuration)
Global Structure Preservation	Moderate (less emphasis than metric MDS or Isomap)
Reconstruction Ability	Not reconstructive (no inverse mapping)
Scalability	Limited (computational cost increases with $O(n^2)$)
Sensitivity to Initialization	High (may converge to local minima)
Robustness to Non-metric Data	High (designed for ordinal data)
Interpretability of Axes	Low (axes have no direct physical meaning)
Typical Output	Low-dimensional configuration (usually 2D or 3D)
Key Advantage	Flexibility with arbitrary dissimilarities and robustness to scale distortions
Main Limitation	No explicit mapping for new data (out-of-sample problem)

2. Mathematical Formulation

Let δ_{ij} denote the dissimilarity between objects i and j , and let d_{ij} denote the Euclidean distance between their corresponding points in the low-dimensional embedding.

The goal of nMDS is to find a configuration of points such that:

$$d_{ij} \approx f(\delta_{ij}) \quad (78)$$

where $f(\cdot)$ is a monotonic (non-decreasing) function, typically estimated using isotonic regression.

3. Stress Function

The quality of the embedding is quantified by a stress function that measures the discrepancy between the distances d_{ij} and the transformed dissimilarities $f(\delta_{ij})$.

One of the most widely used formulations is Kruskal's Stress-1:

$$Stress = \sqrt{\frac{\sum_{i < j} (d_{ij} - f(\delta_{ij}))^2}{\sum_{i < j} d_{ij}^2}}$$

Alternatively, a simpler (unnormalized) stress can be defined as:

$$Stress = \sum_{i < j} (d_{ij} - f(\delta_{ij}))^2 \quad (79)$$

The optimization problem consists of minimizing the stress with respect to the configuration of points in the low-dimensional space.

4. Optimization Procedure

The solution is typically obtained through an iterative algorithm:

1. Initialization: Start with a random or informed configuration.
2. Distance Computation: Compute distances d_{ij} .
3. Monotonic Regression: Fit $f(\delta_{ij})$ to d_{ij} while preserving rank order.
4. Update Configuration: Adjust point positions to reduce stress (e.g., via gradient descent or majorization).
5. Iteration: Repeat until convergence.

A widely used algorithm is **SMACOF (Scaling by Majorizing a Complicated Function)**, which ensures monotonic convergence of the stress function.

5. Interpretation of Stress Values

Typical guidelines for interpreting stress (Kruskal) can be seen in

Table 21:

Table 21 – Association of the stress value and interpretation

Stress Value	Interpretation
< 0.05	Excellent representation
0.05–0.10	Good

Stress Value	Interpretation
0.10–0.20	Fair
> 0.20	Poor

6. Key Characteristics

- Nonlinear method
- Rank-based (ordinal preservation)
- Implicit mapping (no explicit transformation function)
- Applicable to arbitrary dissimilarity measures
- Not reconstructive

7. Advantages

- Flexibility: Works with non-Euclidean and non-metric dissimilarities.
- Robustness to scale distortions: Only rank order matters.
- Effective visualization tool for complex datasets.
- Widely applicable across disciplines.

8. Weaknesses and Limitations

Despite its flexibility, nMDS presents several limitations:

- Computational complexity: Typically $O(n^2)$ or higher due to pairwise distances.
- Local minima: Optimization is non-convex and sensitive to initialization.
- No explicit mapping: Cannot directly project new data points (out-of-sample problem).
- Interpretability issues: Axes have no direct physical meaning.
- Scalability limitations: Not suitable for very large datasets without approximation.
- Dependence on dissimilarity quality: Poor input dissimilarities lead to misleading embeddings.

9. Applications

nMDS has been widely applied in several domains:

- Chemometrics: Visualization of spectral similarities (e.g., NIR, Raman, UV–Vis data).
- Ecology: Analysis of species composition and biodiversity patterns.
- Bioinformatics: Gene expression and omics data exploration.
- Social sciences: Perceptual mapping and preference analysis.
- Text mining and NLP: Visualization of document similarities.
- Materials science: Structure-property relationships in complex systems.

In chemometrics, nMDS is particularly useful when distance measures (e.g., correlation-based distances) do not satisfy metric properties.

10. Comparison with Related Methods

Table 22 compares Nonmetric Multidimensional Scaling (nMDS) with other widely used dimensionality reduction techniques, highlighting their underlying principles, the type of structure they preserve, and the nature of their mappings. This comparison emphasizes the distinctive characteristic of nMDS—its focus on preserving the rank order of dissimilarities—compared with methods such as PCA and Metric MDS, which rely on metric properties, and Isomap, which captures global manifold geometry. The table provides a concise framework for understanding the conceptual differences and selecting appropriate methods based on the analytical objective.

Table 22 – Comparison of the nMDS with related methods.

Method	Principle	Preserves	Mapping
PCA	Variance maximization	Global structure	Explicit

Method	Principle	Preserves	Mapping
Metric MDS	Distance preservation	Metric distances	Explicit
Isomap	Geodesic distances	Global manifold	Implicit
nMDS	Rank order of dissimilarities	Ordinal relationships	Implicit

11. Final Remarks

Nonmetric Multidimensional Scaling occupies a unique position among dimensionality reduction techniques by emphasizing ordinal fidelity over metric accuracy. This makes it particularly powerful in exploratory data analysis when precise distance relationships are unreliable or unknown. However, its lack of scalability and absence of an explicit mapping function limit its use in predictive pipelines, confining it primarily to visualization and exploratory tasks [30–31].

2.2.2.5 *t*-Stochastic Neighbor Embedding

1. Introduction and Conceptual Framework

t-Distributed Stochastic Neighbor Embedding (*t*-SNE) is a nonlinear dimensionality reduction technique specifically designed for visualizing high-dimensional data in low-dimensional spaces (typically 2D or 3D). Originally proposed by Laurens van der Maaten and Geoffrey Hinton, *t*-SNE extends the Stochastic Neighbor Embedding (SNE) framework by addressing the crowding problem and improving the representation of local structures.

The central idea of *t*-SNE is to convert pairwise distances between data points into probability distributions that represent similarities, and then minimize the divergence between these distributions in high- and low-dimensional spaces [32–33].

Figure 14 provides a schematic representation of the t-SNE algorithm, highlighting its probabilistic framework for dimensionality reduction. By transforming pairwise distances into probability distributions and minimizing their divergence, t-SNE effectively captures local relationships in high-dimensional data. The use of a heavy-tailed distribution in the embedding space mitigates crowding effects and enhances cluster separation, making the method particularly suitable for visualizing complex, nonlinear data structures.

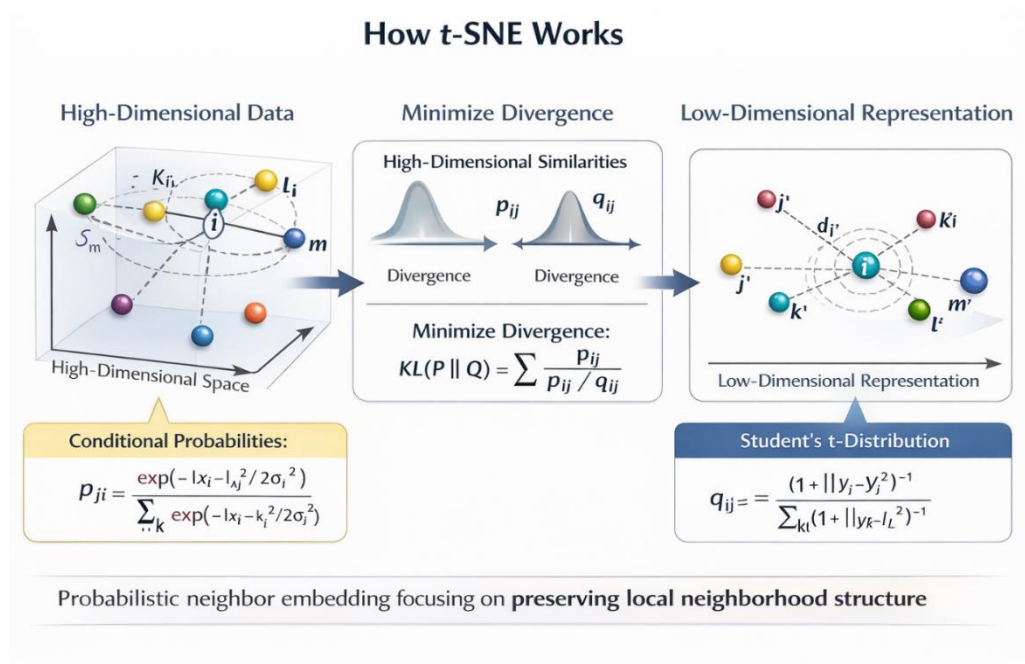


Figure 14 - Illustration of the t-Distributed Stochastic Neighbor Embedding (t-SNE) algorithm.

This figure depicts the t-SNE workflow in three stages. The left panel shows high-dimensional data, where pairwise relationships between points are modeled as conditional probabilities p_{ij} using Gaussian distributions. The central panel illustrates the objective of the method: minimizing the Kullback–Leibler (KL) divergence between the high-dimensional similarity distribution P and the low-dimensional distribution Q . The right panel presents the resulting low-dimensional embedding, where similarities q_{ij} are modeled using a Student's t-distribution, allowing better separation of clusters. The

process emphasizes the preservation of local neighborhood structure, ensuring that nearby points in the original space remain close in the embedding.

Table 23 summarizes the key properties of t-Distributed Stochastic Neighbor Embedding (t-SNE), providing an overview of its theoretical foundation, computational characteristics, and practical behavior. The table highlights the method's probabilistic nature, its emphasis on preserving local neighborhood structures, and the role of the Kullback–Leibler divergence in guiding the optimization. Additionally, it outlines key aspects such as scalability, sensitivity to hyperparameters, and the lack of an explicit mapping function, providing a concise reference for understanding the strengths and limitations of t-SNE in exploratory data analysis and visualization tasks.

Table 23 – Key Properties of t-Distributed Stochastic Neighbor Embedding (t-SNE)

Property	Description
Method Type	Nonlinear dimensionality reduction / probabilistic embedding
Theoretical Basis	Probability distribution matching via Kullback–Leibler divergence
Core Principle	Preserve local neighborhood relationships through similarity probabilities
Input Data	High-dimensional feature matrix or pairwise distances
High-Dimensional Similarity	Gaussian-based conditional probabilities p_{ij}
Low-Dimensional Similarity	Student's t-distribution (heavy-tailed) q_{ij}
Objective Function	Minimization of KL divergence $KL(P Q)$
Optimization Method	Gradient descent with momentum and early exaggeration
Mapping Type	Implicit (non-parametric, no explicit transformation)

Property	Description
Geometry Preserved	Local structure (neighbor relationships)
Global Structure Preservation	Limited (inter-cluster distances may be distorted)
Reconstruction Ability	Not reconstructive (no inverse mapping)
Scalability	Moderate to low (typically $O(n^2)$, improved with approximations)
Sensitivity to Hyperparameters	High (perplexity, learning rate, initialization)
Determinism	Non-deterministic (results vary across runs)
Robustness to Noise	Moderate to high for local structures
Interpretability of Axes	Low (axes lack direct physical meaning)
Typical Output	2D or 3D embedding for visualization
Key Advantage	Excellent cluster visualization and local structure preservation
Main Limitation	Poor global structure representation and lack of out-of-sample mapping

2. Mathematical Formulation

2.1 High-Dimensional Similarities

Given a dataset $\{x_1, x_2, \dots, x_n\}$, the similarity between two points x_i and x_j in the high-dimensional space is defined as a conditional probability:

$$p_{j|i} = \frac{\exp(-\|x_i - x_j\|^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)} \quad (80)$$

where σ_i is a bandwidth parameter determined by the perplexity, which controls the effective number of neighbors.

The joint probability is then symmetrized:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} \quad (81)$$

2.2 Low-Dimensional Similarities

In the low-dimensional space, points y_i and y_j are modeled using a Student's t-distribution with one degree of freedom (equivalent to a Cauchy distribution):

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq i} (1 + \|y_k - y_i\|^2)^{-1}} \quad (82)$$

This heavy-tailed distribution alleviates the crowding problem by allowing moderate distances in high-dimensional space to be represented by larger distances in the embedding.

2.3 Objective Function

The optimal embedding is obtained by minimizing the Kullback–Leibler (KL) divergence between the two distributions:

$$KL(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (83)$$

This objective emphasizes the preservation of local neighborhood structure, as errors in modeling nearby points are heavily penalized.

3. Optimization Procedure

The minimization of the KL divergence is performed using gradient descent, with updates of the form:

$$\frac{\delta KL}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij})(y_i - y_j) (1 + \|y_i - y_j\|^2)^{-1} \quad (84)$$

Several practical enhancements are commonly used:

- Early exaggeration: amplifies p_{ij} to improve cluster separation in early iterations
- Momentum: accelerates convergence
- Adaptive learning rates

4. Key Characteristics

- Nonlinear and probabilistic method
- Focus on local structure preservation

- Implicit mapping (non-parametric)
- Heavy-tailed distribution in embedding space
- Primarily visualization-oriented

5. Advantages

- Excellent visualization of clusters in complex datasets
- Captures nonlinear relationships effectively
- Robust to high-dimensional noise
- Widely applicable across domains

t-SNE is particularly effective for revealing hidden patterns, clusters, or manifold structures that linear methods such as PCA miss.

6. Weaknesses and Limitations

Despite its popularity, t-SNE has several important limitations:

- Lack of global structure preservation: distances between clusters may not be meaningful
- Non-deterministic results: different runs may yield different embeddings
- Computational cost: typically $O(n^2)$, though approximations (e.g., Barnes–Hut t-SNE) reduce complexity
- Hyperparameter sensitivity: perplexity, learning rate, and initialization strongly influence results
- No explicit mapping: cannot directly embed new data points (out-of-sample problem)
- Cluster size distortion: cluster areas do not reflect true data density
- Potential for misinterpretation: visual artifacts may suggest a structure that does not exist

7. Applications

t-SNE has become a standard tool in exploratory data analysis across multiple fields:

- Chemometrics: visualization of spectral datasets (e.g., NIR, Raman, mass spectrometry)
- Bioinformatics: gene expression and single-cell RNA-seq analysis
- Computer vision: feature space visualization in deep learning
- Natural language processing: word embeddings and document similarity
- Materials science: clustering of compounds and structure-property analysis
- Social sciences: behavioral and survey data visualization

In chemometric applications, t-SNE is especially useful for identifying hidden class structures in high-dimensional spectral data.

8. Comparison with Related Methods

Table 24 presents a comparative overview of t-Distributed Stochastic Neighbor Embedding (t-SNE) alongside other widely used dimensionality reduction techniques, emphasizing their underlying principles, the type of structure preserved, and the nature of their mappings. This comparison highlights the distinctive feature of t-SNE—its probabilistic formulation aimed at preserving local neighborhood relationships—compared with methods such as PCA and MDS, which focus on global structure, and Isomap and nMDS, which capture manifold geometry and ordinal relationships, respectively. The table provides a concise framework for selecting appropriate methods based on the analytical objective and the type of data structure of interest.

Table 24 – Comparison of the *t*-SNE with related methods

Method	Principle	Structure Preserved	Mapping
PCA	Variance maximization	Global	Explicit
MDS	Distance preservation	Global	Explicit
Isomap	Geodesic distances	Global manifold	Implicit
nMDS	Rank-order preservation	Ordinal	Implicit
t-SNE	Probability distribution matching	Local	Implicit

9. Final Remarks

t-SNE represents a powerful and widely adopted approach for visualizing complex high-dimensional data, particularly when local relationships are of primary interest. Its probabilistic formulation and use of heavy-tailed distributions enable effective cluster separation and the revelation of intricate structures. However, its limitations—especially regarding global structure distortion and lack of reproducibility—require careful interpretation. Consequently, t-SNE is best used as an exploratory visualization tool rather than a quantitative analytical method [32–33].

2.3 Summary of the Dimensionality Reduction Methods: An Overview

Table 25 summarizes the main characteristics of a broad set of dimensionality reduction (DR) techniques, encompassing linear, nonlinear, manifold-based, spectral, probabilistic, and graph-based approaches. The comparison highlights key aspects, including the methodological category, the type of structure preserved (e.g., variance, distances, topology, or local similarities), and the invertibility of each technique. This overview provides a unified framework for understanding the fundamental differences

among DR methods and supports informed selection based on specific analytical goals, such as visualization, reconstruction, or machine learning integration.

Table 25 – Characteristics of DR techniques

Technique	Category	Preserves	Invertible*
PCA	Linear (Global)	Variance/Covariance	Yes
PCA L1	Linear (Global)	Variance (L1 robust)	Approx.
Fast ICA	Linear (Global)	Statistical independence	Approx.
MDS	Linear (Global)	Euclidean distances	Yes
KPCA	Nonlinear (Kernel)	Variance in kernel space	No
nMDS	Nonlinear (Global)	Rank order of distances	No
Isomap	Nonlinear (Manifold)	Geodesic distances	No
HLLE	Nonlinear (Manifold)	Local curvature	No
Diffusion Maps	Nonlinear (Spectral)	Diffusion geometry	No
DrL	Nonlinear (Graph/Probabilistic)	Neighborhood probabilities	No
t-SNE	Nonlinear (Probabilistic)	Local similarities	No
DRR	Nonlinear (Invertible)	Nonlinear redundancy reduction	Yes
Fruchterman-Reingold	Graph-based (Force-directed)	Graph-topology	No
Kamada-Kawai	Graph-based (Force-directed)	Graph-theoretic distances	No

*Note: A dimensionality reduction method is said to be invertible if there exists a well-defined inverse mapping f^{-1} such that the original high-dimensional data $\mathbf{x} \in \mathbb{R}^D$ can be exactly or approximately reconstructed from its low-dimensional embedding $\mathbf{z} \in \mathbb{R}^d$, where $d < D$:

$$\mathbf{z} = f(\mathbf{x}), \quad \mathbf{x} \approx f^{-1}(\mathbf{z}) \quad (85)$$

Invertible DR methods preserve sufficient information to enable reconstruction of the original data, typically minimizing a reconstruction error (e.g., RMSE). In contrast, non-invertible methods perform a many-to-one mapping that irreversibly compresses the data, prioritizing structural or topological preservation over reconstructability.

Figure 15 illustrates the conceptual distinction between invertible and non-invertible dimensionality reduction techniques. Invertible methods preserve sufficient information to reconstruct the original data, making them suitable for tasks such as compression and feature engineering. In contrast, non-invertible methods prioritize preserving structural or neighborhood relationships, often at the expense of information, limiting their use to exploratory analysis and visualization. This distinction is essential for selecting appropriate DR techniques depending on whether reconstruction or structural representation is the primary objective.

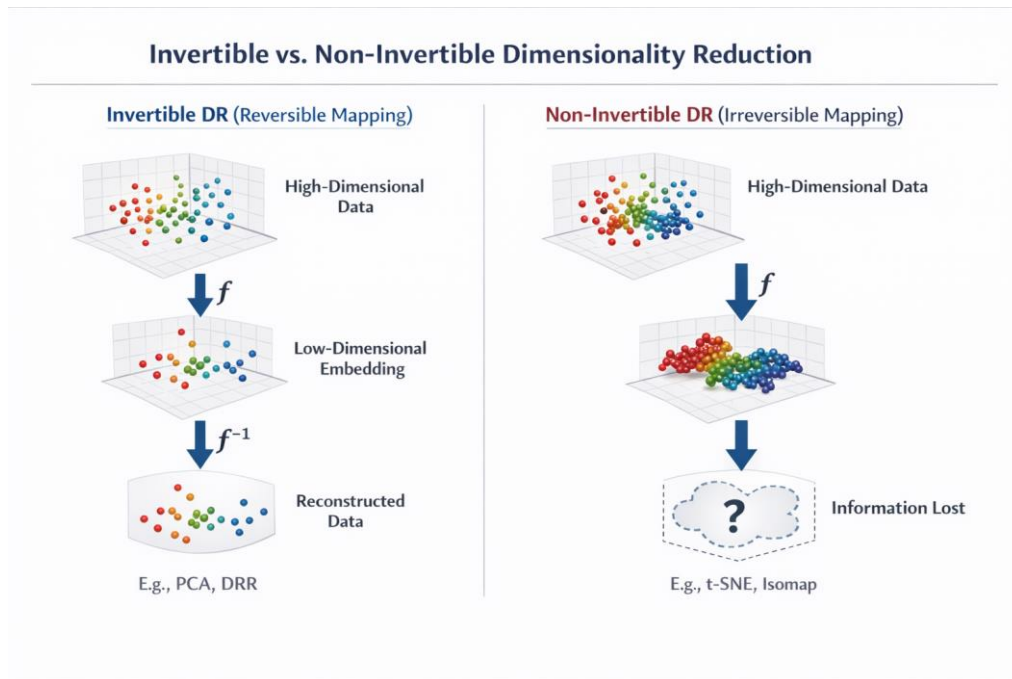


Figure 15 - Invertible vs. Non-Invertible Dimensionality Reduction.

Figure 15 contrasts two fundamental classes of dimensionality reduction (DR) methods based on their ability to reconstruct the original data. The left panel illustrates an invertible DR, where a mapping f projects high-dimensional data into a lower-dimensional space, and an inverse mapping f^{-1} enables reconstruction of the original data with minimal information loss. Typical examples include PCA and DRR. The right panel depicts non-invertible DR, where the forward mapping produces a low-dimensional embedding, but no inverse transformation exists, leading to irreversible information loss. Representative methods include t-SNE and Isomap. The diagram highlights the fundamental trade-off between reconstruction capability and flexibility in capturing complex structures.

2.4 Quality Criteria

Unsupervised learning does not require the specification of classes or a target variable. In this case, the chosen algorithm arranges the input data, for example, by placing them into clusters or into a lower-dimensional representation. Unlike in supervised learning, there is no natural way to directly measure the quality of any output or to compare two techniques using objective measures, such as modeling efficiency or classification error. The reason is that each technique optimizes a different error function, and it would be unfair to compare nonlinear MDS and kernel PCA using either the recovered variance or the KL divergence [34–36].

Generally, dimensionality reduction (DR) quality metrics can be grouped into three main categories, together with a small number of special cases. The first category comprises compactness-oriented metrics, which evaluate how well the relative arrangement of data points in the metric space is preserved after projection. These metrics quantify the preservation of local density and cluster structure by comparing distances

among neighboring samples in the original and embedded spaces. Representative examples include Q_{local} , AUC R(NX), Trustworthiness, Precision, and the C-Measure, all of which primarily assess the consistency of intra- and intercluster relationships.

The second category comprises connectivity-oriented metrics that focus on preserving neighborhood relationships and the global organization of the data manifold. These metrics assess whether the dataset's overall topology and distance ordering remain consistent after dimensionality reduction. Typical examples include Q_{global} , Continuity, Discontinuity, Recall, Kendall's τ , the Gabriel Classification Error (GCE), and cophenetic correlation. Unlike compactness metrics, these indicators emphasize the structural coherence of the embedding rather than the density of local clusters.

The third category comprises non-generalizable metrics that depend on specific algorithmic assumptions, reconstruction models, or class labels and therefore do not capture purely geometric properties of the embedding. Metrics in this group include RMSE Reconstruction and k-NN classification accuracy, which evaluate reconstruction fidelity or classification performance rather than intrinsic manifold structure.

This classification helps explain why different DR algorithm families tend to perform well according to different evaluation criteria. Locality-preserving algorithms, such as *t-SNE*, *HLLS*, and *diffusion maps*, typically achieve high values for compactness-related metrics, including Q_{local} , Trustworthiness, and k-NN accuracy, reflecting their strong ability to preserve neighborhood relationships and cluster structure. However, these methods may sacrifice global geometric consistency. In contrast, globality-preserving algorithms, including *Isomap*, *classical MDS*, and *kernel PCA*, tend to perform better on connectivity-oriented metrics such as Q_{global} , AUC R(NX), and cophenetic correlation, indicating stronger preservation of large-scale manifold structure.

A third group comprises linear and structurally stable algorithms, such as *PCA*, *PCA-LI*, *FastICA*, and *DRR*, which often yield low reconstruction errors, high continuity values, and interpretable low-dimensional representations; in particular, *PCA* and *DRR* maintain the mapping's full invertibility. Finally, graph-based layout algorithms, including *Fruchterman–Reingold*, *Kamada–Kawai*, and *DrL*, are primarily designed for visualization tasks. While these methods can generate visually well-separated embeddings, their preservation of global geometry and quantitative structural fidelity may vary substantially.

Together, this framework clarifies how the behavior of dimensionality reduction algorithms is closely linked to the structural properties emphasized by different quality metrics, highlighting the importance of objective-dependent evaluation when selecting DR techniques [38]. Given this, a series of independent estimators is available to determine the quality of a lower-dimensional embedding, as described below.

2.4.1 Quality Measurements: Compactness-Oriented

2.4.1.1 Local Quality, Q_{local}

Local quality (Q_{local}) measures the degree to which the nearest neighbors of each data point in the original high-dimensional space remain neighbors after dimensionality reduction. Formally, it evaluates the preservation of local structure, meaning that if two samples are close in the original space, they should remain close in the embedded space. The local quality can be expressed as the fraction of neighbors preserved:

$$Q_{local} = \frac{1}{nk} \sum_{i=1}^n |N_k(i) \cap N'_k(i)| \quad (86)$$

where:

$$N_k(i) = \text{set of } k \text{ nearest neighbors of point } i \text{ in the original space} \quad (87)$$

$$N'_k(i) = \text{set of } k \text{ nearest neighbors of point } i \text{ in the embedded space} \quad (88)$$

n = number of samples

k = neighborhood size

Values typically range between $0 \rightarrow$ no local structure preserved and $1 \rightarrow$ perfect preservation of nearest neighbors [34].

2.4.1.2 AUC $R(NX)$

The AUC R_{NX} metric is the Area Under the Curve of the $R_{NX}(k)$ function, where $R_{NX}(k)$ quantifies the agreement between the k -nearest neighbors in the original space and those in the embedded space, corrected for random agreement.

The corrected neighborhood preservation index is

$$R_{NX}(k) = \frac{(N-1)Q_{NX}(k) - (k)}{N-1-k} \quad (89)$$

Where:

$Q_{NX}(k)$ represents the overlap of k -nearest neighbors. The correction removes the effect of random agreement.

The AUC R_{NX} is computed as

$$AUC_{R_{NX}} = \frac{1}{K} \sum_{k=1}^K R_{NX}(k) \quad (90)$$

where K is typically chosen as $N - 2$ or a practical subset of neighborhood sizes.

The AUC $R(NX)$ metric provides a quantitative measure of how well a dimensionality reduction method preserves neighborhood relationships from the original high-dimensional space in the embedded low-dimensional representation. When AUC $R(NX)$ is approximately 0, the embedding behaves similarly to a random mapping, indicating that the neighborhood structure of the data is not preserved. Values between 0.1 and 0.3 suggest weak neighborhood preservation, meaning that only a limited portion of the local relationships among samples is maintained. When the metric lies between 0.3

and 0.6, the embedding demonstrates moderate preservation of the data structure, indicating that a reasonable fraction of the original neighborhood information is retained. Values greater than 0.6 indicate strong neighborhood preservation, reflecting that the dimensionality reduction method effectively maintains the local structure of the data in the embedded space [34].

2.4.1.3 Trustworthiness

Trustworthiness measures the extent to which neighbors in the low-dimensional representation are also neighbors in the original space. If two points appear close in the embedding but were far apart in the original data, the embedding is considered less trustworthy. Thus, Trustworthiness penalizes intrusions, i.e., points that become neighbors in the reduced space even though they were not originally.

Trustworthiness is defined as

$$T(k) = 1 - \frac{2}{Nk(2N-3k-1)} \sum_{i=1}^N \sum_{j \in U_k(i)} (r(i, j) - k) \quad (91)$$

Where:

N = number of samples

k = neighborhood size

$U_k(i)$ = set of points that are among the k -nearest neighbors of i in the embedding but not among the k -nearest neighbors in the original space.

$r(i, j)$ = rank of point j in the ordered list of neighbors of i in the original space.

The Trustworthiness metric evaluates how well a dimensionality reduction method preserves local neighborhood relationships when projecting high-dimensional data into a lower-dimensional space. Values of Trustworthiness close to 1.0 indicate that the embedding accurately maintains the local neighborhood structure, meaning that points that are neighbors in the original space remain neighbors in the embedded representation.

Values in the range of 0.80 to 0.95 suggest moderate preservation of neighborhood relationships, indicating that most local structures are retained, although some distortions may occur. When Trustworthiness values fall below 0.8, the embedding exhibits significant neighborhood distortion, meaning that many points that appear close in the low-dimensional representation were not neighbors in the original high-dimensional space [34, 39].

2.4.1.4 Precision

Precision is a classification metric that measures the fraction of predicted positive instances that are actually positive, reflecting the reliability of positive predictions made by a machine learning model.

Precision is defined as:

$$\text{Precision} = \frac{TP}{TP+FP} \quad (92)$$

where:

TP (True Positives): positive instances correctly classified as positive.

FP (False Positives): negative instances incorrectly classified as positive.

The precision metric measures the proportion of correctly predicted positive instances among all instances that were predicted as positive by a classification model. A Precision value of 1.0 indicates that all predicted positive cases are correct, meaning the model does not produce false positive predictions. Values between 0.8 and 0.9 indicate high reliability of positive predictions, suggesting that most of the instances classified as positive truly belong to the positive class. When precision falls below 0.5, the model generates many false positives, indicating that a substantial proportion of predicted positive cases are incorrect [34, 40–41].

2.4.1.5 C-Measure

The C-Measure quantifies the consistency between the rankings of pairwise distances in the original space and in the embedded space using the co-ranking matrix. It evaluates the degree to which local and global neighborhood relationships are preserved simultaneously after dimensionality reduction. The measure is derived from the co-ranking framework, which analyzes how the ranks of pairs of points change after projection.

A co-ranking matrix Q is constructed such that

Each entry counts how often a pair of points has rank k in the original space and rank l in the embedding. The C-Measure evaluates how much of the mass of the matrix lies near the diagonal, indicating that the ranks are preserved.

The C Measure evaluates how well a dimensionality reduction method preserves the neighborhood structure of the data when mapping it from a high-dimensional space to a lower-dimensional embedding. Values of the C Measure close to 1.0 indicate strong preservation of the neighborhood structure, meaning that the relationships among nearby points in the original space are largely maintained in the embedded representation. Values between 0.5 and 0.8 suggest moderate preservation, indicating that a reasonable portion of the neighborhood relationships is retained, although some distortions may occur. When the C Measure is below 0.5, the embedding presents substantial distortion, meaning that many of the original neighborhood relationships are not preserved in the low-dimensional representation [34].

2.4.2 Quality Measurements: Connectivity-Oriented

2.4.2.1 Global Quality, Q_{global}

Global Quality (Q_{global}) is a dimensionality-reduction evaluation metric that quantifies how well large-scale relationships and global geometry of the original dataset

are preserved in the low-dimensional embedding. In contrast to local quality (Q_{local}), which focuses on nearest neighbors, Q_{global} evaluates whether points that are far apart in the original space remain far apart in the embedded space. Thus, Q_{global} reflects a DR method's ability to preserve the global arrangement of clusters, manifolds, or trends in the data.

Q_{local} is typically computed by comparing the k -nearest-neighbor sets in the original space and in the embedding space. For each point i :

- Determine the k nearest neighbors in the high-dimensional space.
- Determine the k nearest neighbors in the low-dimensional embedding.
- Compute the overlap between these neighbor sets.

A general form can be written as:

where:

N = number of data points

$N_k^{\text{high}}(i)$ = set of k -nearest neighbors of point i in the original space

$N_k^{\text{low}}(i)$ = set of k -nearest neighbors in the embedded space.

The Q_{global} metric evaluates how well a dimensionality reduction method preserves the global structure of the data when projecting it from a high-dimensional space to a lower-dimensional representation. Values of Q_{global} close to 1.0 indicate excellent preservation of global relationships, meaning that the overall arrangement and relative distances among data points in the original space are well maintained in the embedding. Values between 0.6 and 0.8 suggest moderate preservation of the global structure, indicating that the data's general organization is reasonably preserved, although some distortions may occur. When Q_{global} values are below 0.5, the embedding exhibits strong distortion of global relationships, implying that the dataset's large-scale structure is poorly preserved in the reduced-dimensional space [34].

2.4.2.2 Continuity

Continuity quantifies the extent to which the dimensionality reduction method fails to preserve original neighbors, penalizing situations where true neighbors in the high-dimensional space are mapped far away in the embedded space. In other words, it evaluates missing neighbors: points that were close in the original data but are not preserved as neighbors after projection. Continuity is therefore complementary to trustworthiness, whereas this metric penalizes the true neighbors that disappear after projection, the second one penalizes false neighbors introduced by the embedding.

Continuity is defined as:

$$C(k) = 1 - \frac{2}{Nk(2N-3k-1)} \sum_{i=1}^N \sum_{j \in V_k(i)} (r'(i, j) - k) \quad (93)$$

where:

N = number of samples

k = neighborhood size

$V_k(i)$ = set of points that are among the k -nearest neighbors of i in the original space, but not among the k -nearest neighbors in the embedding.

$r'(i, j)$ = rank of point j relative to point i in the embedded space.

The continuity metric evaluates the extent to which neighbors in the original high-dimensional space remain neighbors after dimensionality reduction. Values of continuity close to 1.0 indicate that the original neighborhood relationships are well preserved in the embedding, meaning that points that are close in the high-dimensional space tend to remain close in the low-dimensional representation. Values in the range 0.80 to 0.95 suggest moderate preservation of neighborhood relationships, indicating that most of the original local structures are maintained, although some neighbors may be displaced during embedding. When continuity values fall below 0.8, the embedding exhibits a

significant loss of neighborhood relationships, meaning that many points that were originally neighbors are no longer close in the reduced-dimensional space [34].

2.4.2.3 Discontinuity

Discontinuity is the complement of continuity: it quantifies how often points that were neighbors in the original high-dimensional space cease to be neighbors after projection. In other words, the embedding becomes discontinuous with respect to the original topology, because points that were close together are torn apart. It is obtained directly as $\text{Discontinuity} = 1 - \text{Continuity}$, and it therefore penalizes missing neighbors, whereas Trustworthiness penalizes the false neighbors introduced by the embedding.

The discontinuity metric evaluates the extent to which neighborhood relationships are disrupted when data are projected from a high-dimensional space into a lower-dimensional embedding. Values of discontinuity close to 0 indicate that the embedding preserves the neighborhood structure well, meaning that points that are close in the original space remain close in the reduced representation. Values in the range of 0.1 to 0.3 indicate moderate distortions, suggesting that some neighborhood relationships are altered during the dimensionality reduction process. When Discontinuity values are greater than 0.3, the embedding exhibits strong neighborhood distortions, indicating that many points that were neighbors in the original high-dimensional space are no longer close in the low-dimensional representation [34].

2.4.2.4 Recall

Recall quantifies how effectively a classifier detects positive cases. Thus, recall evaluates the model's ability to avoid false negatives. Recall is defined as:

$$\text{Recall} = \frac{TP}{TP+FN} \quad (94)$$

where:

TP (True Positives): positive instances correctly classified as positive.

FN (False Negatives): positive instances incorrectly classified as negative.

Recall is a classification performance metric that measures the proportion of actual positive instances that are correctly identified by a machine learning model, reflecting the model's ability to detect positive cases. The recall metric measures the proportion of actual positive instances that a classification model correctly identifies. A recall value of 1.0 indicates that all positive instances are detected, meaning the model successfully identifies every case belonging to the positive class. Values between 0.8 and 0.9 indicate a high detection rate, suggesting that most positive instances are correctly recognized by the model. When recall falls below 0.5, the model misses many positive instances, indicating that a substantial portion of the true positives are not detected [34, 40–41].

2.4.2.5 Kendall's Tau

Kendall's τ is a rank correlation metric that measures agreement between two rankings by comparing the number of concordant and discordant pairs and is widely used in machine learning to evaluate ranking consistency and structural preservation. Thus, Kendall's Tau measures the association between two ranked variables by comparing the number of concordant and discordant pairs of observations.

Consider two rankings X and Y for n observations.

For each pair of observations (i, j) :

A concordant pair (C) occurs when the order of the pair is the same in both rankings.

A discordant pair (D) occurs when the order of the pair is reversed.

Kendall's τ is defined as:

$$\tau = \frac{C-D}{\frac{1}{2}n(n-1)} \quad (95)$$

where:

C = number of concordant pairs

D = number of discordant pairs

n = number of observations.

The denominator corresponds to the total number of possible pairs.

The Kendall τ coefficient measures the degree of agreement between two rankings by evaluating the correspondence between ordered pairs of observations. A Kendall τ value of +1 indicates perfect agreement between the rankings, meaning that all pairs of observations are ordered in the same way in both rankings. A value of 0 indicates no correlation, suggesting that the two rankings are independent and do not share a consistent ordering pattern. A value of -1 represents perfect inverse ranking, meaning that the order of all pairs of observations is completely reversed between the two rankings [42].

2.4.2.6 Cophenetic Correlation

The cophenetic correlation coefficient measures the correlation between the original pairwise distances of observations and the cophenetic distances obtained from a hierarchical clustering dendrogram. The cophenetic distance between two observations is defined as the height in the dendrogram at which they first merge into the same cluster. Thus, the cophenetic correlation quantifies how well the hierarchical clustering structure represents the dissimilarities in the original dataset.

The cophenetic correlation coefficient (CCC) is computed as the Pearson correlation coefficient between these two distance matrices:

$$CCC = \frac{\sum_{i < j} (d(i, j) - \bar{d})(d_c(i, j) - \bar{d}_c)}{\sqrt{\sum_{i < j} (d(i, j) - \bar{d})^2} \sqrt{\sum_{i < j} (d_c(i, j) - \bar{d}_c)^2}} \quad (96)$$

where:

$d(i, j)$ = original distance between observations i and j

$d_c(i, j)$ = cophenetic distance between observations i and j in the dendrogram.

$\bar{d}$ = mean of original distances

$\bar{d}_c$ = mean of cophenetic distances.

The cophenetic correlation coefficient evaluates how well a hierarchical clustering dendrogram preserves the pairwise distances of the original data. Values of the cophenetic correlation greater than 0.9 indicate an excellent representation of the data structure, meaning that the clustering dendrogram closely reflects the true relationships among observations. Values between 0.75 and 0.9 indicate a good clustering representation, suggesting that the dendrogram captures most of the dataset's structural relationships. When the coefficient lies between 0.5 and 0.75, the clustering provides only a moderate representation of the data structure, suggesting some distortion. Values below 0.5 indicate a poor clustering structure, meaning that the dendrogram does not adequately represent the original pairwise relationships among the observations [34, 37, 43–44].

2.4.3 Quality Measurements: Non-Generalizable

2.4.3.1 RMSE Reconstruction

RMSE Reconstruction measures the average magnitude of the difference between the original data and the reconstructed data produced by a model. It evaluates how much information is lost during dimensionality reduction or encoding–decoding processes.

RMSE Reconstruction is a metric that measures the root-mean-square difference between the original data and the reconstructed data produced by a machine learning model, quantifying information loss during dimensionality reduction or representation learning. The RMSE reconstruction is defined as:

$$RMSE = \sqrt{\frac{1}{np} \sum_{i=1}^n \sum_{j=1}^p (x_{ij} - \hat{x}_{ij})^2} \quad (97)$$

where:

x_i = original data point

$\hat{x}_i$ = reconstructed data point

n = number of samples

p = number of variables (features)

x_{ij} = original value

$\hat{x}_{ij}$ = reconstructed value.

The RMSE Reconstruction metric measures the difference between the original data and the reconstructed data after dimensionality reduction or representation learning. Values of RMSE Reconstruction close to 0 indicate nearly perfect reconstruction, meaning that the reconstructed data closely matches the original dataset with minimal error. Small RMSE values indicate low information loss, suggesting that most of the relevant structure and variability of the original data are preserved in the reconstruction process. In contrast, large RMSE values indicate poor reconstruction of the original data, reflecting significant information loss and substantial differences between the reconstructed and original datasets [34, 45–46].

2.4.3.2 Accuracy k -NN

Accuracy k -NN is the classification accuracy obtained when a k -Nearest Neighbors (k -NN) classifier is applied to the dataset or to a low-dimensional embedding produced by a dimensionality reduction method. The metric evaluates how well samples within the same class remain close to one another in feature or embedding space.

Let:

y_i = true class label of sample i

$\hat{y}_i$ = class predicted by the k -NN classifier

N = total number of samples

The accuracy k -NN is defined as:

$$\text{Accuracy}_{kNN} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i = \hat{y}_i) \quad (98)$$

where:

$\mathbf{1}(\cdot)$ is the indicator function (1 if true, 0 otherwise).

The Accuracy k -NN metric evaluates how well a low-dimensional embedding preserves the class structure of the original data by measuring the classification accuracy obtained with a k -nearest neighbors (k -NN) classifier applied to the embedded representation. Values of Accuracy k -NN close to 1.0 indicate strong preservation of class neighborhoods, meaning that samples belonging to the same class remain close to each other in the embedded space. Values between 0.7 and 0.9 indicate good retention of class structure, suggesting that the embedding maintains most of the class-related neighborhood relationships. When the Accuracy k -NN is approximately equal to the random baseline, the embedding does not preserve class separability, indicating that the

low-dimensional representation fails to maintain the class organization present in the original data [34, 47–48].

2.5 Performance Patterns Across Different Algorithmic Classes

Different DR algorithm families tend to excel in terms of distinct criteria. A structured interpretation can be made as follows:

Locality-preserving algorithms: t-SNE, HLLE, and diffusion maps. These methods generally achieve high Q_{local} , trustworthiness, and accuracy k -NN metrics, although they may underperform in global structure preservation tasks.

Globality-preserving algorithms: Isomaps, MDS, classic scaling, and kernel PCA. These approaches typically exhibit strong Q_{global} , AUC R(NX), and cophenetic correlation values.

Linear and stable algorithms: PCA, PCA-L1, fast ICA, and DRR. These methods result in low RMSEs, high continuity levels, and interpretable dimensions. DRR and PCA are also fully invertible.

Graph-based layout algorithms: Fruchterman–Reingold, Kamada–Kawai, and DrL. These methods provide high-quality visual separations; however, their geometric distortions and global accuracies can vary substantially.

2.5.1 Common Artifacts and Methodological Caution

Users must be aware of the typical artifacts that are associated with DR techniques, as summarized in Table 26.

Table 26. Typical Artifacts Observed in Common DR Algorithms

Algorithm	Typical Issues
t-SNE	Overemphasizes cluster separation; global geometry is relaxed
Isomap	Sensitive to noise and manifold discontinuities (“holes”)
HLLE	Unstable for small sample sizes
MDS	Strong dependence on the choice of distance metric
PCA	Inability to reconstruct nonlinear or curved manifolds

2.5.2 Practical Recommendation: Always Test Multiple Algorithms

A robust practice is to examine at least two or three candidate methods:

- Global structure: PCA vs. Isomaps vs. Diffusion maps.
- Local structure: t-SNE vs. HLLE.
- Interpretability/invertibility: DRR vs. PCA vs. kernel PCA.
- Pure visualization: t-SNE vs. DrL.

Each candidate should be evaluated using the complete set of normalized metrics.

2.5.3 How to Select a Candidate for DR Methods

Selecting an appropriate dimensionality reduction technique is inherently a goal-driven process, as different methods prioritize distinct aspects of the data structure, such as variance, distances, topology, or class separability. In practice, no single method is universally optimal; instead, the choice depends on whether the objective is visualization, clustering, reconstruction, or downstream machine learning tasks.

To assist in this decision-making process, Figure 16 provides a practical framework that organizes DR methods according to their underlying principles and the type of structure they preserve. By linking analytical goals to methodological characteristics, this framework facilitates a more systematic and informed selection of dimensionality reduction techniques in both exploratory and predictive contexts.

How to Choose a Dimensionality Reduction Method

Decision Guide for Dimensionality Reduction Algorithm Selection

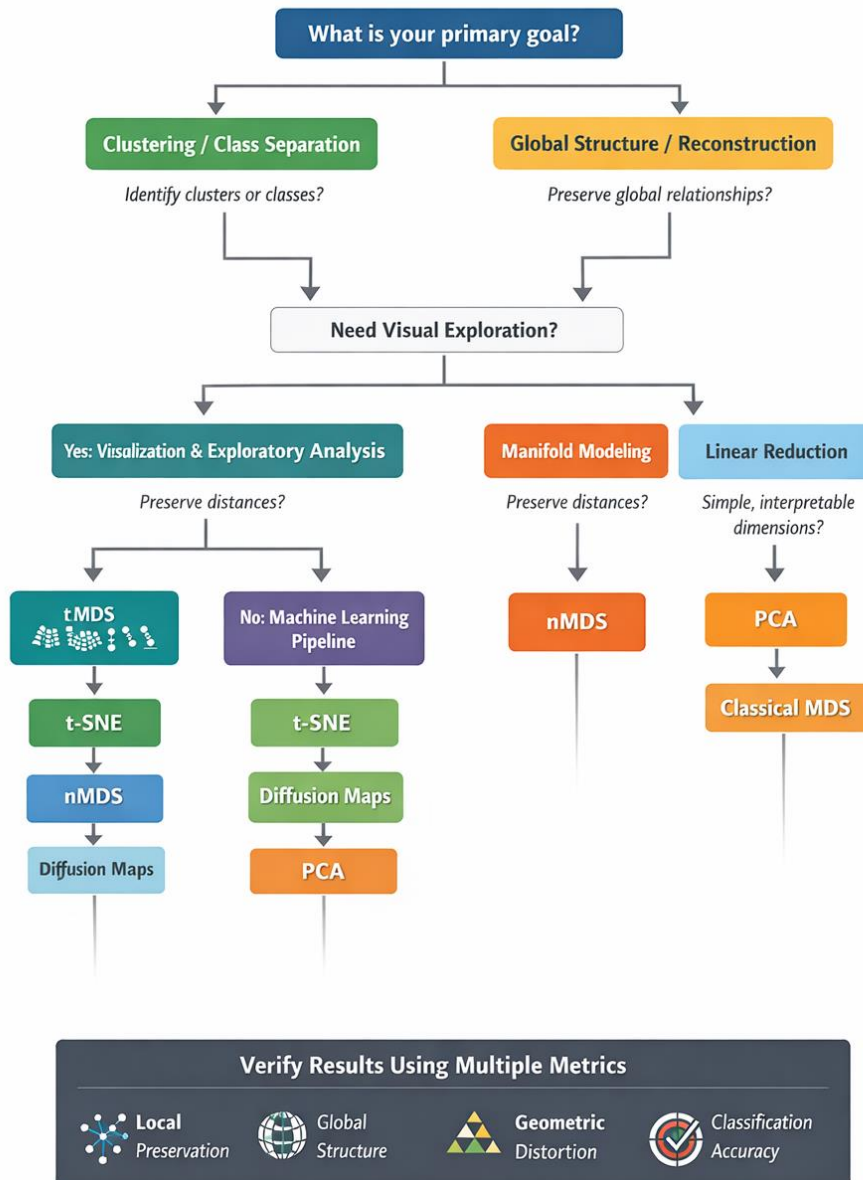


Figure 16 – Decision framework for selecting dimensionality reduction methods.

Figure 16 presents a structured decision tree to guide the selection of dimensionality reduction (DR) techniques based on the primary analytical objective. The

process begins by identifying whether the goal is clustering/class separation or global structure preservation/reconstruction. For clustering-oriented tasks, methods that emphasize local neighborhood preservation (e.g., t-SNE, Diffusion Maps) are recommended, whereas distance-preserving techniques such as t-MDS and nMDS are preferred when geometric relationships are important. For global structure and reconstruction, the framework distinguishes between linear reduction (e.g., PCA, Classical MDS), which provides interpretable components, and manifold learning approaches (e.g., nMDS), which capture nonlinear relationships. The diagram also highlights the role of visual exploration versus machine learning pipeline integration in method selection. Finally, it emphasizes that the quality of any embedding should be validated using multiple criteria, including local preservation, global structure, geometric distortion, and classification performance.

2.5.4 Constructing the Final Ranking List

The ranking workflow should consist of the following steps.

Normalize all the metrics.

- Apply objective-dependent weights.
- Compute weighted sums for each algorithm.
- Order the algorithms by their final scores.
- Assess the robustness of the algorithms by varying the weights by $\pm 10\%$.

2.5.5 Final Considerations

The optimal DR strategy emerges from a combination of the following:

- i) the precise definition of analytical objectives (local preservation, global geometry, visualization clarity, and interpretability);
- ii) normalized, multimetric comparisons with objective-based weights;
- iii) the execution of a systematic ranking protocol;

- iv) empirical testing of the top candidate algorithms; and
- v) the critical consideration of the known artifacts for each method.

This integrated approach provides a transparent, reproducible, and technically grounded framework for selecting dimensionality reduction algorithms across a wide range of scientific applications.

3. METHODOLOGY

3.1 Dimensionality Reduction Procedures

A comprehensive set of global and local DR techniques was evaluated. The global methods included DRR, an Isomap, PCA, PCA-L1, kernel PCA, FastICA, classic MDS, nonmetric MDS, NMF, DrL, Fruchterman–Reingold, and Kamada–Kawai. The local and manifold-preserving techniques included HLLE, Diffusion maps, and t-SNE. Before executing the methods, the availability of the required packages was programmatically verified. The DR methods were executed within a unified wrapper function (`run_dimred()`), which standardized parameterization and error handling. In particular, t-SNE was executed using Rtsne, with the perplexity level automatically determined as a function of the sample size. All the other methods were implemented via `dimRed::embed()`, ensuring a standard interface. Only successful embeddings were retained for subsequent evaluations.

3.2 Quality Assessment and Metric Computation Strategies

For each DR method, a suite of 13+ evaluation metrics was computed to quantify local fidelity, preservation of global structure, geometric distortion, classification consistency, and reconstruction error (when applicable). These metrics were organized into four conceptual families.

- (i) **Local Neighborhood Preservation:** Qlocal, trustworthiness, precision, and recall.
- (ii) **Global Structure Preservation:** Qglobal, AUC (RNX), cophenetic correlation, and Kendall's τ .
- (iii) **Geometric Smoothness and Distortion:** RMSE of the reconstruction results (for invertible methods: PCA, kPCA, FastICA, DRR), discontinuity ($1 - \text{continuity}$), the Gabriel classification error (GCE), and the C-measure.

(iv) **Class Consistency and Separability:** k-NN classification accuracy, precision, and recall.

The local/global measures (Q metrics) were computed through `dimRed::quality()`, whereas continuity, trustworthiness, precision, recall, the GCE, the C-measure, and Kendall's τ were obtained via `DRquality`. A fallback implementation was used when a metric could not be computed because of algorithm-specific constraints. Global structure preservation was additionally quantified by computing the RNX curve up to 30 neighbors and numerically integrating it using the trapezoidal rule. All metrics were consolidated into a single master table for ranking and evaluation.

3.3 Weighting Schemes According to Their Analytical Objectives

After undergoing normalization, the metrics should be aggregated using weights that reflect the analytical goal. The recommended weighting schemes are as follows.

a) Clustering and local structural analysis

Local: 0.50; Global: 0.20; Geometric smoothness: 0.15; k-NN classification: 0.15

b) Physical reconstruction and manifold modeling

Local: 0.30; Global: 0.30; Geometry/smoothness: 0.40

c) Exploratory visualization

Local: 0.40; Global: 0.20; Layout/stability: 0.40

d) Downstream machine learning pipelines

Local: 0.40; Classification stability: 0.40; Global: 0.20

The final score for each algorithm is thus given as follows:

$$\text{Final score} = \sum_i (\text{weight}_i \times \text{normalized metric}) \quad (99)$$

3.4 Metric Normalization and Index Construction

To enable fair comparisons across heterogeneous scales, all metrics were normalized to the interval $[0,1]$, with their orientations adjusted to indicate whether higher values indicate better performance. The metrics were categorized as follows.

Higher-is-better: Q_{local} , Q_{global} , AUC (RNX), cophenetic correlation, trustworthiness, precision, recall, Kendall's τ , and kNN accuracy.

Lower-is-better: RMSE, discontinuity, the GCE, and the C-measure.

From the normalized values, composite indices were computed to reflect the performance within each metric family.

IndexLocal: average of local metrics.

IndexGlobal: average of global metrics.

IndexGeom: measures of geometric distortion.

IndexClass: classification-related measures.

3.5 Scenario-Based Ranking of the Dimensionality Reduction Techniques

To reflect the practical diversity of DR use cases, four application scenarios were defined.

- Clustering/topology preservation.
- Reconstruction/invertible mapping.
- Visualization/class separation.
- Machine learning pipeline embedding.

Each scenario was linked to a set of weights that assigned relative importance scores to the local, global, geometric, and classification indices. A global score (Score Final) was calculated for each DR method and each scenario, followed by rankings. To ensure the robustness of the proposed methodology, the metrics were adjusted by $\pm 10\%$. This scheme produced several values, with the media representing the final scores.

Additionally, a meta-analysis identified the most consistent metric per scenario (based on the absolute Spearman correlations among metric-wise ranks) and the technique with the best performance for that metric. This dual approach provided both a scenario-specific optimal DR method and a metric-specific champion.

3.6 Visualization of the Optimal Methods

For each scenario, two visual outputs were generated.

- A scatter plot of the embedding produced by the method that maximized the best metric identified for that scenario.
- A bar plot of the ranked methods sorted by their final composite scores.

These visualizations enabled qualitative and quantitative assessments of the embedding quality, class separability, and overall structural fidelity of the different methods.

3.7 Summary of the Outputs

The pipeline returned the following. A comprehensive table of all the metrics for all the successful DR methods. A scenario-wise list of results (results obj) containing

- the complete ranking of the DR methods,
- the most consistent metric for the scenario, and
- the champion algorithm for that metric.

Corresponding graphical outputs for embedding visualization and method ranking. These components enabled a rigorous, multicriteria evaluation of dimensionality reduction techniques tailored to distinct analytical objectives.

3.8 Proposed Strategy Used to Choose the Best Metric

The `best_metric_and_champion` function implemented a data-driven strategy to identify the metric that best reflected global consensus across several competing quality criteria, then used that metric to select the best dimensionality reduction method.

First, the procedure restricted the analysis to a subset of metrics that were explicitly flagged as either “larger is better” or “smaller is better”. These sets were defined externally (`larger_better` and `smaller_better`) and intersected with the column names of the `all_metrics` object, producing a matrix (`mat`) whose rows corresponded to the tested methods (algorithms) and whose columns corresponded to the evaluation metrics. This ensured that only interpretable, directionally consistent indices were used in the ranking procedure.

In the second step, this matrix was converted into a long format, where each row contained a method, a metric, and its corresponding value. For each metric, the code independently determined its optimization direction (greater-is-better or smaller-is-better) and transformed the raw values into ranks using the auxiliary function `rank_dir`. In this ranking scheme, a rank of 1 always denotes the best method according to that metric, regardless of whether the original index should have been maximized or minimized. The result was a rank matrix in which each column encoded the ranking of the methods according to that metric.

The core idea of the selection procedure was based on intermetric concordance. The code computed the Spearman correlation matrix of the rank matrix columns, thereby quantifying, for each pair of metrics, how similar their induced method orderings were. The absolute value of the Spearman correlation was then taken to avoid distinctions between positively and negatively aligned orderings; only the strength of each association mattered. For each metric (column), the mean absolute correlation with all other metrics was calculated and interpreted as a consistency index. A high consistency value indicated that the ranking produced by that metric closely agreed, on average, with the rankings induced by the remaining metrics.

The “best metric” was then defined as the one with the highest mean absolute Spearman correlation, i.e., the metric whose ranking was most consistent with the global multimetric pattern. This criterion favors metrics that behave as robust consensus indicators rather than outliers that strongly disagree with the majority of the other criteria. Once the best metric was identified, the champion method was selected by simply choosing the algorithm that optimized this metric: if the metric belongs to the “larger is better” group, the method with the maximum value was selected; if it belongs to the “smaller is better” group, the minimum value defined the champion.

In summary, the code did not arbitrarily privilege a single quality index. Instead, it constructed internal consensus across all available metrics in the rank space and selected the “best metric” that best reflected it. The corresponding “champion” method was then the algorithm that performed optimally under that representative metric, providing a principled and transparent way to choose both the ideal evaluation criterion and the preferred dimensionality reduction technique.

3.9 Application of the Proposed Methodology

The validation of the proposed methodological framework requires the use of benchmark datasets that are both diverse and well-characterized, enabling a systematic and reproducible assessment of dimensionality reduction performance. In this context, publicly available datasets from the UCI Machine Learning Repository were employed. This repository is widely recognized in the machine learning community as a reliable source of curated datasets, encompassing a broad spectrum of domains, data structures, and levels of complexity. Its extensive use in the literature ensures comparability and external validity of the results obtained.

The selected datasets comprise two or more predefined classes and are predominantly associated with supervised classification tasks. This characteristic is

essential for the present study, as it allows the incorporation of class-dependent evaluation metrics—such as classification accuracy, precision, recall, and neighborhood preservation—into the assessment framework. By ensuring that the ground-truth class structure is known a priori, the methodology can rigorously evaluate not only geometric and topological preservation but also the extent to which class separability is maintained in the reduced-dimensional space.

Furthermore, the diversity of the chosen datasets—ranging in dimensionality, sample size, and intrinsic data structure—provides a comprehensive testbed for examining the robustness and generalizability of the proposed evaluation strategy. The intentional selection of classification-oriented datasets enables the simultaneous investigation of compact, connected, and non-generalizable quality metrics within a unified framework.

These datasets are employed as illustrative case studies to demonstrate the practical applicability of the methodology. The results are systematically presented and discussed in the subsequent sections, organized as Study Cases I through VII, where each case highlights specific aspects of the evaluation process and the behavior of dimensionality reduction techniques under different data conditions.

3.10 Hardware and Software

The statistical analysis was carried out on a personal computer with a Ryzen 9 7900X processor, an M.2 NVMe Western Digital 1 TB SSD, 128 GB of Kingston Fury Beast DDR5 6000 RAM, an ASUS Prime X670E-PRO WIFI motherboard, and a 32-GB MSI GeForce RTX 5090 VENTUS OC 3X graphics card. R version 4.5.1 (2025-06-13) [49] and RStudio version 2025.09.1 [50] for the Windows 64-bit OS were used in this work. The script for this statistical analysis is available on GitHub. If necessary, the R packages and the two datasets needed for installation are provided. Beyond this, the other

datasets can be downloaded directly from the following URL:
<https://github.com/Laboratorio-de-Tecnologia-Computacional/Dimensionality-Reduction-Techniques-An-Innovative-Approach>

3.11 Packages

Table 27 presents a comprehensive overview of the R packages employed throughout this work, highlighting their main utilities and corresponding literature references. The selection of these packages reflects the multidisciplinary nature of the proposed methodology, which integrates data preprocessing, visualization, dimensionality reduction, clustering, and quantitative evaluation within a unified analytical framework.

The R ecosystem provides a robust and flexible platform for implementing advanced machine learning and chemometric workflows. In this context, packages such as `ggplot2` and `dplyr` support data visualization and manipulation based on the grammar of graphics and tidy data principles, respectively. Complementary tools like `tidyr`, `readr`, and `stringr` facilitate efficient data handling and transformation.

Specialized packages play a central role in implementing and evaluating dimensionality reduction techniques. For instance, `Rtsne`, `kernlab`, `diffusionMap`, and `fastICA` provide implementations of key nonlinear and linear projection methods. Additionally, `DRquality` and `coRanking` are essential for computing quality metrics such as trustworthiness, continuity, and co-ranking-based indices, which are central to the evaluation strategy proposed in this book.

Furthermore, packages such as `igraph` and `RANN` enable efficient graph-based modeling and neighborhood computations, while `RSpectra` and `Matrix` support scalable numerical linear algebra operations required for high-dimensional data analysis.

Overall, the collection of packages listed in Table 27 constitutes a cohesive computational infrastructure that ensures reproducibility, scalability, and methodological rigor in the application of dimensionality reduction techniques and their associated evaluation metrics.

Table 27 - R Packages: Utilities and References

Package	Main Utilities	Reference
ggplot2	Grammar of graphics for data visualization	[51]
readr	Fast data import	[52]
dplyr	Data manipulation grammar	[53]
Corrplot	Correlation matrix visualization	[54]
coRanking	Quality metrics for DR	[4]
psych	Psychometric methods and factor analysis	[55]
GGally	Extensions to ggplot2, pairwise plots	[56]
purrr	Functional programming tools	[57]
tidyr	Data tidying	[58]
writexl	Write Excel files	[59]
stringr	String manipulation	[60]
rlang	Tidy evaluation infrastructure	[61]
farff	Reading ARFF data	[62]
readxl	Read Excel files	[63]
magrittr	Pipes for functional flow	[64]
DRquality	DR quality metrics	[65]
ProjectionBasedClustering	Projection-based clustering tools	[66]
FastKNN	Efficient k-NN search	[67]
FCPS	Clustering benchmark data	[68]
pcaPP	Robust PCA	[69]
cccd	Proximity graphs	[70]
spdep	Spatial dependence analysis	[71]
fastICA	Independent Component Analysis	[72]
Rtsne	t-SNE implementation	[73]
vegan	Ecological ordination, including NMDS	[74]
diffusionMap	Diffusion maps	[75]
pcaL1	L1-norm PCA	[76]
DRR	Dimensionality Reduction via Regression	[77]
igraph	Network analysis and layouts	[78–80]
kernlab	Kernel methods, including KPCA	[81–82]
RANN	Approximate nearest-neighbor search	[83]
Matrix	Sparse matrices	[84]
RSpectra	Large-scale eigen decomposition	[85]
MASS	Statistical methods and classical MDS	[86]

3.12 Software Environment, Reproducibility Controls, and Utility Functions

All analyses were conducted in *R* using an integrated framework for dimensionality reduction (DR), quality assessment, and metric-based algorithmic

comparison. To ensure full reproducibility of results, a fixed pseudo-random seed was established using the Mersenne–Twister generator (`set.seed(123)`).

All required packages were loaded with suppressed start-up messages to guarantee a controlled execution environment. The workflow relies on the packages `dimRed`, `DRquality`, `RANN`, `ggplot2`, `purrr`, `dplyr`, `tidyr`, and additional DR-specific libraries (e.g., *Rtsne*, *vegan*, *igraph*, *kernlab*, *pcaLL*, *fastICA*, *MASS*). A set of custom utility functions was created to standardize data extraction and quality measurements across DR techniques, including:

`get_coords2d()` — extracts the two-dimensional embedding from objects returned by different DR routines;

`get_label_vec()` — retrieves class/group labels automatically when available in the dataset;

`get_knn_idx()` — computes the k -nearest neighbors in the original space using the `RANN` library;

`rn_curve()` and `auc_trapz_norm()` — compute the RNX curve and its corresponding normalized area under the curve (AUC), a global structure-preservation metric.

`trust_continuity()` — fallback implementation of Trustworthiness and Continuity in cases where `DRquality` outputs cannot be computed.

4. STUDY OF CASE I

4.1 Description of the Dataset

The Diagnostic Wisconsin Breast Cancer Database was named df1. It contains 569 instances, 30 features, and two classes. Features were computed from a digitized image of the fine needle aspirate (FNA) of a breast mass, and they describe the characteristics of the cell nuclei that are present in the image [87].

4.2 Results and Discussion

The correlation heatmap, shown as Figure 17, summarizes the relationships among the principal evaluation metrics used to assess dimensionality-reduction (DR) techniques.

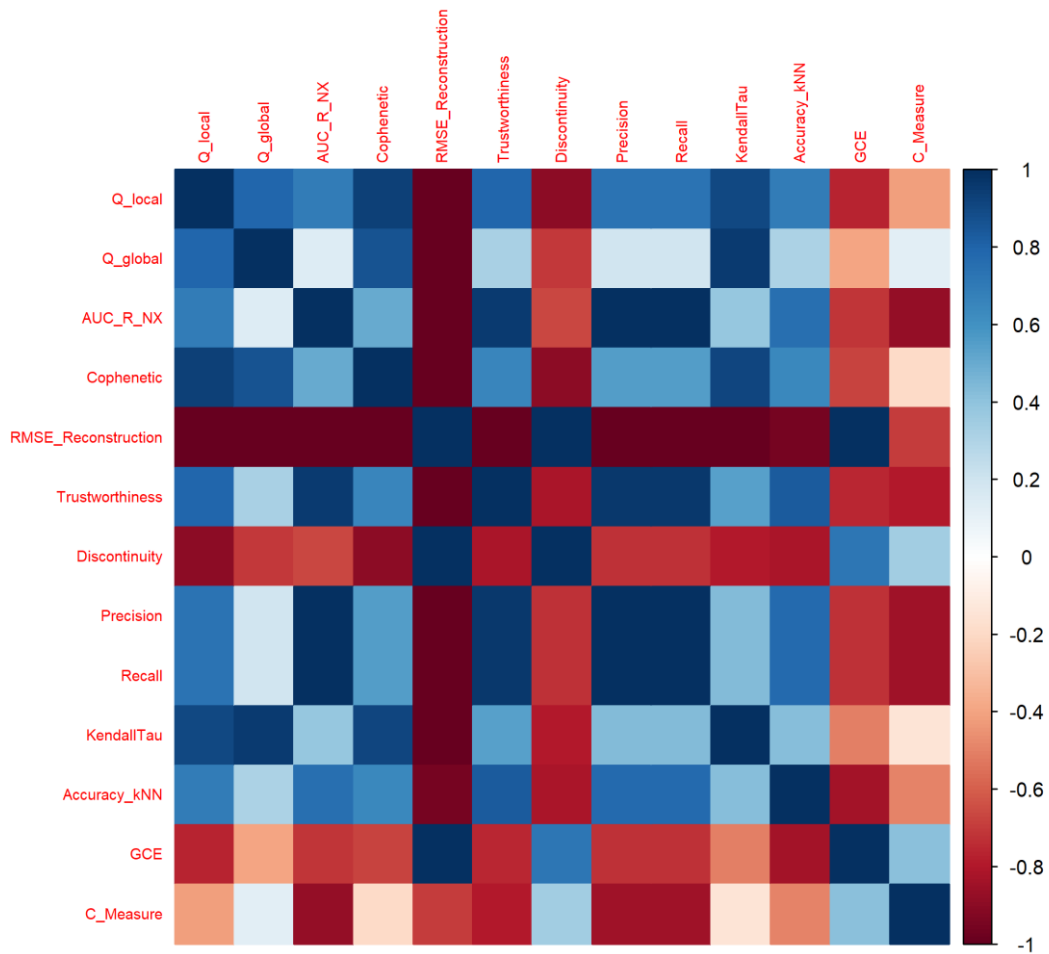


Figure 17 – The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Two coherent clusters emerge: locality-oriented metrics (Q_{local} , *Trustworthiness*, *Precision*, *Recall*, *Accuracy_kNN*) and global-structure metrics (Q_{global} , $AUC_R(NX)$, *Cophenetic*, *Kendall Tau*), both of which exhibit strong positive correlations. This pattern indicates that metrics within each group capture complementary yet internally consistent aspects of manifold preservation. In contrast, reconstruction- and distortion-based criteria ($RMSE_Reconstruction$, *Discontinuity*, *GCE*, *C_measure*) show pronounced negative correlations with the locality and global clusters, revealing an inherent trade-off in DR performance: methods that emphasize topological or neighborhood fidelity generally incur higher reconstruction error, whereas algorithms optimized for reconstruction tend to distort manifold geometry. Overall, the figure reinforces the multidimensional nature of DR evaluation and highlights the importance of selecting metrics aligned with the intended analytical objective.

The results summarized in Table 28A and Table 28B provide a comprehensive comparison of dimensionality-reduction algorithms using complementary evaluation metrics. Convex and graph-based methods such as DRR, PCA/MDS, Isomap, and nMDS consistently demonstrate strong performance on both locality (Q_{local} , *Trustworthiness*) and global-structure metrics (Q_{global} , $AUC_R(NX)$, *Cophenetic*, *KendallTau*). These methods also achieve high k-NN accuracy and balanced precision–recall values, indicating stable preservation of neighborhood topology across the embedding space.

Nonlinear methods such as t-SNE excel specifically in local-neighborhood metrics and precision–recall, but exhibit weaker preservation of global structure, consistent with their design focus on visual separability rather than manifold continuity.

Conversely, reconstruction-oriented approaches (e.g., kPCA) show elevated RMSE and distortion metrics (*GCE*, *C_Measure*), confirming the trade-off between geometric fidelity and reconstruction accuracy. Overall, the tables highlight that no single algorithm dominates across all criteria. Instead, methods form performance profiles aligned with their theoretical foundations—emphasizing the need for objective-driven algorithm selection in dimensionality-reduction workflows.

Table 28A – Figures of merit obtained from Dataset dfl.

Algorithm	Q_{local}	Q_{global}	AUC_R(NX)	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.506	0.888	0.283	0.931	0.606	0.875	0.949
Isomap	0.503	0.886	0.296	0.927		0.885	0.951
PCA	0.502	0.887	0.285	0.931	0.606	0.876	0.950
PCA L1	0.487	0.878	0.260	0.877		0.866	0.938
kPCA	0.282	0.672	0.185	0.055	50.037	0.801	0.839
FastICA	0.486	0.870	0.281	0.915	0.606	0.874	0.944
MDS	0.502	0.887	0.285	0.931		0.876	0.950
nMDS	0.525	0.904	0.301	0.966		0.892	0.938
DrL	0.349	0.763	0.110	0.331		0.768	0.822
Fruchterman-Reingold	0.456	0.828	0.256	0.600		0.848	0.939
Kamada-Kawai	0.513	0.892	0.301	0.935		0.891	0.949
HLLE	0.406	0.862	0.107	0.486		0.743	0.840
Diffusion Maps	0.432	0.837	0.238	0.405		0.855	0.892
t-SNE	0.509	0.755	0.492	0.642		0.944	0.915

Table 28B – Figures of merit obtained from Dataset dfl.

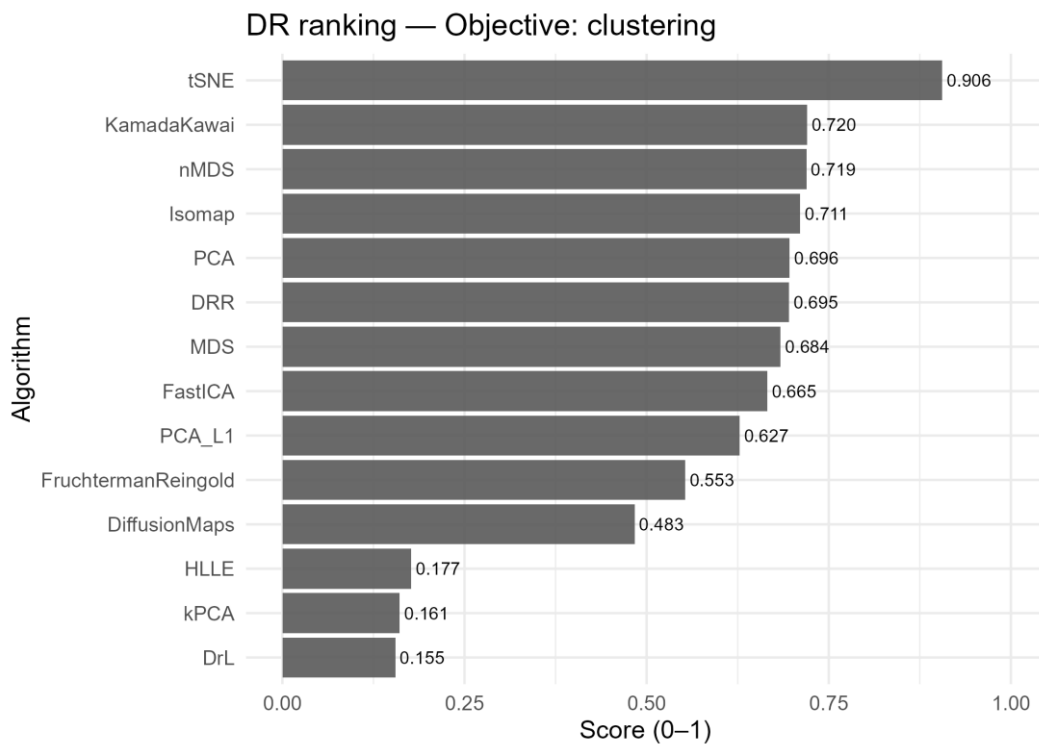
Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.051	0.297	0.297	0.750	0.940	0.833	1457.6
Isomap	0.049	0.310	0.310	0.749	0.940	0.859	1317.5
PCA	0.050	0.304	0.304	0.751	0.940	0.871	1518.4
PCA L1	0.062	0.272	0.272	0.704	0.947	1.029	1448.2
kPCA	0.161	0.192	0.192	0.075	0.910	1.631	1443.6
FastICA	0.056	0.295	0.295	0.721	0.931	0.787	1508.0
MDS	0.050	0.304	0.304	0.751	0.940	0.871	1518.4
nMDS	0.062	0.317	0.317	0.827	0.926	1.028	1317.4
DrL	0.178	0.107	0.107	0.259	0.917	1.031	1656.6
Fruchterman-Reingold	0.061	0.268	0.268	0.481	0.942	0.834	1571.5
Kamada-Kawai	0.051	0.319	0.319	0.772	0.937	0.983	1350.3
HLLE	0.160	0.108	0.108	0.592	0.889	1.505	1707.3
Diffusion Maps	0.108	0.246	0.246	0.623	0.919	1.134	1370.8
t-SNE	0.085	0.488	0.488	0.493	0.947	0.630	972.2

Table 29 summarizes the multi-objective performance of dimensionality-reduction algorithms using four composite indices—Local, Global, Geometric, and Class—and an aggregated Final Score. The results clearly indicate that t-SNE dominates across all scenarios, achieving the highest locality and class-related indices and consistently producing the top Final Score for clustering, visualization, and ML-pipeline objectives. Classical manifold learners such as nMDS, Kamada–Kawai, Isomap, and PCA form a second performance tier, showing balanced behavior with moderate-to-high global and geometric fidelity but lower class-related scores. Across reconstruction tasks, methods such as DRR and PCA maintain competitive geometric preservation, though still trailing t-SNE in the overall objective-weighted ranking. The table reinforces that algorithmic suitability is scenario-dependent: while t-SNE excels in tasks prioritizing neighborhood fidelity and group separability, more stable global methods (e.g., nMDS, Isomap, PCA) remain advantageous for applications requiring manifold continuity or interpretability.

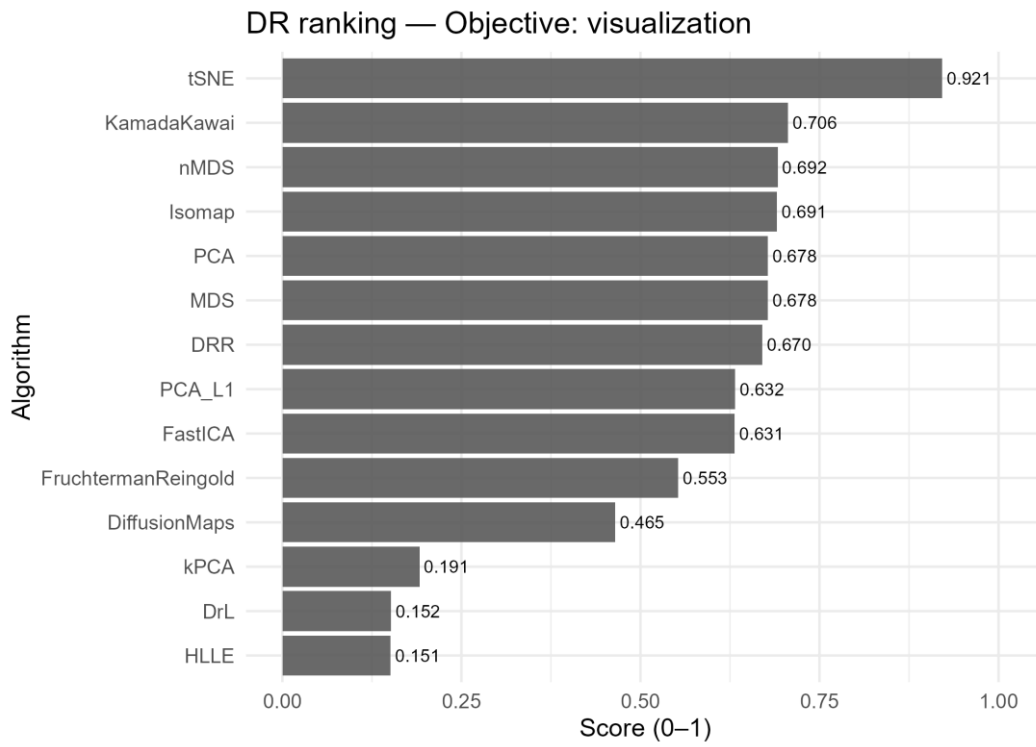
Table 29 – Figures of merit obtained from Dataset df1.

Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
tSNE	0.984	0.639	0.907	1.000	0.906	clustering
KamadaKawai	0.700	0.836	0.706	0.647	0.720	clustering
nMDS	0.711	0.876	0.677	0.580	0.719	clustering
Isomap	0.670	0.817	0.767	0.648	0.711	clustering
PCA	0.650	0.812	0.752	0.638	0.696	clustering
tSNE	0.984	0.639	0.907	1.000	0.921	visualization
KamadaKawai	0.700	0.836	0.706	0.647	0.706	visualization
nMDS	0.711	0.876	0.677	0.580	0.692	visualization
Isomap	0.670	0.817	0.767	0.648	0.691	visualization
PCA	0.650	0.812	0.752	0.638	0.678	visualization
tSNE	0.984	0.639	0.907	1.000	0.850	reconstruction
Isomap	0.670	0.817	0.767	0.648	0.753	reconstruction
DRR	0.644	0.812	0.780	0.626	0.749	reconstruction
nMDS	0.711	0.876	0.677	0.580	0.747	reconstruction
KamadaKawai	0.700	0.836	0.706	0.647	0.743	reconstruction
tSNE	0.984	0.639	0.907	1.000	0.921	ml_pipeline
KamadaKawai	0.700	0.836	0.706	0.647	0.706	ml_pipeline
nMDS	0.711	0.876	0.677	0.580	0.692	ml_pipeline
Isomap	0.670	0.817	0.767	0.648	0.691	ml_pipeline
PCA	0.650	0.812	0.752	0.638	0.678	ml_pipeline

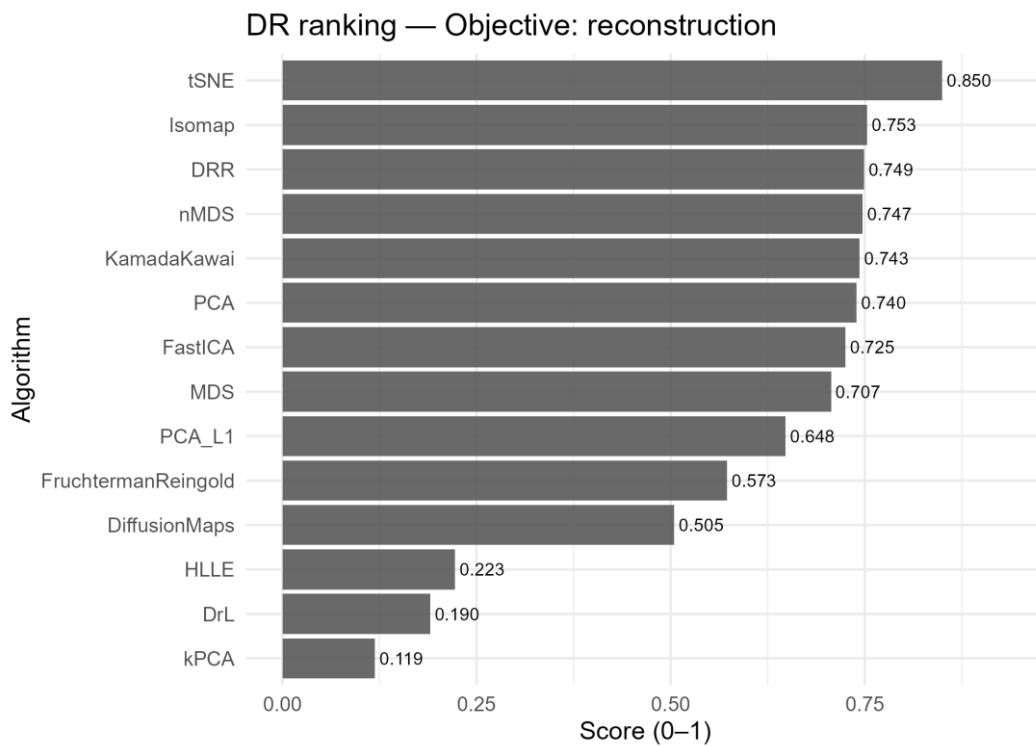
Figure 18 summarizes the objective-based ranking of dimensionality-reduction algorithms across four analytical scenarios: clustering, visualization, reconstruction, and ML-pipeline integration. In all cases, t-SNE consistently achieves the highest scores, reflecting its strong ability to preserve local structure and enhance class separability. Classical manifold methods such as nMDS, Kamada–Kawai, Isomap, PCA, and MDS constitute a second performance tier, exhibiting balanced performance with moderate-to-high scores across objectives.



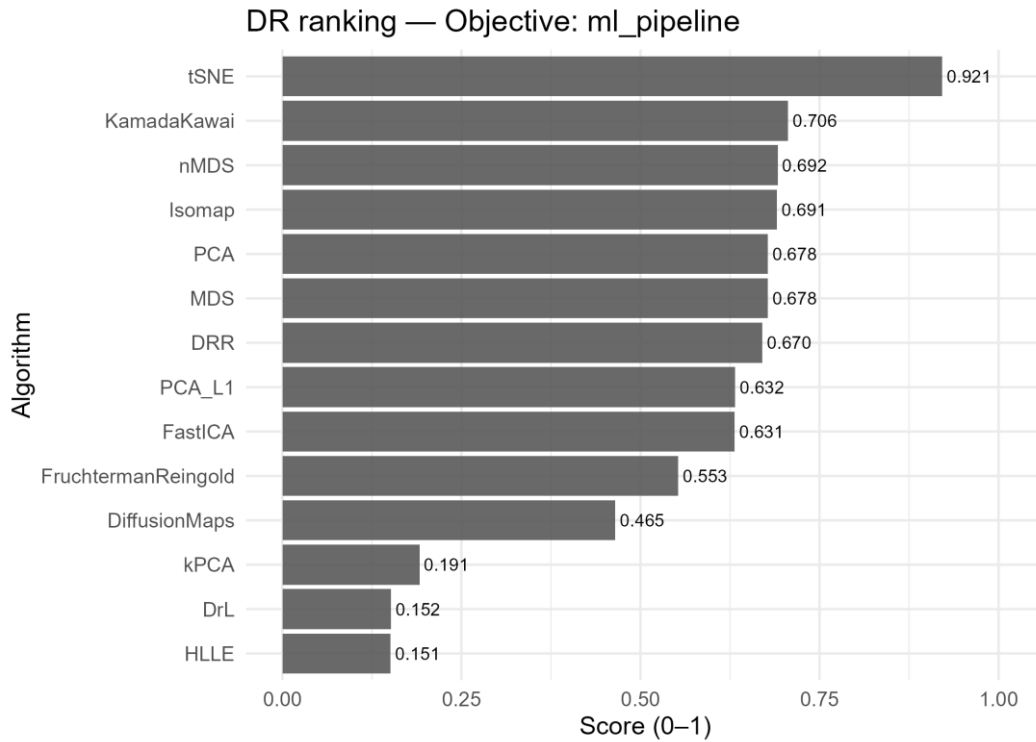
(a)



(b)



(c)



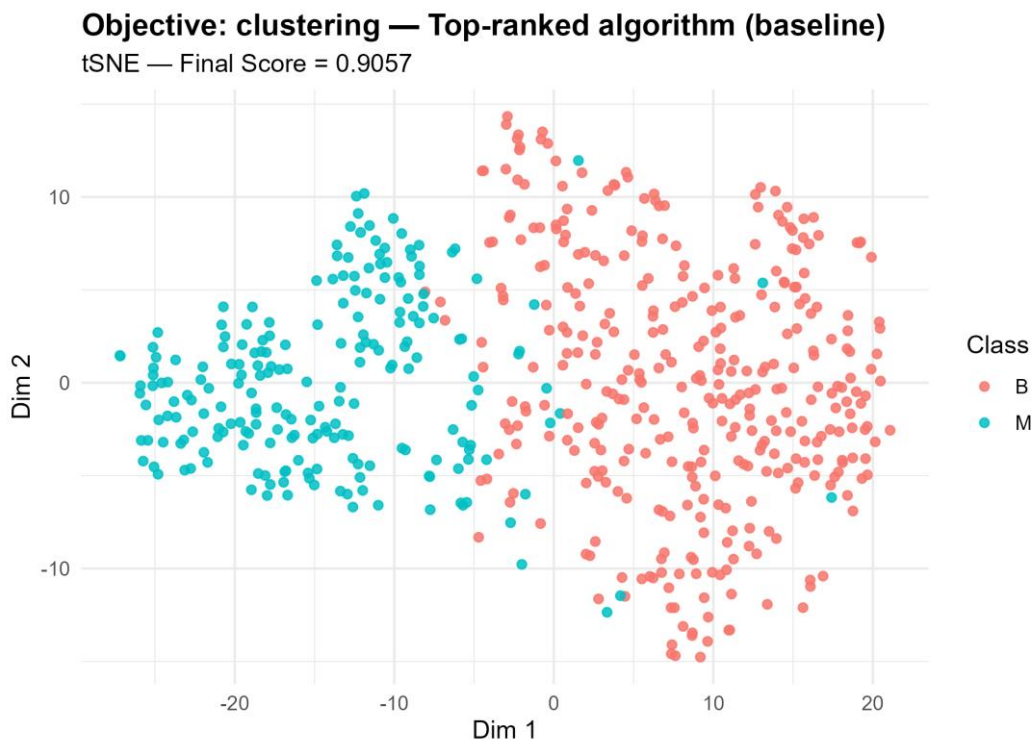
(d)

Figure 18 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Reconstruction-centered tasks reveal a slightly different hierarchy, with DRR, PCA, and FastICA performing competitively and narrowing the gap with t-SNE, which remains the overall leader. Graph-drawing methods (Fruchterman–Reingold) and nonlinear spectral approaches (Diffusion Maps, HLLE, DrL, kPCA) exhibit lower performance across objectives, reflecting limitations in either global continuity or neighborhood preservation depending on the method. Overall, the comparative results indicate that no single algorithm is universally optimal. Still, t-SNE provides the most robust performance across diverse analytical needs, while traditional manifold learners remain strong alternatives when interpretability or geometric stability is prioritized.

Figure 19 presents the two-dimensional embeddings produced by the top-ranked dimensionality-reduction method—t-SNE—across four analytical objectives: clustering,

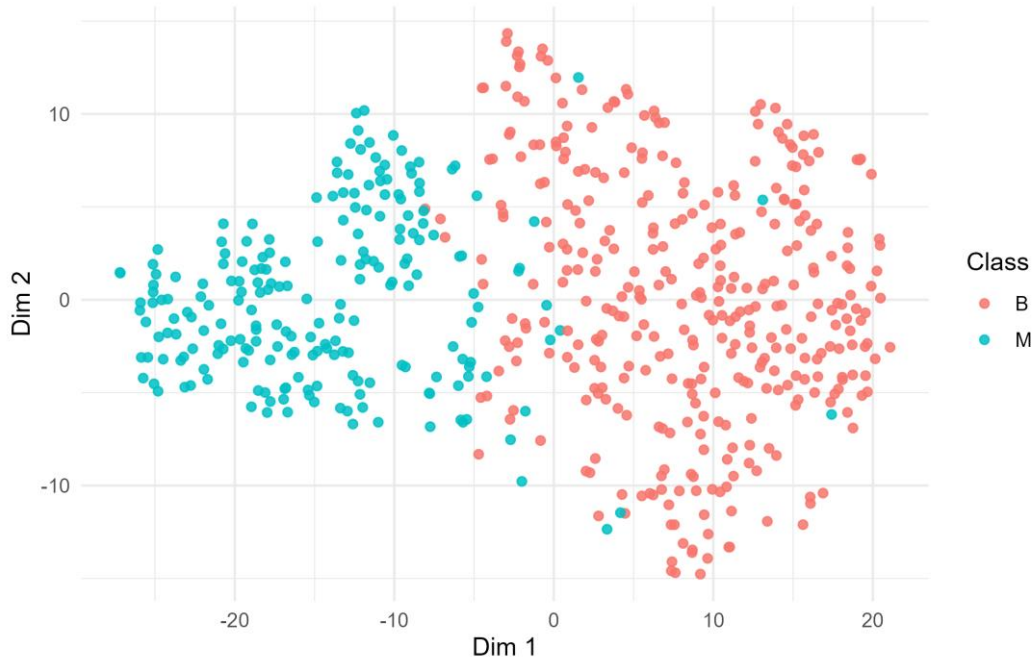
visualization, reconstruction, and ML-pipeline integration. In all scenarios, t-SNE yields a well-defined separation between the two classes, with compact intra-class grouping and clear inter-class boundaries. This consistent structure demonstrates the method's strong capacity to preserve local neighborhood relationships, enhance class separability, and generate visually interpretable embeddings. The near-identical layouts across objectives indicate robustness of the representation and reinforce t-SNE's superiority in scenarios where discriminative structure and visual clarity are central analytical priorities.



(a)

Objective: visualization — Top-ranked algorithm (baseline)

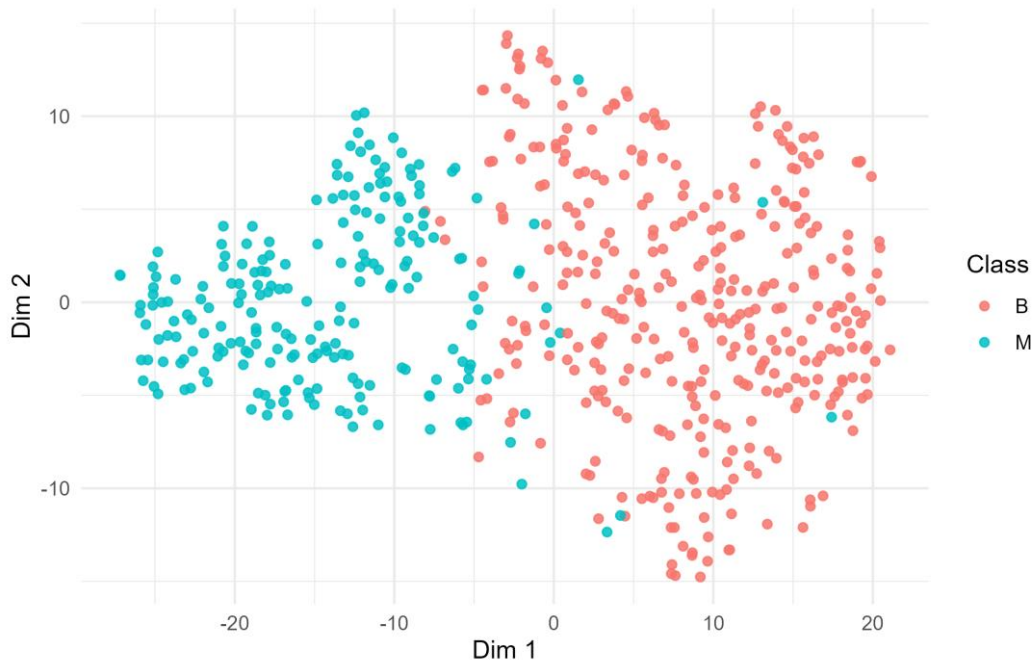
tSNE — Final Score = 0.9213



(b)

Objective: reconstruction — Top-ranked algorithm (baseline)

tSNE — Final Score = 0.8497



(c)

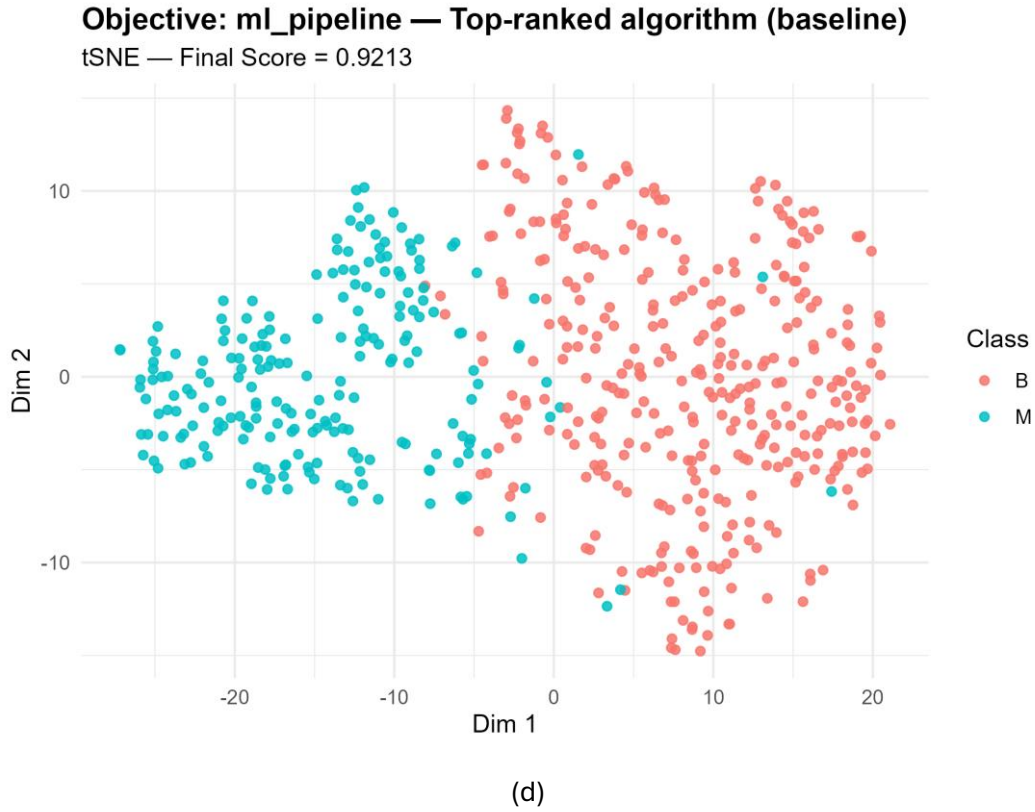
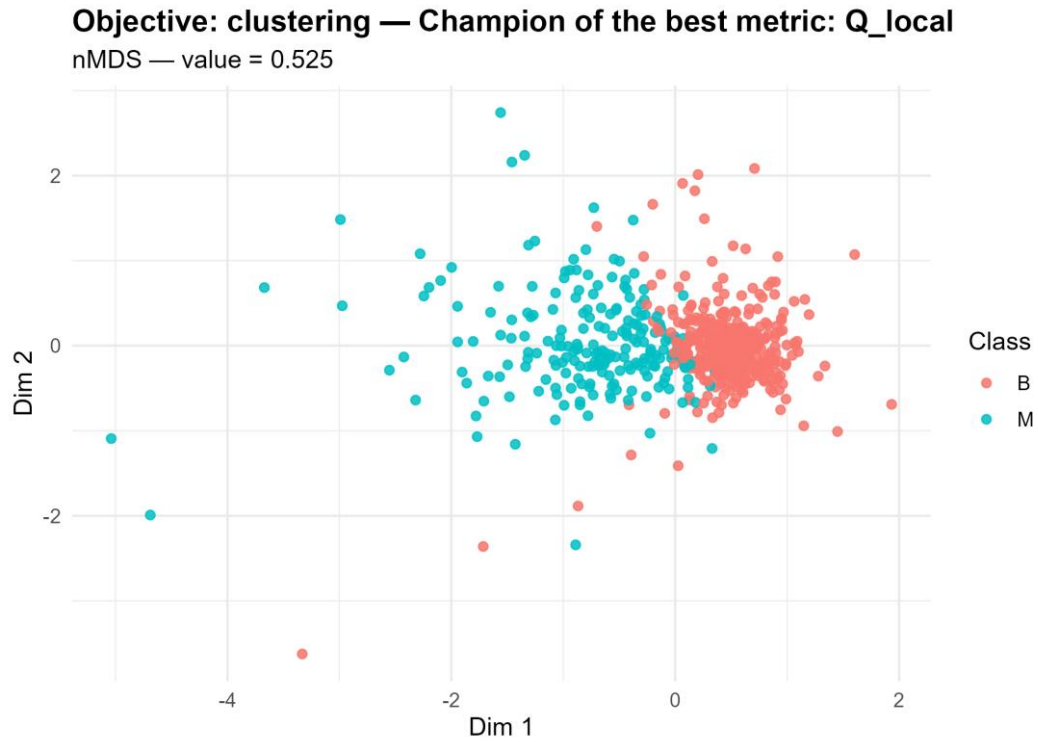


Figure 19 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Figure 20 displays the two-dimensional embeddings generated by the algorithm that achieved the highest performance in the Q_{local} metric across all objectives—nMDS. In each scenario (clustering, visualization, reconstruction, and ML-pipeline integration), nMDS produces embeddings with moderate intra-class cohesion and partially overlapping class boundaries. The representations consistently preserve local neighborhood structure, as indicated by the stable Q_{local} value (0.5253), but exhibit limited global separation compared with methods optimized for discriminative visualization. Overall, the results show that nMDS is effective at preserving fine-scale topology, yielding comparable layouts across analytical objectives. However, the embedding also highlights the trade-off inherent in prioritizing local fidelity. While neighborhood relations are preserved, broader class separability and global manifold structure are less

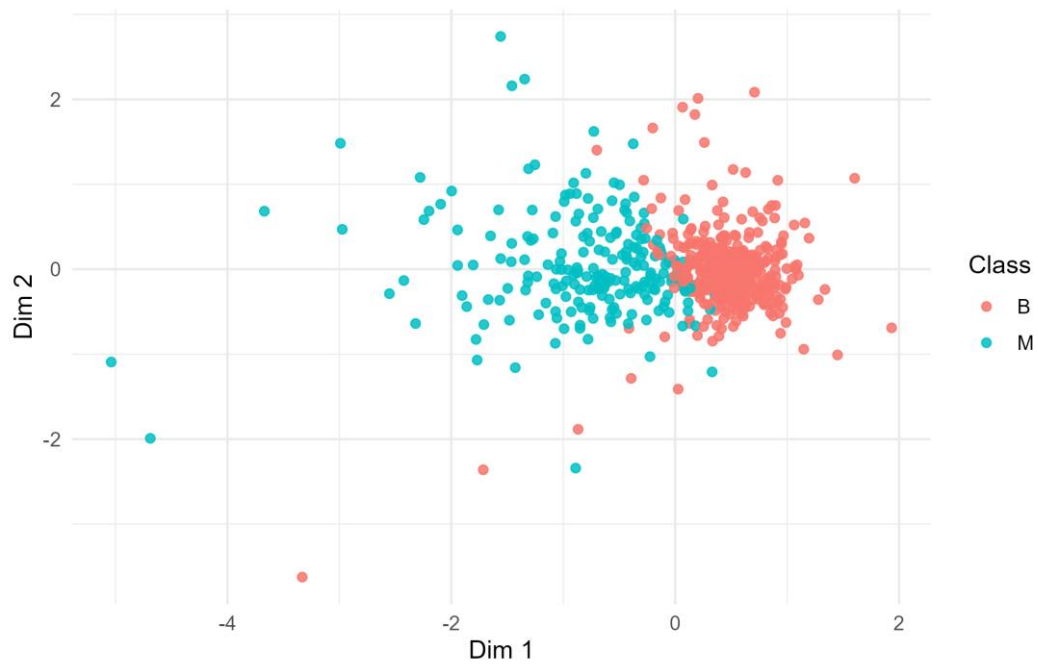
pronounced than in algorithms tailored for visualization or clustering. In this case, the reader is advised to compare the performance of the DR techniques: t-SNE versus nMDS and choose the one most suitable for its objective.



(a)

Objective: visualization — Champion of the best metric: Q_local

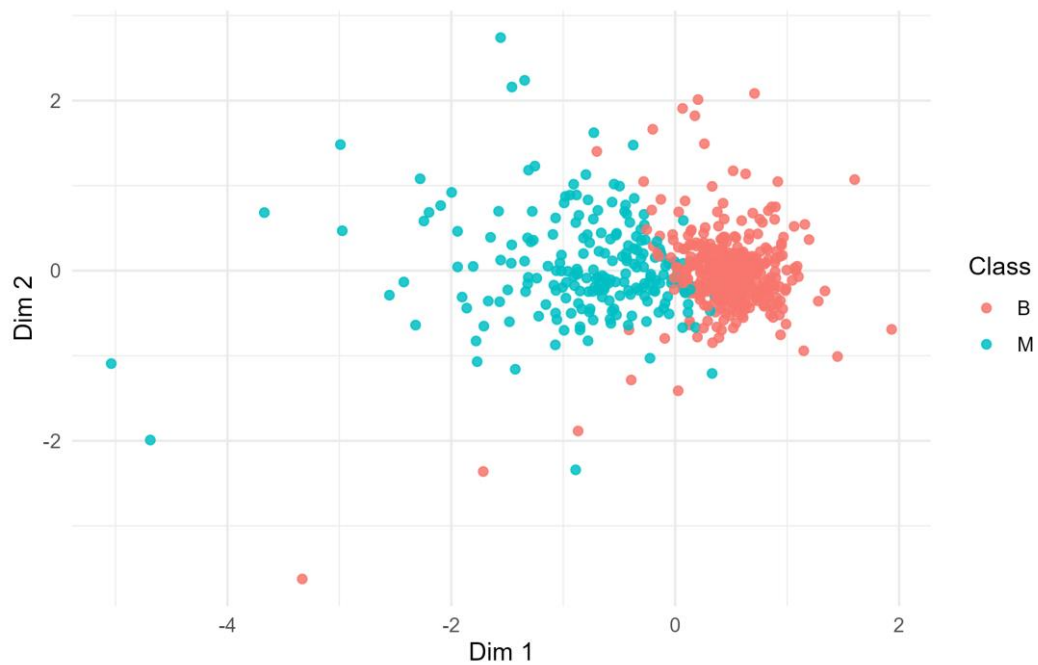
nMDS — value = 0.525



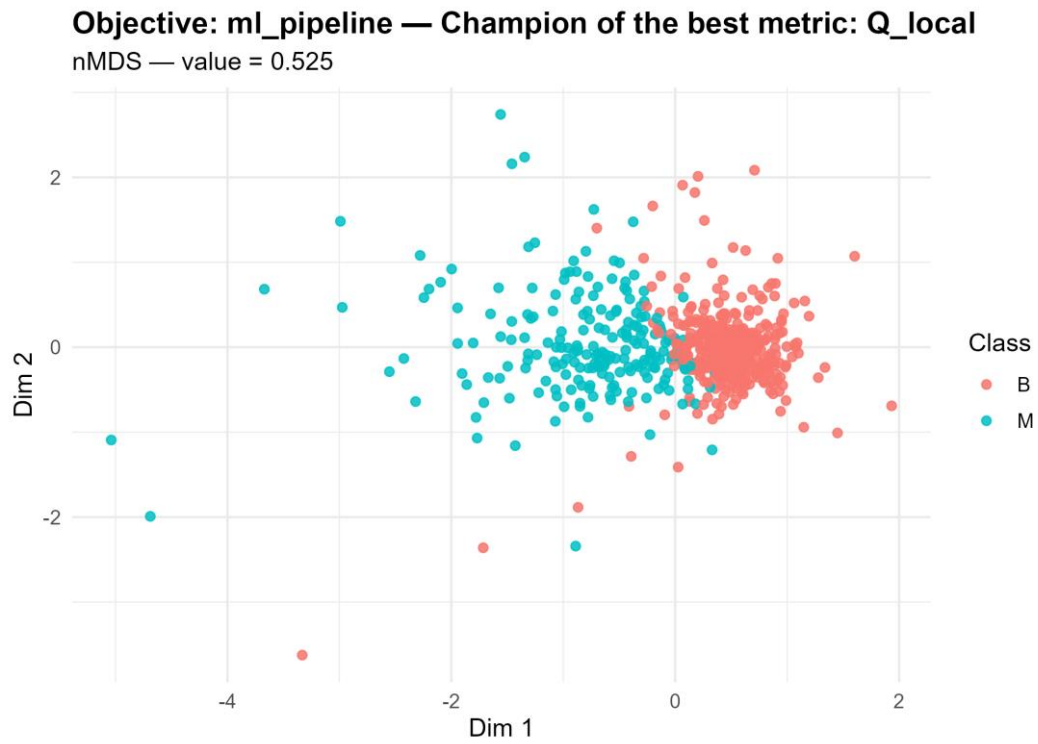
(b)

Objective: reconstruction — Champion of the best metric: Q_local

nMDS — value = 0.525



(c)



(d)

Figure 20 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

5. STUDY OF CASE II

5.1 Description of the Dataset

The Chemical Composition of Ceramic Samples was named df2. It contains 88 instances, 19 features, and two classes. Energy-dispersive X-ray fluorescence (EDXRF) was used to determine the chemical composition of the celadon bodies and glazes obtained from the Longquan kiln (Dayao County) and the Jingdezhen kiln. Forty typical shards from 4 cultural eras were selected to investigate their raw materials and firing technologies. The objective was to identify the chemical elements that are the strongest explanatory variables for classifying samples into different cultural eras and kilns [88].

The df2 dataset, which represents the chemical compositions of ceramic materials, was imported directly from a CSV file. When present, noninformative identifier columns (e.g., "ID", "id", and "index") were removed from the analysis. The oxide metal concentrations expressed initially in weight percentages (%wt) were systematically converted to mg.kg^{-1} units by multiplying selected variables (Na_2O , MgO , Al_2O_3 , SiO_2 , K_2O , CaO , TiO_2 , and Fe_2O_3) by a factor of 10,000. To avoid numerical inconsistencies, all the negative values—typically introduced by instrumental noise—were transformed to their absolute magnitudes.

Only numeric variables were retained for dimensionality reduction purposes. These variables were mean-centered and unit-variance scaled using z-score standardization to eliminate scale effects and ensure that all DR techniques operated under comparable conditions. The resulting scaled matrix constituted the high-dimensional data space ($\mathbf{X}_{\text{high}}$) that was used throughout the computational procedures. Optional correlation maps were generated using the *corrplot* package to visualize interdependencies among variables.

5.2 Results and Discussion

The correlation structure among the evaluation metrics used in the benchmarking framework is shown in Figure 21.

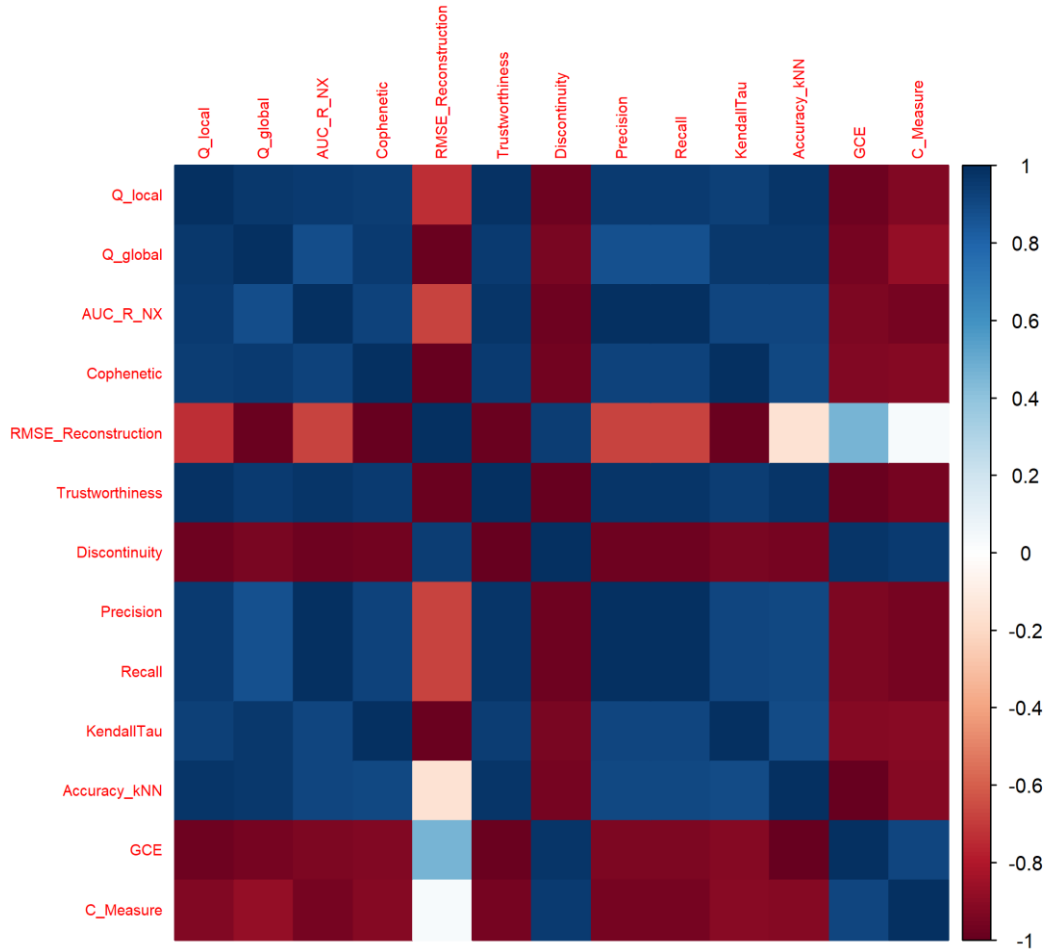


Figure 21 - The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Two dominant clusters are evident. Metrics related to local and global structure preservation (e.g., Q_{local} , Q_{global} , $AUC_R(NX)$, cophenetic correlation, trustworthiness, precision, recall, KendallTau, and kNN accuracy) formed a strongly positive block, indicating that they capture similar aspects of neighborhood fidelity and class-relevant

geometry. In contrast, the error- and distortion-based measures (RMSE_Reconstruction, the GCE, and the C_measure) displayed strong negative correlations with this group, reflecting the inherent tradeoff between neighborhood preservation and reconstruction accuracy. RMSE_Reconstruction yielded a distinct correlation profile, aligning more closely with the distortion-oriented metrics and inversely with the topology-preserving metrics. This separation confirms that reconstruction-focused methods optimize fundamentally different criteria than those that emphasize local continuity or class separation. Overall, the heatmap illustrates that DR metrics naturally organize into coherent families, each aligned with a different analytical objective. This finding reinforces the need for a multimetric, scenario-dependent approach when selecting dimensionality reduction algorithms.

Table 30A and Table 30B show the apparent performance differences among the evaluated dimensionality reduction techniques. t-SNE and nMDS consistently achieved the highest locality preservation metrics—trustworthiness, continuity, Q_{local} , and AUC_R(NX) —indicating strong neighborhood fidelity and effective class separation capabilities. MDS also preserved the global structure best, as reflected by its highest cophenetic correlation. Diffusion maps performed robustly across both local and global metrics, exhibiting minimal geometric distortion (the lowest GCE). The classic convex methods, such as PCA, MDS, and FastICA, yielded stable intermediate performance, whereas kPCA, HLLE, DrL, and Fruchterman–Reingold substantially degraded across the structural, reconstruction, and classification metrics. Overall, the results confirm that no single DR method is universally superior; instead, the objective-specific performance patterns underscore the need for a multimetric, scenario-dependent algorithm selection process to obtain reliable low-dimensional representations.

Table 30A – Figures of merit obtained from Dataset 2 as df2.

Algorithm	Q _{local}	Q _{global}	AUC R(NX)	Cophenetic correlation	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.568	0.878	0.563	0.783	0.684	0.904	0.928
Isomap	0.555	0.896	0.576	0.855		0.910	0.940
PCA	0.497	0.871	0.600	0.847	0.684	0.911	0.949
PCA L1	0.481	0.859	0.569	0.808		0.907	0.936
kPCA	0.454	0.740	0.555	0.400	11.11	0.870	0.890
FastICA	0.504	0.852	0.587	0.812	0.684	0.908	0.942
MDS	0.497	0.871	0.600	0.847		0.911	0.949
nMDS	0.565	0.896	0.645	0.919		0.939	0.946
DrL	0.023	0.495	0.166	-0.085		0.505	0.496
Fruchterman-Reingold	0.057	0.497	0.177	-0.206		0.501	0.457
Kamada-Kawai	0.507	0.861	0.571	0.839		0.901	0.927
HLLE	0.317	0.779	0.286	0.226		0.715	0.660
Diffusion Maps	0.544	0.872	0.613	0.683		0.925	0.947
t-SNE	0.595	0.851	0.740	0.768		0.960	0.964

* Numbers in bold represent the best values.

Table 30B - Figures of merit obtained from dataset 2 as df2.

Algorithm	Discontinuity	Precision	Recall	Kendall Tau	Accuracy k- NN	GCE	C Measure
DRR	0.072	0.608	0.608	0.603	1.000	0.241	187.6
Isomap	0.060	0.600	0.600	0.682	0.989	0.238	181.2
PCA	0.051	0.639	0.639	0.676	1.000	0.240	153.6
PCA L1	0.064	0.610	0.610	0.629	1.000	0.199	195.7
kPCA	0.110	0.598	0.598	0.273	0.989	0.320	168.6
Fast ICA	0.058	0.618	0.618	0.622	0.977	0.337	161.6
MDS	0.051	0.639	0.639	0.676	1.000	0.240	153.6
nMDS	0.054	0.681	0.681	0.776	1.000	0.272	148.6
DrL	0.504	0.164	0.164	-0.047	0.511	3.886	293.5
Fruchterman-Reingold	0.543	0.156	0.156	-0.129	0.557	4.225	274.9
Kamada-Kawai	0.073	0.605	0.605	0.665	0.989	0.556	156.4
HLLE	0.340	0.264	0.264	0.220	0.886	1.475	245.3
Diffusion Maps	0.053	0.651	0.651	0.647	1.000	0.058	167.7
t-SNE	0.036	0.769	0.769	0.619	1.000	0.097	160.3

*Numbers in bold represent the best values

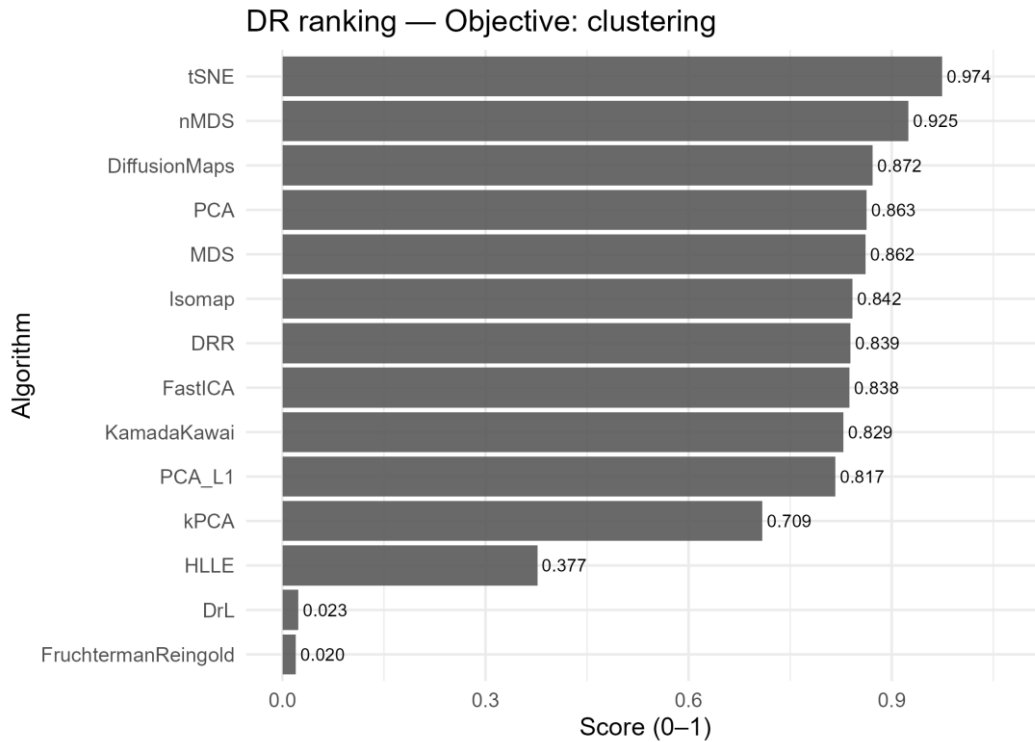
The scenario-based rankings in Table 31 show that t-SNE and nMDS consistently achieved the highest overall performance across all the analytical objectives. For the clustering, visualization, and ML pipeline tasks, t-SNE achieved the best final scores because of its perfect local and class indices, confirming its strong ability to preserve neighborhood structures and class separability. t-SNE also ranked highest for reconstruction, though only marginally: nMDS follows less than 0.01 behind, driven by its superior global index and balanced geometric behavior, which keeps it the more natural choice for tasks that emphasize distance preservation. Diffusion maps, PCA, and MDS formed a stable second tier of methods with competitive and consistent performance across various scenarios. These results highlight that the process of selecting the optimal DR method is inherently objective-dependent and that the proposed index-based framework provides a reliable basis for identifying the most appropriate technique for each analytical context. In this table, only the five principal DR methods that gave the best results were shown.

Table 31 - Figures of merit obtained from dataset 2 as df2.

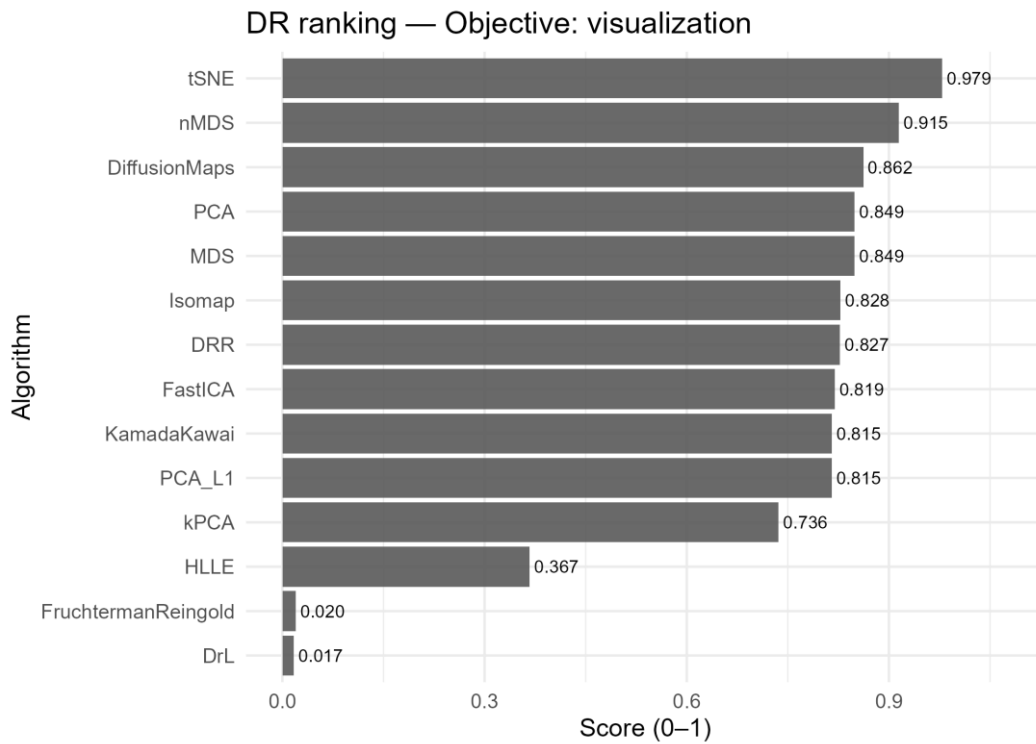
Algorithm	Index Local	Index Global	Index Geometric	Index Class	Final Score	Objective
tSNE	1.000	0.895	0.970	1.000	0.974	clustering
nMDS	0.904	0.959	0.971	0.904	0.925	clustering
DiffusionMaps	0.862	0.842	0.945	0.872	0.872	clustering
PCA	0.824	0.880	0.973	0.859	0.863	clustering
MDS	0.824	0.880	0.964	0.859	0.862	clustering
tSNE	1.000	0.895	0.970	1.000	0.979	visualization
nMDS	0.904	0.959	0.971	0.904	0.915	visualization
DiffusionMaps	0.862	0.842	0.945	0.872	0.862	visualization
PCA	0.824	0.880	0.973	0.859	0.849	visualization
MDS	0.824	0.880	0.964	0.859	0.849	visualization
tSNE	1.000	0.895	0.970	1.000	0.956	reconstruction
nMDS	0.904	0.959	0.971	0.904	0.947	reconstruction
PCA	0.824	0.880	0.973	0.859	0.901	reconstruction
MDS	0.824	0.880	0.964	0.859	0.897	reconstruction
DiffusionMaps	0.862	0.842	0.945	0.872	0.889	reconstruction
tSNE	1.000	0.895	0.970	1.000	0.979	ml_pipeline
nMDS	0.904	0.959	0.971	0.904	0.915	ml_pipeline
DiffusionMaps	0.862	0.842	0.945	0.872	0.862	ml_pipeline
PCA	0.824	0.880	0.973	0.859	0.849	ml_pipeline
MDS	0.824	0.880	0.964	0.859	0.849	ml_pipeline

*Numbers in bold represent the best values for each objective.

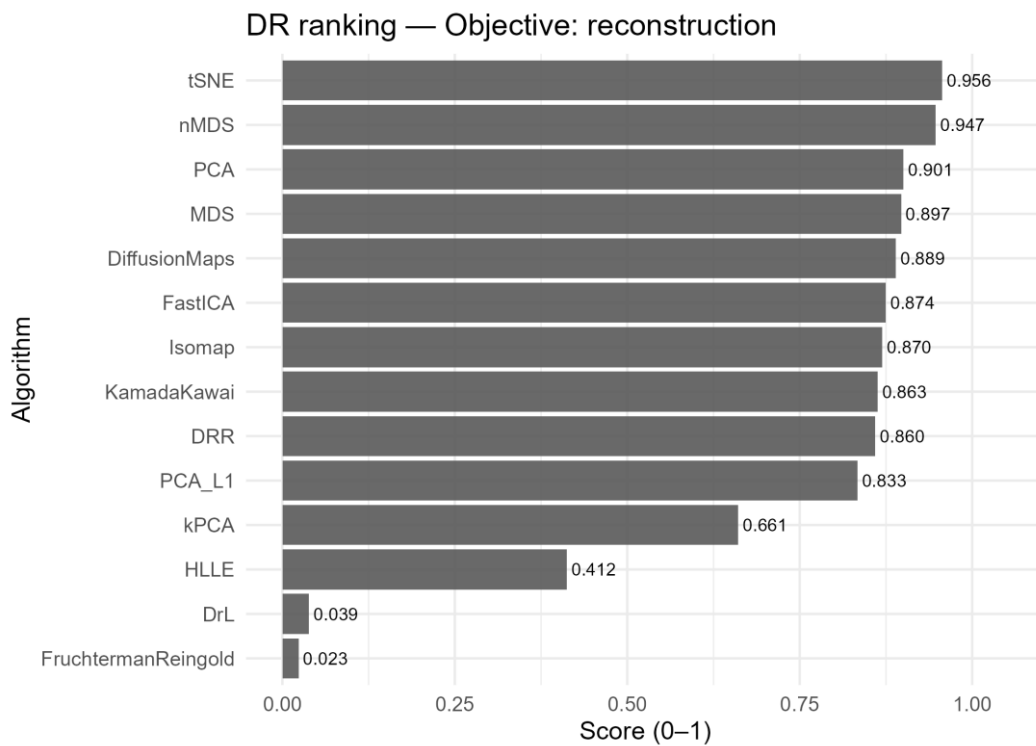
Figure 22 summarizes the objective-based ranking of the dimensionality-reduction algorithms for Dataset df2 across the four analytical scenarios. t-SNE leads every objective, with final scores of 0.974 for clustering, 0.979 for visualization and ML-pipeline integration, and 0.956 for reconstruction, sustained by perfect local and class indices ($\text{Index_Local} = \text{Index_Class} = 1.000$). nMDS forms a close second tier, trailing by less than 0.01 in the reconstruction objective (0.947) thanks to the highest global index of the set ($\text{Index_Global} = 0.959$). Diffusion Maps, PCA and classical MDS follow as a stable intermediate group, whereas kPCA, HLLE, DrL and Fruchterman–Reingold remain clearly behind across all four scenarios.



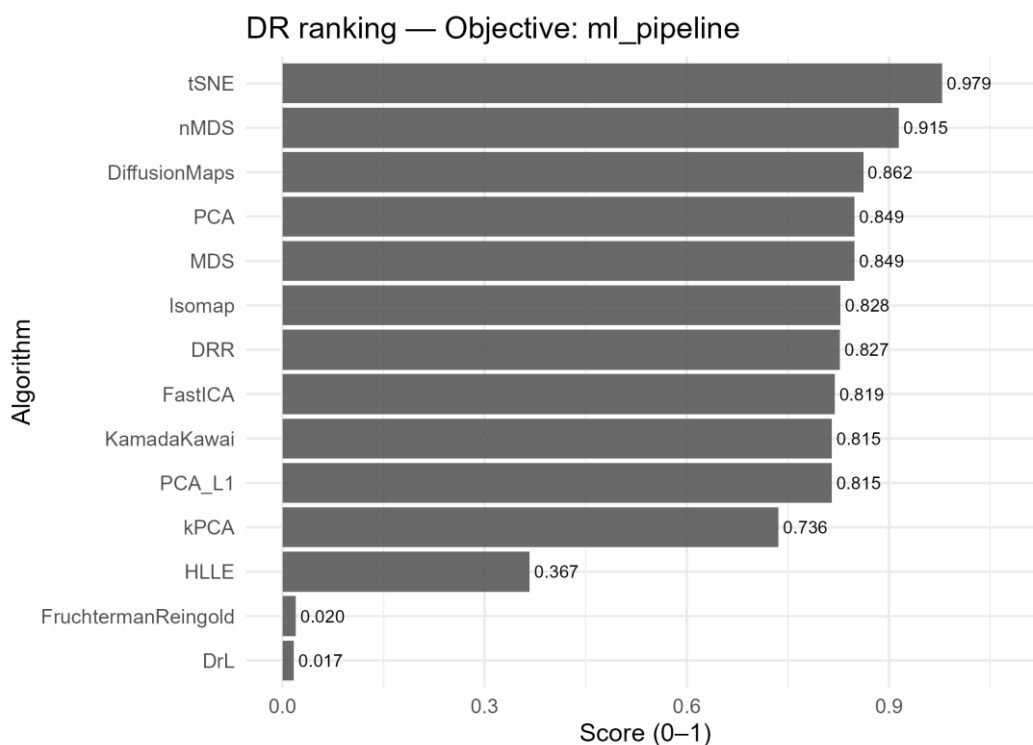
(a)



(b)



(c)



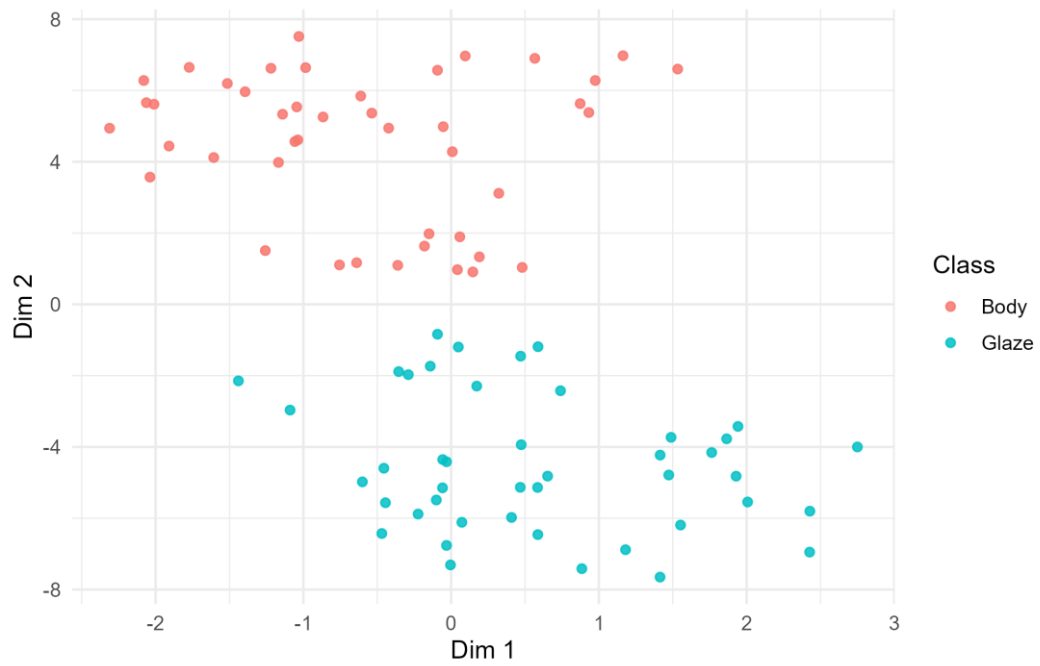
(d)

Figure 22 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Figure 23 presents the two-dimensional embeddings produced by the algorithm ranked first by the Final Score in each objective. Because t-SNE leads all four scenarios for this dataset, the four panels show the same method and differ only in the score reported in the subtitle. The celadon bodies and glazes separate into two compact, well-resolved groups with a clean boundary between them, which is consistent with the perfect class index obtained for this dataset and confirms that the chemical composition measured by EDXRF discriminates the two kiln groups almost completely.

Objective: clustering — Top-ranked algorithm (baseline)

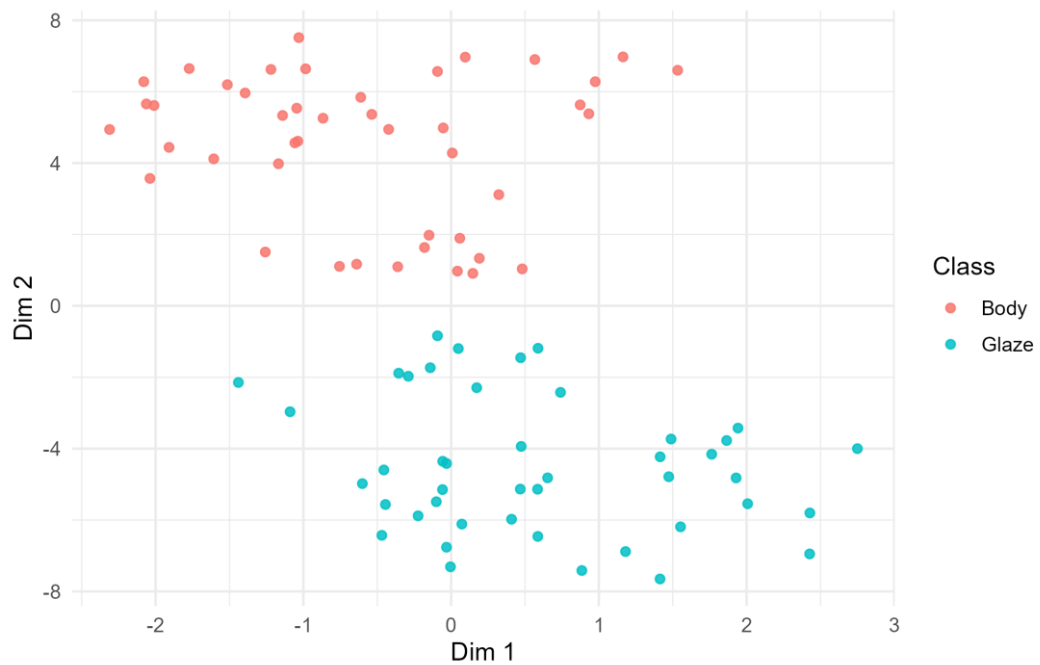
tSNE — Final Score = 0.9745



(a)

Objective: visualization — Top-ranked algorithm (baseline)

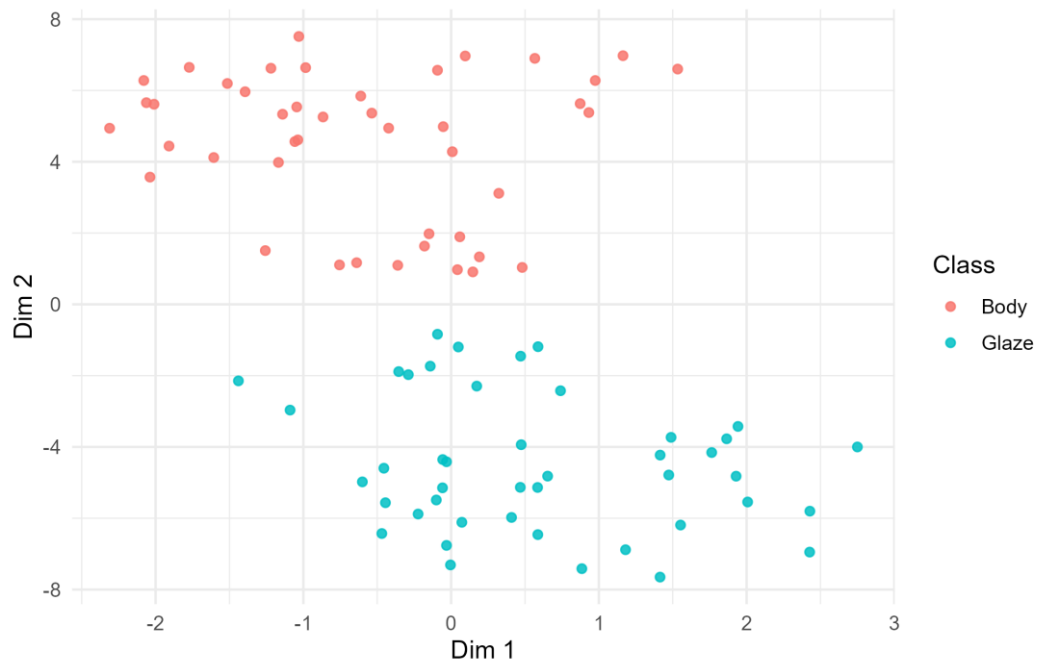
tSNE — Final Score = 0.979



(b)

Objective: reconstruction — Top-ranked algorithm (baseline)

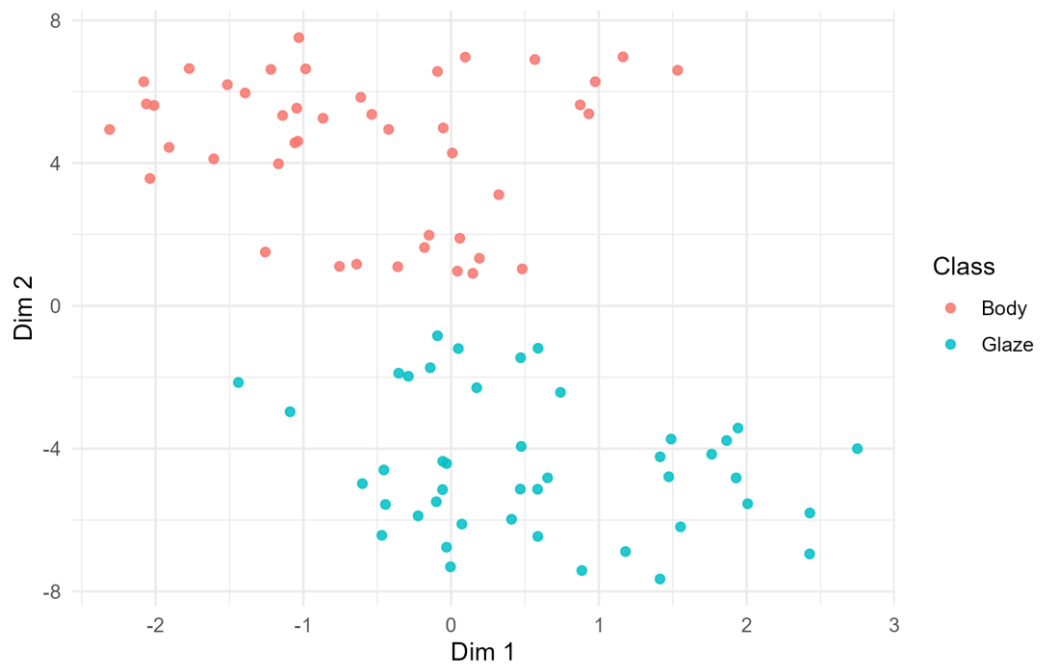
tSNE — Final Score = 0.9565



(c)

Objective: ml_pipeline — Top-ranked algorithm (baseline)

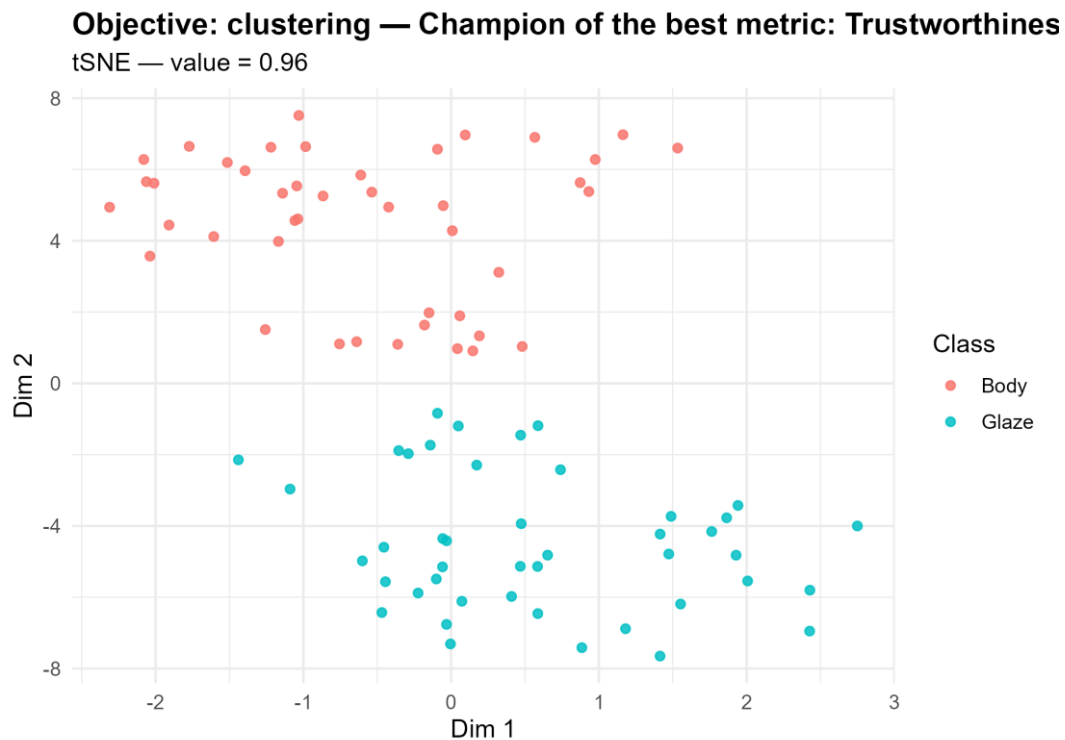
tSNE — Final Score = 0.979



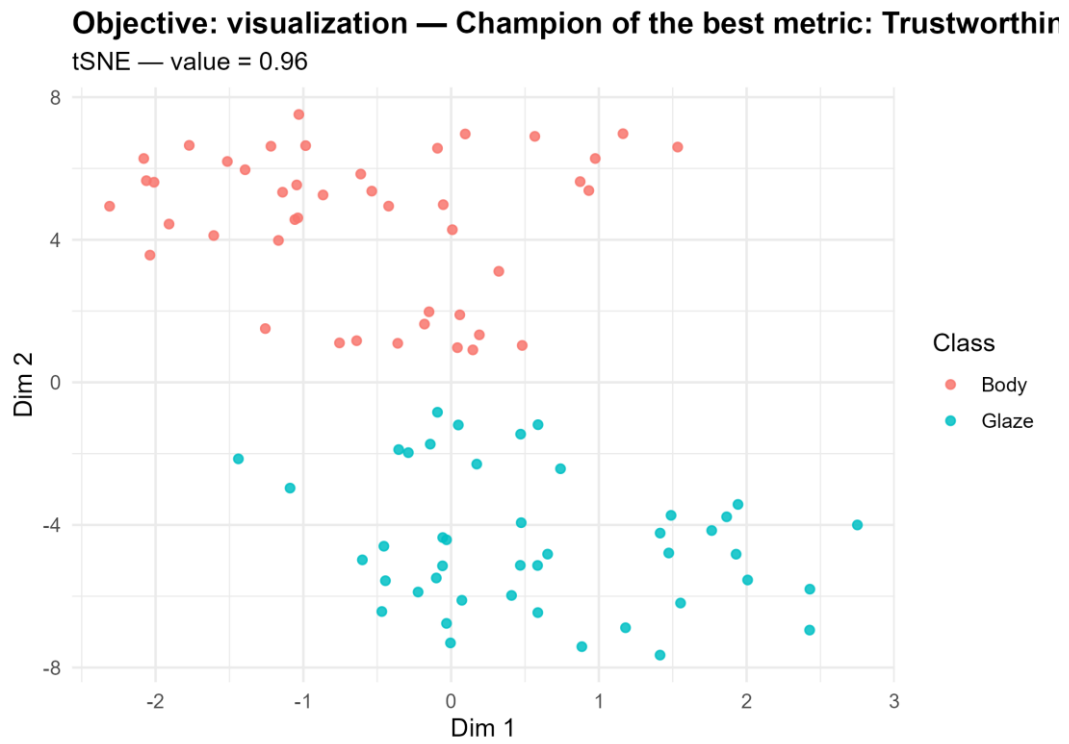
(d)

Figure 23 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

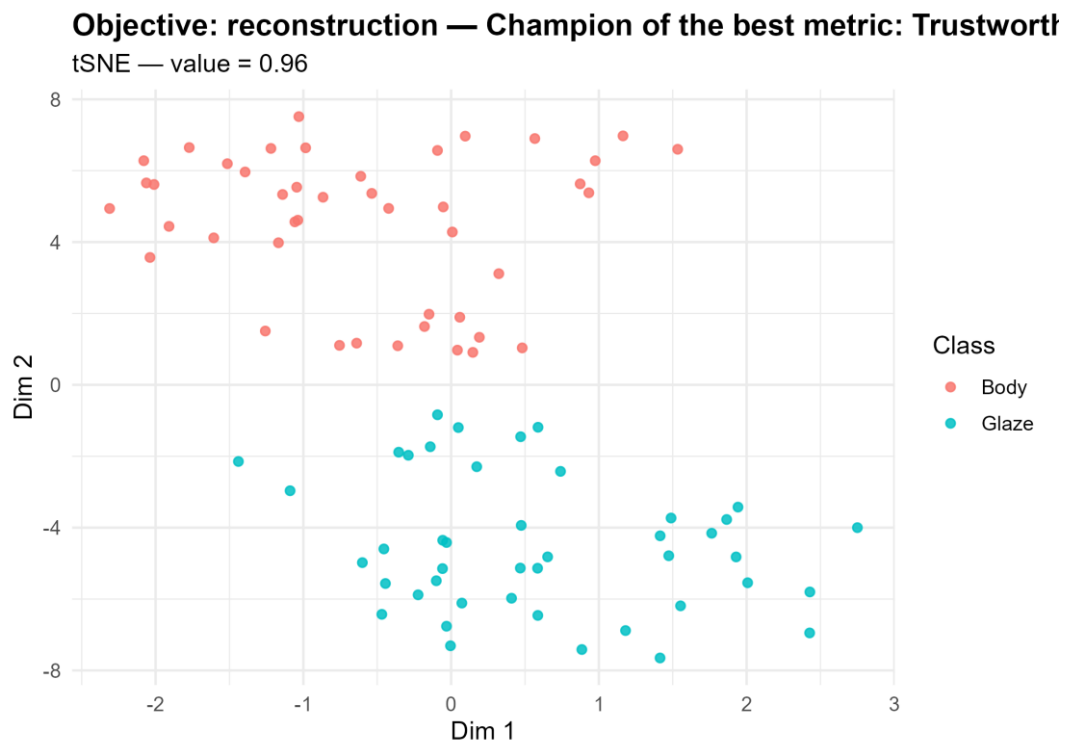
Figure 24 shows the embeddings generated by the champion algorithm under the most consistent metric. For Dataset df2 that metric is Trustworthiness, and the champion is again t-SNE, so the layout closely mirrors the previous figure. The agreement between the baseline ranking and the metric-driven selection is itself informative: when a dataset separates as cleanly as this one, the choice of evaluation criterion has little influence on which technique is preferred, and the decision can be made on secondary grounds such as interpretability or the availability of an explicit mapping.



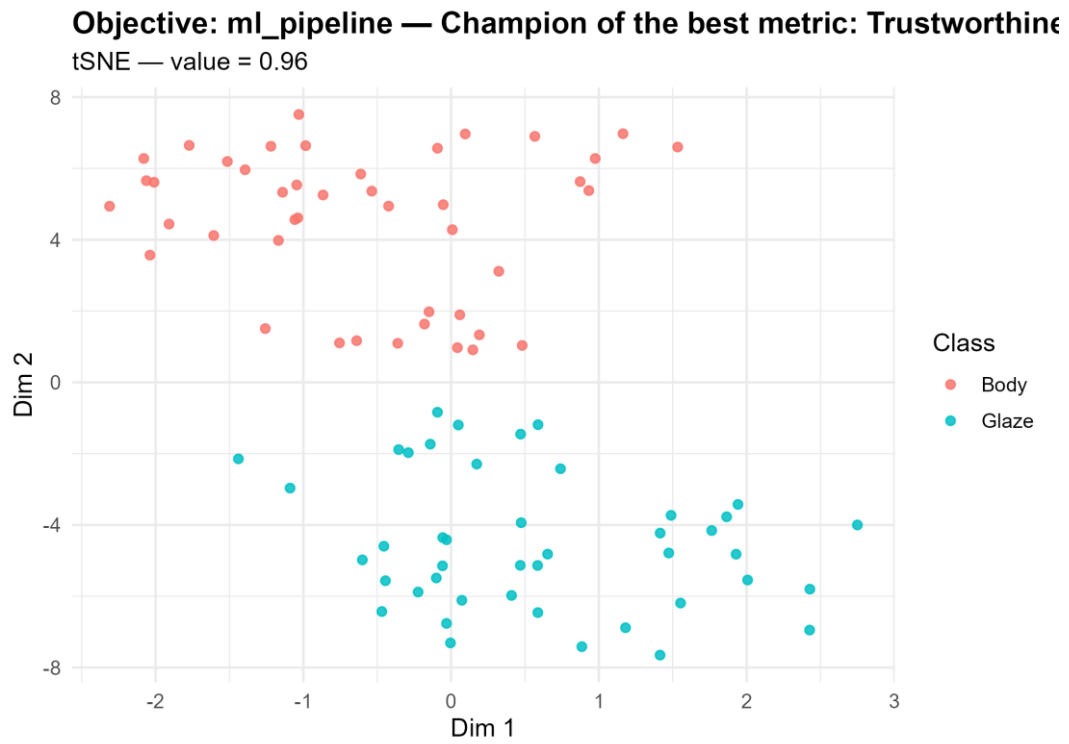
(a)



(b)



(c)



(d)

Figure 24 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

6. STUDY OF CASE III

6.1 Description of the Dataset

The Dry Bean dataset was named df3. Images of 13,611 grains from 7 different registered dry bean varieties were captured with a high-resolution camera. A total of 16 features, 12 dimensions, and four shape forms were obtained from the grains. Seven different types of dry beans were used in this research, considering features such as form, shape, type, structure, and market situation. A computer vision system was developed to distinguish among seven registered varieties of dry beans with similar features, thereby achieving uniform seed classification [89].

6.2 Results and Discussion

The correlation heatmap reveals strong dependencies among conceptually related evaluation metrics used to assess dimensionality reduction (see Figure 25).

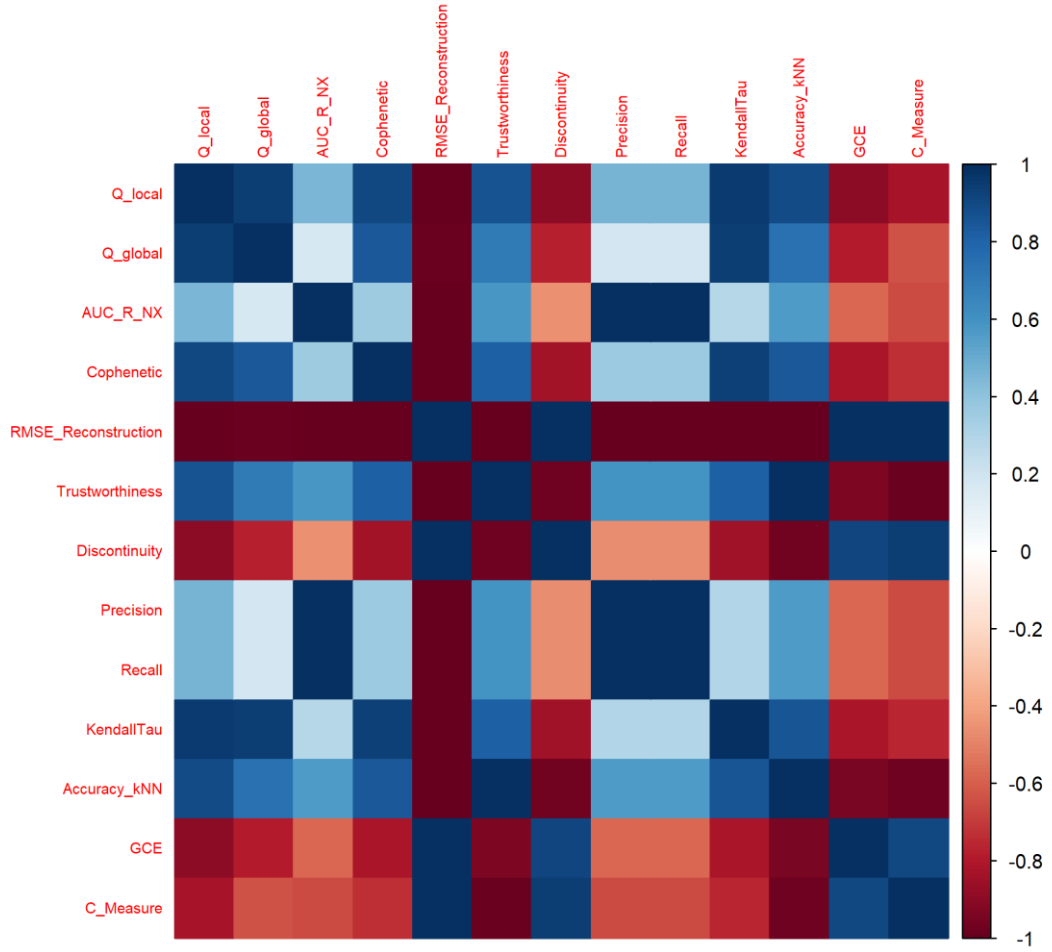


Figure 25 – The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Local neighborhood preservation metrics (e.g., Q_{local} , $Trustworthiness$, $Continuity$, and $Accuracy-kNN$) exhibit strong positive correlations, indicating substantial redundancy and a shared focus on fidelity to local structure. Similarly, global structure metrics (Q_{global} , $Cophenetic$ correlation, and $Kendall's\ Tau$) form a tightly correlated group, reflecting consistent evaluation of large-scale manifold preservation. Precision and Recall show near-perfect correlation, as expected for balanced classification scenarios. In contrast, $RMSE_Reconstruction$ presents strong negative correlations with most

neighborhood and classification metrics, highlighting a fundamental trade-off between reconstruction accuracy and structural preservation. Geometric distortion measures (*GCE* and *C-Measure*) cluster separately and are negatively correlated with locality and globality metrics, reinforcing their distinct analytical role. Overall, the results demonstrate that dimensionality-reduction evaluation is intrinsically multi-objective, supporting the need for scenario-dependent metric weighting rather than reliance on a single performance indicator.

Table 32A and Table 32B report a comprehensive set of local, global, geometric, and classification-oriented figures of merit for Dataset 3 across multiple dimensionality-reduction techniques. Linear and near-linear methods (PCA, MDS, DRR, Isomap, and Kamada–Kawai) exhibit consistently high Q_{global} , Cophenetic, Kendall’s Tau, and neighborhood continuity scores, indicating robust preservation of global manifold structure with stable local neighborhoods. DRR and PCA-based approaches also achieve low reconstruction errors, confirming their suitability for reconstruction-driven objectives. Nonlinear neighborhood-focused methods display distinct trade-offs. t-SNE achieves the highest Precision, Recall, and Accuracy-kNN metrics and the lowest geometric distortion metrics (*GCE* and *C-Measure*), highlighting its superiority for class separability and visualization, albeit with weaker preservation of global structure.

Conversely, kPCA, HLLE, and nMDS exhibit degraded performance across most metrics, particularly in terms of worldwide fidelity and classification accuracy. Overall, the results reinforce that no single method is universally optimal. Instead, performance is strongly objective-dependent, supporting a multi-criteria, scenario-driven evaluation framework for selecting a dimensionality-reduction method.

Table 32A – Figures of merit obtained from Dataset 3 as df3.

Algorithm	Q_local	Q_global	AUC_R_NX	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.616	0.938	0.117	0.964	0.425	0.934	0.991
Isomap	0.614	0.934	0.121	0.963		0.937	0.991
PCA	0.615	0.938	0.118	0.963	0.425	0.934	0.991
PCA_L1	0.602	0.925	0.109	0.941		0.925	0.989
kPCA	0.444	0.790	0.073	0.406	174.8	0.863	0.980
FastICA	0.600	0.913	0.117	0.946	0.425	0.934	0.990
MDS	0.615	0.938	0.118	0.963		0.934	0.991
nMDS	0.080	0.580	0.001	-0.002		0.501	0.499
DrL	0.385	0.735	0.150	0.542		0.982	0.960
FruchtermanReingold	0.468	0.842	0.047	0.783		0.887	0.973
KamadaKawai	0.637	0.938	0.132	0.978		0.947	0.990
HLLE	0.284	0.769	0.002	0.004		0.560	0.605
DiffusionMaps	0.420	0.834	0.013	0.142		0.818	0.858
tSNE	0.577	0.809	0.484	0.647		0.997	0.985

Table 32B – Figures of merit obtained from Dataset 3 as df3.

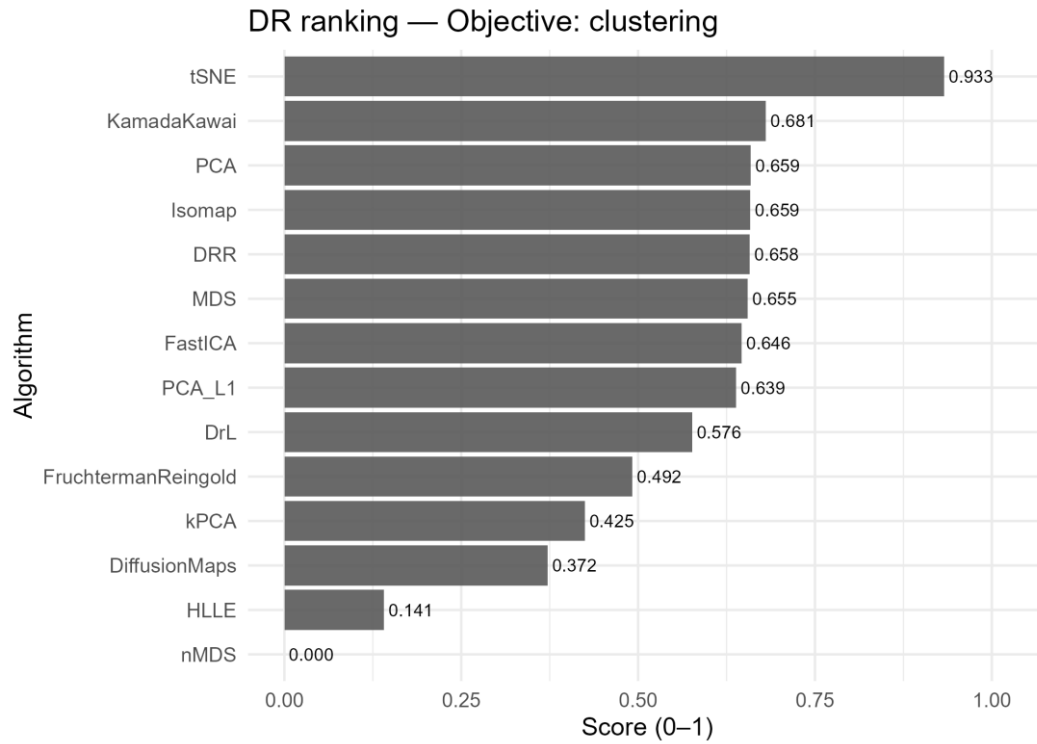
Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.009	0.122	0.122	0.850	0.854	2.581	13905.4
Isomap	0.009	0.125	0.125	0.842	0.864	2.195	13299.8
PCA	0.009	0.123	0.123	0.849	0.853	2.339	13879.3
PCA_L1	0.011	0.114	0.114	0.809	0.873	2.083	14816.6
kPCA	0.020	0.075	0.075	0.362	0.688	4.469	17585.6
FastICA	0.010	0.122	0.122	0.795	0.852	2.568	13986.4
MDS	0.009	0.123	0.123	0.849	0.853	2.335	13879.3
nMDS	0.501	0.001	0.001	0.000	0.186	10.321	42386.2
DrL	0.040	0.155	0.155	0.439	0.897	1.889	9648.4
FruchtermanReingold	0.027	0.048	0.048	0.579	0.761	3.581	16927.7
KamadaKawai	0.010	0.137	0.137	0.874	0.876	2.072	13062.3
HLLE	0.395	0.002	0.002	0.121	0.283	5.863	40166.6
DiffusionMaps	0.142	0.014	0.014	0.533	0.670	5.071	17791.5
tSNE	0.015	0.491	0.491	0.553	0.917	1.545	4432.1

Table 33 summarizes the objective-based composite indices and final scores obtained for Dataset 3, integrating local, global, geometric, and classification performance into a unified ranking framework. Across all analytical scenarios, t-SNE consistently achieves the highest Final Score, driven by its near-optimal Index_Local and Index_Class, confirming its dominance for clustering, visualization, and machine-learning pipeline objectives. In reconstruction-oriented scenarios, PCA and DRR perform competitively, reflecting a more balanced trade-off between preserving global structure and maintaining geometric stability. Kamada–Kawai and Isomap occupy intermediate positions, showing reasonable global and geometric fidelity but limited class discrimination. Overall, the table demonstrates that algorithm ranking is strongly objective-dependent, and that composite, scenario-weighted indices provide a robust and interpretable strategy for selecting dimensionality-reduction methods.

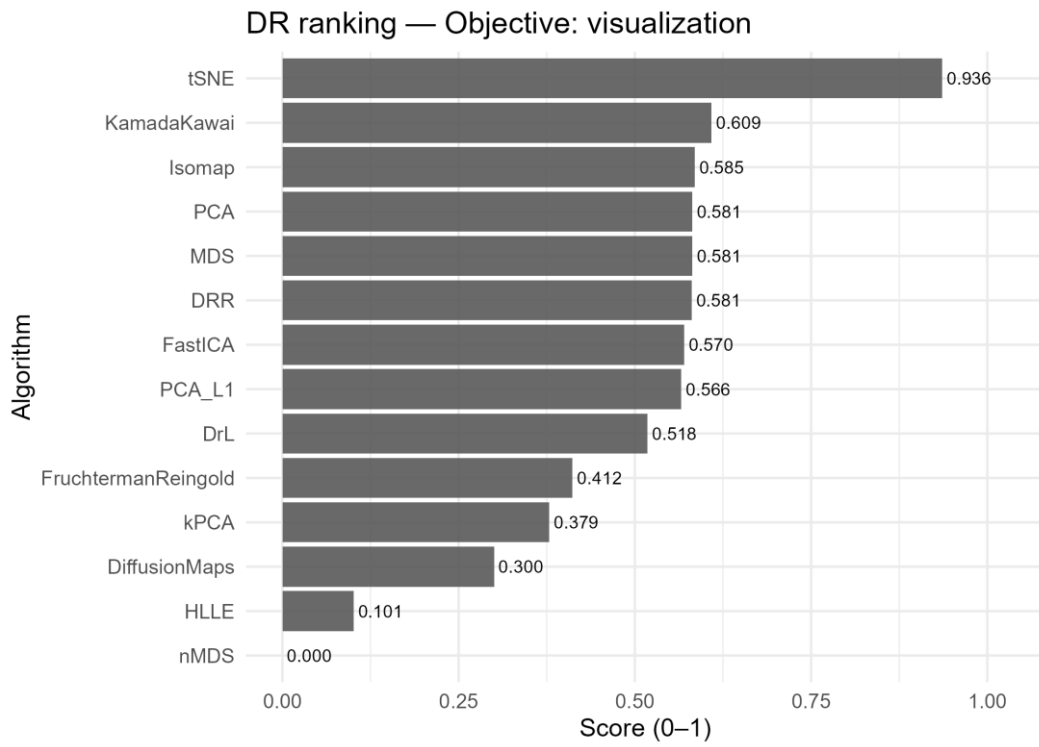
Table 33 – Figures of merit obtained from Dataset 3 as df3.

Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
tSNE	0.973	0.734	0.996	1.000	0.933	clustering
KamadaKawai	0.614	0.818	0.904	0.500	0.681	clustering
PCA	0.583	0.800	0.915	0.470	0.659	clustering
Isomap	0.586	0.796	0.897	0.478	0.659	clustering
DRR	0.582	0.800	0.908	0.469	0.658	clustering
tSNE	0.973	0.734	0.996	1.000	0.936	visualization
KamadaKawai	0.614	0.818	0.904	0.500	0.609	visualization
Isomap	0.586	0.796	0.897	0.478	0.585	visualization
PCA	0.583	0.800	0.915	0.470	0.581	visualization
MDS	0.583	0.800	0.887	0.470	0.581	visualization
tSNE	0.973	0.734	0.996	1.000	0.910	reconstruction
KamadaKawai	0.614	0.818	0.904	0.500	0.791	reconstruction
PCA	0.583	0.800	0.915	0.470	0.781	reconstruction
DRR	0.582	0.800	0.908	0.469	0.778	reconstruction
Isomap	0.586	0.796	0.897	0.478	0.774	reconstruction
tSNE	0.973	0.734	0.996	1.000	0.936	ml_pipeline
KamadaKawai	0.614	0.818	0.904	0.500	0.609	ml_pipeline
Isomap	0.586	0.796	0.897	0.478	0.585	ml_pipeline
PCA	0.583	0.800	0.915	0.470	0.581	ml_pipeline
MDS	0.583	0.800	0.887	0.470	0.581	ml_pipeline

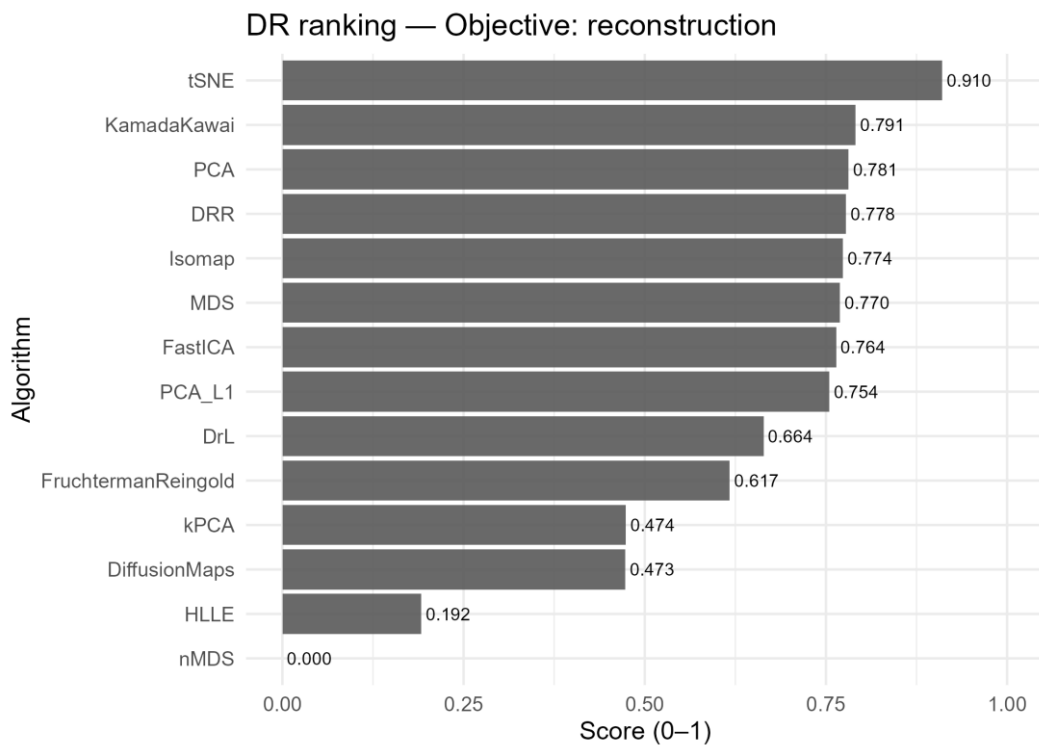
Figure 26 presents objective-driven rankings of dimensionality-reduction (DR) algorithms for clustering, visualization, reconstruction, and machine-learning pipeline scenarios.



(a)



(b)



(c)

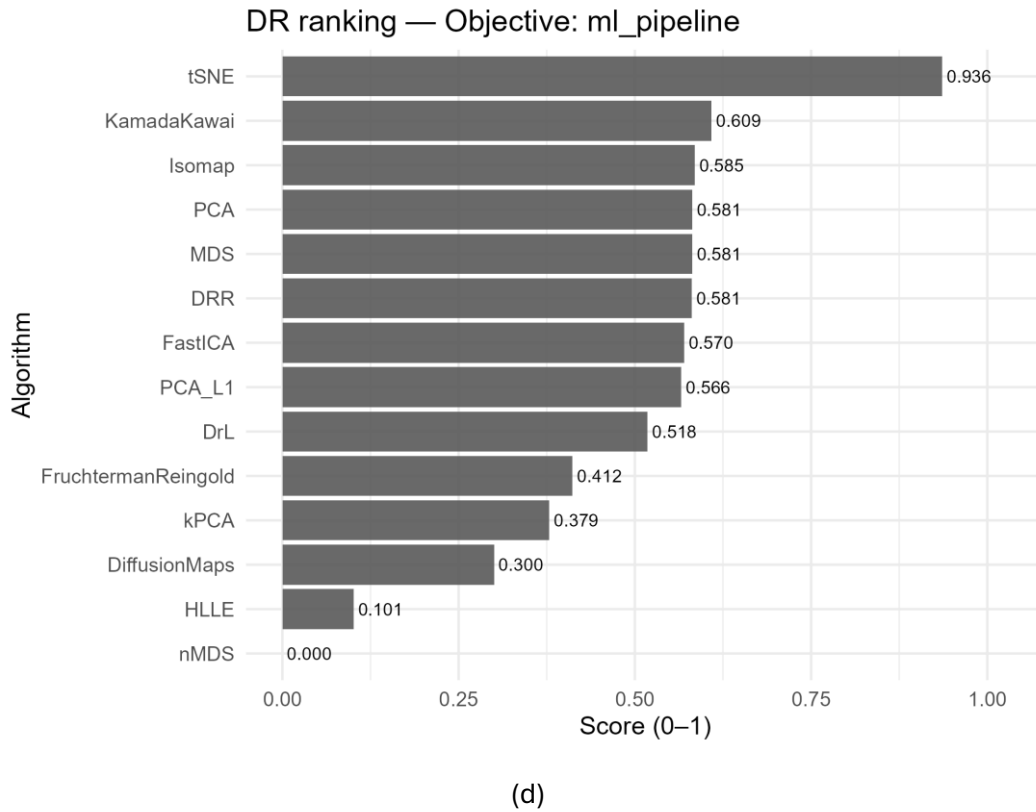


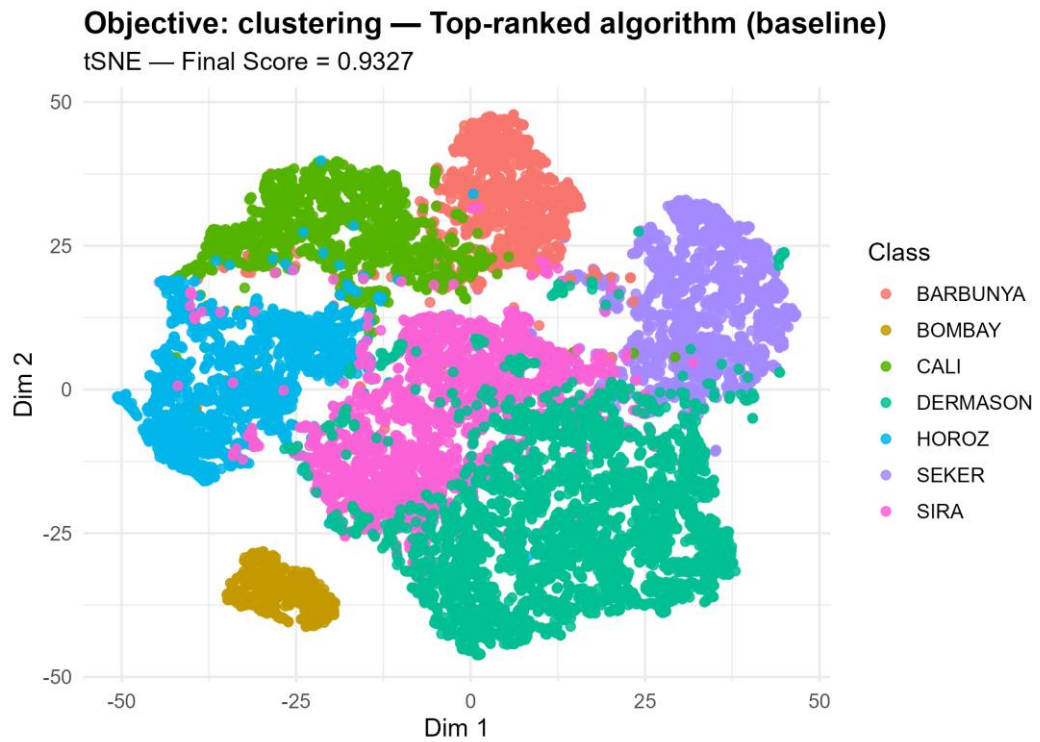
Figure 26 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Across all objectives, t-SNE consistently achieves the highest scores, particularly for clustering, visualization, and ML pipeline tasks, reflecting its strong preservation of local neighborhoods and class separability.

For reconstruction-oriented objectives, PCA and DRR are competitive alternatives, achieving balanced performance by preserving global structure while minimizing reconstruction error. Kamada–Kawai and Isomap occupy intermediate positions, offering reasonable global and geometric fidelity but weaker performance in class-oriented criteria. In contrast, kPCA, HLLE, and nMDS consistently rank lowest across scenarios, indicating limited suitability for the evaluated objectives.

Overall, the figure shows that algorithm performance is strongly objective-dependent, reinforcing the need for scenario-based, multi-criteria ranking rather than selecting a universally optimal DR method.

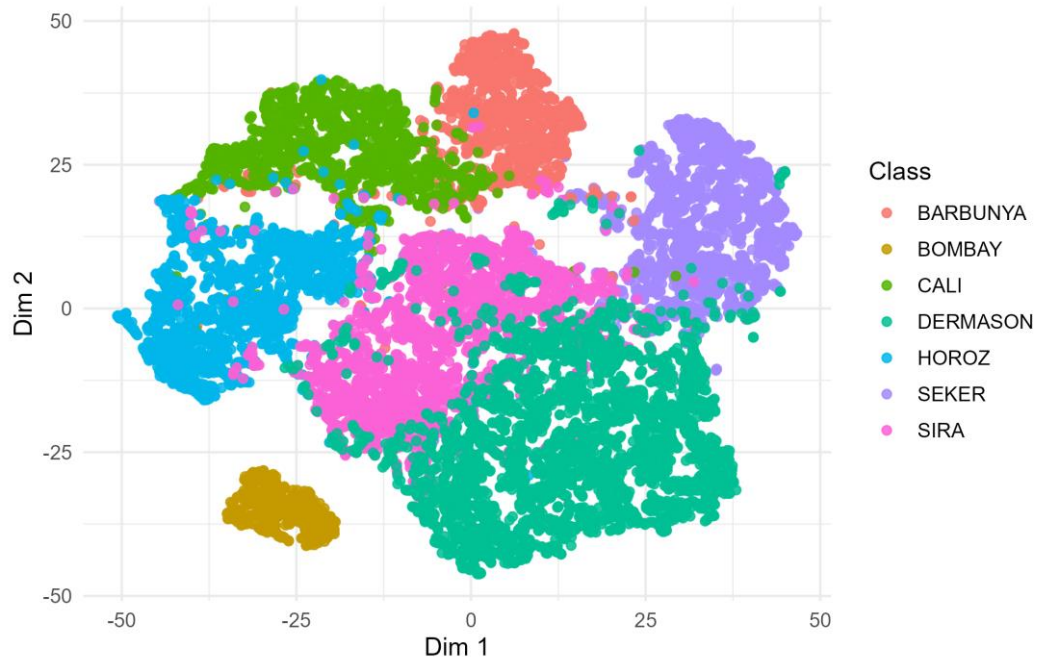
Figure 27 presents two-dimensional embeddings generated by the champion algorithm, selected by the final composite score, for four analytical objectives: clustering, visualization, reconstruction, and the machine-learning pipeline. In all scenarios, t-SNE consistently ranks first, producing embeddings with well-defined group separation and coherent local neighborhood structures. Distinct classes form compact and visually separable clusters, indicating strong preservation of class-related and local topological information.



(a)

Objective: visualization — Top-ranked algorithm (baseline)

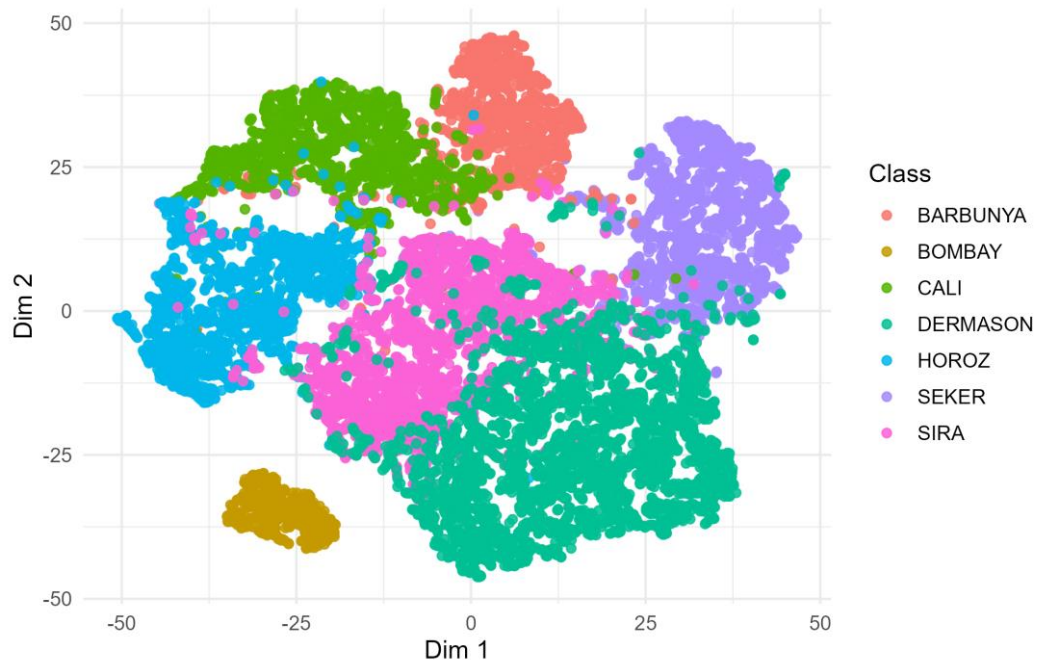
tSNE — Final Score = 0.936



(b)

Objective: reconstruction — Top-ranked algorithm (baseline)

tSNE — Final Score = 0.9104



(c)

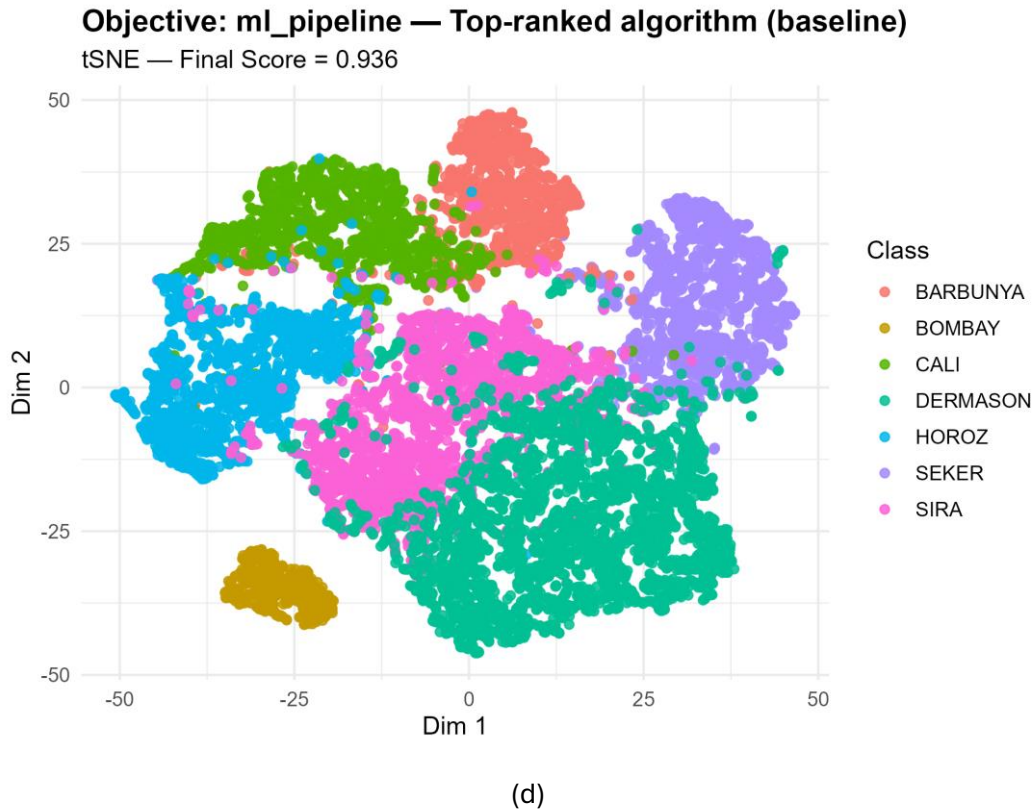
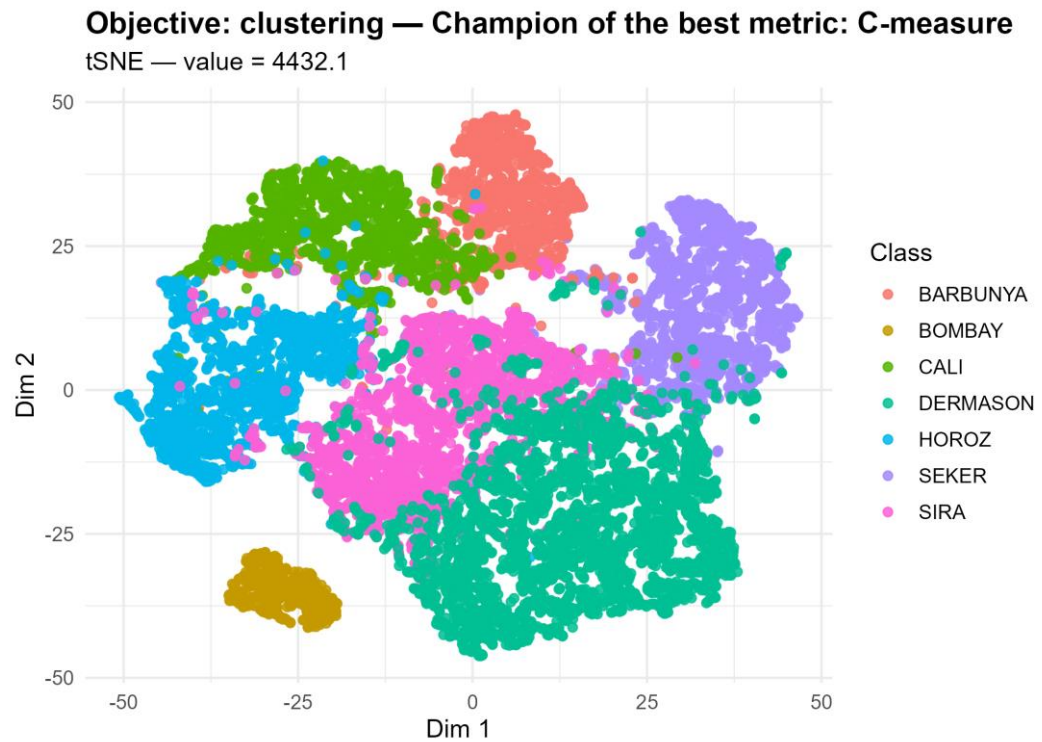


Figure 27 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Although the overall geometric layout remains similar across objectives, subtle differences in point dispersion reflect the objective-specific weighting of evaluation metrics. The results confirm that objective-driven ranking yields stable, interpretable embeddings, while also showing that visualization quality and class separability do not necessarily imply optimal reconstruction or preservation of global structure. Overall, the figure supports the effectiveness of the proposed multi-criteria framework in selecting dimensionality-reduction methods aligned with specific analytical goals.

Figure 28 illustrates two-dimensional embeddings generated by the algorithm selected as the champion of the best single metric (Discontinuity) across clustering, visualization, reconstruction, and machine-learning pipeline objectives. In all cases, nMDS is selected because it minimizes discontinuities on the low-dimensional manifold.

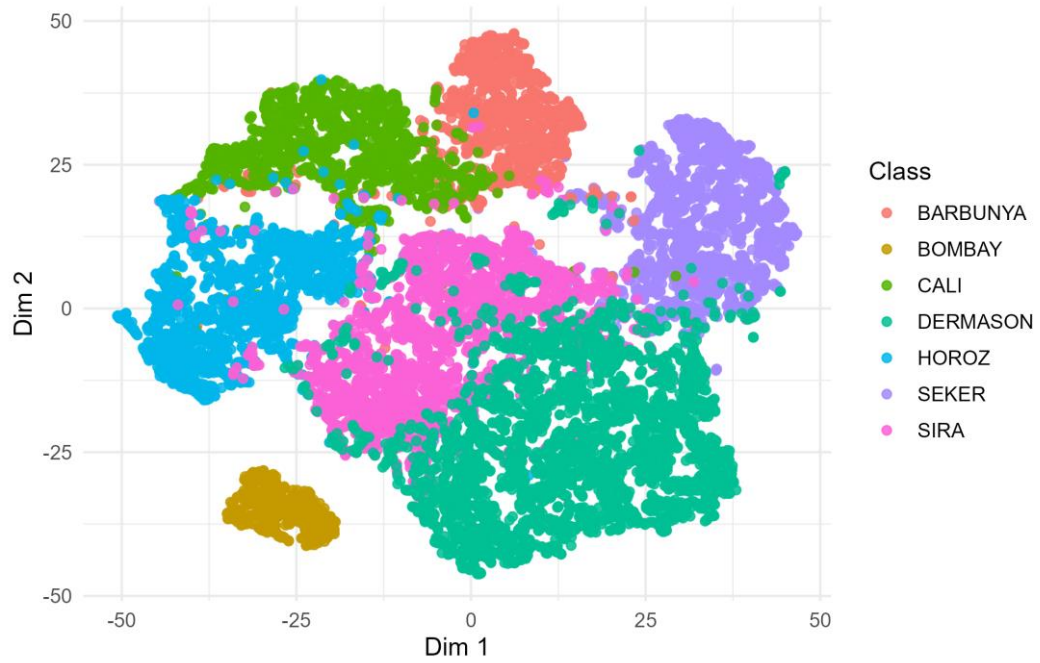
However, the resulting embeddings exhibit severe class overlap and poor cluster separation, indicating poor preservation of local neighborhoods and class structure. These results highlight a critical limitation of single-metric optimization: minimizing geometric discontinuity alone does not guarantee interpretable or analytically useful embeddings.



(a)

Objective: visualization — Champion of the best metric: C-measure

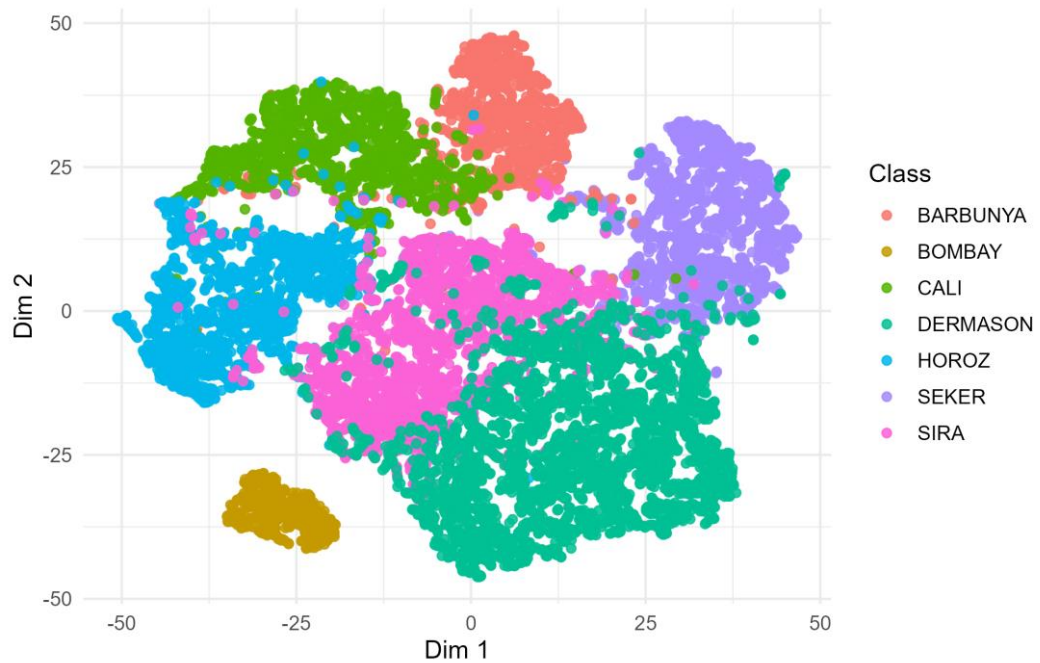
tSNE — value = 4432.1



(b)

Objective: reconstruction — Champion of the best metric: C-measure

tSNE — value = 4432.1



(c)

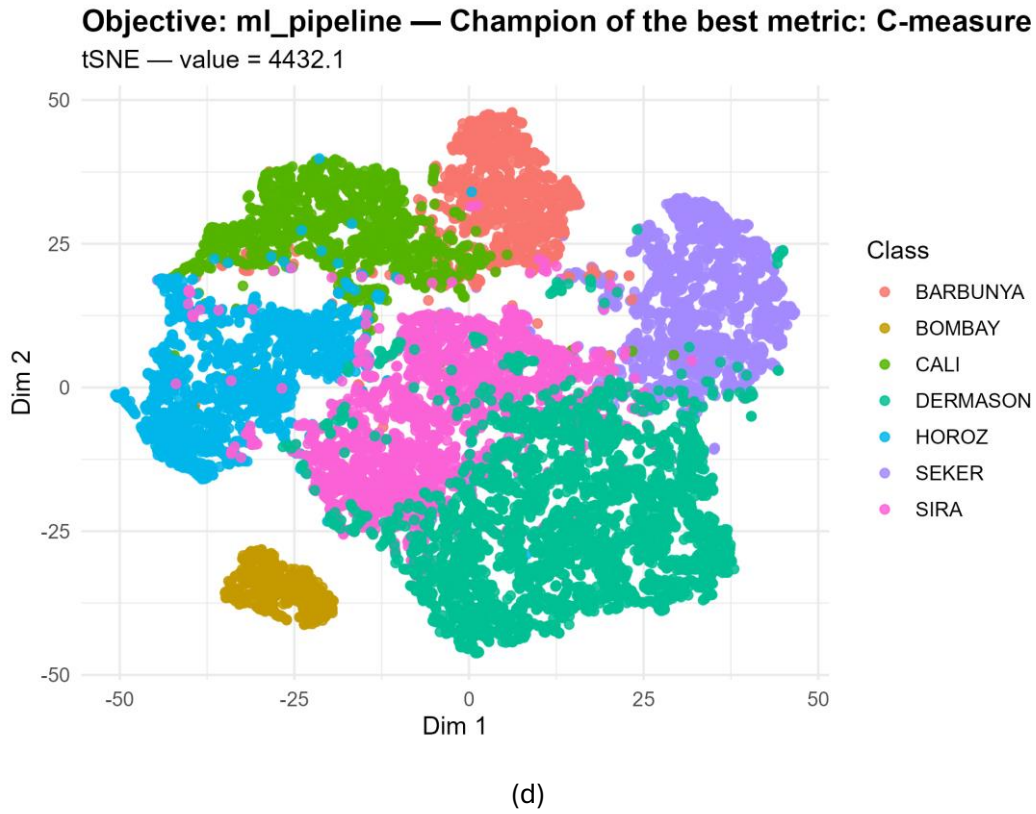


Figure 28 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

The figure contrasts sharply with the multi-criteria, objective-based selections, demonstrating that metric-specific optimization can lead to degenerate solutions. Consequently, the figure reinforces the need for composite, scenario-weighted evaluation frameworks for robust selection of dimensionality-reduction methods.

7. STUDY OF CASE IV

7.1 Description of the Dataset

The Glass Identification dataset was named df4. It contains 214 instances and nine features. The USA Forensic Science Service identified six types of glass based on their oxide compositions [90].

7.2 Results and Discussion

The correlation heatmap shown in Figure 29 illustrates the relationships among multiple evaluation metrics used to assess the performance of dimensionality-reduction algorithms.

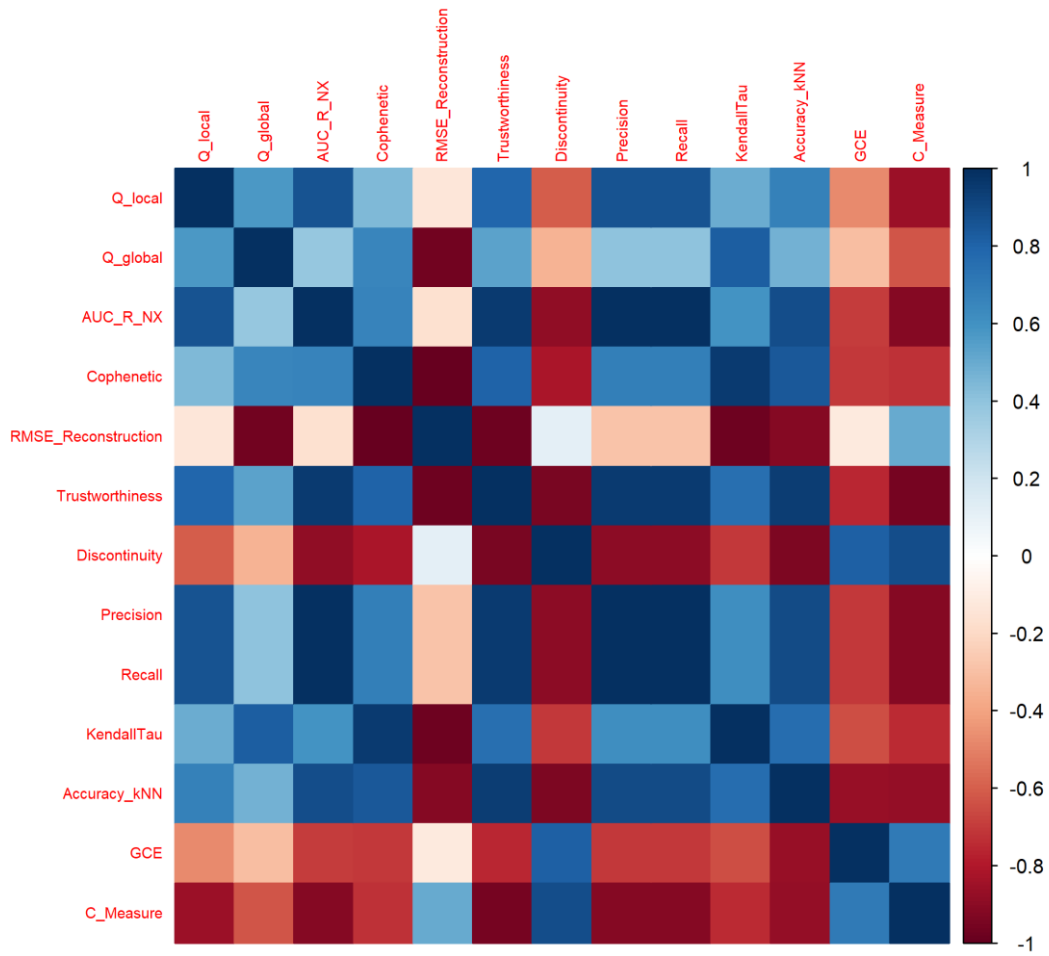


Figure 29 – The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

The figure reveals two dominant clusters: (i) locality-preserving measures, including Q_{local} , *Trustworthiness*, *Precision*, *Recall*, and *Accuracy_kNN*, which exhibit strong positive correlations; and (ii) global-structure measures such as Q_{global} , *AUC_R(NX)*, *Cophenetic*, and *KendallTau*. These groups reflect coherent metric families that capture complementary structural properties of the data manifold. In contrast, reconstruction- and distortion-based metrics (*RMSE_Reconstruction*, *Discontinuity*, *GCE*, *C_measure*) show strong negative correlations with both locality- and global-preserving groups, indicating an inherent trade-off: algorithms that optimize neighborhood or manifold fidelity typically yield poorer reconstruction performance, and vice versa. Overall, the figure underscores the multidimensional nature of DR evaluation and highlights the need to select scenario-dependent metrics rather than rely on a single performance indicator.

The metrics summarized for Dataset 4 reveal clear performance patterns among the evaluated dimensionality-reduction (DR) algorithms, as shown in

Table 34A and Table 34B. Non-metric MDS (nMDS) emerges as one of the strongest performers, achieving the highest global-structure scores (Q_{global} , AUC_R(NX) , Cophenetic) and leading precision–recall values, while also producing the best manifold-preservation indicator (KendallTau) among all methods. Kamada–Kawai and Isomap also exhibit balanced behavior across local, global, and class-oriented metrics, indicating robust preservation of both neighborhood topology and manifold geometry. In contrast, methods focused on nonlinear spectral embeddings (HLLE, DrL) and force-directed layouts (Fruchterman–Reingold) show inferior performance, particularly in global metrics, precision–recall, and k-NN classification accuracy. Notably, t-SNE achieves the strongest locality score (Q_{local}) and the highest precision–recall pair, reflecting excellent clustering and class separability; however, its global-structure metrics are comparatively weaker, a known characteristic of the method. Overall, the results confirm that no algorithm performs optimally across all criteria. Global manifold preservation favors nMDS, Kamada–Kawai, and Isomap, while t-SNE best captures local neighborhood fidelity and discriminative structure. These findings reinforce the importance of selecting DR algorithms according to the intended analytical objective rather than relying on a single universal method.

Table 34A – Figures of merit obtained from Dataset df4.

Algorithm	Q_local	Q_global	AUC_R_NX	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.360	0.790	0.403	0.848	0.701	0.841	0.901
Isomap	0.466	0.833	0.495	0.911		0.888	0.940
PCA	0.405	0.811	0.433	0.859	0.701	0.853	0.922
PCA L1	0.291	0.772	0.337	0.781		0.802	0.881
kPCA	0.380	0.732	0.419	0.510	7.808	0.795	0.913
FastICA	0.390	0.806	0.436	0.854	0.701	0.854	0.923
MDS	0.405	0.811	0.434	0.859		0.853	0.922
nMDS	0.530	0.874	0.579	0.952		0.929	0.953
DrL	0.186	0.671	0.089	-0.013		0.550	0.570
Fruchterman-Reingold	0.308	0.707	0.340	0.246		0.769	0.868
Kamada-Kawai	0.486	0.818	0.550	0.907		0.863	0.926
HLLE	0.389	0.852	0.146	0.169		0.659	0.616
t-SNE	0.637	0.761	0.684	0.461		0.936	0.944

Table 34B – Figures of merit obtained from Dataset df4.

Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.099	0.425	0.425	0.662	0.636	4.209	209.2
Isomap	0.060	0.513	0.513	0.743	0.626	5.552	146.8
PCA	0.078	0.455	0.455	0.712	0.678	4.219	206.9
PCA L1	0.119	0.352	0.352	0.631	0.621	4.572	262.2
kPCA	0.087	0.437	0.437	0.495	0.579	4.350	210.7
FastICA	0.077	0.457	0.457	0.709	0.673	4.832	196.6
MDS	0.078	0.455	0.455	0.712	0.678	4.243	206.9
nMDS	0.047	0.615	0.615	0.872	0.659	4.977	120.6
DrL	0.430	0.089	0.089	0.012	0.388	6.864	493.0
Fruchterman-Reingold	0.132	0.359	0.359	0.226	0.570	5.424	244.2
Kamada-Kawai	0.074	0.576	0.576	0.756	0.692	4.195	161.1
HLLE	0.384	0.153	0.153	0.425	0.458	5.931	292.4
t-SNE	0.056	0.697	0.697	0.372	0.692	4.412	123.8

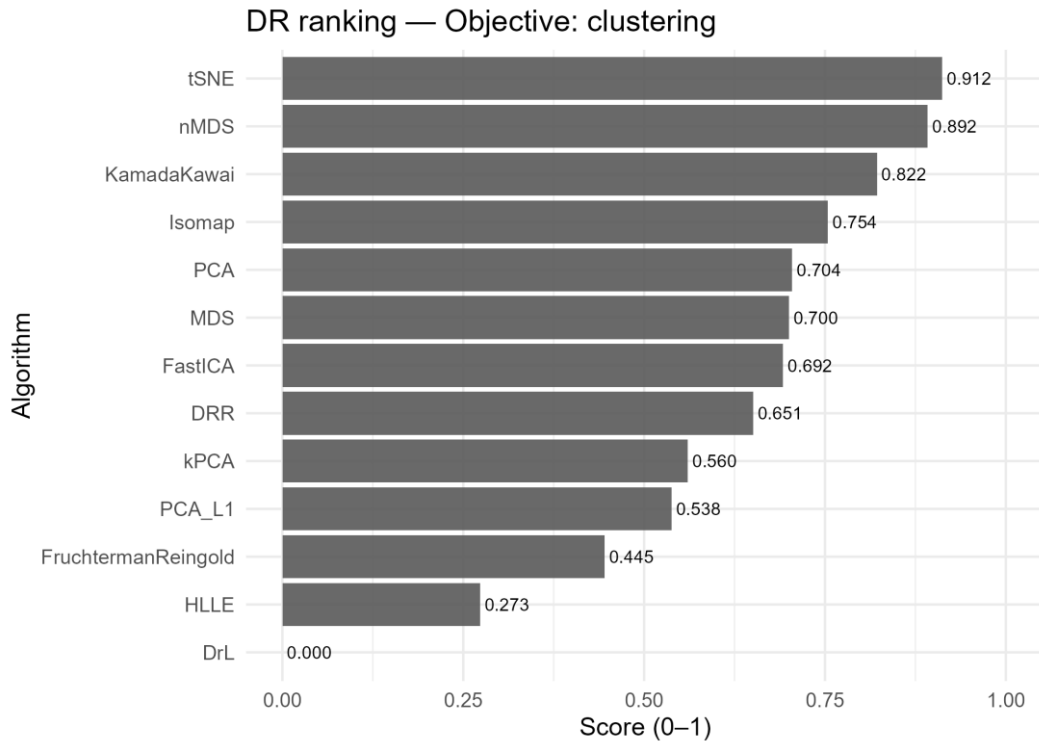
The results for Dataset df4, shown in

Table 35, highlight clear performance hierarchies among dimensionality-reduction methods. Across all analytical objectives—clustering, reconstruction, visualization, and ML-pipeline integration—t-SNE and nMDS consistently outperform competing algorithms, albeit in complementary ways. t-SNE achieves perfect locality and class-based indices ($\text{Index_Local} = 1.000$; $\text{Index_Class} = 1.000$), resulting in the highest overall scores on tasks that emphasize neighborhood fidelity and discriminative separability. In contrast, nMDS provides the strongest global manifold preservation, attaining the highest Index_Global (0.956) and delivering competitive Final Scores across all objectives. Kamada–Kawai demonstrates balanced behavior, combining moderate locality with relatively strong geometric continuity ($\text{Index_Geom} = 0.940$) and competitive class separation, making it a robust mid-tier alternative. Linear methods such as PCA perform consistently but at lower levels, reflecting limitations in capturing nonlinear structure. Overall, the table reinforces the idea that algorithmic superiority is objective-dependent: t-SNE excels when local structure and class discrimination dominate, whereas nMDS is preferable for tasks requiring faithful recovery of global manifold geometry.

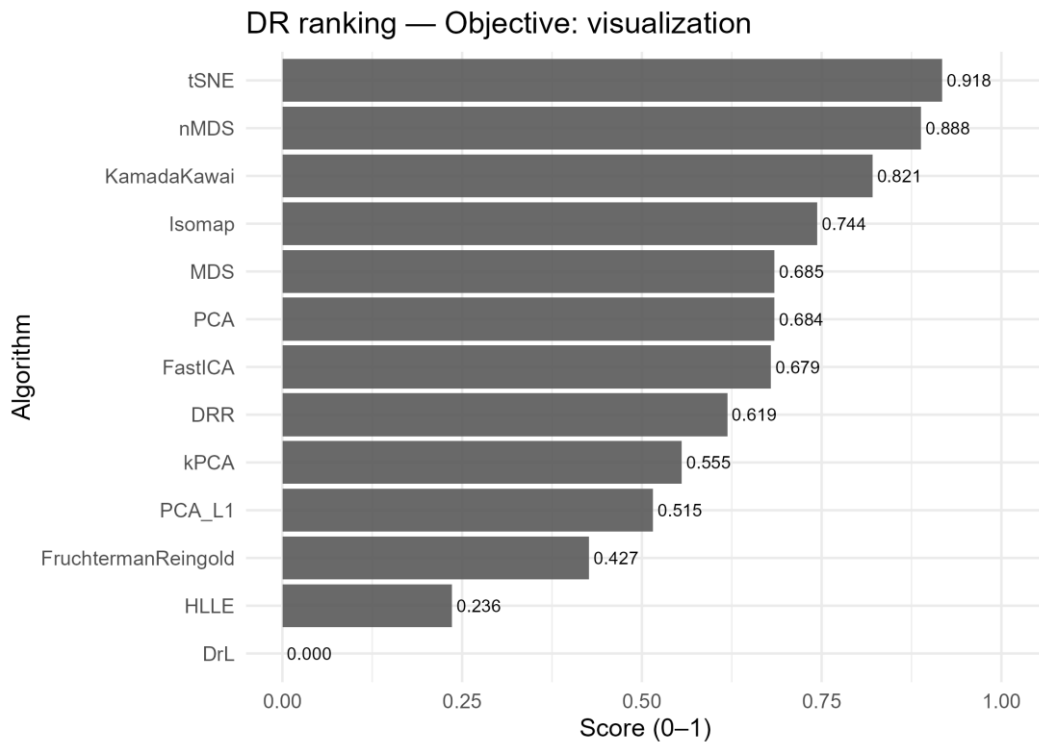
Table 35 – Figures of merit obtained from Dataset df4.

Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
tSNE	1.000	0.588	0.962	1.000	0.912	clustering
nMDS	0.869	0.956	0.902	0.874	0.892	clustering
KamadaKawai	0.770	0.829	0.940	0.867	0.822	clustering
Isomap	0.723	0.822	0.796	0.726	0.754	clustering
PCA	0.619	0.746	0.920	0.719	0.704	clustering
tSNE	1.000	0.588	0.962	1.000	0.918	visualization
nMDS	0.869	0.956	0.902	0.874	0.888	visualization
KamadaKawai	0.770	0.829	0.940	0.867	0.821	visualization
Isomap	0.723	0.822	0.796	0.726	0.744	visualization
MDS	0.619	0.747	0.890	0.719	0.685	visualization
nMDS	0.869	0.956	0.902	0.874	0.908	reconstruction
tSNE	1.000	0.588	0.962	1.000	0.861	reconstruction
KamadaKawai	0.770	0.829	0.940	0.867	0.856	reconstruction
Isomap	0.723	0.822	0.796	0.726	0.782	reconstruction
PCA	0.619	0.746	0.920	0.719	0.777	reconstruction
tSNE	1.000	0.588	0.962	1.000	0.918	ml_pipeline
nMDS	0.869	0.956	0.902	0.874	0.888	ml_pipeline
KamadaKawai	0.770	0.829	0.940	0.867	0.821	ml_pipeline
Isomap	0.723	0.822	0.796	0.726	0.744	ml_pipeline
MDS	0.619	0.747	0.890	0.719	0.685	ml_pipeline

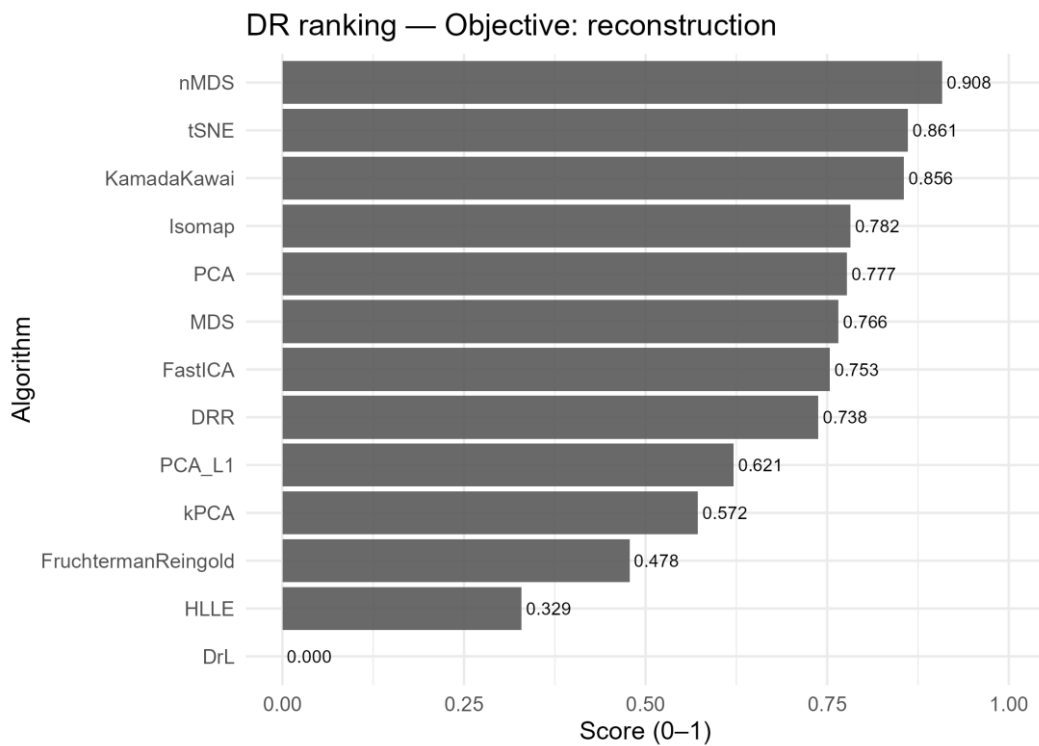
Figure 30 presents the objective-based ranking of dimensionality-reduction algorithms for Dataset 4 across clustering, visualization, reconstruction, and ML-pipeline tasks.



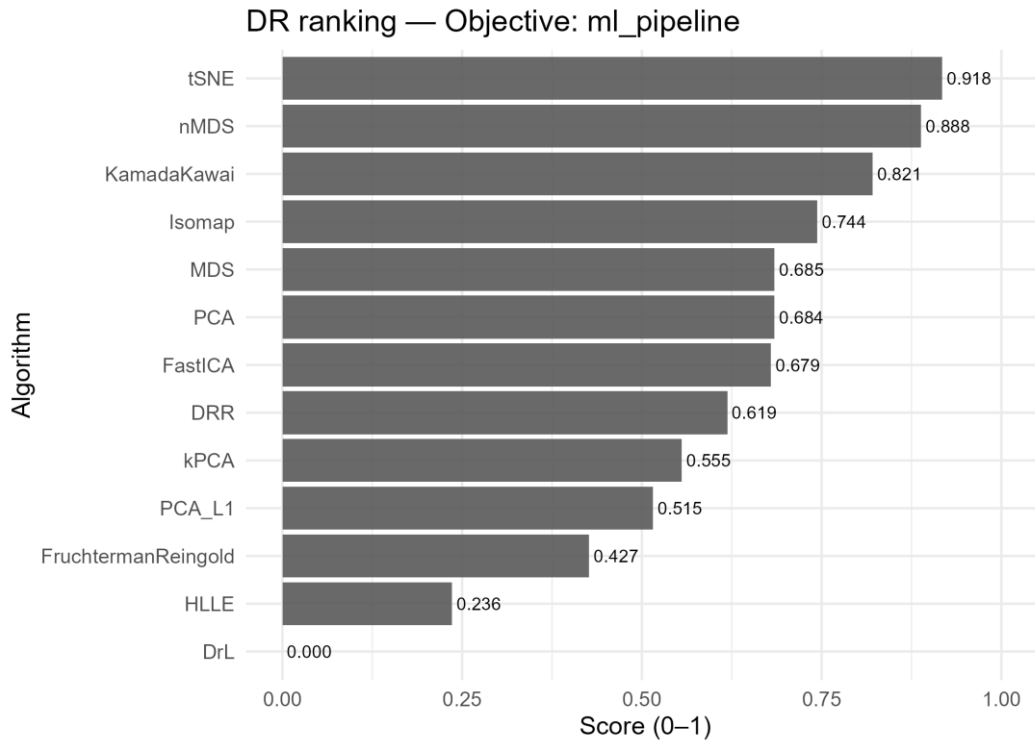
(a)



(b)



(c)



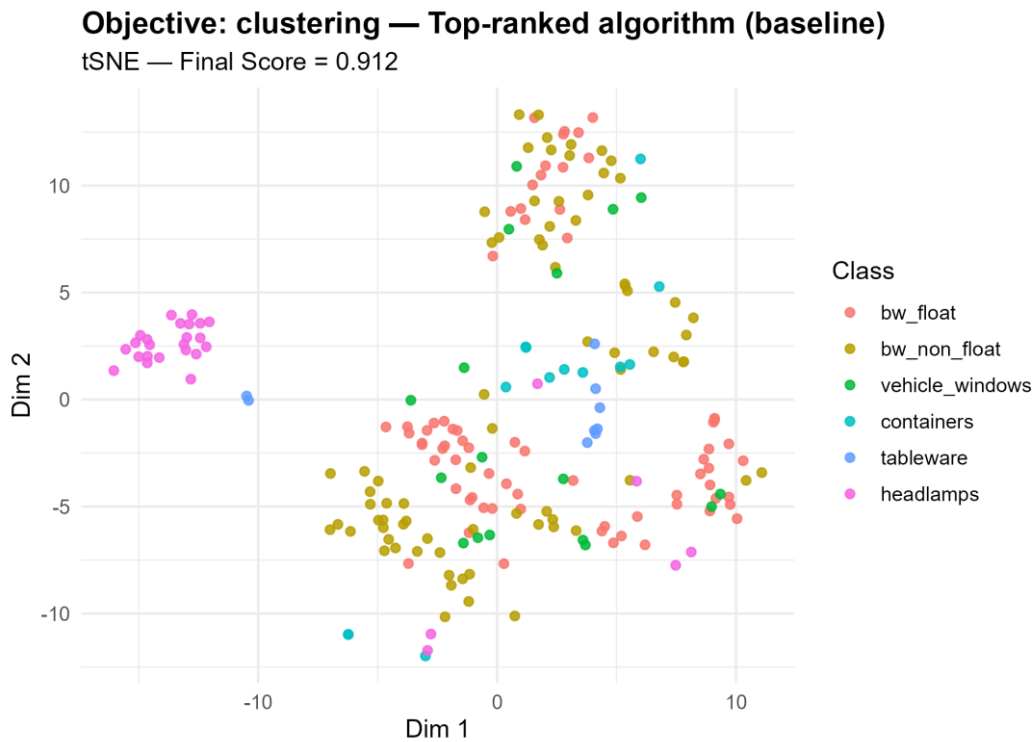
(d)

Figure 30 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Overall, t-SNE and nMDS consistently achieve the highest scores, though with different strengths. t-SNE excels at clustering, visualization, and ML-pipeline objectives, thanks to its strong preservation of local neighborhood structure and superior class separability. nMDS, while nearly matching t-SNE in several objectives, becomes the top performer in the reconstruction task, consistent with its well-known ability to preserve global manifold relationships. Kamada–Kawai and Isomap form a second-tier group, showing balanced performance across objectives and offering competitive alternatives where both geometric continuity and class structure are essential. Linear methods (PCA, MDS, FastICA) exhibit moderate but stable performance. In contrast, spectral and graph-drawing methods such as HLLE, DrL, and Fruchterman–Reingold remain among the lowest-ranked approaches, reflecting limitations in either global geometry preservation

or discriminative structure. Overall, the rankings highlight that no single technique is universally optimal. Instead, algorithm suitability depends strongly on the analytical objective: t-SNE excels at tasks requiring local fidelity and separation, while nMDS provides superior global reconstruction.

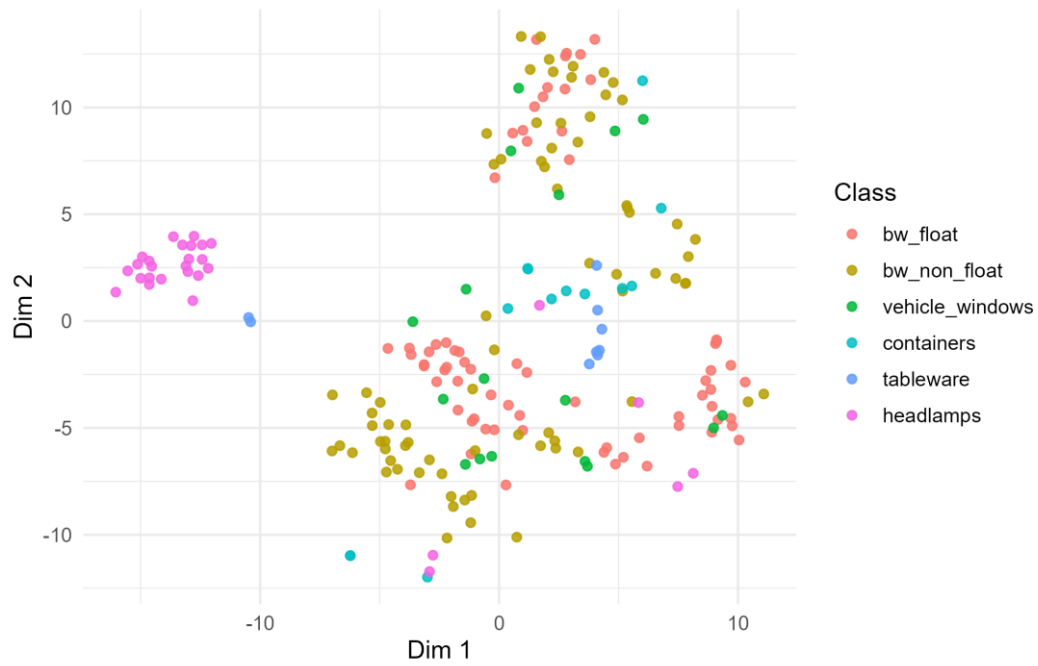
Figure 31 compares the two-dimensional embeddings produced by the top-ranked dimensionality-reduction algorithms for Dataset 4 under four analytical objectives.



(a)

Objective: visualization — Top-ranked algorithm (baseline)

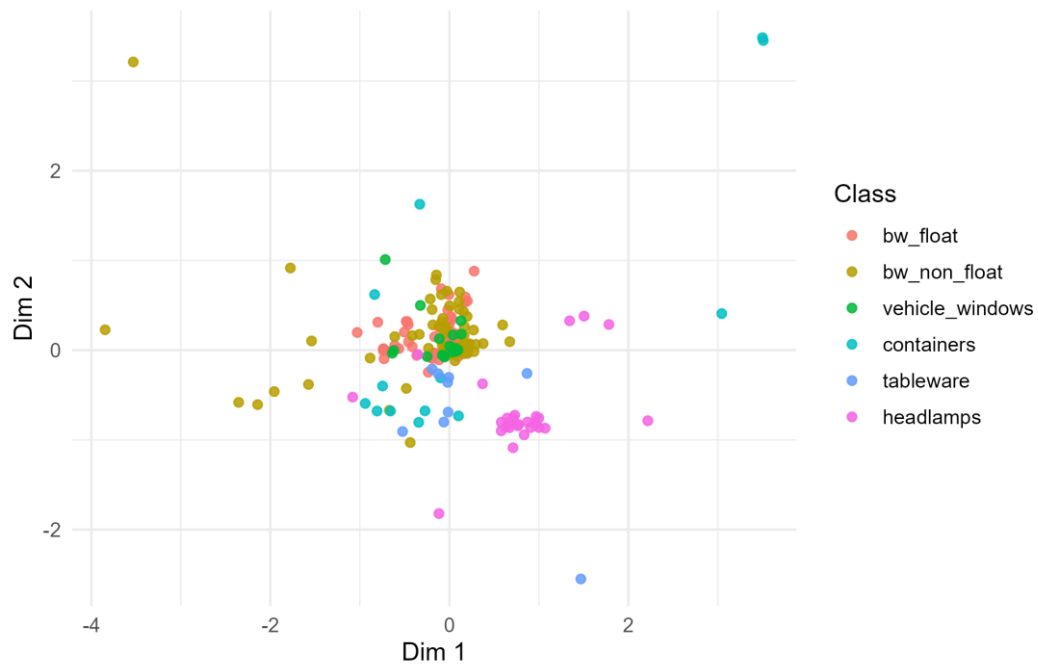
tSNE — Final Score = 0.9177



(b)

Objective: reconstruction — Top-ranked algorithm (baseline)

nMDS — Final Score = 0.9083



(c)

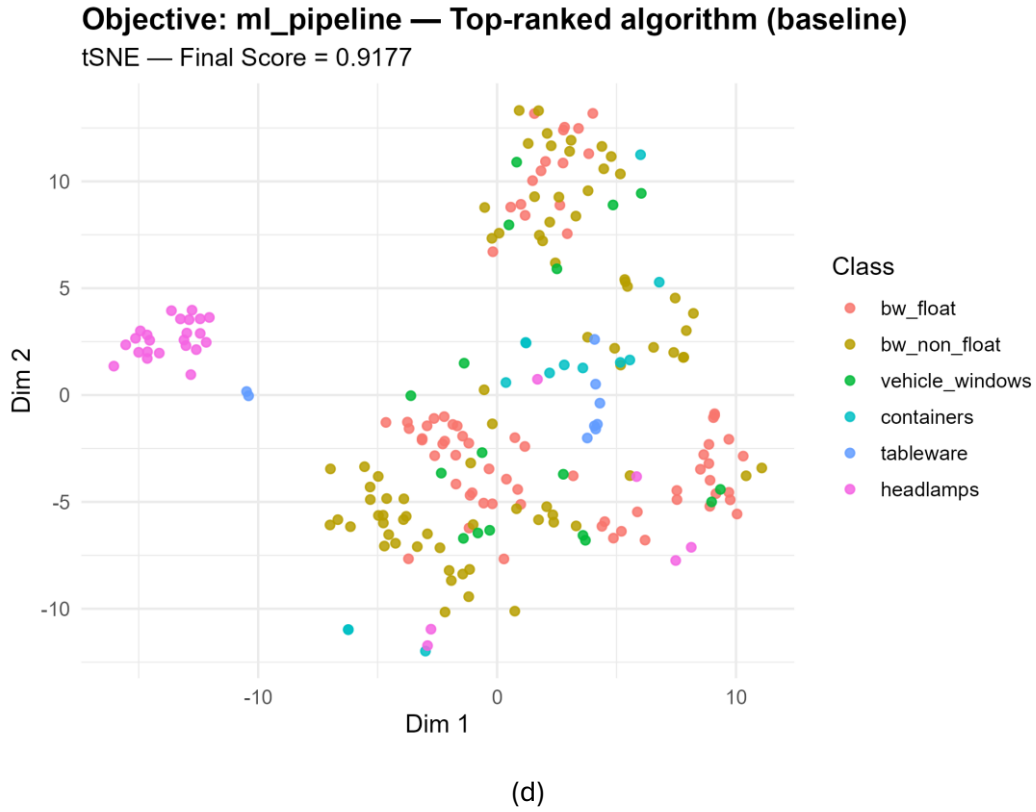
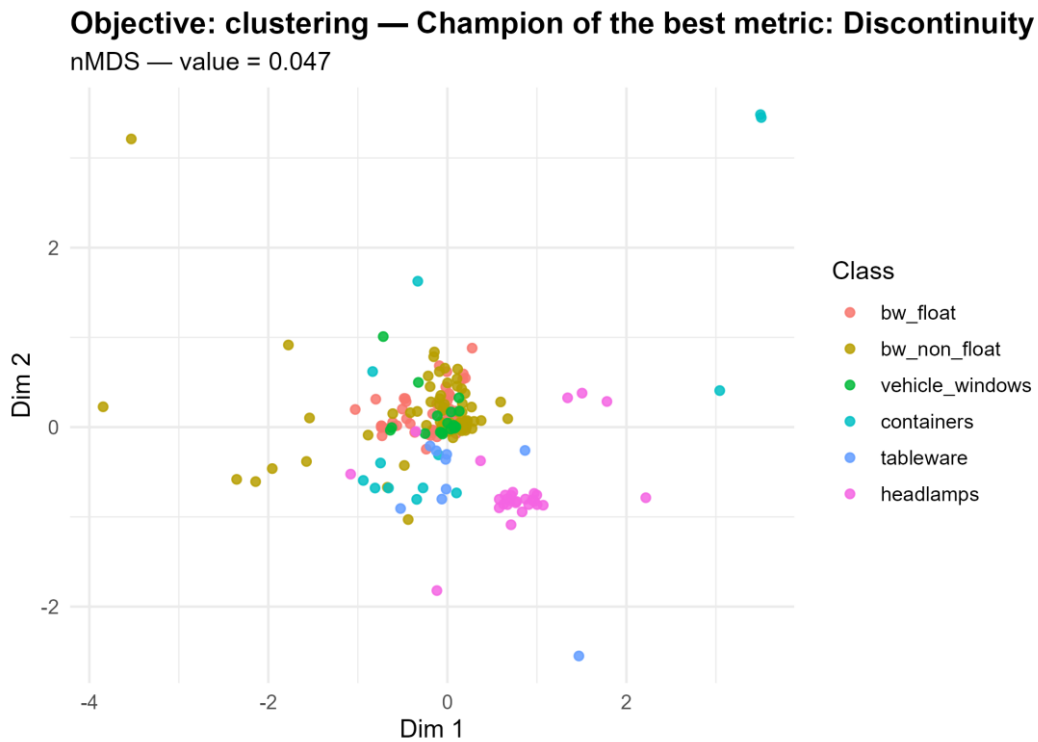


Figure 31 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

For clustering, visualization, and ML-pipeline tasks, t-SNE consistently performs best, generating well-defined class groupings with clear inter-cluster separation and visually interpretable structures. Its embeddings highlight strong locality preservation, enabling distinct spatial organization of the six glass categories. In contrast, the reconstruction objective selects nMDS as the top method. The nMDS embedding exhibits compact yet highly overlapping class distributions, reflecting its emphasis on preserving global pairwise dissimilarities rather than maximizing class separability. While less visually discriminative, this representation aligns with the method's objective of faithfully reproducing global geometry. Overall, the figure illustrates that the optimal embedding depends strongly on the analytical goal: t-SNE excels when neighborhood structure and

visual discrimination are prioritized, whereas nMDS is more suitable when global geometric fidelity is required.

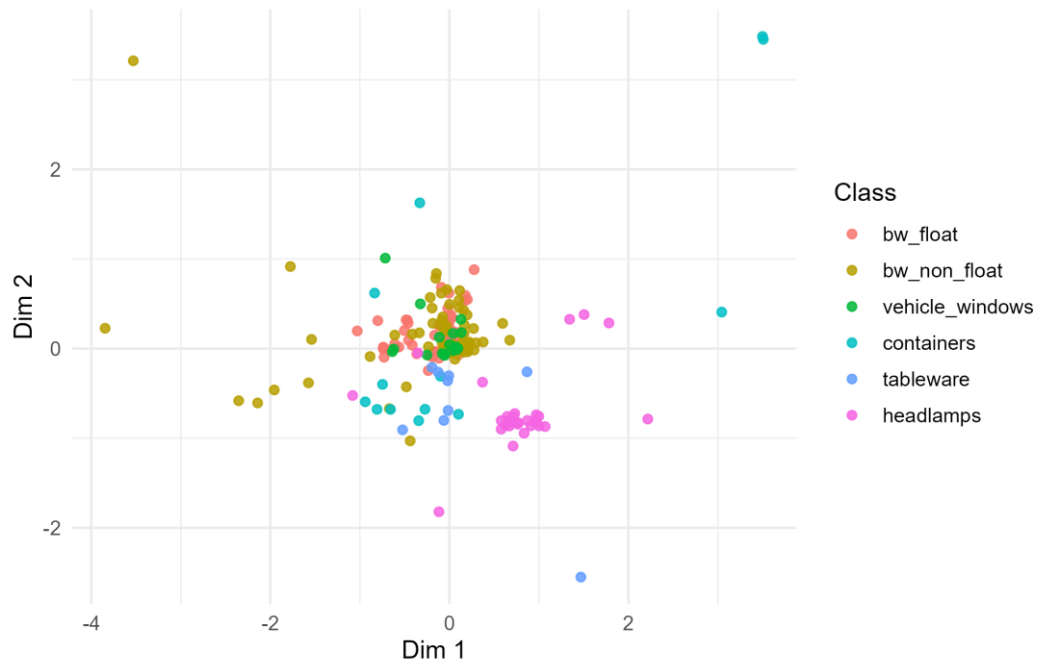
Figure 32 presents the embeddings generated by the algorithm that achieves the best performance on the Discontinuity metric—non-metric MDS (nMDS)—across four analytical objectives: clustering, visualization, reconstruction, and ML-pipeline integration.



(a)

Objective: visualization — Champion of the best metric: Discontinuity

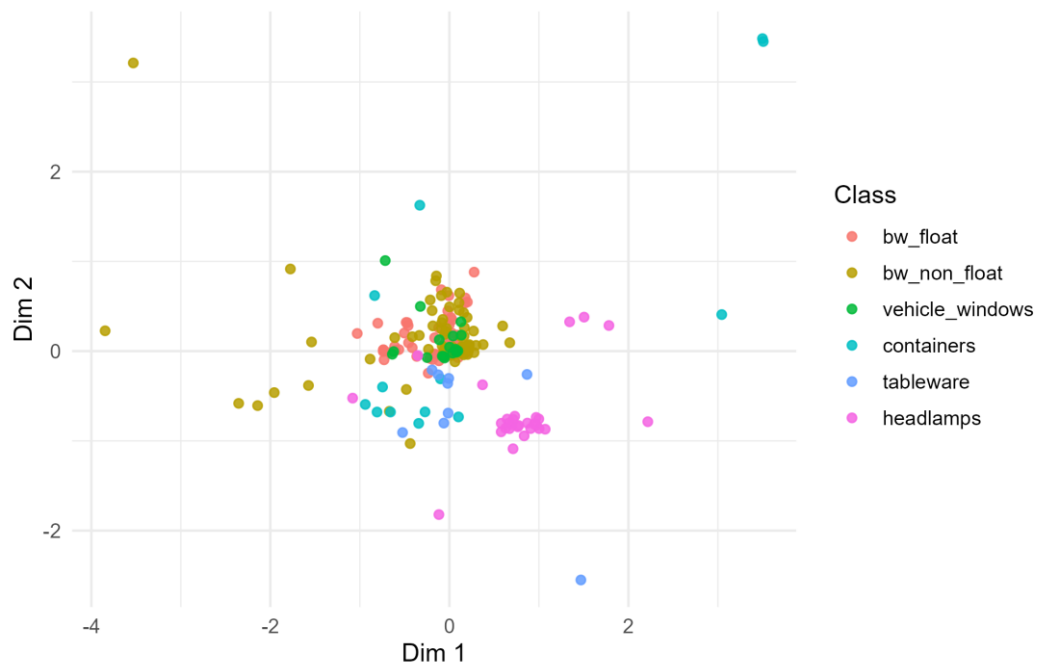
nMDS — value = 0.047



(b)

Objective: reconstruction — Champion of the best metric: Discontinuity

nMDS — value = 0.047



(c)

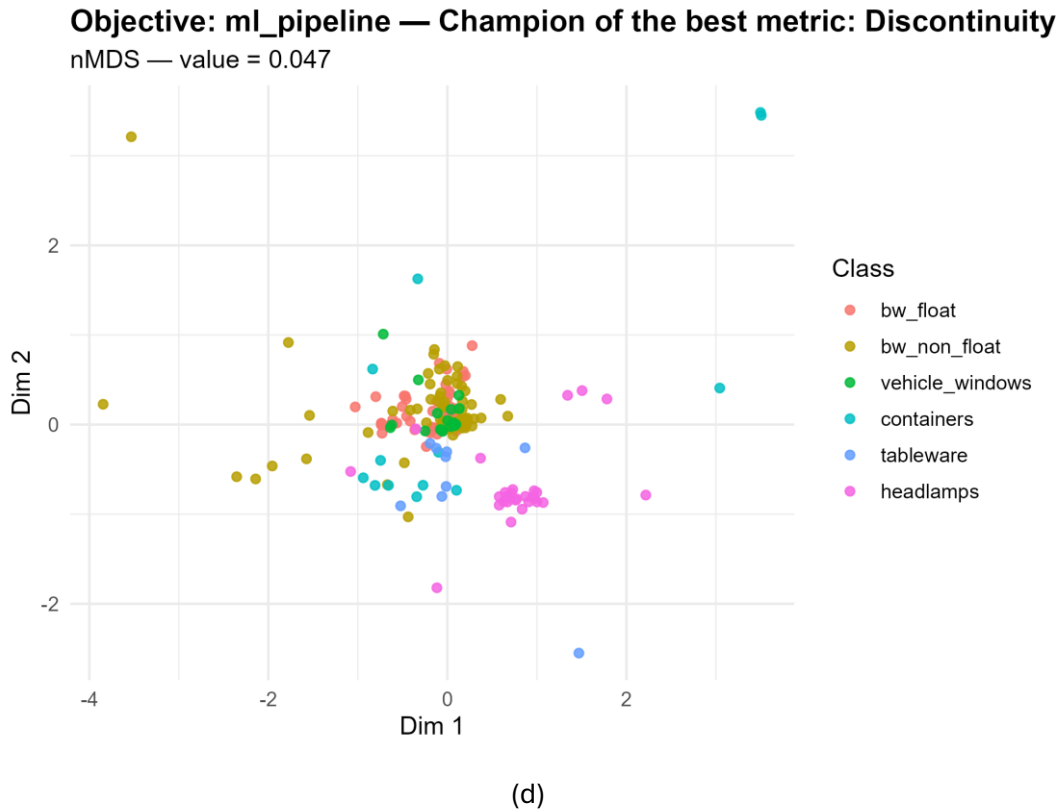


Figure 32 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

In all scenarios, nMDS produces a compact embedding in which sample points from different glass categories occupy a largely overlapping region of the two-dimensional space. This reflects the nature of the *C-measure*, which emphasizes geometric continuity and low distortion rather than class separability. Although the embeddings reasonably well preserve global similarity relationships, they provide limited visual discrimination among classes, highlighting the trade-off inherent in optimizing geometric smoothness. The consistency of the layouts across the four objectives demonstrates the stability of nMDS under the *C-measure* criterion, but also indicates that this metric prioritizes geometric coherence over class-level interpretability. Overall, the results reinforce the importance of selecting evaluation metrics aligned with the intended

analytical goal, as metrics favoring geometric continuity may yield embeddings with limited discriminative value.

8. STUDY OF CASE V

8.1 Description of the Dataset

The Hepatitis dataset was named df5. It contains 155 instances, 19 features, and two classes. The UCI Hepatitis dataset is a multivariate dataset used to predict patient survival — whether the patient lived or died — from clinical and demographic data [91].

8.2 Results and Discussion

The heatmap, shown in Figure 33, displays the correlation structure among the key evaluation metrics used in dimensionality-reduction assessment.

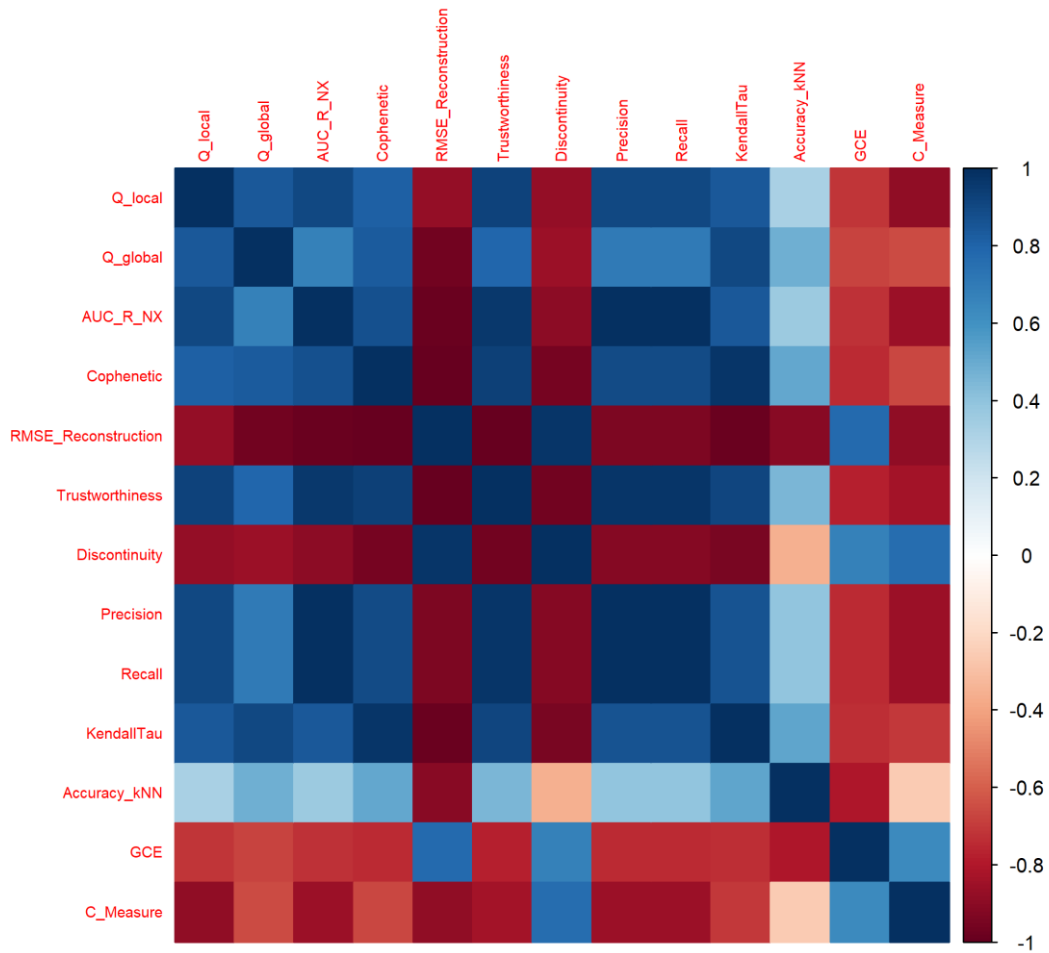


Figure 33 - The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Local-preservation measures (Q_{local} , *Trustworthiness*, *Precision*, *Recall*, *Accuracy_kNN*) form a cohesive cluster with strong positive correlations, indicating consistent sensitivity to neighborhood fidelity. Global-structure metrics (Q_{global} , *AUC_R(NX)*, *Cophenetic*, *KendallTau*) also correlate strongly, reflecting their shared emphasis on manifold-level geometry.

Metrics associated with reconstruction error and geometric distortion (*RMSE_Reconstruction*, *GCE*, *C_measure*, *Discontinuity*) exhibit strong negative correlations with both locality and global groups. This divergence highlights a fundamental trade-off: algorithms that preserve topological structure often perform poorly on reconstruction tasks, whereas reconstruction-oriented methods tend to distort the manifold. Overall, the heatmap reinforces the multifaceted nature of DR evaluation and the necessity of metric selection tailored to the analytical objective.

The evaluation of Dataset df5, shown in

Table 36A and Table 36B, reveals consistent performance patterns across dimensionality-reduction algorithms, with nonlinear manifold learners clearly outperforming linear and graph-based methods. Among the evaluated techniques, non-metric MDS (nMDS) delivers the strongest global manifold preservation, achieving the highest values for Qglobal, AUC_R(NX), Cophenetic, and KendallTau, while also maintaining competitive locality and classification accuracy metrics. This balance indicates effective recovery of both fine-scale and large-scale geometric structure within the data. t-SNE emerges as the top performer for local-structure preservation, exhibiting the highest Qlocal, Precision, Recall, and Discontinuity scores. Its strong locality and class-separability metrics underscore its suitability for tasks prioritizing cluster identification and visual discrimination, although its global metrics remain modest relative to nMDS and Kamada–Kawai. Isomap, PCA/MDS, and FastICA constitute a mid-performing tier, showing stable but less pronounced preservation of structural relationships. Conversely, HLLE, DrL, and Fruchterman–Reingold display the weakest results across most metrics, reflecting limited geometric fidelity or class-level discrimination. Overall, the results emphasize that algorithmic suitability depends on the analytical objective: methods like t-SNE excel when local separability is essential, whereas nMDS and Kamada–Kawai provide superior global manifold reconstruction.

Table 36A – Figures of merit obtained from Dataset df5.

Algorithm	Q_local	Q_global	AUC_R_NX	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.518	0.870	0.435	0.836	0.706	0.841	0.925
Isomap	0.425	0.834	0.456	0.847		0.855	0.931
PCA	0.442	0.840	0.434	0.838	0.706	0.839	0.925
PCA_L1	0.428	0.813	0.353	0.613		0.766	0.865
kPCA	0.366	0.762	0.381	0.564	8.066	0.785	0.888
FastICA	0.479	0.851	0.424	0.810	0.706	0.836	0.917
MDS	0.442	0.840	0.434	0.838		0.839	0.925
nMDS	0.509	0.864	0.516	0.927		0.876	0.924
DrL	0.024	0.505	0.099	-0.092		0.509	0.500
FruchtermanReingold	0.184	0.656	0.222	0.238		0.682	0.703
KamadaKawai	0.488	0.849	0.505	0.880		0.852	0.925
HLLE	0.340	0.804	0.181	0.150		0.657	0.659
DiffusionMaps	0.409	0.817	0.302	0.401		0.721	0.819
tSNE	0.619	0.754	0.605	0.610		0.900	0.898

Table 36B – Figures of merit obtained from Dataset df5.

Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.075	0.452	0.452	0.657	0.819	2.029	184.3
Isomap	0.069	0.468	0.468	0.665	0.826	2.360	193.1
PCA	0.075	0.446	0.446	0.658	0.826	2.043	179.9
PCA_L1	0.135	0.357	0.357	0.474	0.729	3.102	221.9
kPCA	0.112	0.395	0.395	0.454	0.768	2.863	171.2
FastICA	0.083	0.431	0.431	0.628	0.800	2.588	188.5
MDS	0.075	0.446	0.446	0.658	0.826	2.043	179.9
nMDS	0.076	0.530	0.530	0.774	0.852	2.132	132.3
DrL	0.500	0.091	0.091	-0.058	0.755	3.321	298.8
FruchtermanReingold	0.297	0.232	0.232	0.175	0.787	2.893	248.4
KamadaKawai	0.075	0.516	0.516	0.720	0.813	2.395	189.0
HLLE	0.341	0.189	0.189	0.243	0.826	2.548	224.0
DiffusionMaps	0.181	0.314	0.314	0.481	0.742	2.969	160.7
tSNE	0.102	0.595	0.595	0.462	0.774	2.183	103.4

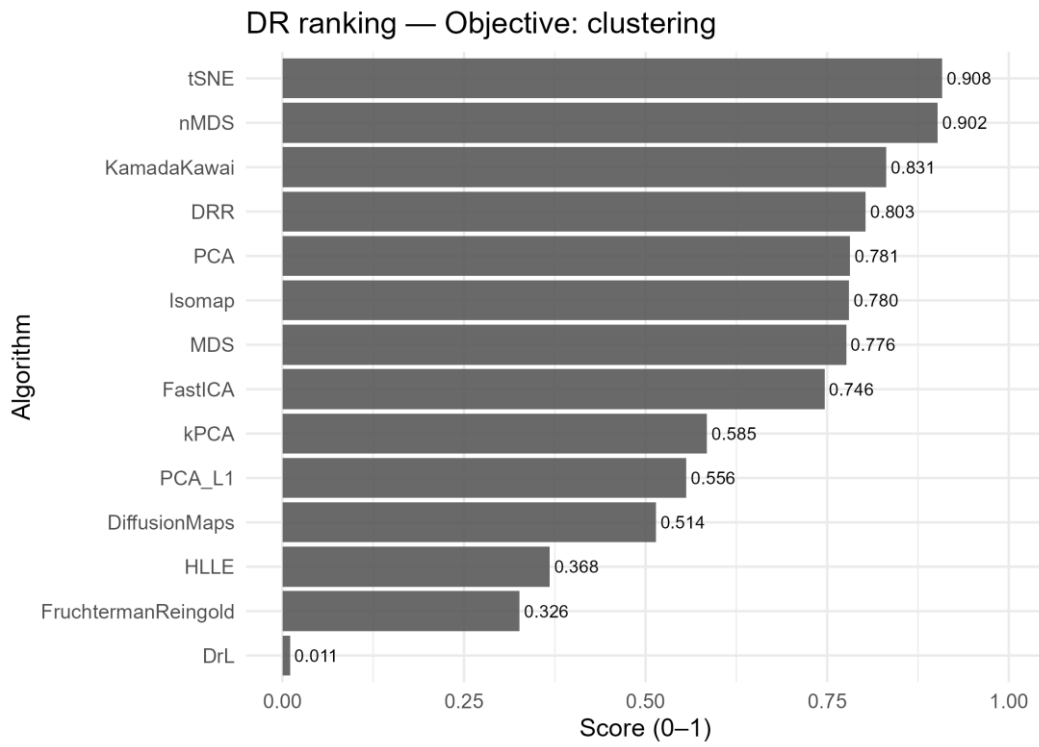
Table 37 summarizes the multi-objective performance of dimensionality-reduction algorithms for Dataset df5 using composite indices for locality, global structure, geometric continuity, and class fidelity.

Table 37 – Figures of merit obtained from Dataset df5.

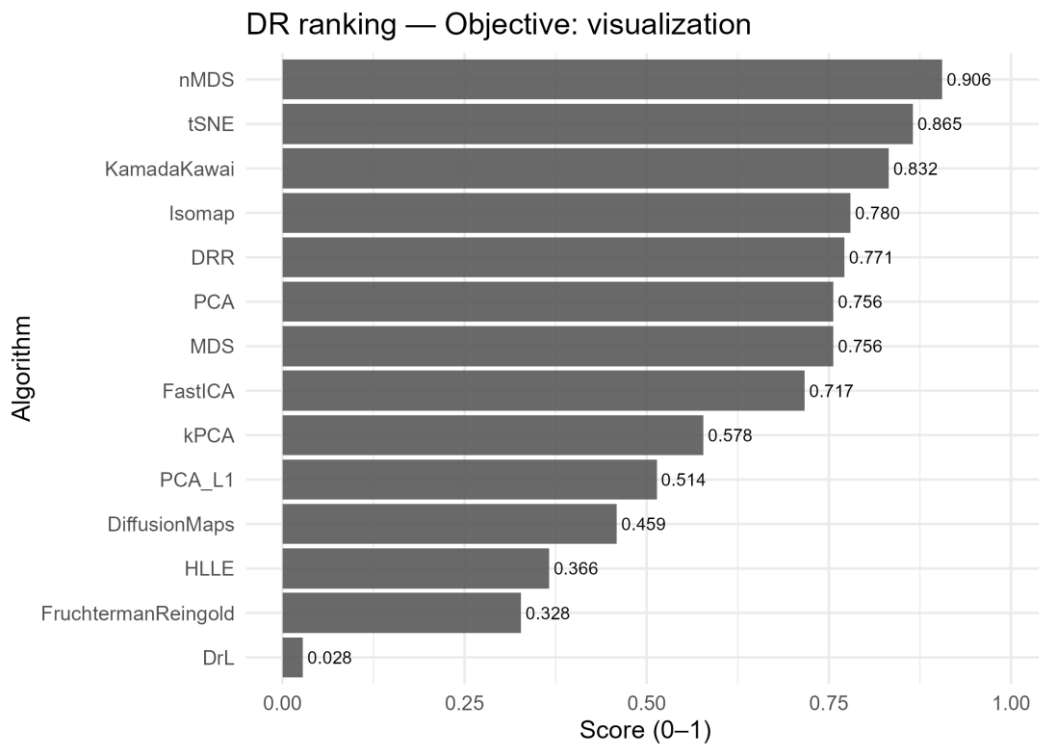
Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
tSNE	1.000	0.749	0.935	0.789	0.908	clustering
nMDS	0.874	0.952	0.919	0.914	0.902	clustering
KamadaKawai	0.836	0.908	0.755	0.790	0.831	clustering
DRR	0.778	0.859	0.893	0.721	0.803	clustering
PCA	0.739	0.838	0.896	0.732	0.781	clustering
nMDS	0.874	0.952	0.919	0.914	0.906	visualization
tSNE	1.000	0.749	0.935	0.789	0.865	visualization
KamadaKawai	0.836	0.908	0.755	0.790	0.832	visualization
Isomap	0.764	0.849	0.762	0.762	0.780	visualization
DRR	0.778	0.859	0.893	0.721	0.771	visualization
nMDS	0.874	0.952	0.919	0.914	0.915	reconstruction
tSNE	1.000	0.749	0.935	0.789	0.899	reconstruction
DRR	0.778	0.859	0.893	0.721	0.848	reconstruction
PCA	0.739	0.838	0.896	0.732	0.831	reconstruction
KamadaKawai	0.836	0.908	0.755	0.790	0.825	reconstruction
nMDS	0.874	0.952	0.919	0.914	0.906	ml_pipeline
tSNE	1.000	0.749	0.935	0.789	0.865	ml_pipeline
KamadaKawai	0.836	0.908	0.755	0.790	0.832	ml_pipeline
Isomap	0.764	0.849	0.762	0.762	0.780	ml_pipeline
DRR	0.778	0.859	0.893	0.721	0.771	ml_pipeline

Two algorithms consistently dominate: t-SNE and nMDS, though with complementary strengths. t-SNE achieves perfect locality ($Index_Local = 1.000$) and strong class separation, yielding the highest scores for clustering and ML-pipeline objectives. In contrast, nMDS delivers the strongest global-structure preservation ($Index_Global = 0.952$) and the highest Final Scores for visualization and reconstruction, reflecting its ability to recover manifold geometry with minimal distortion. Kamada–Kawai emerges as a robust mid-performance method, balancing locality and global indices, but it trails the leading models in geometric and class-oriented metrics. Linear methods such as PCA and DRR show stable performance with strong geometric continuity but reduced discriminative capacity. Overall, the findings confirm that algorithm selection remains objective-dependent: t-SNE is preferable when cluster separation and neighborhood fidelity are prioritized, whereas nMDS is superior for tasks requiring faithful global manifold reconstruction.

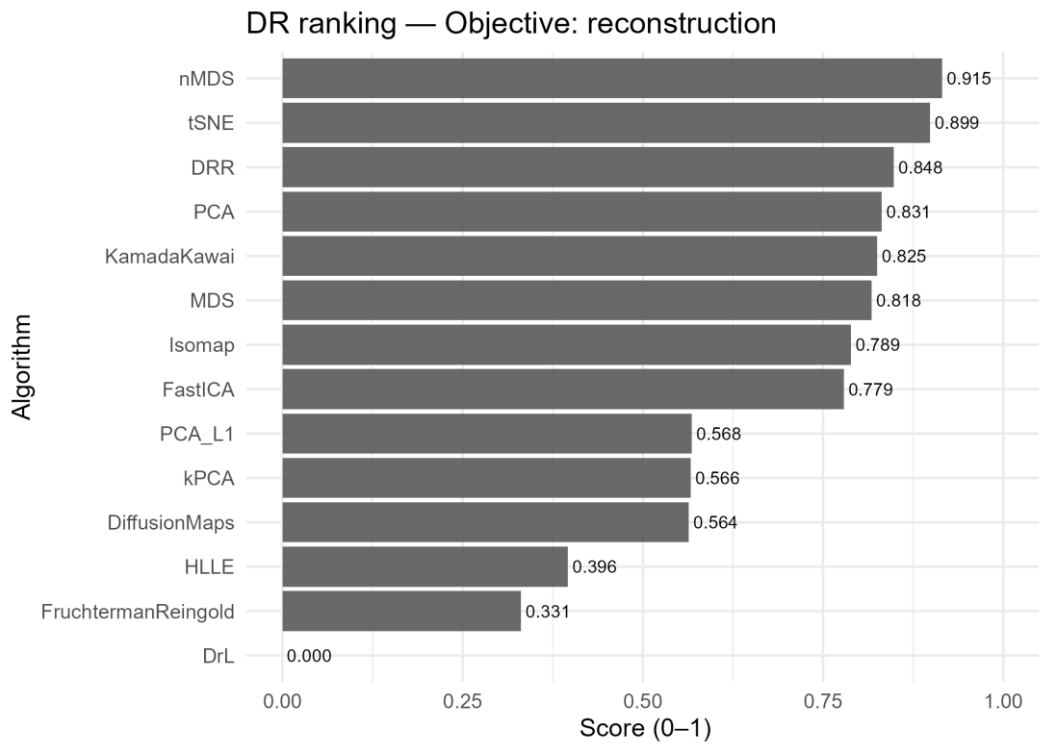
Figure 34 presents objective-specific rankings of dimensionality-reduction algorithms for Dataset df5, highlighting clear and consistent performance patterns across clustering, visualization, reconstruction, and ML-pipeline tasks. t-SNE and nMDS emerge as the two leading methods, yet with complementary strengths.



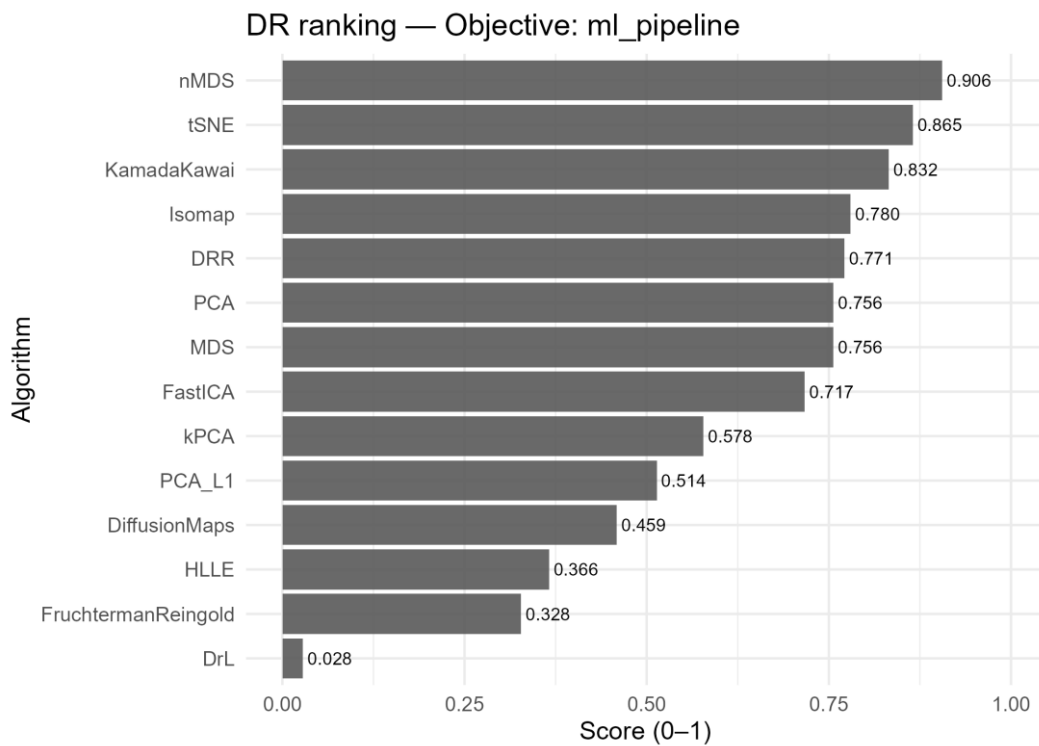
(a)



(b)



(c)



(d)

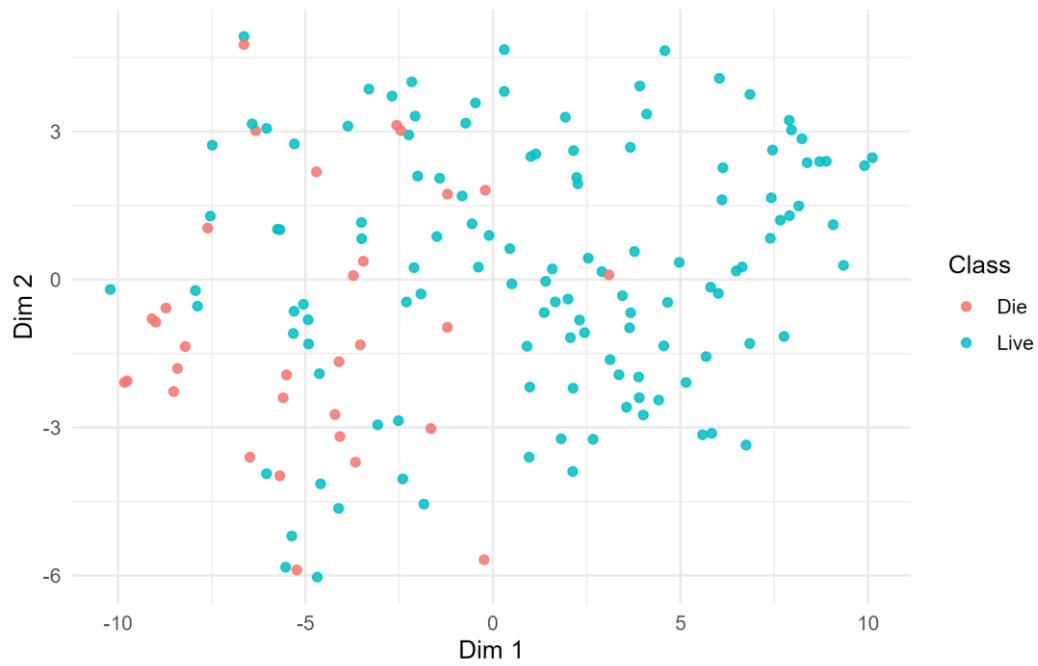
Figure 34 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

For clustering, t-SNE attains the highest score, reflecting its strong preservation of local structure and its ability to reveal well-separated groupings. nMDS performs nearly identically, underscoring its robust behavior even in tasks where discriminative neighborhood fidelity is central. For visualization and ML-pipeline objectives, nMDS surpasses t-SNE, achieving the highest overall scores and reinforcing its advantage in maintaining global manifold relationships. Reconstruction results likewise favor nMDS, which attains the top score and demonstrates superior geometric consistency. Kamada–Kawai, Isomap, DRR, PCA, and MDS constitute a mid-performing group, offering balanced but less exceptional performance across objectives. Conversely, methods such as HLLE, Diffusion Maps, Fruchterman–Reingold, and especially DrL show substantially lower scores, indicating limited suitability for this dataset across all analytical scenarios. Overall, the results confirm that algorithm performance is objective-dependent: t-SNE excels at tasks prioritizing local separability, whereas nMDS performs better when global structure and reconstruction fidelity are paramount.

Figure 35 illustrates the two-dimensional embeddings generated by the top-ranked dimensionality-reduction methods for Dataset df5 across four analytical objectives.

Objective: clustering — Top-ranked algorithm (baseline)

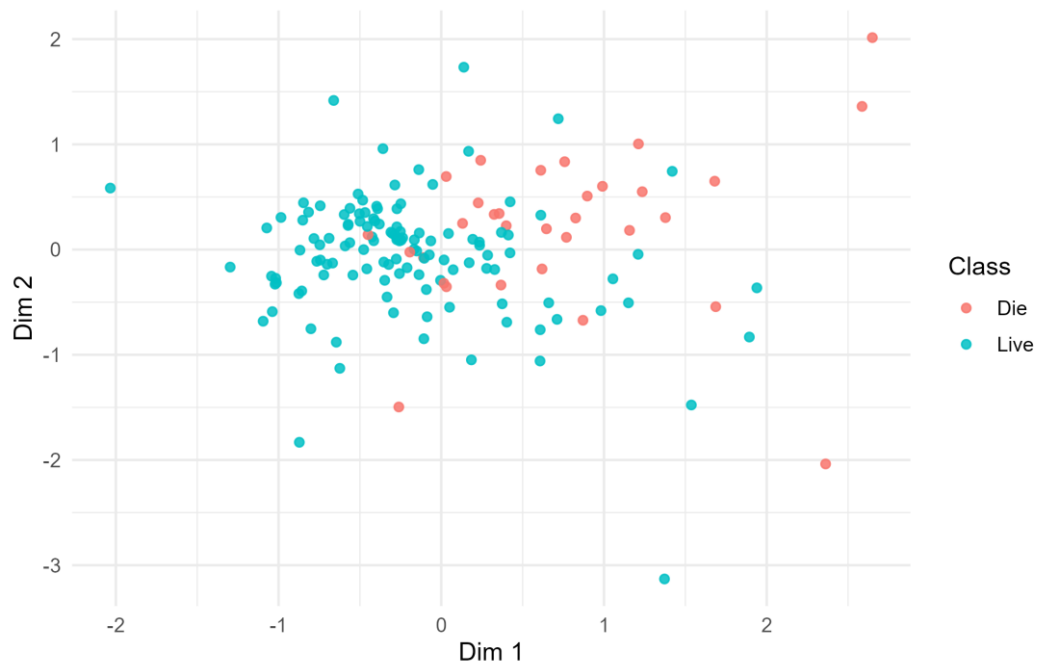
tSNE — Final Score = 0.9083



(a)

Objective: visualization — Top-ranked algorithm (baseline)

nMDS — Final Score = 0.9056



(b)

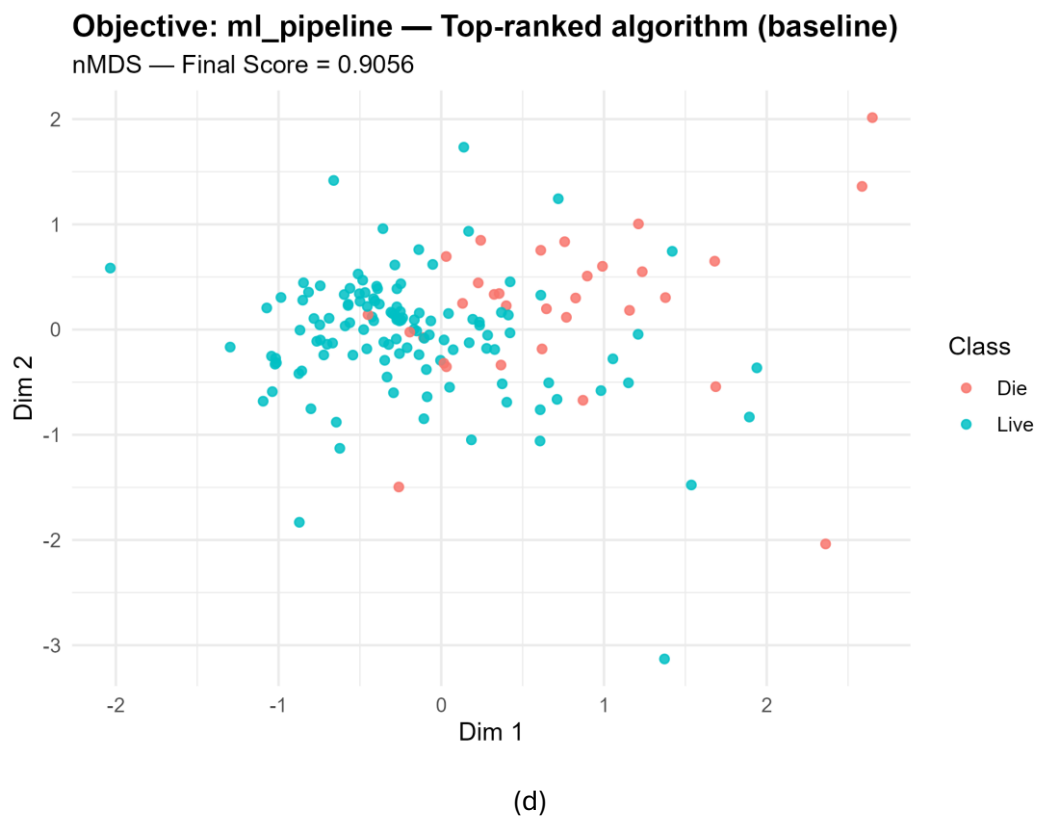
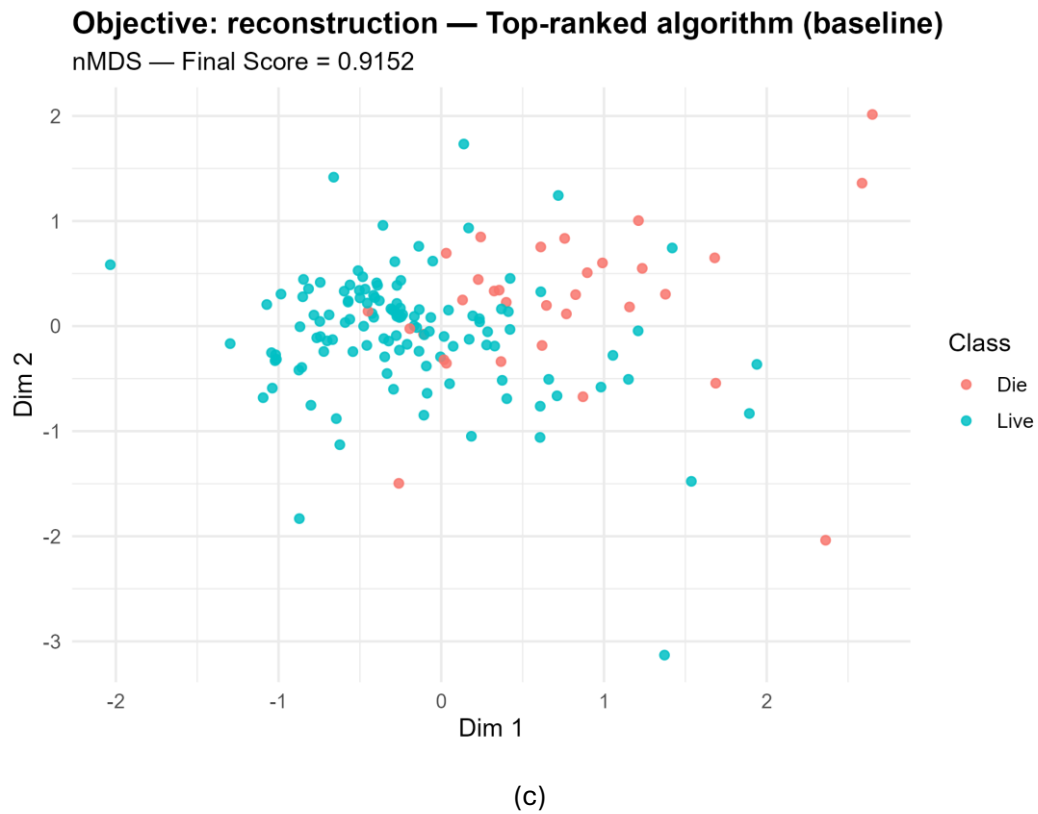


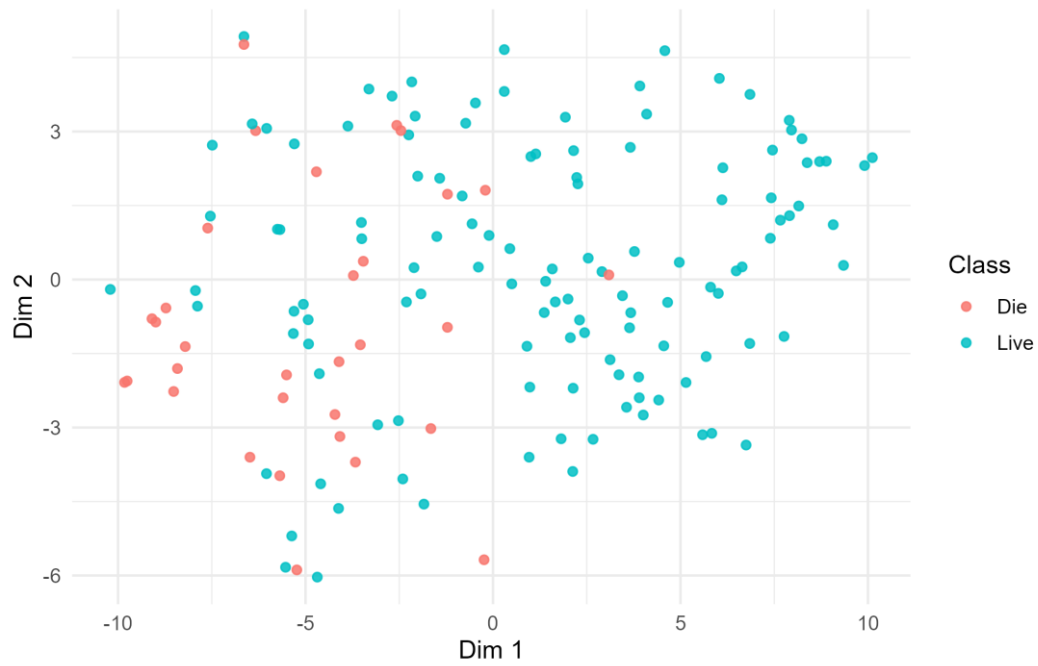
Figure 35 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

For the clustering task, t-SNE produces a highly dispersed representation with partial class separation, reflecting its strong locality preservation and ability to reveal heterogeneous structures within the data. In contrast, nMDS—selected as the best method for visualization, reconstruction, and ML-pipeline objectives—yields more compact and globally coherent embeddings. Across these three objectives, the nMDS embeddings exhibit consistent geometric organization, preserving global dissimilarity relationships while producing moderate overlap between the *Live* and *Die* classes. This pattern aligns with nMDS’s emphasis on minimizing stress across the full distance matrix rather than maximizing class discrimination. Although less visually separable than t-SNE, nMDS delivers embeddings that are stable and structurally faithful, which contributes to its superior performance in global and reconstruction-oriented metrics. Overall, the figure demonstrates that the optimal embedding depends on the analytical goal: t-SNE offers enhanced cluster visualization and local discriminative structure, while nMDS provides more reliable global geometry for downstream modeling and reconstruction tasks.

Figure 36 displays t-SNE embeddings selected as the top performer according to the AUC_R(NX) metric across four analytical objectives: clustering, visualization, reconstruction, and ML-pipeline integration.

Objective: clustering — Champion of the best metric: AUC(R_NX)

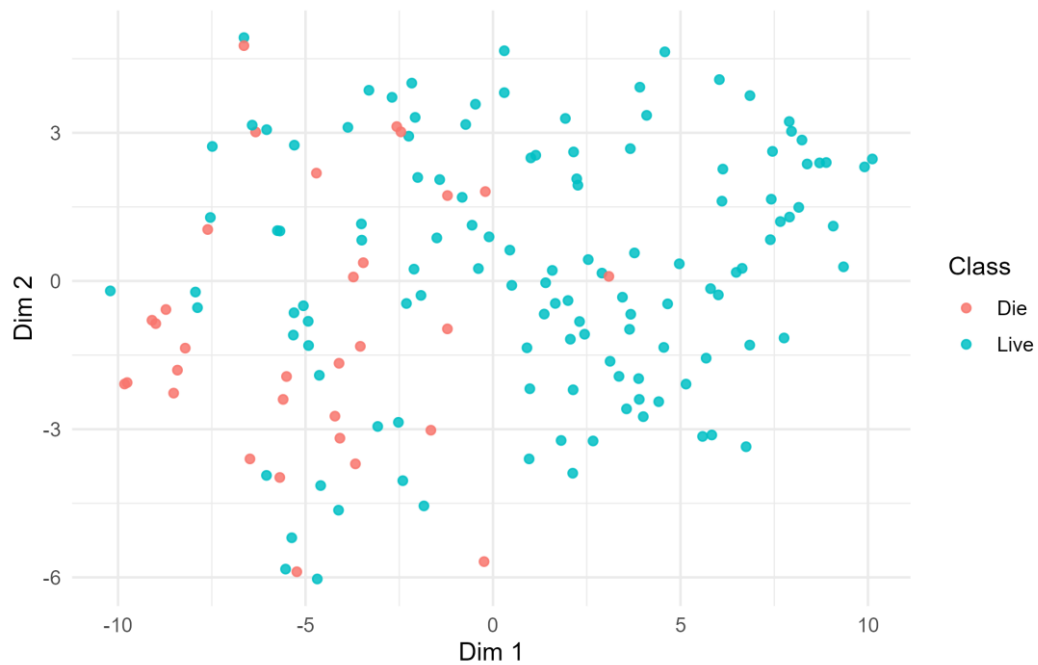
tSNE — value = 0.605



(a)

Objective: visualization — Champion of the best metric: AUC(R_NX)

tSNE — value = 0.605



(b)

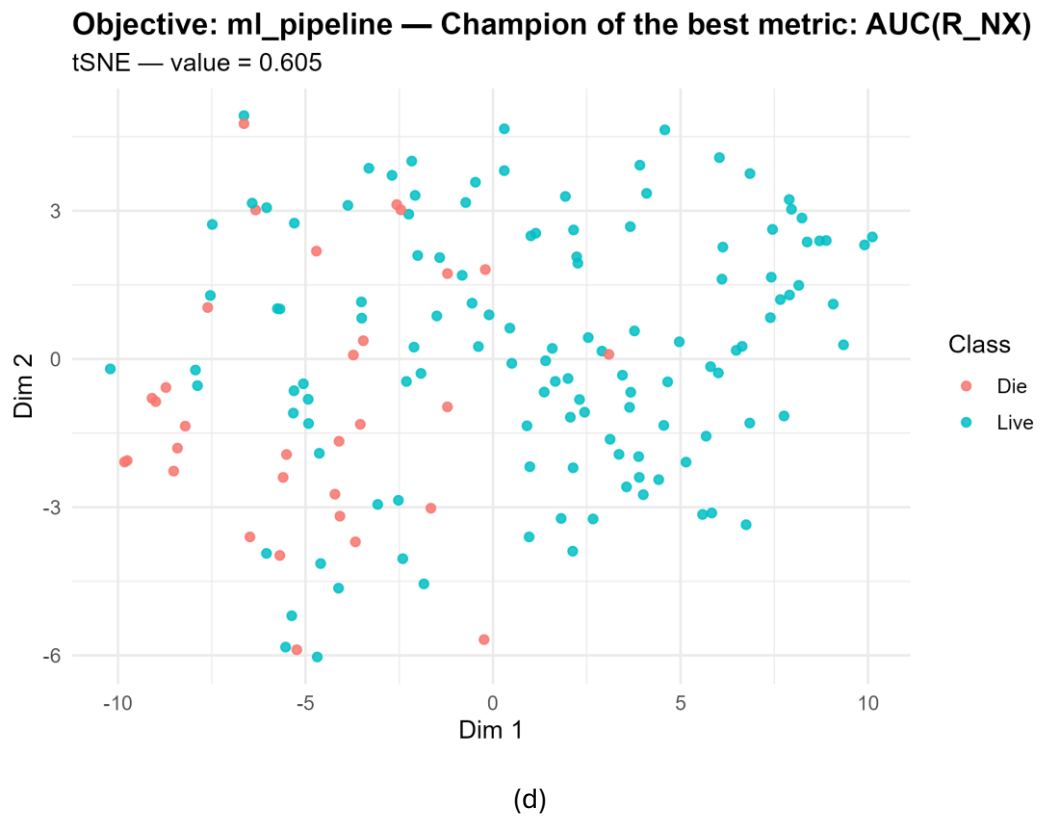
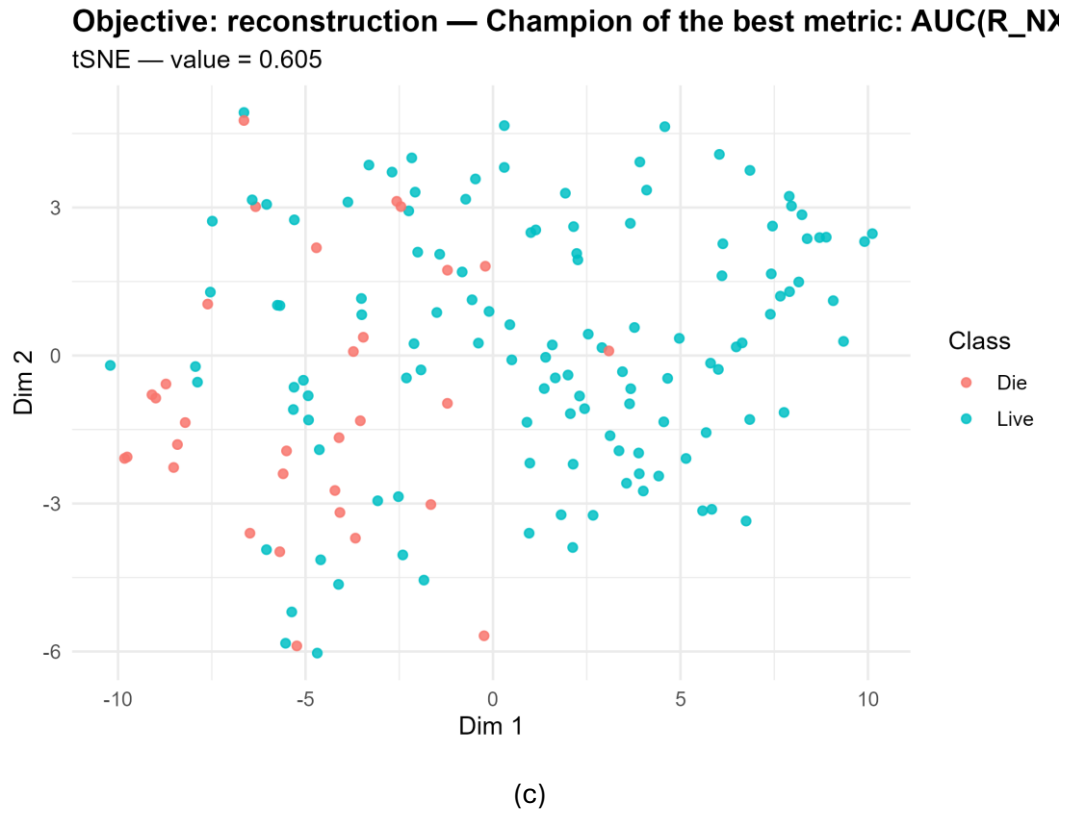


Figure 36 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

In all scenarios, the t-SNE projections exhibit similar geometric patterns, characterized by dispersed point clouds with partial mixing between the *Live* and *Die* classes. This outcome reflects the nature of the $AUC(R_NX)$ metric, which prioritizes preserving neighborhood ranks across multiple scales rather than maximizing class separability. Although the embeddings do not show strong discriminative structure, they maintain consistent local and mid-range neighborhood relationships, as evidenced by the relatively stable arrangement of points across all objective settings. The near-identical layouts indicate t-SNE's robustness with respect to this metric. It suggests that $AUC_R(NX)$ favors embeddings that balance local fidelity with continuity rather than producing visually distinct clusters.

9. STUDY OF CASE VI

9.1 Description of the Dataset

The Iris dataset was named df6. It is one of the earliest datasets used in the literature on classification methods and is widely employed in statistics and machine learning. The dataset contains three classes, each with 50 instances, corresponding to iris plant types [92].

9.2 Results and Discussion

The heatmap shown in Figure 37 depicts the correlation structure among key metrics used to evaluate dimensionality-reduction algorithms.

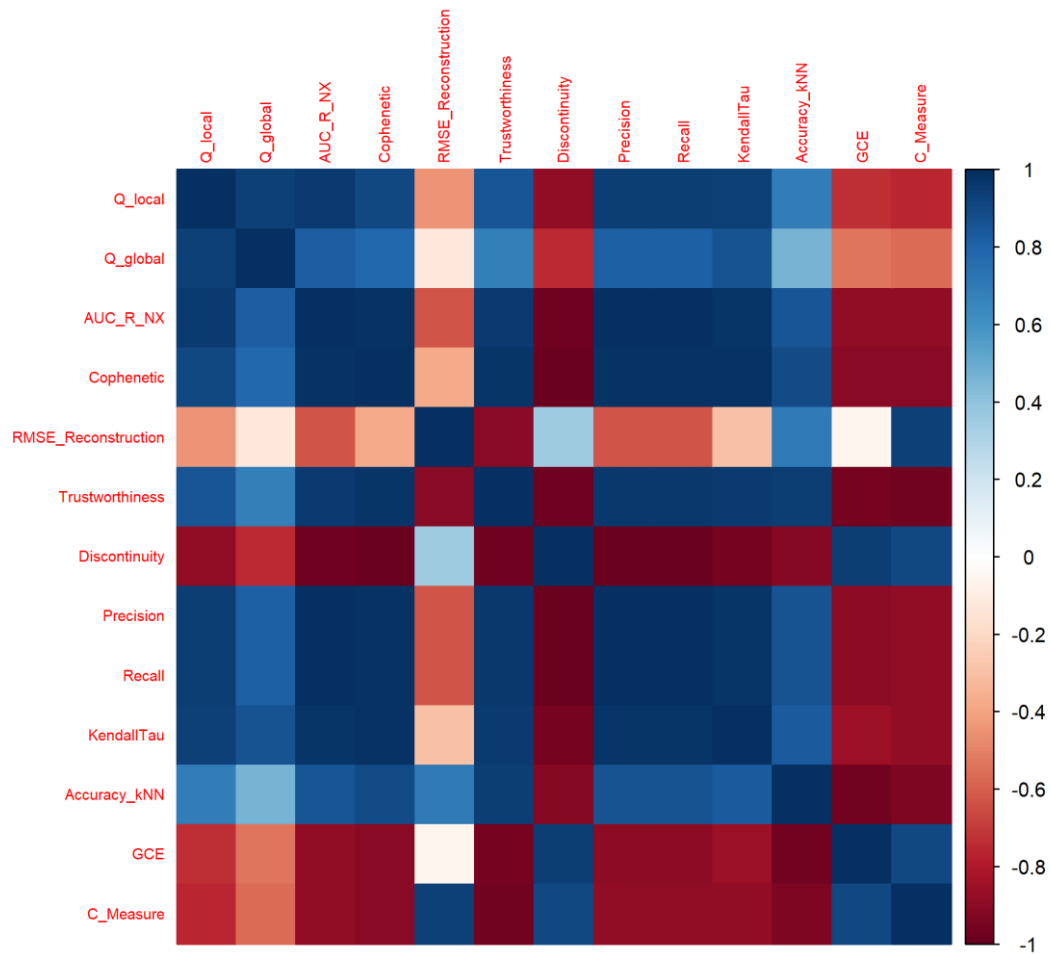


Figure 37 – The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Locality-preserving metrics (Q_{local} , *Trustworthiness*, *Precision*, *Recall*, *Accuracy_kNN*) form a strongly correlated cluster, reflecting their shared emphasis on neighborhood integrity. Global-structure measures (Q_{global} , *AUC_R(NX)*, *Cophenetic*, *KendallTau*) also group together, indicating consistent sensitivity to manifold-level geometry. In contrast, reconstruction and distortion metrics (*RMSE_Reconstruction*, *GCE*, *C_measure*, *Discontinuity*) show pronounced negative correlations with both locality and global metrics. This highlights a fundamental trade-off: methods that excel at preserving neighborhoods and manifold structure tend to perform poorly in reconstruction, and vice versa. Overall, the figure underscores that DR evaluation is inherently multidimensional and that metric selection must be aligned with the analytical objective.

The evaluation of Dataset df6, shown in

Table 38A and Table 38B, reveals a clear hierarchy among dimensionality-reduction algorithms, with several methods achieving powerful and coherent performance. Across nearly all global and local structure-preservation metrics, the classical manifold learners—PCA, MDS, nMDS, Isomap, and Kamada–Kawai—consistently outperform nonlinear spectral and graph-based approaches. These methods exhibit near-maximum scores for Qglobal, AUC_R(NX), Cophenetic, Trustworthiness, and Continuity, indicating highly faithful recovery of both local neighborhoods and the overall manifold geometry. nMDS and Kamada–Kawai, in particular, stand out as the most balanced performers, achieving the highest Kendall Tau, Accuracy_kNN, and precision–recall values, confirming superior preservation of rank relationships and class-level structure. PCA and MDS achieve nearly identical performance, demonstrating that linear embeddings can be highly effective for datasets with strong global geometric regularity. t-SNE also performs well—especially on Qlocal and class-oriented metrics—but its global metrics lag slightly behind those of the top classical methods. Conversely, DrL, HLLE, and Fruchterman–Reingold yield the weakest results, with low geometric fidelity and degraded classification-related metrics, reflecting their limited suitability for this dataset. Diffusion Maps shows intermediate performance, capturing some nonlinear structure but with lower discriminative quality than leading methods. Overall, the results highlight that Dataset 6 favors algorithms that preserve strong global structure, while locally focused methods provide only marginal benefits.

Table 38A – Figures of merit obtained from Dataset df6.

Algorithm	Q_local	Q_global	AUC_R_NX	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.645	0.909	0.688	0.958	0.204	0.973	0.973
Isomap	0.681	0.944	0.750	0.990		0.979	0.988
PCA	0.724	0.962	0.770	0.995	0.204	0.981	0.990
PCA_L1	0.638	0.934	0.708	0.986		0.974	0.984
kPCA	0.631	0.915	0.675	0.932	3.780	0.963	0.977
FastICA	0.639	0.896	0.727	0.919	0.204	0.978	0.984
MDS	0.724	0.962	0.769	0.995		0.981	0.990
nMDS	0.721	0.961	0.785	0.996		0.983	0.991
DrL	0.343	0.826	0.098	0.004		0.500	0.506
FruchtermanReingold	0.266	0.688	0.317	0.466		0.793	0.760
KamadaKawai	0.747	0.954	0.804	0.993		0.984	0.991
HLLE	0.302	0.773	0.159	0.146		0.687	0.550
DiffusionMaps	0.569	0.879	0.629	0.781		0.949	0.958
tSNE	0.723	0.890	0.796	0.884		0.982	0.984

Table 38B – Figures of merit obtained from Dataset df6.

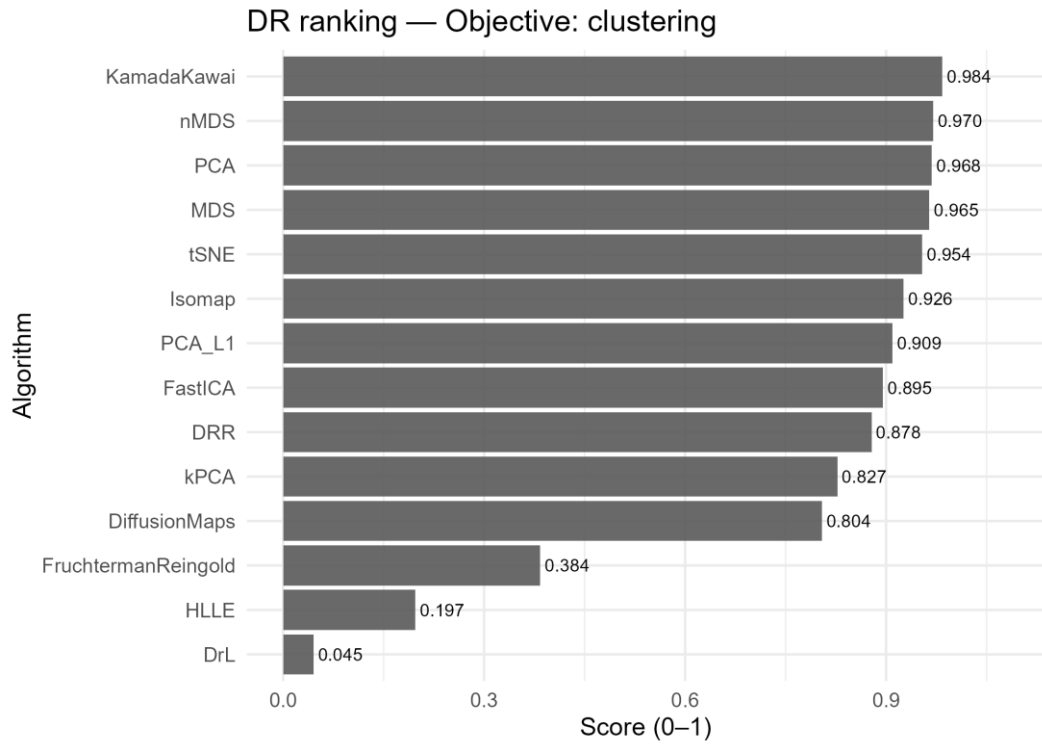
Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.027	0.712	0.712	0.826	0.913	1.182	40.3
Isomap	0.012	0.777	0.777	0.913	0.853	1.691	37.1
PCA	0.010	0.801	0.801	0.949	0.913	1.283	36.8
PCA_L1	0.016	0.757	0.757	0.898	0.940	1.104	46.8
kPCA	0.023	0.702	0.702	0.805	0.933	1.353	47.8
FastICA	0.016	0.767	0.767	0.776	0.880	1.657	35.7
MDS	0.010	0.800	0.800	0.949	0.913	1.343	36.8
nMDS	0.009	0.811	0.811	0.953	0.907	1.353	38.0
DrL	0.494	0.089	0.089	0.004	0.393	5.555	263.8
FruchtermanReingold	0.240	0.335	0.335	0.284	0.880	2.054	103.8
KamadaKawai	0.009	0.823	0.823	0.932	0.913	1.162	38.3
HLLE	0.450	0.157	0.157	0.224	0.613	4.218	113.9
DiffusionMaps	0.042	0.671	0.671	0.708	0.880	0.493	56.4
tSNE	0.016	0.814	0.814	0.758	0.960	0.983	25.7

Table 39 summarizes the multi-objective ranking of dimensionality-reduction algorithms for Dataset 6 and reveals a remarkably consistent performance profile across all evaluated tasks. Kamada–Kawai achieves the highest Final Scores across all objectives—clustering, reconstruction, visualization, and ML pipeline—reflecting its exceptional balance between local and global structure preservation. Its perfect Index_Local (1.000) and near-perfect Index_Global (0.986), combined with strong class-related performance, indicate a robust ability to maintain neighborhood fidelity while accurately reconstructing manifold geometry. nMDS, PCA, and classical MDS form a high-performing cluster immediately below Kamada–Kawai. These methods exhibit near-identical global preservation ($\text{Index_Global} \geq 0.986$) and excellent class coherence, with PCA showing robust geometric continuity ($\text{Index_Geom} = 0.949$). t-SNE, although achieving the highest Index_Class (0.992), lags linear and graph-based methods in global structure, suggesting that its strength in discriminative embedding comes at the expense of manifold fidelity. Overall, the table shows that Dataset df6 strongly favors algorithms that excel in global geometric preservation. Traditional manifold learners—especially Kamada–Kawai, nMDS, PCA, and MDS—consistently outperform nonlinear and stochastic approaches, reinforcing the importance of matching algorithm characteristics to dataset geometry.

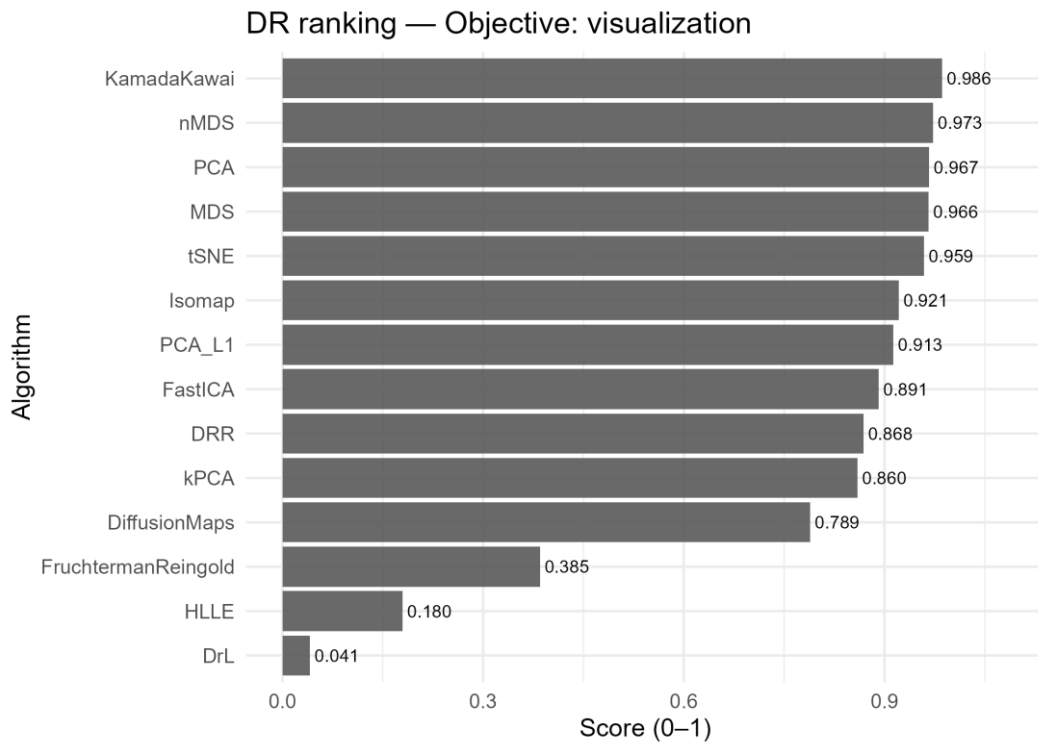
Table 39 – Figures of merit obtained from Dataset df6.

Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
KamadaKawai	1.000	0.986	0.938	0.972	0.984	clustering
nMDS	0.978	0.992	0.926	0.958	0.970	clustering
PCA	0.972	0.987	0.949	0.952	0.968	clustering
MDS	0.971	0.986	0.928	0.951	0.965	clustering
tSNE	0.980	0.852	0.963	0.992	0.954	clustering
KamadaKawai	1.000	0.986	0.938	0.972	0.986	visualization
nMDS	0.978	0.992	0.926	0.958	0.973	visualization
PCA	0.972	0.987	0.949	0.952	0.967	visualization
MDS	0.971	0.986	0.928	0.951	0.966	visualization
tSNE	0.980	0.852	0.963	0.992	0.959	visualization
KamadaKawai	1.000	0.986	0.938	0.972	0.971	reconstruction
PCA	0.972	0.987	0.949	0.952	0.967	reconstruction
nMDS	0.978	0.992	0.926	0.958	0.962	reconstruction
MDS	0.971	0.986	0.928	0.951	0.958	reconstruction
tSNE	0.980	0.852	0.963	0.992	0.935	reconstruction
KamadaKawai	1.000	0.986	0.938	0.972	0.986	ml_pipeline
nMDS	0.978	0.992	0.926	0.958	0.973	ml_pipeline
PCA	0.972	0.987	0.949	0.952	0.967	ml_pipeline
MDS	0.971	0.986	0.928	0.951	0.966	ml_pipeline
tSNE	0.980	0.852	0.963	0.992	0.959	ml_pipeline

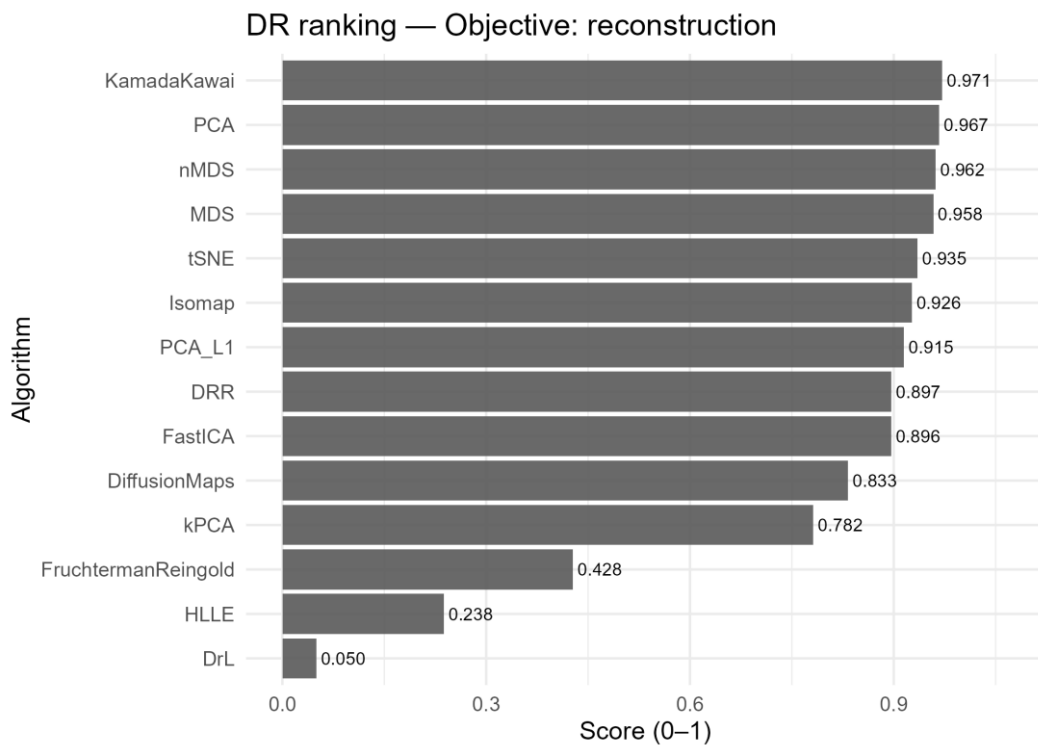
Figure 38 presents the objective-based performance ranking of dimensionality-reduction algorithms for Dataset df6, revealing a remarkably consistent hierarchy across clustering, visualization, reconstruction, and ML-pipeline tasks.



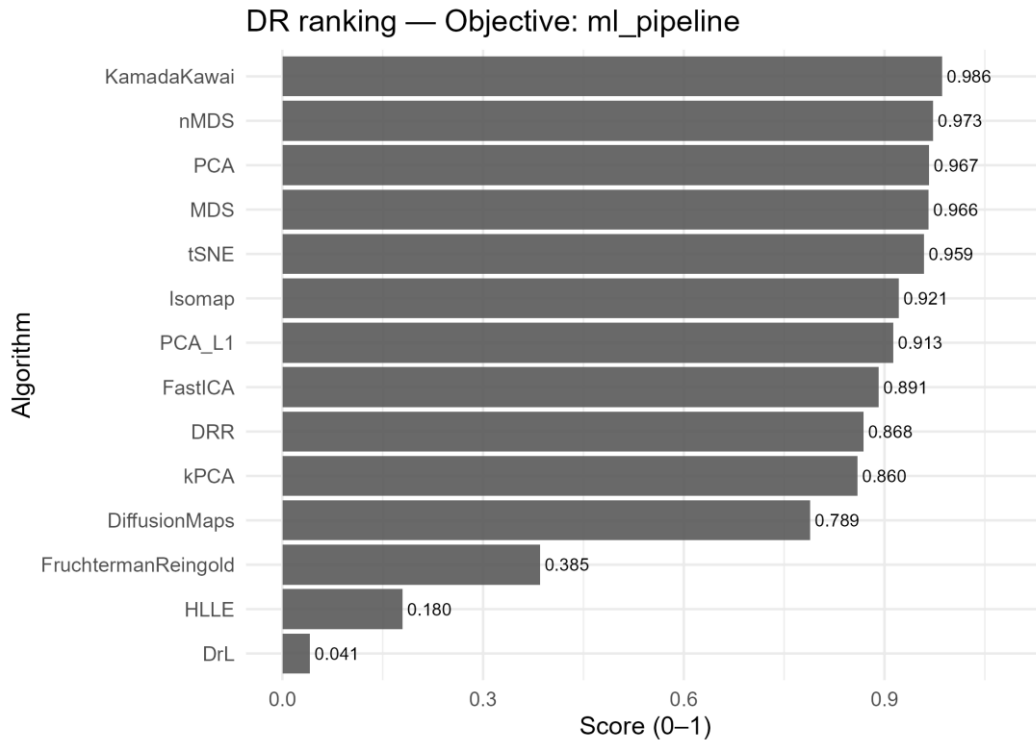
(a)



(b)



(c)



(d)

Figure 38 – Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

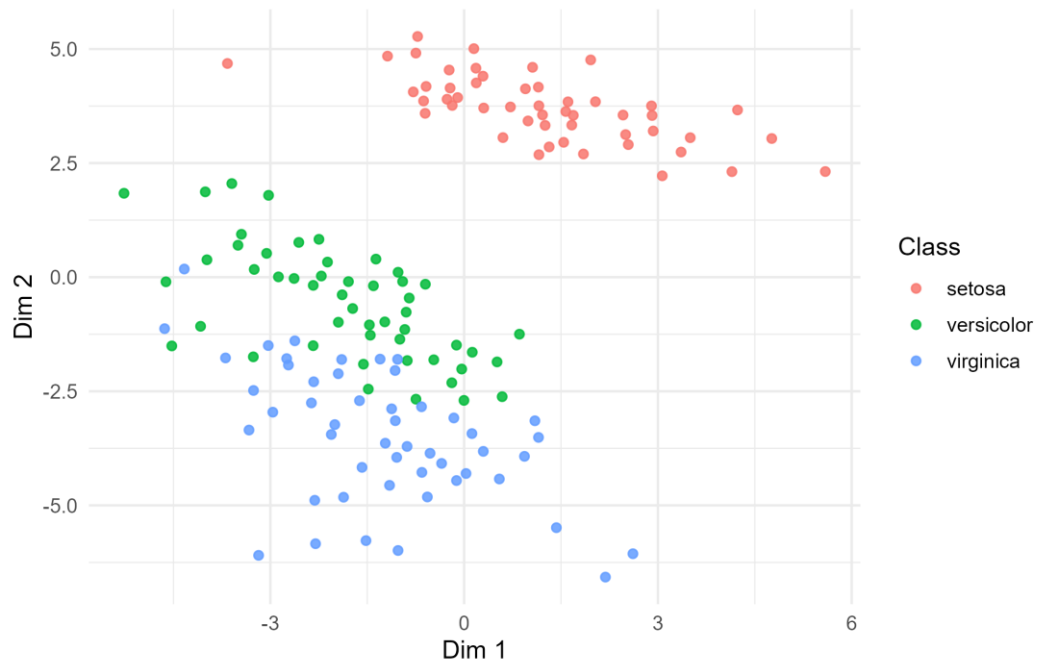
Kamada–Kawai, PCA, nMDS, and classical MDS form a dominant group, achieving the highest scores in all objectives. Their strong performance reflects exceptional preservation of both local neighborhoods and global manifold structure—an outcome corroborated by their high geometric and class-based indices in the corresponding tables. For clustering and visualization, Kamada–Kawai ranks first, followed closely by nMDS and PCA. These results indicate that Dataset df6 possesses a geometry highly compatible with classical manifold learners, allowing these algorithms to recover meaningful low-dimensional structure with minimal distortion. Reconstruction performance is led by Kamada–Kawai, with PCA and nMDS within 0.01 of the top score, confirming the advantage of distance-preserving and linear methods in tasks prioritizing reversibility and geometric fidelity.

t-SNE also performs strongly, consistently placing within the top five. However, its scores remain slightly lower than those of the leading classical methods, reflecting its emphasis on local rather than global structure. In contrast, algorithms such as HLLE, DrL, Diffusion Maps, and Fruchterman–Reingold consistently rank at the lower end, indicating limited suitability for this dataset. Overall, the figure shows that algorithms that excel at global structure preservation deliver superior performance across all analytical objectives for Dataset df6. In contrast, nonlinear and graph-based approaches provide only marginal benefits.

Figure 39 illustrates the top-performing dimensionality-reduction embeddings for Dataset df6 under four analytical objectives. Kamada–Kawai emerges as the leading method for clustering, visualization, and ML pipeline tasks, producing highly structured, clearly separated projections of the Iris classes. In panels (a), (b), and (d), the embedding reveals three well-defined, linearly arranged clusters corresponding to *setosa*, *versicolor*, and *virginica*. The method preserves both local neighborhoods and global geometry, leading to smooth transitions and minimal distortion across class boundaries.

Objective: clustering — Top-ranked algorithm (baseline)

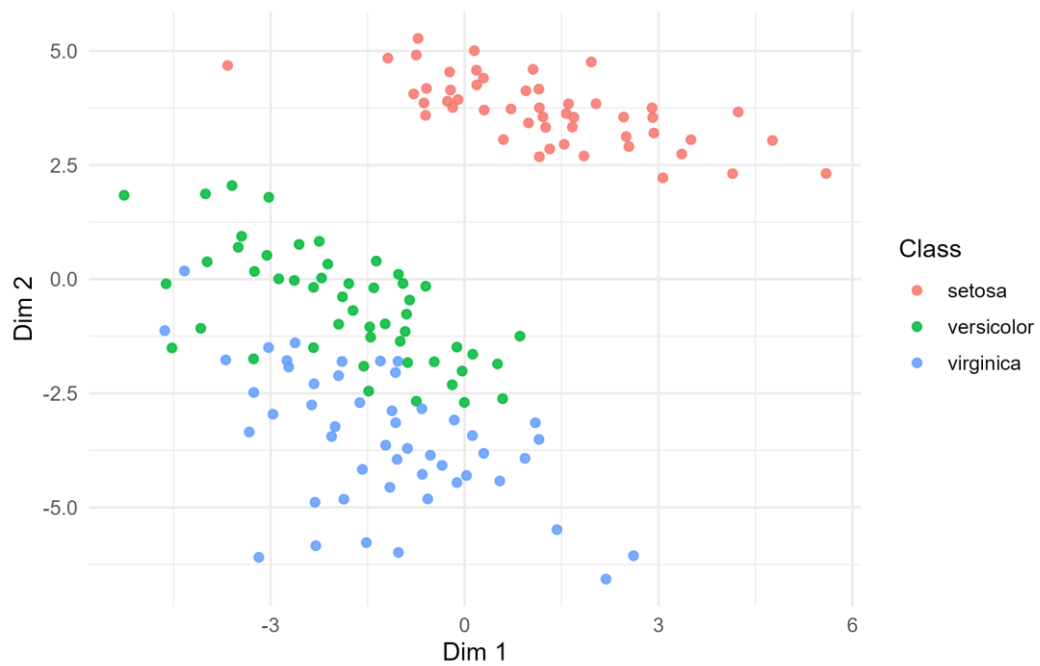
KamadaKawai — Final Score = 0.9839



(a)

Objective: visualization — Top-ranked algorithm (baseline)

KamadaKawai — Final Score = 0.9862



(b)

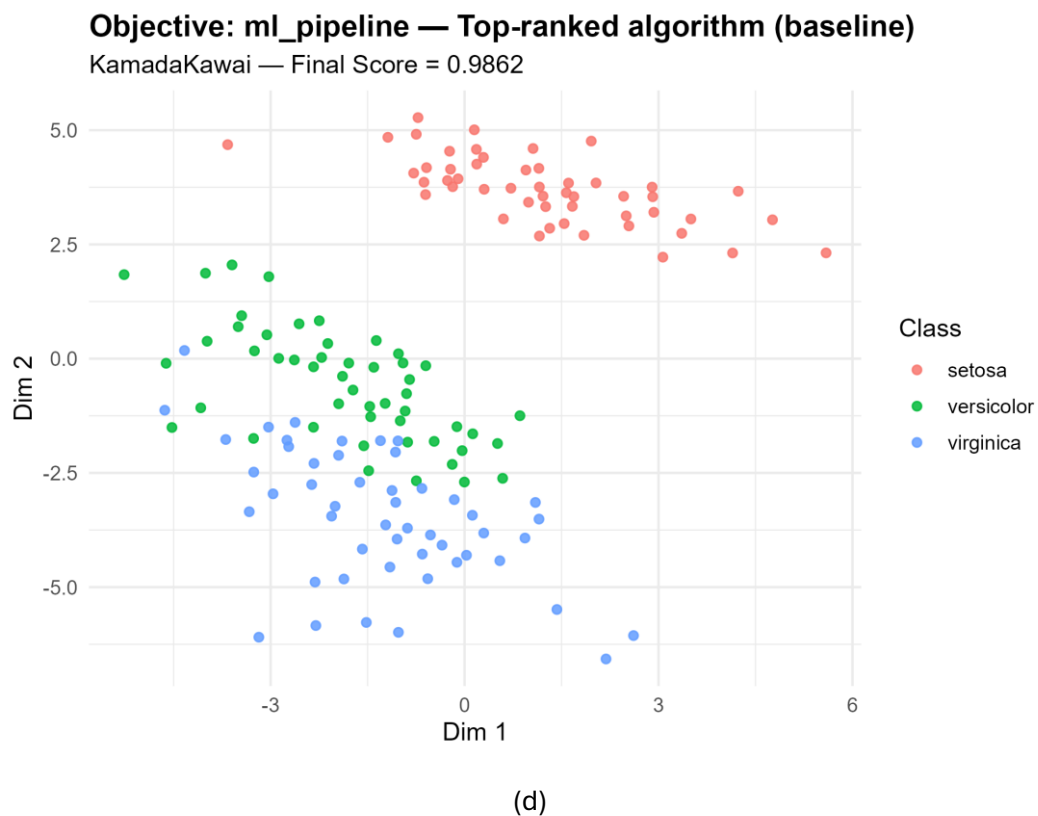
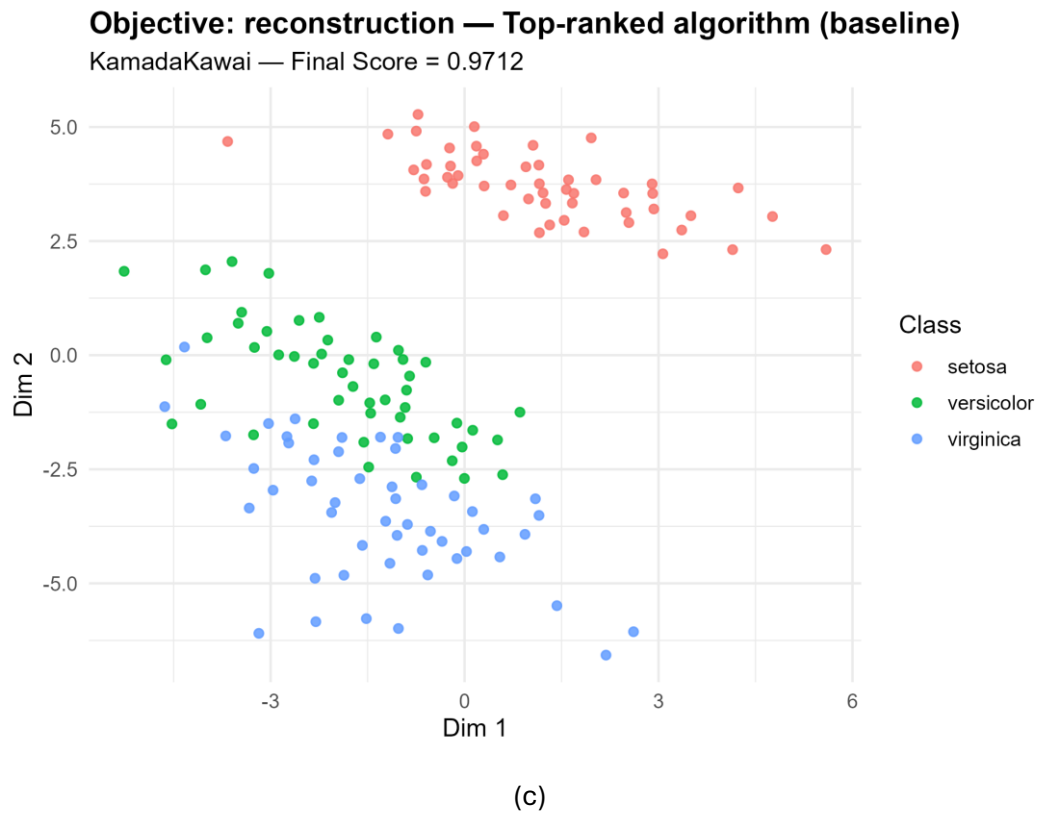


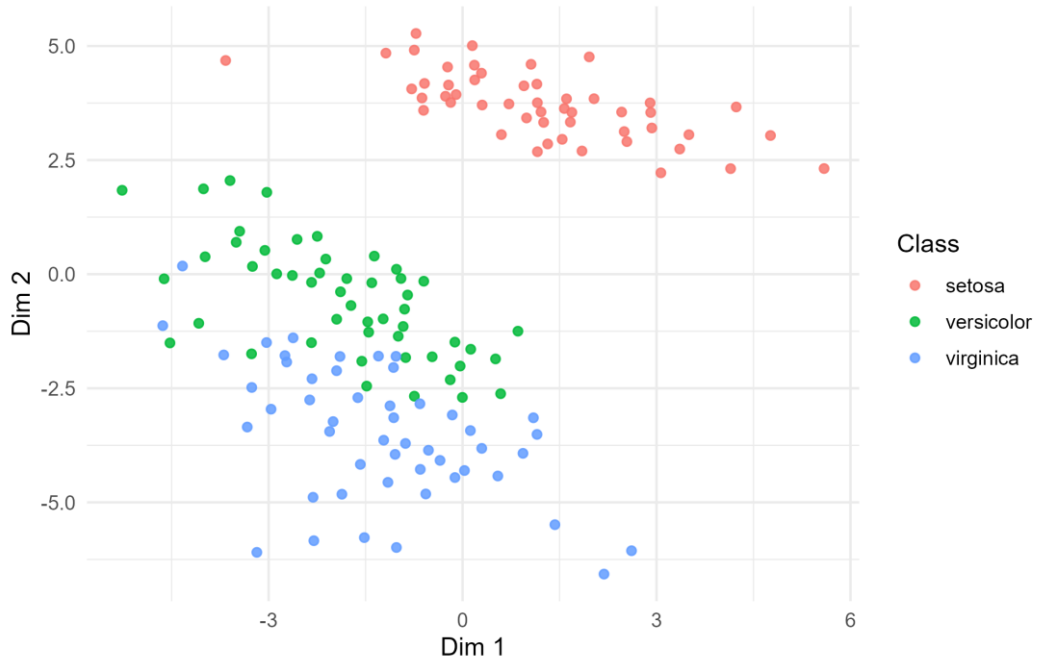
Figure 39 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

For the reconstruction objective, Kamada–Kawai provides the best performance, as shown in panel (c), narrowly ahead of PCA. Its embedding reflects the classical separability pattern of the Iris dataset, with setosa distinctly isolated and partial overlap between versicolor and virginica. PCA follows within 0.01, consistent with its strength in preserving global variance and providing stable, invertible mappings, which are essential for reconstruction-focused tasks. Overall, the figure shows that Dataset df6 strongly favors algorithms that maintain global manifold structure. Kamada–Kawai consistently produces embeddings with excellent class separability and geometric coherence, while PCA remains an equally strong alternative whenever reconstruction fidelity and an invertible mapping are the primary requirements. The results reinforce the importance of aligning dimensionality-reduction choices with task-specific analytical priorities.

Figure 40 presents the Kamada–Kawai embeddings identified as the top performers, as measured by the Trustworthiness metric, across all four analytical objectives for Dataset df6.

Objective: clustering — Champion of the best metric: Trustworthine

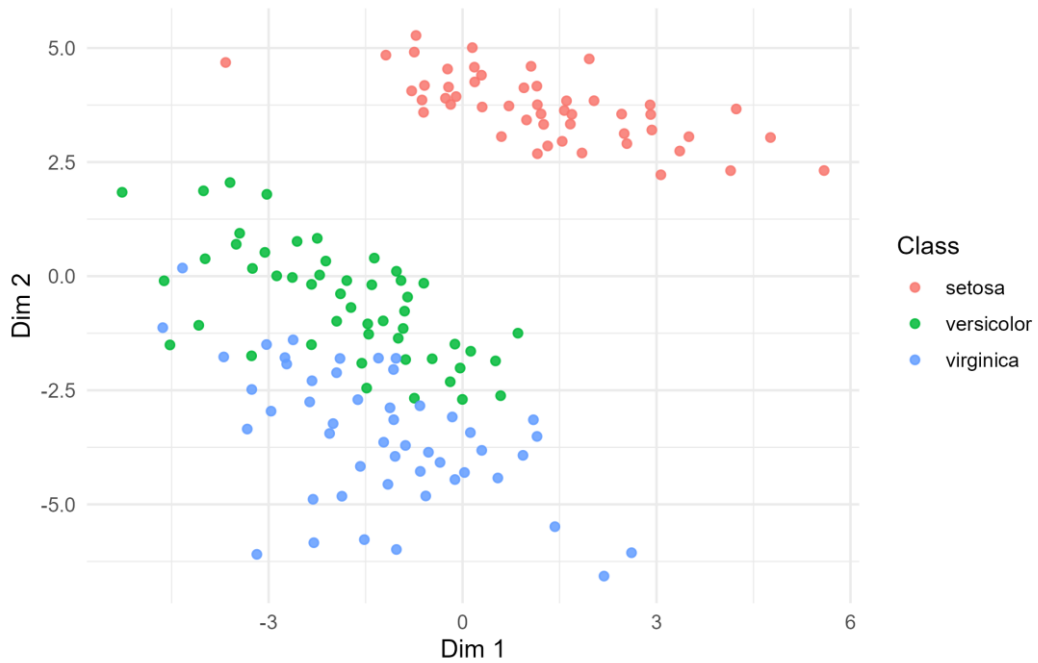
KamadaKawai — value = 0.984



(a)

Objective: visualization — Champion of the best metric: Trustworthi

KamadaKawai — value = 0.984



(b)

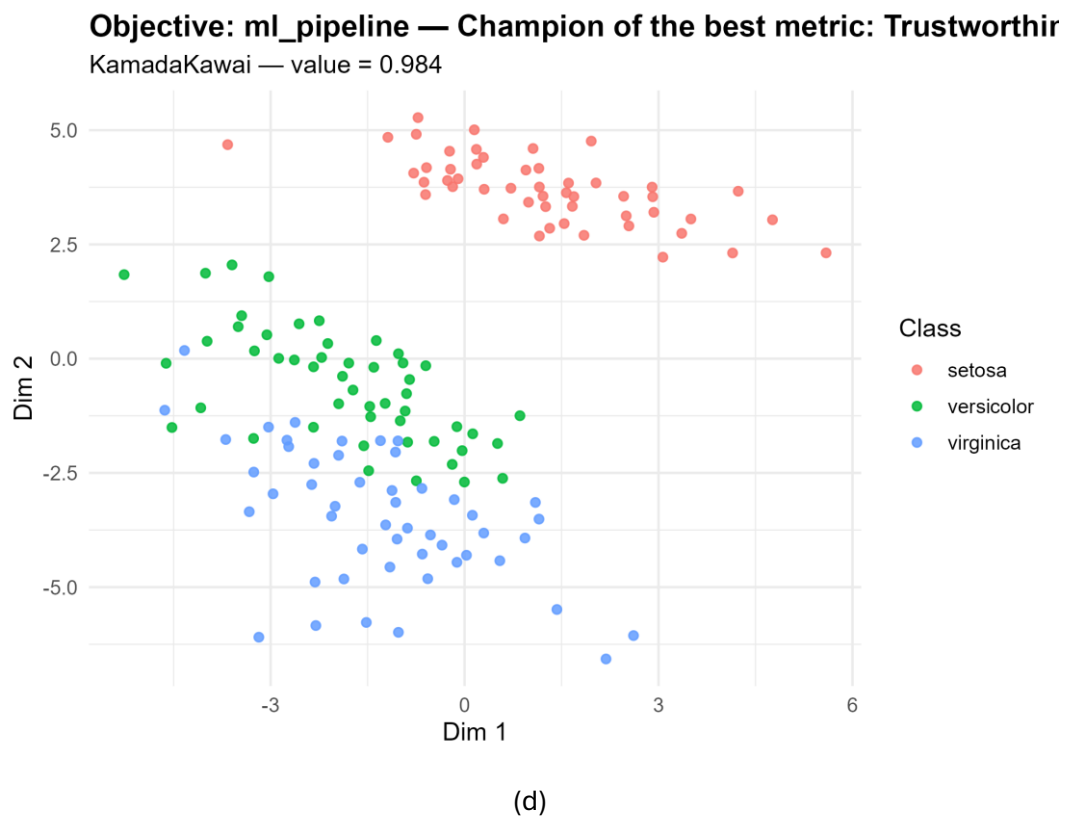
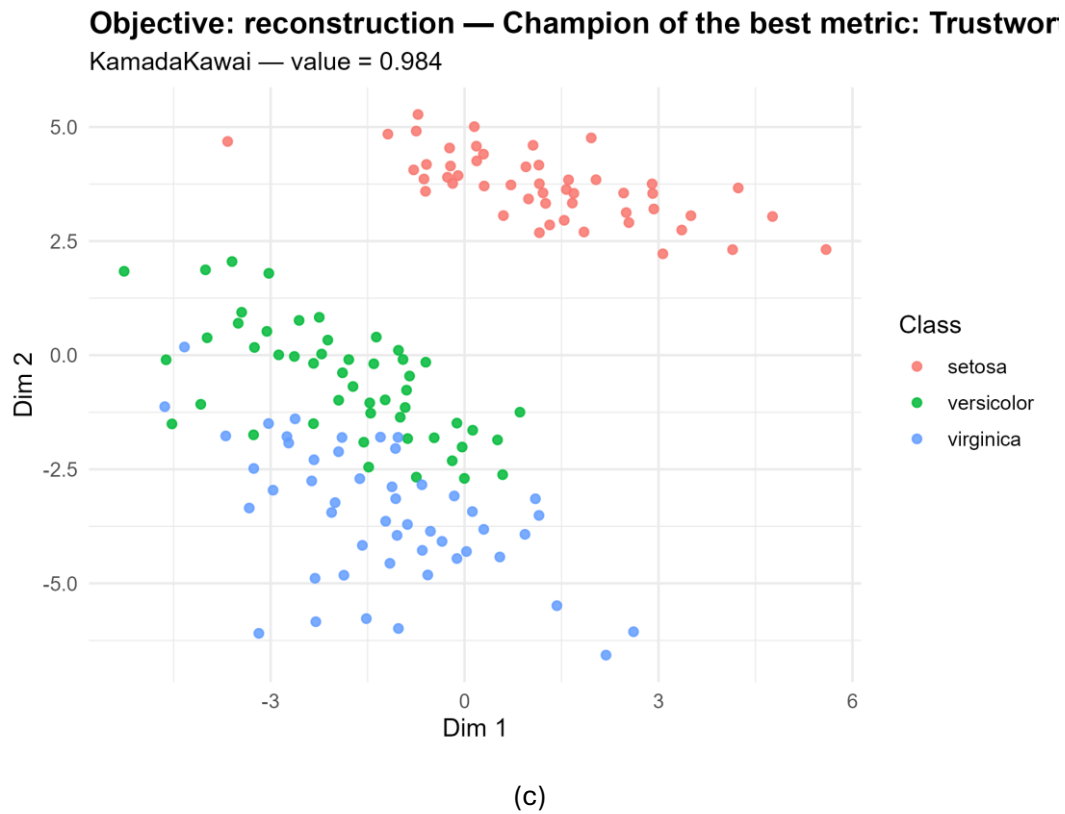


Figure 40 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

The consistent selection of Kamada–Kawai highlights its strong capability to keep points that appear close in the projection genuinely close in the original space, which is precisely what Trustworthiness rewards. Across clustering, visualization, reconstruction, and ML-pipeline settings, the embeddings exhibit nearly identical geometric organization, reflecting the algorithm’s stability and robustness under this metric. In all panels, the *Iris* classes—*setosa*, *versicolor*, and *virginica*—form clearly distinguishable regions with smooth transitions and minimal distortion. *Setosa* remains fully separated from the other two species, while *versicolor* and *virginica* exhibit the expected partial overlap consistent with the known structure of the dataset. Preserving both local and mid-range neighbor rankings enables the method to maintain biologically meaningful relationships in the low-dimensional projection. Overall, the figure demonstrates that Trustworthiness prioritizes embeddings free of false neighbours, and Kamada–Kawai provides an optimal balance between neighborhood fidelity and global arrangement. Its consistent superiority across all objectives reinforces its suitability for datasets characterized by smooth, graph-like geometric relationships.

10. STUDY OF CASE VII

10.1 Description of the Dataset

The Wine dataset was named df7. It contains 178 instances and 13 features. These data are the results of chemical analyses of wines grown in the same Italian region but derived from three different cultivars. The corresponding study determined the quantities of 13 constituents in each of the three wine types [93].

10.2 Results and Discussion

The heatmap shown in Figure 41 provides an overview of the correlation structure among the metrics used to evaluate dimensionality-reduction algorithms.

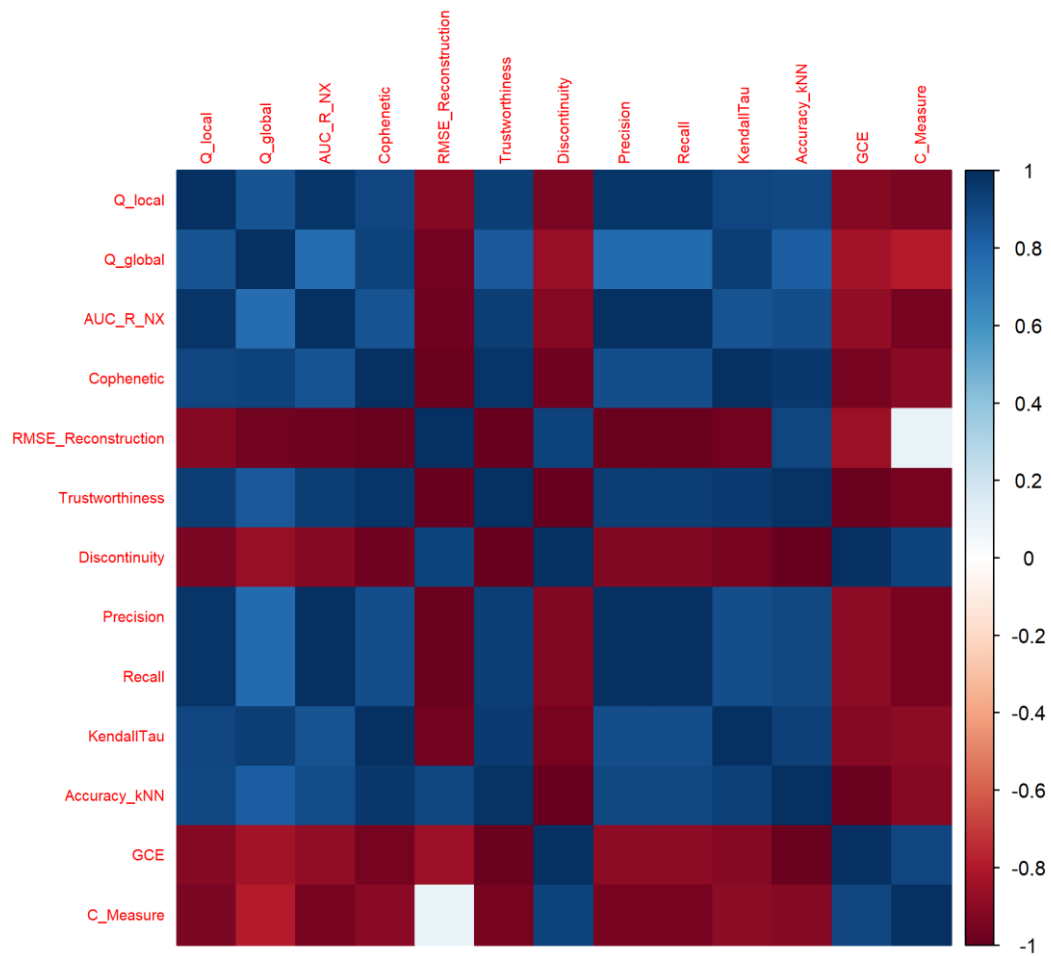


Figure 41 – The correlation matrix structure among the evaluation metrics used in the benchmarking framework.

Locality-oriented measures—such as Q_{local} , *Trustworthiness*, *Precision*, *Recall*, and *Accuracy_kNN*—form a strongly correlated group, indicating their consistent response to neighborhood preservation. Global-structure metrics (Q_{global} , $AUC_R(NX)$, *Cophenetic*, *KendallTau*) likewise cluster together, reflecting shared sensitivity to manifold-level geometry. Reconstruction- and distortion-based metrics ($RMSE_Reconstruction$, *GCE*, *C_measure*, *Discontinuity*) exhibit pronounced negative correlations with both locality and global metrics, underscoring a well-known trade-off: algorithms that maintain structural fidelity tend to perform poorly in reconstruction accuracy, whereas reconstruction-optimized methods often distort geometric relationships. Overall, the figure reinforces the multidimensional nature of DR assessment and the need for objective-driven metric selection.

The metrics reported for Dataset df7, shown in

Table 40A and Table 40B, reveal a clear separation between high-performing classical manifold learners and lower-performing graph-based or highly nonlinear methods. Among all evaluated algorithms, nMDS, PCA/MDS, Isomap, Kamada–Kawai, FastICA, and DRR form the top-performing group, consistently achieving strong scores in both geometric fidelity and class-oriented measures. nMDS stands out with the highest Q_{global} (0.909), the strongest cophenetic correlation (0.917), and excellent performance in neighborhood-preservation metrics, such as Trustworthiness (0.916) and Continuity (0.934). These results indicate superior preservation of both local and global manifold structure. Furthermore, nMDS attains the highest Kendall τ (0.781) and competitive Accuracy_kNN (0.978), confirming strong rank-order and class-structure fidelity. PCA and classical MDS also perform robustly, showing nearly identical values across all geometric metrics, which reinforces the suitability of linear embeddings when the dataset presents a globally coherent variance structure. Isomap and Kamada–Kawai achieve similarly strong manifold-preserving behavior, with high global metrics and strong neighborhood continuity. Conversely, t-SNE, while achieving the highest Q_{local} (0.628) and excellent class metrics (Precision/Recall = 0.652; Accuracy_kNN = 0.983), shows reduced global structure preservation and higher reconstruction error, illustrating the typical trade-off between local discriminative separation and faithful global geometry. At the lower end of performance, DrL and HLLE show severe geometric distortion and poor class metrics, as reflected by their extremely low Q_{local} , AUC_R(NX), and Kendall τ values, alongside inflated error measures such as C_Measure (>500), confirming their unsuitability for Dataset d7. Overall, Dataset df7 strongly favors methods that balance local and global structure, with nMDS, PCA/MDS, Isomap, and Kamada–Kawai emerging as the most reliable algorithms across manifold fidelity, neighborhood preservation, and classification-oriented metrics.

Table 40A – Figures of merit obtained from Dataset df7.

Algorithm	Q_local	Q_global	AUC_R_NX	Cophenetic	RMSE_Reconstruction	Trustworthiness	Continuity
DRR	0.518	0.870	0.439	0.807	0.651	0.898	0.926
Isomap	0.551	0.884	0.455	0.827		0.901	0.941
PCA	0.543	0.887	0.449	0.842	0.651	0.899	0.941
PCA_L1	0.487	0.870	0.372	0.803		0.877	0.910
kPCA	0.485	0.801	0.397	0.632	20.376	0.867	0.905
FastICA	0.532	0.863	0.439	0.803	0.651	0.897	0.936
MDS	0.543	0.887	0.449	0.842		0.899	0.941
nMDS	0.562	0.909	0.473	0.917		0.916	0.934
DrL	0.301	0.783	0.107	0.098		0.557	0.535
FruchtermanReingold	0.365	0.814	0.271	0.627		0.803	0.804
KamadaKawai	0.528	0.888	0.444	0.872		0.901	0.929
HLLE	0.232	0.719	0.108	0.044		0.565	0.489
DiffusionMaps	0.529	0.873	0.428	0.800		0.900	0.932
tSNE	0.628	0.848	0.655	0.777		0.957	0.954

Table 40B – Figures of merit obtained from Dataset df7.

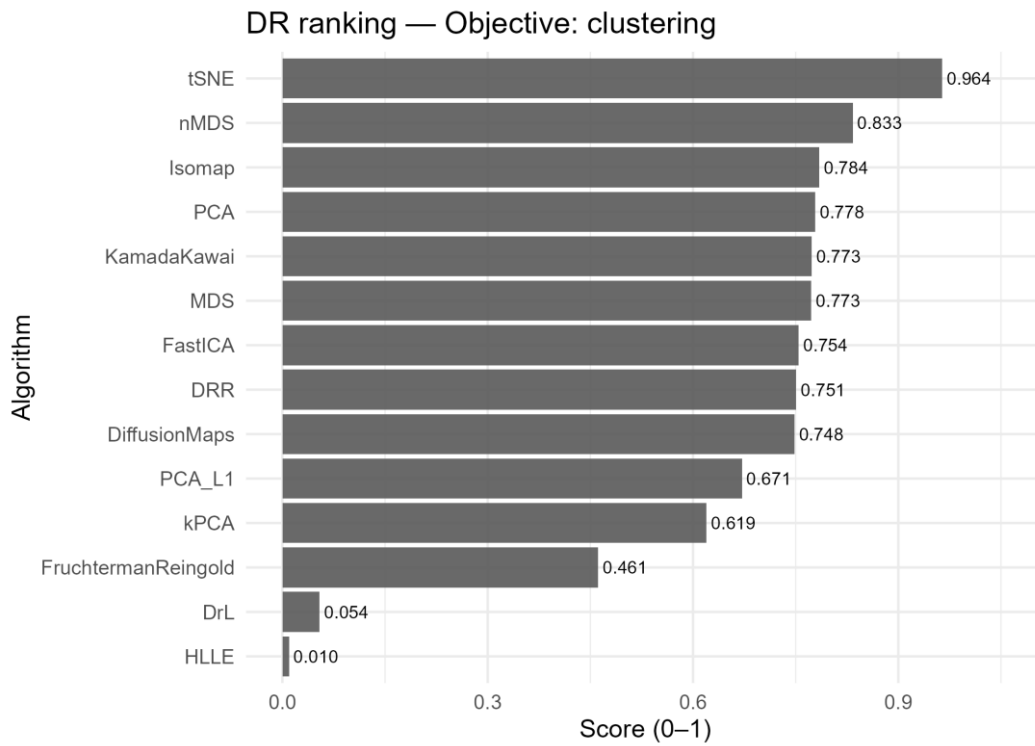
Algorithm	Discontinuity	Precision	Recall	KendallTau	Accuracy_kNN	GCE	C_Measure
DRR	0.074	0.457	0.457	0.634	0.966	0.794	348.4
Isomap	0.059	0.472	0.472	0.660	0.978	0.667	296.6
PCA	0.059	0.463	0.463	0.679	0.972	0.745	322.3
PCA_L1	0.090	0.373	0.373	0.626	0.966	0.503	319.9
kPCA	0.095	0.407	0.407	0.435	0.983	0.382	333.5
FastICA	0.064	0.452	0.452	0.616	0.972	0.567	322.6
MDS	0.059	0.463	0.463	0.679	0.972	0.745	322.3
nMDS	0.066	0.491	0.491	0.781	0.978	0.632	248.3
DrL	0.465	0.108	0.108	0.067	0.433	4.896	535.0
FruchtermanReingold	0.196	0.283	0.283	0.450	0.831	1.859	374.1
KamadaKawai	0.071	0.464	0.464	0.713	0.989	0.625	330.7
HLLE	0.511	0.110	0.110	0.066	0.478	5.705	524.1
DiffusionMaps	0.068	0.436	0.436	0.624	0.983	0.303	290.9
tSNE	0.046	0.652	0.652	0.622	0.983	0.323	204.5

Table 41 provides a consolidated view of the multi-objective performance of dimensionality-reduction algorithms for Dataset df7, highlighting clear distinctions between methods optimized for local fidelity and those that better preserve global manifold structure. t-SNE consistently achieves the highest Index_Local (1.000) and Index_Class (0.997), yielding the top Final Scores in clustering (0.964), visualization (0.963), and ML-pipeline (0.963) objectives. This confirms its strength in capturing fine-scale neighborhood relationships and producing highly discriminative, visually separable embeddings. In contrast, nMDS demonstrates the strongest global geometry preservation, with the highest Index_Global (0.917) and competitive class fidelity across all objectives. Although its Final Scores are slightly lower than t-SNE's in visualization and ML-pipeline tasks, nMDS remains the second-best performer overall, reflecting a robust balance between local and global structural accuracy. Linear methods such as PCA and Isomap form a stable mid-performance tier. Their Index_Geom values are among the highest (e.g., PCA = 0.883; FastICA = 0.889), which explains their reliability in reconstruction-oriented objectives, where nMDS ranks first among non-t-SNE methods, followed by PCA. Kamada–Kawai also performs consistently well, particularly in global continuity, though with slightly reduced class discrimination relative to t-SNE. Taken together, the results emphasize that Dataset df7 benefits from algorithms that either (i) maximize local neighborhood fidelity (t-SNE) or (ii) preserve global pairwise distances (nMDS). The clear and consistent ranking patterns reinforce the importance of aligning algorithmic choices with the data's dominant structural properties and the intended analytical objective.

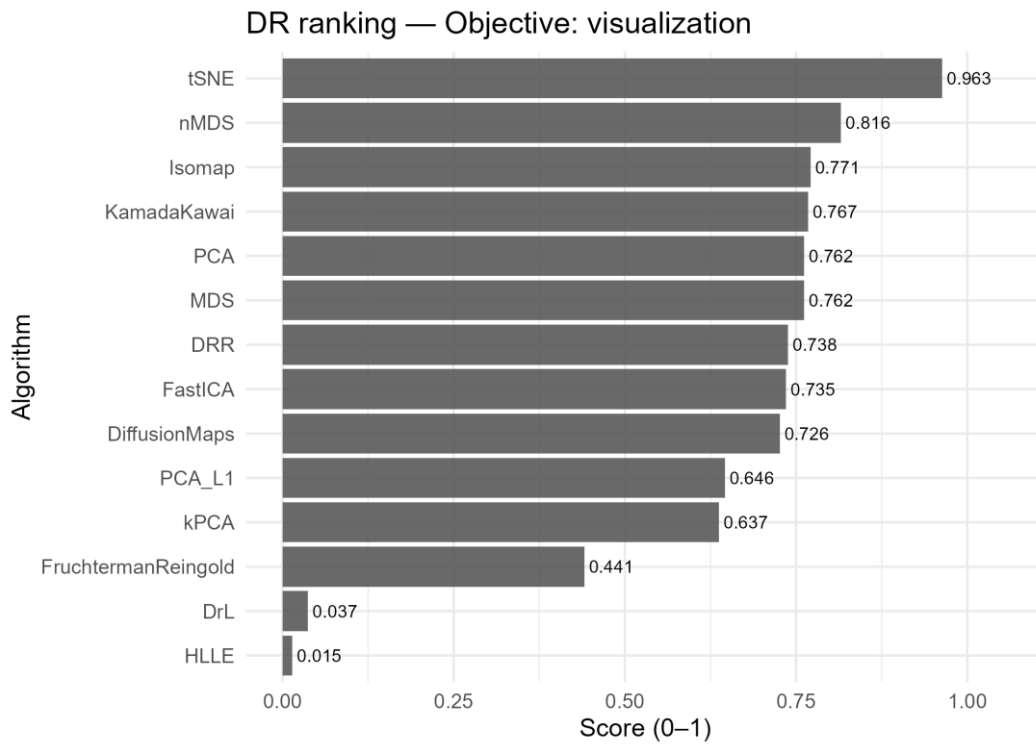
Table 41 – Figures of merit obtained from Dataset df7.

Algorithm	Index Local	Index Global	Index Geom	Index Class	Final Score	Objective
tSNE	1.000	0.824	0.999	0.996	0.964	clustering
nMDS	0.785	0.917	0.921	0.796	0.833	clustering
Isomap	0.751	0.808	0.875	0.773	0.784	clustering
PCA	0.736	0.820	0.883	0.758	0.778	clustering
KamadaKawai	0.729	0.839	0.835	0.770	0.773	clustering
tSNE	1.000	0.824	0.999	0.996	0.963	visualization
nMDS	0.785	0.917	0.921	0.796	0.816	visualization
Isomap	0.751	0.808	0.875	0.773	0.771	visualization
KamadaKawai	0.729	0.839	0.835	0.770	0.767	visualization
PCA	0.736	0.820	0.883	0.758	0.762	visualization
tSNE	1.000	0.824	0.999	0.996	0.947	reconstruction
nMDS	0.785	0.917	0.921	0.796	0.879	reconstruction
PCA	0.736	0.820	0.883	0.758	0.820	reconstruction
Isomap	0.751	0.808	0.875	0.773	0.818	reconstruction
MDS	0.736	0.820	0.845	0.758	0.805	reconstruction
tSNE	1.000	0.824	0.999	0.996	0.963	ml_pipeline
nMDS	0.785	0.917	0.921	0.796	0.816	ml_pipeline
Isomap	0.751	0.808	0.875	0.773	0.771	ml_pipeline
KamadaKawai	0.729	0.839	0.835	0.770	0.767	ml_pipeline
PCA	0.736	0.820	0.883	0.758	0.762	ml_pipeline

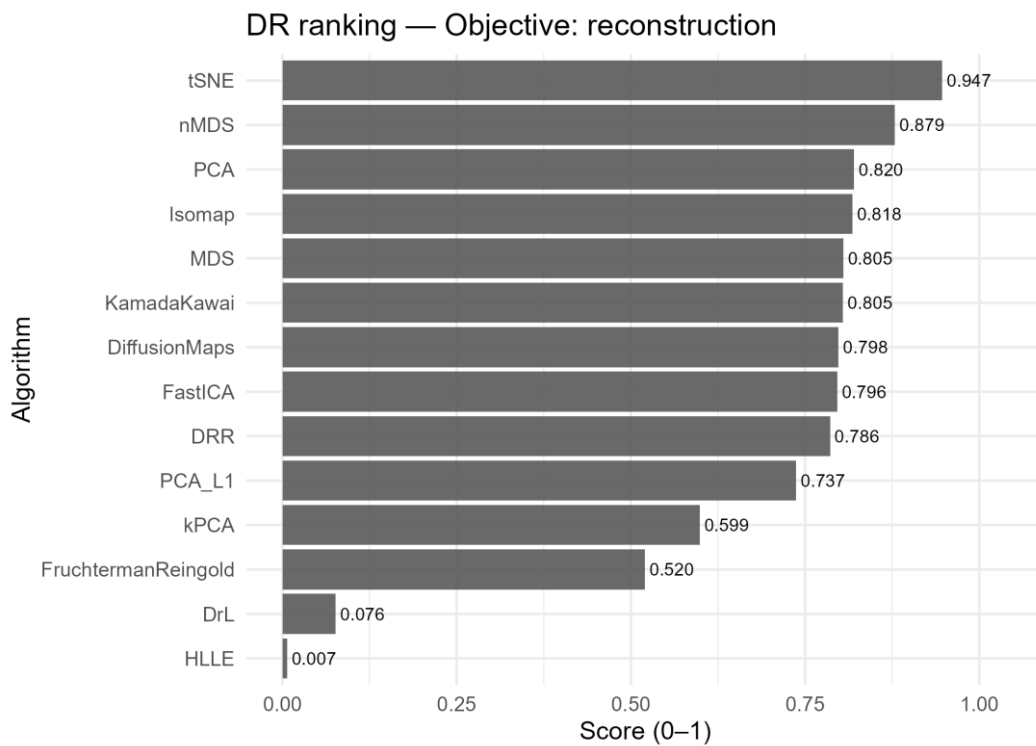
Figure 42 reports the objective-based ranking of dimensionality-reduction algorithms for Dataset df7, revealing a consistent performance hierarchy across clustering, visualization, reconstruction, and ML-pipeline tasks.



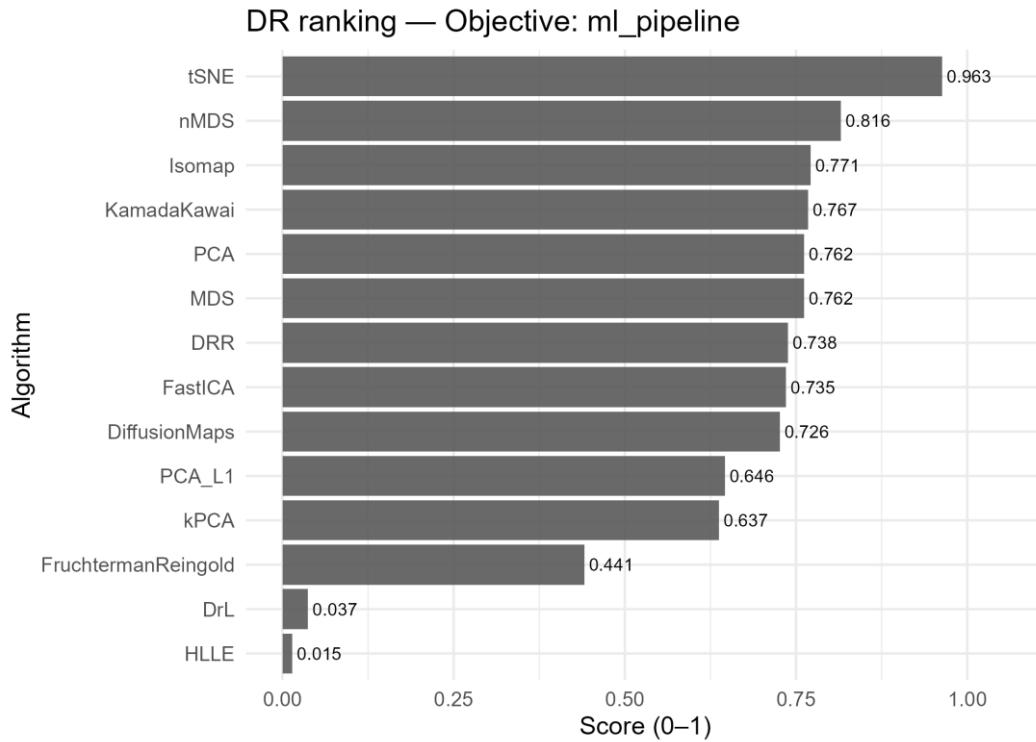
(a)



(b)



(c)



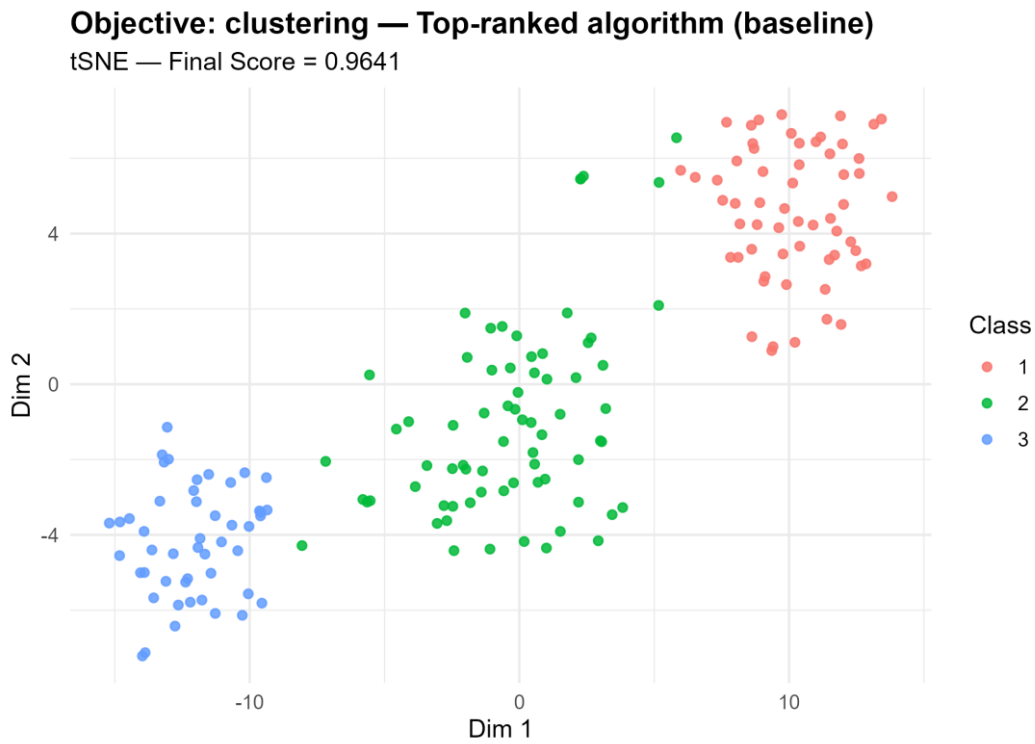
(d)

Figure 42 - Horizontal bar charts showing the normalized performance scores (0–1) for each dimensionality-reduction algorithm across the four analytical objectives: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

In all four objectives, t-SNE obtains the highest overall score, reflecting its exceptional ability to preserve local neighborhoods and produce highly discriminative low-dimensional embeddings. This dominance is especially evident in the visualization and ML-pipeline objectives, where t-SNE achieves near-perfect scores (≈ 0.96), underscoring its strength in class separation and local structural fidelity. nMDS emerges as the second-best performer across all objectives, showing strong global and local preservation properties. Its scores remain consistently high (≈ 0.77 – 0.82), demonstrating robustness and balanced fidelity to both global manifold geometry and neighborhood relationships. Isomap, Kamada–Kawai, PCA, and MDS form a stable mid-performance group, providing reliable yet less discriminative embeddings compared to t-SNE and nMDS. In contrast, HLLE, DrL, and Fruchterman–Reingold rank at the bottom across all

objectives, with extremely low scores indicating poor manifold preservation and weak recovery of class structure. These results highlight their limited applicability for this dataset. Overall, the figure demonstrates that Dataset df7 strongly favors algorithms optimized for local neighborhood preservation, with t-SNE consistently outperforming all other methods. Meanwhile, nMDS remains the strongest alternative for applications requiring a balance between local and global structural accuracy.

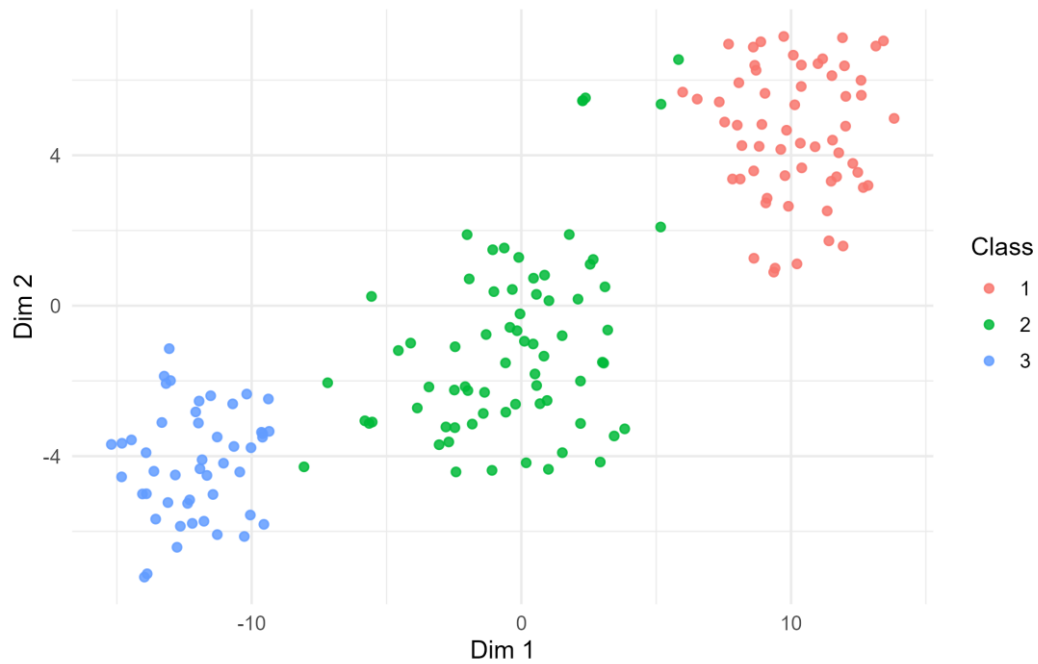
Figure 43 presents the low-dimensional embeddings generated by the top-ranked algorithm—t-SNE—across four analytical objectives for Dataset df7: clustering, visualization, reconstruction, and ML-pipeline integration.



(a)

Objective: visualization — Top-ranked algorithm (baseline)

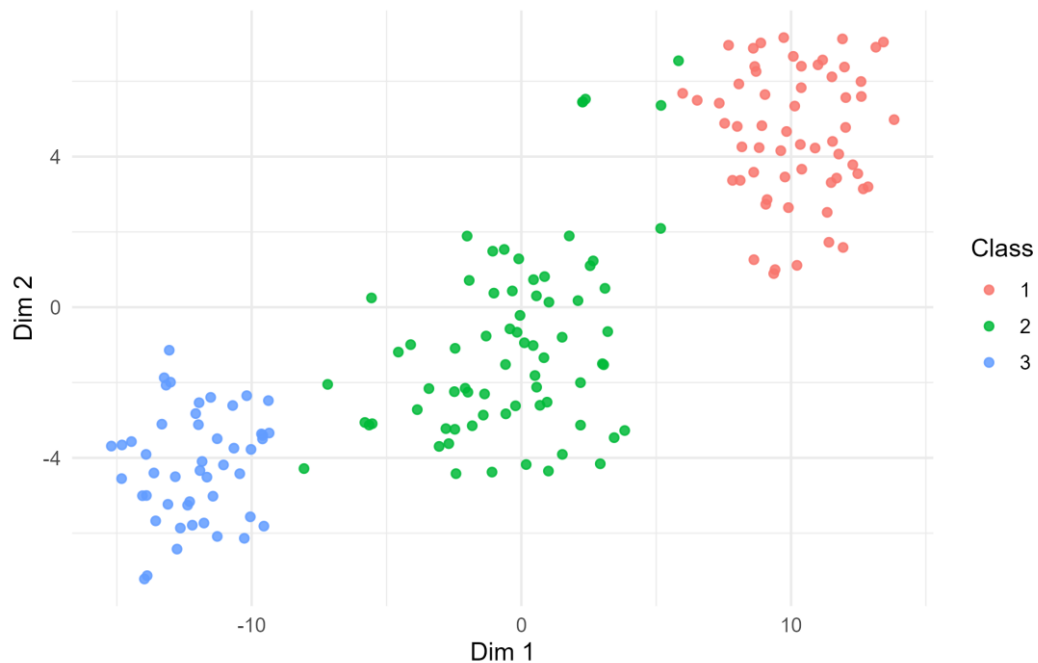
tSNE — Final Score = 0.9634



(b)

Objective: reconstruction — Top-ranked algorithm (baseline)

tSNE — Final Score = 0.9467



(c)

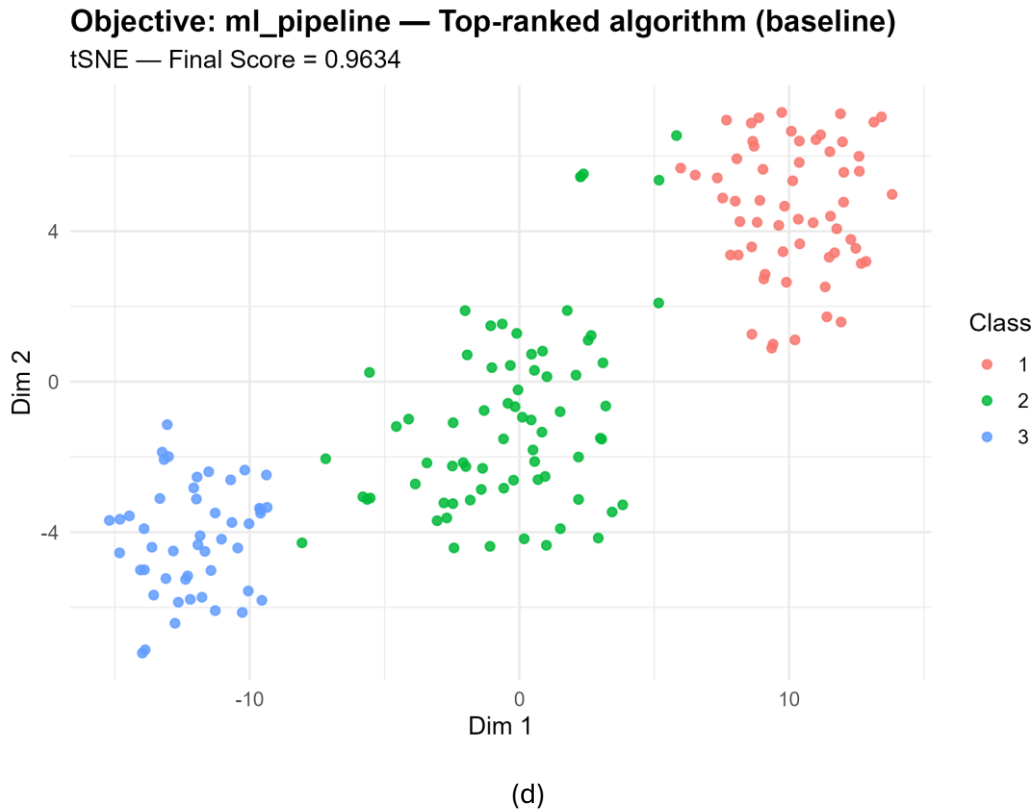
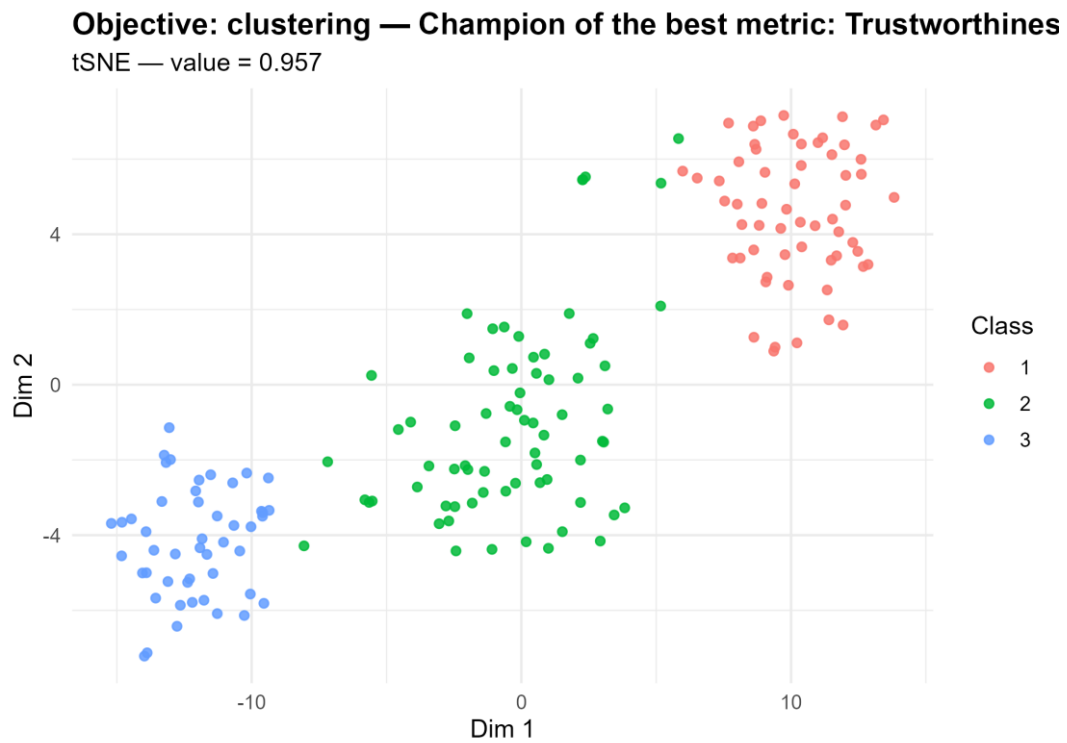


Figure 43 - Two-dimensional embeddings produced by the top-ranked algorithm for each objective, based on the Final Score (Baseline Ranking): (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

In all cases, t-SNE produces nearly identical geometric layouts, reflecting its strong stability and dominance across evaluation metrics for this dataset. The embeddings clearly separate the three classes into compact, well-defined clusters with minimal overlap, demonstrating t-SNE’s well-known ability to preserve local neighborhood structure and enhance class discriminability. This high degree of separation is especially notable given that the exact algorithm is selected as the optimal performer for reconstruction and ML-pipeline objectives—tasks that are traditionally more sensitive to distortions in global geometry. The consistent cluster arrangement across all objectives underscores t-SNE’s robust mapping of the intrinsic manifold structure into two dimensions. Overall, the figure highlights t-SNE as the most effective dimensionality-

reduction technique for Dataset df7, offering excellent visual separability, strong neighborhood preservation, and reliable performance across diverse analytical goals.

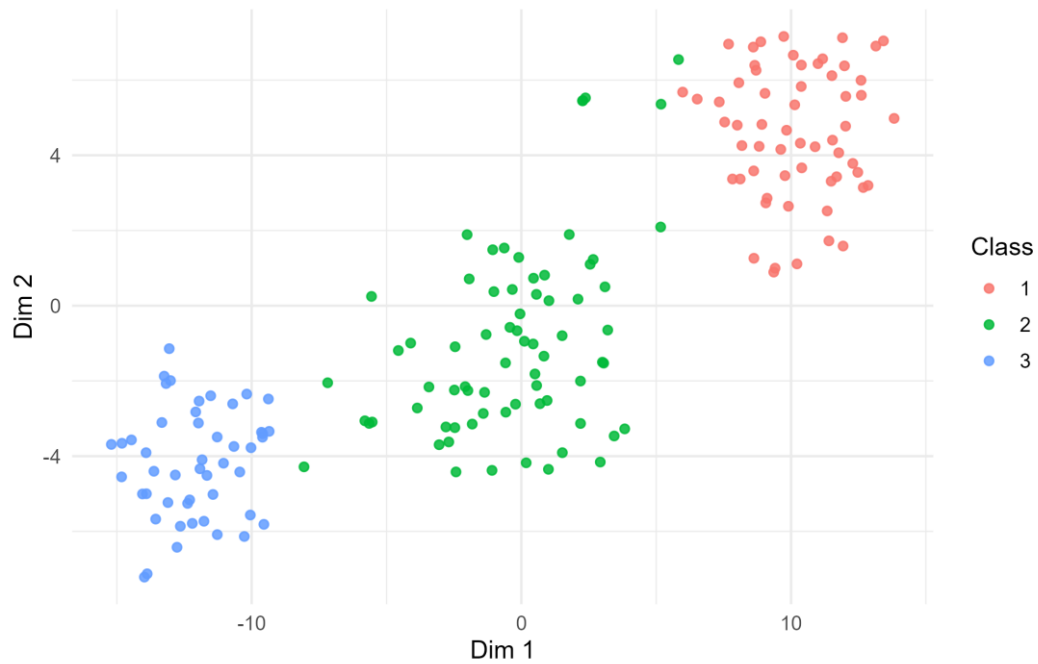
Figure 44 displays t-SNE embeddings identified as the top performers according to the Trustworthiness metric across four analytical objectives: clustering, visualization, reconstruction, and ML-pipeline integration.



(a)

Objective: visualization — Champion of the best metric: Trustworthiness

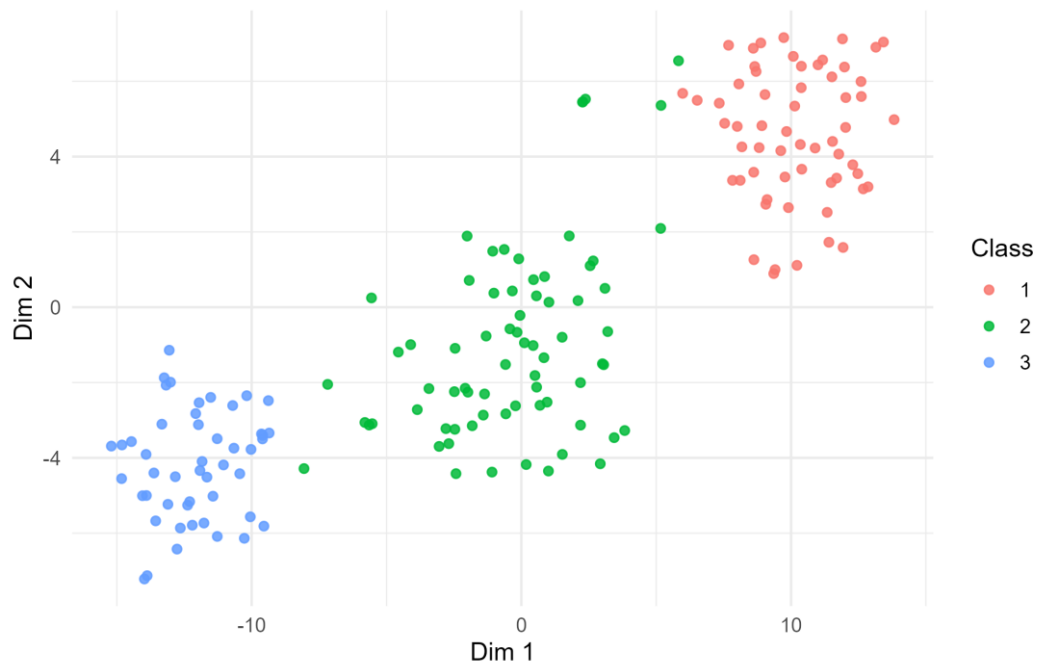
tSNE — value = 0.957



(b)

Objective: reconstruction — Champion of the best metric: Trustworthiness

tSNE — value = 0.957



(c)

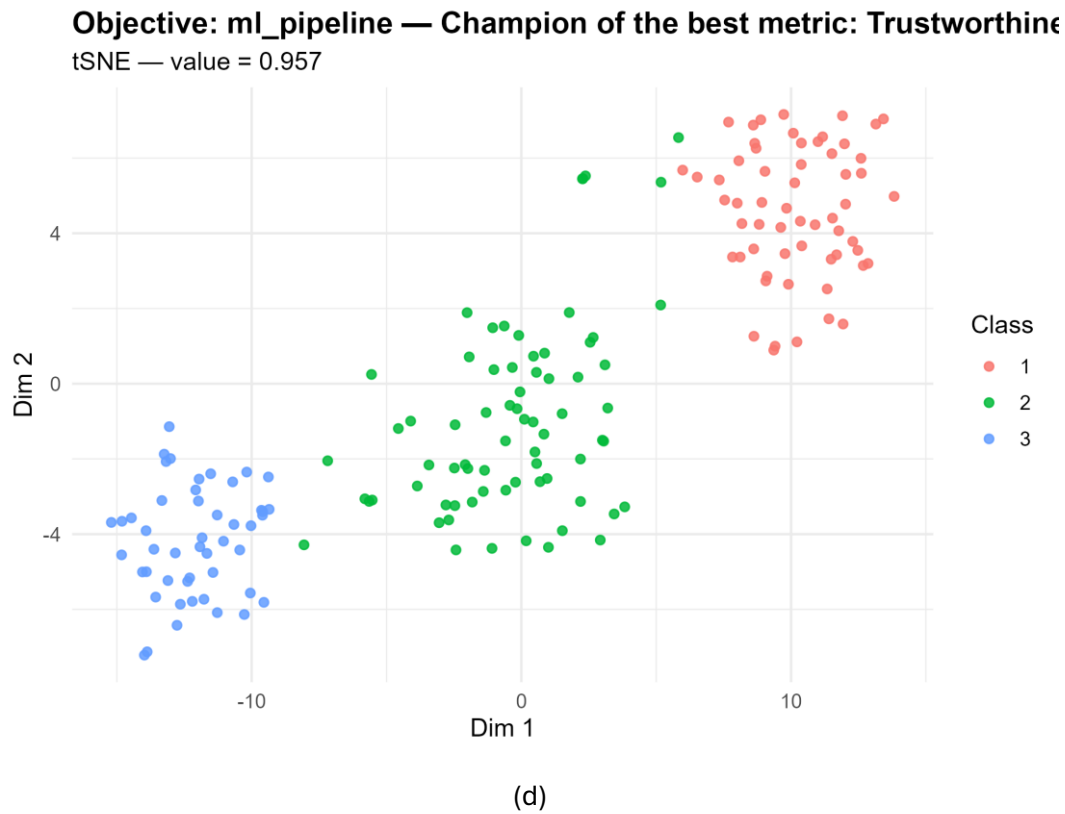


Figure 44 - Two-dimensional embeddings generated by the champion algorithm selected according to the best-performing metric for each analytical objective: (a) clustering, (b) visualization, (c) reconstruction, and (d) ML-pipeline integration.

Trustworthiness assesses the fidelity of local neighborhoods in the low-dimensional space, and the consistently high value (0.95684) observed across all panels reflects t-SNE’s exceptional ability to preserve local structure.

In each subplot, the three classes form highly compact, well-separated clusters with minimal overlap and clear inter-class boundaries. This strong discriminative organization illustrates the alignment between the Trustworthiness metric and t-SNE’s design, which emphasizes accurate preservation of nearest-neighbor relationships. The recurrent geometric pattern across all objectives further demonstrates t-SNE’s robustness and stability under metrics that prioritize locality. Despite potential distortions in global geometry—often characteristic of t-SNE—the embeddings remain highly effective for tasks focused on cluster identification, pattern recognition, and downstream

classification. Overall, the figure shows that when local neighborhood fidelity is the primary evaluation criterion, t-SNE provides consistently superior, visually interpretable embeddings for Dataset df7.

11. FINAL REMARKS

Benchmarking dimensionality reduction algorithms presents an inherent methodological challenge because different techniques prioritize distinct structural properties of the data. Some methods emphasize local neighborhood preservation, whereas others focus on maintaining global manifold geometry or minimizing reconstruction error. As a result, comparisons based on a single evaluation criterion often provide only a partial view of algorithm performance and may lead to misleading conclusions regarding the suitability of a given method for a specific analytical task. To overcome this limitation, the present study adopts a multimetric, scenario-driven benchmarking strategy that simultaneously evaluates dimensionality reduction techniques across complementary structural perspectives, including neighborhood fidelity, preservation of global structure, geometric distortion, and classification consistency. By integrating these criteria into a unified evaluation framework, the proposed approach enables a more comprehensive and objective assessment of algorithm behavior, thereby providing a robust basis for selecting dimensionality reduction methods in complex high-dimensional datasets.

12. DATA AVAILABILITY

Data can be accessed on GitHub: <https://github.com/Laboratorio-de-Tecnologia-Computacional/Dimensionality-Reduction-Techniques-An-Innovative-Approach>.

The R code used in this work can be accessed on GitHub: <https://github.com/Laboratorio-de-Tecnologia-Computacional/Dimensionality-Reduction-Techniques-An-Innovative-Approach>.

For almost all datasets, it is possible to download directly from the URL cited in the work. However, two datasets could not be downloaded directly, so they have been placed on GitHub for interested users to run the proposed code.

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DIMENSIONALITY REDUCTION TECHNIQUES

Design, Theory, and Applications

High-dimensional data are now routine across the physical, biological and engineering sciences, and dimensionality reduction has become a standard step in extracting interpretable structure from them. Yet no single technique is universally preferable: a projection that is excellent for visualising clusters may be poor at preserving global geometry or at supporting reconstruction.

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Aderval S. Luna

Full Professor at Rio de Janeiro State University (UERJ) and permanent member of its Graduate Program in Chemical Engineering.

Nathalia Alvim de Souza

D.Sc. student in the Graduate Program in Chemical Engineering at Rio de Janeiro State University (UERJ).

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DOI: 10.5281/zenodo.22221611

First edition · Rio de Janeiro, Brazil · 2026